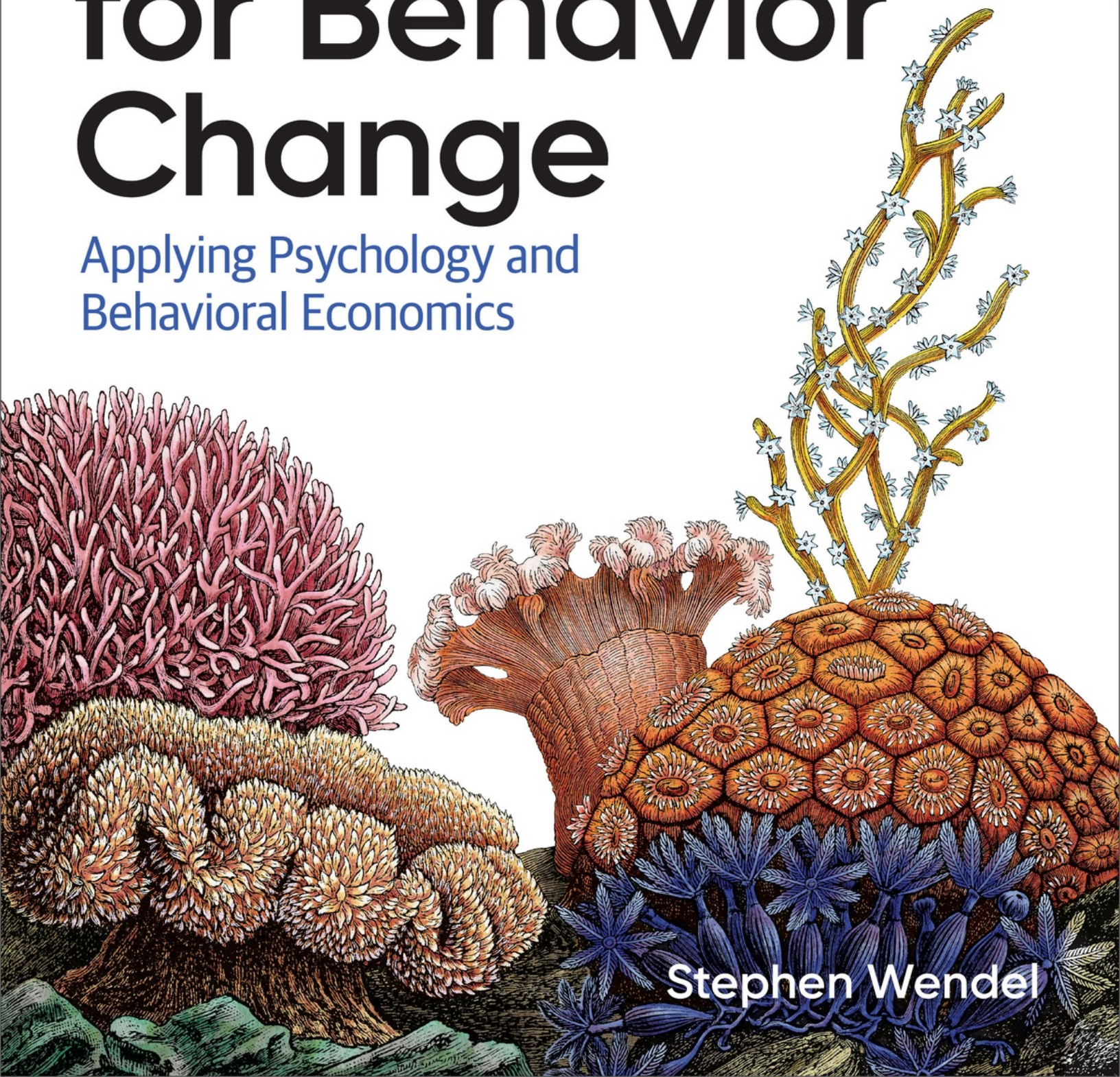


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Second
Edition

Designing for Behavior Change

Applying Psychology and
Behavioral Economics



Stephen Wendel

Designing for Behavior Change

SECOND EDITION

Applying Psychology and
Behavioral Economics

Stephen Wendel

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Designing for Behavior Change

by Stephen Wendel

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Preface

Eight years ago, when I started writing the first edition of this book, behavioral science consisted of a few prominent researchers and PhD students scattered across business schools and psych departments, along with some humorous books about the mistakes people make. Very few companies were trying to bridge the gap between academic theory and real products that helped real people in their lives. Personally, I had to scrounge to find companies where anyone was even familiar with the research, let alone formally trained in it. The idea of intentionally designing products to change user behavior was strange or even alarming to many people. There were pockets of innovative work—from the UK government’s Nudge Unit to BJ Fogg’s Persuasive Technology Lab to the behavioral consulting firm ideas42—but they were not well known outside of their communities.

That’s all changed now. In partnership with the Action Design Network and the Behavioral Science Policy Association—two nonprofit organizations dedicated to fostering the practical application of behavioral science, which didn’t exist eight years ago—we recently surveyed the landscape of applied behavioral science. Over two hundred teams, representing behavioral science teams with over 1,500 members, responded—and we know there are many more out there.

There are now dedicated teams applying behavioral science to develop new products, communications, and policies that serve their users across the world: from Qatar to Spokane, Washington. While Silicon Valley firms like Google and Uber are well represented, so are stalwart mainline companies like Walmart and Aetna, with dozens of small consulting shops scattered around the US, Europe, and beyond.

What are they doing? While each effort is unique, each of these groups is trying to develop products, communications, or policies that cause their users to do *something different* in their lives. In other words, they are designing for behavior change. Whereas traditional design is fundamentally

about solving a user need, behavioral design is about solving a user need where doing so entails, in a sense, “solving the user” too: changing the person in order to solve the problem.

And that’s where this book comes in. This is a guidebook on how to do it yourself: how to identify behavioral problems your users face, develop clever solutions to help them overcome those obstacles, iteratively learn from the process, build a team that does this, and make it successful in an organization. And, underneath it all, how to think about this work in an ethical, thoughtful way and avoid the serious mistakes and abuses that some practitioners are facing now—and that threaten our field as a whole.

Along the way, I’ll try to teach you the fundamental understanding of the mind that many behavioralists like me have: of a quirky, elegant, but necessarily imperfect decision-making process that guides your users’ decisions and actions. I’ll give you the core lessons and a framework to understand the research literature, but this book isn’t fundamentally about behavioral science theory (for me, Nobel Laureate Daniel Kahneman’s *Thinking, Fast and Slow* is still the best introductory book on behavioral science out there—I highly recommend it). Instead, our focus will be on *action*: what you can do, in your work, right now.

How This Book (and This New Edition) Came About

In 2019, O'Reilly asked if I could do a second edition of the book, updating it for tremendous growth in the field since it first came out. I was happy to do so—if only to celebrate that growth and help support its further advancement in my small way.

The ideas and process I discuss here come, foremost, from my own experience working in this field over the last 11 years: first at the personal finance company HelloWallet, and now at the investment research company Morningstar. It is enriched by countless conversations with other practitioners in the field: those with existing teams and those just seeking to enter the field. I'm particularly grateful to the Action Design Network, the nonprofit organization that I helped start in 2013, which has grown far beyond my dreams or certainly my intentions, with events all across North America on applied behavioral science. I will also be weaving in lessons from the survey I mentioned before, which we believe is the largest and most comprehensive survey of teams designing for behavior change ever conducted (though the way the field is growing, it won't be for long!).

Personally, I started doing applied behavioral science while completing my PhD and working at HelloWallet. HelloWallet began as many start-ups do: with a great deal of energy, a clear problem to solve, and a fundamentally misguided notion of how to solve it. We wanted to help everyday Americans improve their finances—and wrote an application that encouraged them to set up budgets and save for the future. There was just one little problem: everyone already knew how to do that. The problem people faced wasn't a lack of understanding; it was that they didn't act on it. There was a tremendous gap between people's intentions and their actions.

Thankfully, in addition to having a great (but misguided) idea, we had the ability to measure whether we were being successful. We looked at whether people used our app. They didn't. We also looked at whether it was helping the few people who used it to improve their finances. It wasn't. I'm

thankful for those hard lessons because it's better to know *early* when something isn't working and to be able to learn from that and do better—which will be a recurring theme throughout this book.

At HelloWallet, we set up an engine for behavioral experimentation: building scientific experiments into the underlying platform of the application. My motto was to make it “easier to test than to argue”—i.e., where product managers or designers disagreed about which version of a feature would more effectively change spending behavior, it was better to test both approaches in the field than to argue about which *might* be right. And by and large we succeeded—both in making it easy to run tests and in having the impact we sought.

We found that we could help people change their bank habits, to cut the money they lost to ATM and other banks fees by 25%; for lower-income families that equated to nearly a day's wages per month. We found that we could successfully nudge people to put aside money for savings by giving them a simple and easy-to-understand display of their finances, compared to that of their peers. And we also found that some of our efforts—like congratulating people on their budgeting success—actually backfired: they spent more! We quickly discontinued that feature. The first edition of *Designing for Behavior Change* was, in many ways, a description of what we were learning as we learned it. It detailed the process we were using in practice. In fact, the book arose out of an internal guidebook that I'd written for my teammates on applied behavioral science.

Since then, I've gained a great deal more experience with the challenges of designing products that help people change behavior. But, I've also verified that the fundamental approach was sound; the four-step process laid out in the book is what I still use today—though I've started using different names for the steps. I've modified the process around the edges, making it more efficient and flexible, but it is still tremendously useful in my own work. After the first edition, I also subsequently learned that many other teams had independently developed a similar iterative approach.

About five years ago, I moved to Morningstar (along with some of my teammates from HelloWallet) and was given the opportunity to set up a much larger behavioral science team. We work across the company, on issues ranging from effective investing to retirement saving and spending to challenges of internal decision making. We conduct applied research both with our in-house team and with leading behavioral scientists from academia.

This edition draws upon my more recent work at Morningstar, formal and informal consulting I've done with other organizations in our space, and especially work through the Action Design Network. The Action Design Network has grown from our initial monthly meetup in DC to a broad base of volunteers organizing events across 15 North American cities. Through it, I was able to learn from the challenges many other teams face, the approach they use, and their solutions.

In this new edition of the book, I've sought to bring together these diverse streams of insight to a coherent guidebook for practitioners to help you learn from the best of our knowledge in the field so you can design your own products to more effectively help users change their own behavior.

Who This Book Is For

As you can probably tell by now, this book is aimed at practitioners—the people who design and build products or communications with specific behavioral goals. Teams that design for behavior change should generally include the following roles, and individuals in each of these roles will find practical, how-to instructions in this book:

- Interaction designers, information architects, user researchers, human factors experts, human–computer interaction (HCI) practitioners, and other UX folks
- Product managers, product owners, and project managers
- Marketing and communications professionals
- Behavioral scientists (including behavioral economists, psychologists, and judgment and decision-making experts) interested in products and communications that apply the research literature

The person doing the work of designing for behavior change could be any one of these people. At Morningstar, we have each of these roles, but most of my team is composed of behavioral researchers. This work can be, and often is, done wonderfully by UX folks. They are closest to the look and feel of the product and have its success directly in their hands. This approach enriches their current practice by adding an extra theoretical layer to design hypotheses and tests.

Product owners and managers are also well positioned to seamlessly integrate the skills of designing for behavior change to make their products effective. Finally, there are other behavioral scientists (like me) working in applied product development and consulting at organizations like ideas42 and the Center for Advanced Hindsight. So, the people designing for behavior change probably wear other hats as well.

In addition, this book is for entrepreneurs and managers. If you've ever read *Nudge*, *Blink*, or *Predictably Irrational*,¹ and wondered how you could

apply them to your own product and users, read on. While the book is about helping users take action in their lives, that doesn't mean that designing for behavior change is incompatible with a for-profit business model.

Businesses make a profit; that's how they exist. So, you'll find suggestions for building a successful business model on voluntary behavior change. If in addition to making a profit, you are helping your users take action and change their behavior, this book can help you do it.

Nonprofit entities and some government agencies often explicitly focus on helping users change their behavior; *Designing for Behavior Change* can help. For example, the UK's **Behavioural Insights Team** is widely applying behavioral research to public policy and services around the world. Where relevant, I'll note parts that are particularly important for nongovernmental organizations (NGOs) and government agencies. Because it's more compact to write, I'll refer primarily to "companies" here. In almost all cases, I really mean companies, organizations, and relevant government agencies.

Finally, my expertise is software development, so I'll use the terminology that I use in my day-to-day life—applications, software, and programs. You don't need to be in software development to find this relevant to you. In fact, some of the most innovative work in persuasive design, one of the fields that this book draws inspiration from, is in the design of everyday objects.² As you apply *Designing for Behavior Change* to your work, whether in software or beyond, I'd love to talk with you and share notes! You can reach me at steve@behavioraltechnology.co.

Combining Research, Data, and Product Expertise

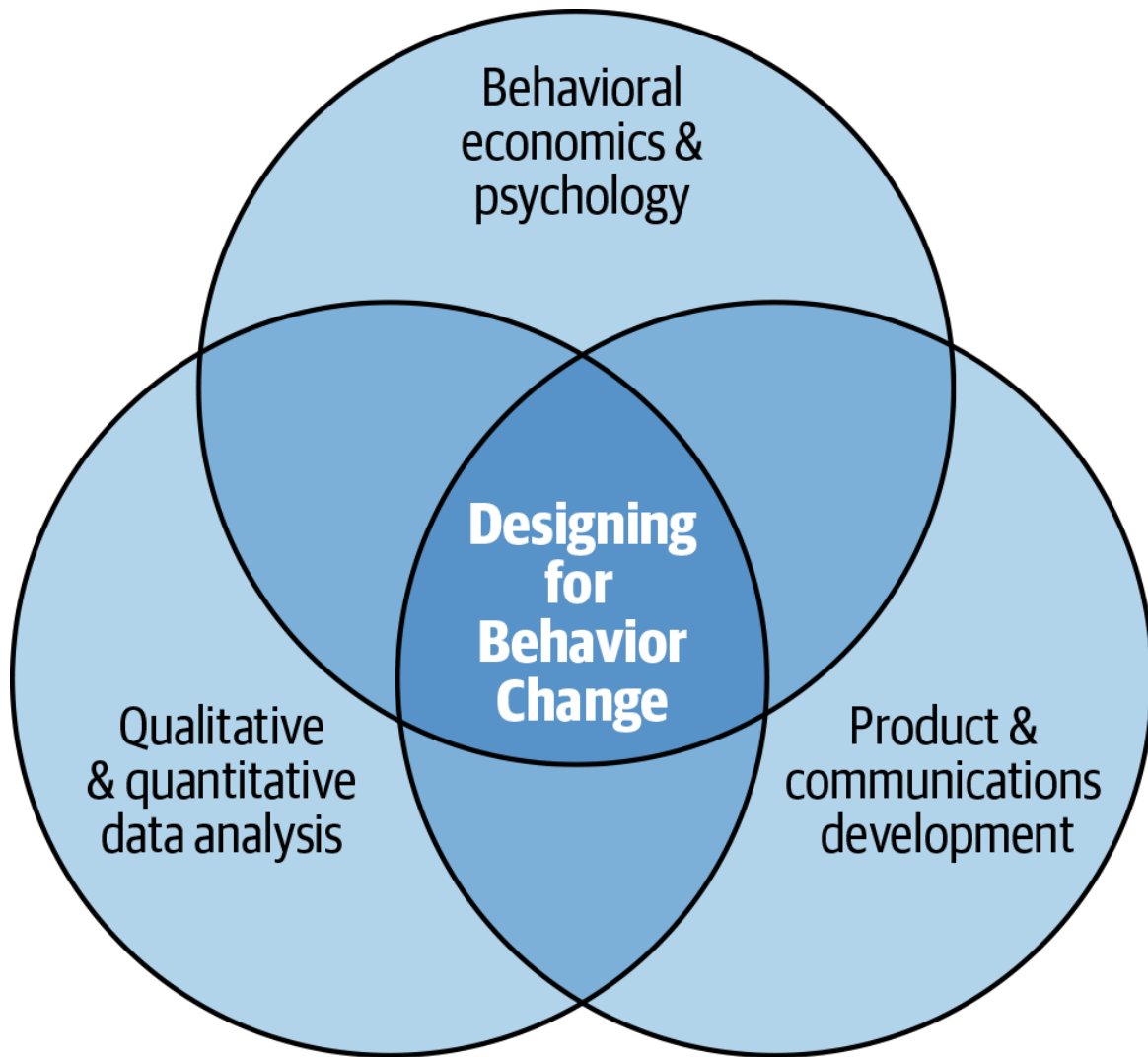
One of the book's recurring themes is that understanding how the mind works is not enough to build behaviorally effective products.

In addition to behavioral science research, we need two sets of skills to support the process. First, we need to plan for data analysis (both qualitative and quantitative) and for refinement and iteration based on that data. That means adding metrics to the application and conducting user research to

understand individual behavior, analyzing the data, and making improvements over time based on it.

Second, we need to build products that people actually enjoy using. I know it sounds obvious, but it's something that's often forgotten as we build products designed to educate, motivate, or otherwise help our users. We tend to focus on the behavior change (and how important it is) and forget the fact that people still have to *choose* to use our products. Users avoid boring, frustrating, ugly applications, so we should remember the lessons of good product design, from identifying user needs and frustrations to designing an intuitive, beautiful user interface.

When you bring these raw ingredients together—behavioral research, product design or marketing expertise, and data analysis—you have what's needed to design for behavior change:



What You Need to Know to Benefit from This Book

This book gives you enough knowledge in each of these three areas to get oriented and to start working on concrete products and communications. It covers most of the behavioral research you'll need to finish the products as well, but at some point along the way, you'll need people who are experts in qualitative or quantitative data and in product design. [Chapter 17](#) provides a detailed look at the skills required for a team that designs for behavior change, including where you can find (or develop) them.

If you *are* an expert in one of these areas, all the better. The book will show you how designing for behavior change builds upon and complements your existing expertise. You'll find out how to leverage your existing skills to

play a leading role in the development of behaviorally effective products and communications within your organization.

What Types of Behaviors This Can Help With

The techniques I'll talk about here assume that the product will support an action that people aspire to but have had difficulty undertaking. Learning a language. Sticking to a diet. Meeting new people. This may seem like it applies only to a narrow set of products, but I've found that there are two big groups of behaviors that fit these criteria:

- Behaviors that users want to change *within their daily lives*
- Behaviors *within the product itself* that are part of using the product effectively

Or, from the perspective of the company making the product, behavior change is either:

- The core value of the product for users
- Required for users to extract the value of the product

In the first case, users have some behavioral problem in their daily lives and buy the product to help them with it. In the second case, users have some other need that the product solves, but they must adapt and change their behavior in order for the product to deliver on its promise.

The first case includes:

- Controlling diabetes
- Paying off credit card debts
- Getting back in shape
- Getting involved in their communities

Often these behaviors relate to big-picture social issues like health and wellness. When we design products that support these behaviors, we help the individual and impact our society at the same time. Oracle Utilities's Opower and Google's Nest, for example, are products that help individuals decrease individual energy usage: saving people money and helping the environment at the same time. Other products that change behavior in this way are Fitbit (exercise) and Weight Watchers (diet).

As I send this book off for final production, the coronavirus COVID-19 is spreading across the globe. We've seen a rapid mobilization in the healthcare community, including work by behavioral scientists to help promote social distancing and handwashing behavior.³ Researchers are mounting large-scale, rapid turnaround studies to test techniques to keep people safe—by applying behavioral science to the design of communications and products.⁴ It's an impressive display of the power of behavioral science to help people take action that's in the best interest of the individual and society as a whole.⁵

The second type of behavioral change is far more mundane. Individuals often seek to change behavior as a means to an end. Let's say a user wants to learn a new language and gets a software package to help do it. Simply learning how to effectively use the software can take some substantial changes in behavior *within the product*—building new habits to log in daily and practice the language, overcoming fears about looking foolish while doing it, and so on. The user wants to take an action (learning the language) but struggles. A well-designed product can help the user make those personal adjustments.

This second type, and the products that require it, is much broader than the first. It covers the sweep of voluntary changes in behavior that people might make to benefit from products they've already chosen to use. It touches upon a huge swath of the consumer product space, from Yelp to Square to Rosetta Stone. Some examples of actions that occur *within software products* that one might try to improve include:

- Organizing email contacts

- Drawing decent flowcharts
- Formatting documents

As with many other behavioral scientists and designers, I believe that no design is neutral.⁶ Anything we design that interacts with other people—communications, products, services, etc.—has an impact on their behavior and ultimately their lives and outcomes. Here, we talk about how to make that process intentional and, hopefully, beneficial.

For both types of behavior change, the goal is to develop products that help users take action and to deliver the value that the company offers. This voluntary, transparent support for behavior change helps companies be successful as well.

What This Book Is Not About

If you're looking for a book on how to make people do something *you* want them to do (even if they don't want to do it), you're in the wrong place. I'm not necessarily judging your motives or aims; I just can't help you much.

In particular, this book isn't designed to help with persuasion. There are many ethical and thoughtful uses of persuasion—and we all seek to persuade each other in our daily lives. While there are many similarities with designing for *voluntary* behavior change, other issues arise such as educating users about a product, building a convincing argument, rhetorical delivery, building rapport over time, and more. We won't cover them here. The topic of voluntary (i.e., pre-persuasion) behavior change is big enough as it is!

In addition, this book isn't intended to help with trickery or coercion (for both practical and ethical reasons). Sadly, there are many companies trying to do just that, though—and it's dangerous for our field. I'll delve into the details of what's happening across our industry and how we're starting to face well-deserved backlash from both government regulators and thoughtful technologists in [Chapter 4](#).

The Chapters Ahead

In the following pages, I walk you through each of the skills you need to design for behavior change, starting with a firm foundation in how the mind makes decisions. Then, I show each step that's required to develop a new product: moving from discovery to design to implementation and refinement. I introduce each concept where it is first needed. In [Part III](#), I step back and give some additional information on the scope of the industry (and where the jobs are), how to build a behavioral team at your organization, and likely problems you'll face along the way. That's it.

However, if you're looking for a more formal chapter outline, here you go:

[Part I: How the Mind Works](#)

Chapter 1 arms you with your first skill: an understanding of how the mind makes decisions. You'll get an overview of the current literature on decision making, as well as a dozen key lessons and their implications for the design process.

Chapter 2 then describes the six high-level factors that must come together at the same time for a person to take action. They form the CREATE Action Funnel—Cue, Reaction, Evaluation, Ability, Timing, and Experience—which shows you what needs to be addressed in your product and where users usually drop off.

Chapter 3 looks at the reverse problem: how to help your users stop unwanted habits or improve poor decisions.

Chapter 4 delves into detail about the ethical challenge facing the field: how too many practitioners have used these techniques to manipulate users into buying (or overusing) their products, how we're fooling ourselves if we think we wouldn't do the same under the right (or wrong) conditions, and how we can tackle these problems head on.

Part II: A Blueprint for Behavior Change

Chapter 5 introduces you to the larger process of designing for behavior change—how your new knowledge about the mind can be used in practice—using the acronym DECIDE: Define the problem, Explore the context, Craft the intervention, Implement within the product, Determine its impact, and Evaluate next steps.

Chapter 6 starts charting your course by defining the problem: the overall outcomes you hope the product will deliver and who it seeks to help. From there, it demonstrates how to elicit a potential idea for behavior change—how, specifically, your users will get fit or take control of their finances, for example.

Chapter 7 explores the context in which your users will act with a behavioral map: a narrative of how the product team envisions user interaction with the product and how users will change their behavior. It then evaluates the various behaviors they *could* change in light of their

needs, interests, and prior experience. It refines your initial plan to finalize the specific behavior the product will support.

Chapter 8 introduces you to the meat of the DECIDE process: designing the intervention itself. We look, at a high level, at the main strategies you can use to help users change behavior, using the story of a fish stranded on the beach.

In Chapters **9** and **10**, we look at how to craft specific interventions, based on behavioral research, to support your users to take action.

Chapter 9 presents interventions that are appropriate when users face problems of Cue, Reaction, or Evaluation; next, **Chapter 10** looks at interventions for Ability, Timing, and Experience.

Chapter 11 wraps up the discussion of crafting an intervention with two extensions: how you can handle multi-step complex interventions and how to help the user hinder negative unwanted actions.

Chapter 12 presents some tips for implementing the intervention within the product itself. Applied behavioral science doesn't require a particular development approach or technology; it does, however, require high-quality data on outcomes. We discuss how to build those metrics, and the ability to measure changes in them, in the initial product development—rather than as a hacked-up afterthought.

Chapter 13 focuses on determining the product's impact, its success or failure. It starts with the most powerful tool there is for impact measurement: the humble A/B test. Since most readers are likely familiar with A/B tests already, this chapter dives into the details of how to use it effectively to gather rigorous data and the common pitfalls that practitioners face.

Chapter 14 looks at other ways to determine impact, when A/B tests or other randomized control trials aren't available. I cover how to utilize statistical models to gain insight. I talk about the challenges of gauging the causal impact of software on real-world behavior and how to overcome them.

Chapter 15 concludes the DECIDE blueprint for behavioral design by helping you Evaluate next steps after your implementation. It looks at how to find problems that limit impact, including how qualitative and quantitative analyses are both needed, and work hand in hand.

Part III: Build Your Team and Make It Successful

Chapter 16 provides the results from the Behavioral Teams survey in detail. Case studies are scattered throughout the book, but here we dig into the numbers: how large the field is, where the jobs are, and what challenges and successes other teams are having.

Chapter 17 looks at what it takes to start applying behavioral science at your organization with a team of 1 or a team of 20. We look at how to make the case to stakeholders in-house and the skillset you or other team members will need.

Finally, **Chapter 18** wraps things up with a quick review of the designing for behavior change process and key takeaways on how to make it happen in your organization. It also covers many of the questions that can arise when putting these lessons into practice.

At the end of the book, there's information for those who are looking to dive even deeper:

- **Glossary of Terms:** a glossary of key terms, like *behavioral map* and *data bridge*
- **Bibliography:** a comprehensive list of the works cited in this book

In the first edition of this book, there was a list of online resources (Appendix B). We've moved that **online**. Each of the core chapters in **Part I** ends with "A Short Summary of the Ideas" if you only have a few minutes. They are a useful wrap-up that also give advice if you're just looking for an informal process to sketch things out before you jump in head first.

The core chapters of **Part II** wrap up with "Putting It into Practice"—sections with concrete examples and exercises to help you put each

chapter's ideas to work.

Let's Talk

My hope with this book is to further the conversation about voluntary behavior change and help build up the tools needed to develop behaviorally effective products. However, even after writing two editions (and likely even after 20) I have no illusions about the completeness of this work—there's still a tremendous amount to figure out. We're all going to learn as we go along.

Personally, I'm always looking to learn, share, and collaborate, so don't hesitate to reach out if you have a cool story to share, an idea for a research project that would further develop the field, or an idea for a behavior-changing project that you'd like to bounce off someone. You can find me under *sawendel* on Twitter, LinkedIn, and AngelList; my contact information is on my website, <http://about.me/sawendel>.

If you think there's something that can be improved in this book or find something that is inaccurate, please reach out to me and tell me about it. One of the many benefits of working with O'Reilly Media as a publisher is that a lot of you will be reading this book in an electronic format and can quickly get an updated version of the book if corrections need to be made. For those who are reading this in paper form, I'll keep a list of corrections, additions, and other updates online at behavioraltechnology.co.

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Since the first publication of *Designing for Behavior Change*, I've continued to build upon and refine the theoretical and practical tools presented in the book. The tools that I use with my own team, that I've trained other companies and organizations with, are extensions of those initial ideas (and indeed, the first edition itself was a description of our practice and approach at HelloWallet at the time). In this edition, I've brought together a number of those artifacts and lessons from applying this approach over that last six years. In particular:

- The exercises in the “Putting It in Practice” sections of Chapters 6–14 incorporate parts of a **workbook** I developed for my team at HelloWallet to apply these concepts.
- **Chapter 13** incorporates sections from a guide to experiments I wrote for my team at Morningstar.
- Chapters 1, 3, and 4 incorporate part of an introduction to behavioral science that I wrote for a book exploring the applications of behavioral science to one's personal and spiritual life.
- **Chapter 16** describes a survey I conducted in 2019 with the nonprofit organizations the Behavioral Science Policy Association and Action Design Network, and have published separately on our websites.

In each case, the source material was used as a starting point, often with significant adaptations and changes. All materials are used with permission.

Acknowledgments

With the first edition of this book, I had many people to thank for their help in developing the ideas and process presented within. The list has only grown.

I continue to talk with and gain inspiration from people I started working with on the first edition—from Rob Pinkerton and Katy Milkman to the Action Design Team, especially Zarak Khan, Matthew Ray and Erik Johnson. Here, I'd like to call attention to additional people I've learned from since that first edition. Top of the list would be my intellectual sparring partner, Ryan Murphy, and my team at Morningstar: Heymika Bhatia, Sarwari Das, Jatin Jain, Sam Lamas, Sagneet Kaur, Alistair Murray, Sarah Newcomb, Shwetabh Sameer, Stan Treger, and Leon Zeng. Many thanks to Ray Sin as well, for your thoughtful research. Outside of Morningstar, I'm particularly grateful to Paul Adams, Julián Arango, Florent Buisson, May C., Jesse Dashefsky, Clay Delk, Barbara Doulas, Darrin Henein, Fumi Honda, Peter Hovard, Anne-Marie Léger, Jens Oliver Meiert, Brian Merlob, Shafi Rehman, Neela Saldanha, Nelson Taruc, and Mark Wyner for their comments on the draft.

In terms of the rich stream of behavioral research that I draw upon here, there are simply too many people than can be thanked. Even though it is not the current fashion to do so in popular-press books, I make a point to cite authors of the specific studies I reference throughout the body of the book, though I don't have space to offer a full literature review. For some readers, this may seem overly academic or, frankly, boring. That's certainly not my goal! Rather, it is the simple acknowledgment of the many thoughtful researchers who have done great work and that both I and you, as readers of this book, benefit from. I want never to overstate my ingenuity or to dim the glow of their bright light. My primary error, I fear, is not giving enough credit to the amazing work of other researchers. Please accept my apologies for any citations that are missing.

And as always, I'd like to thank Alexia, Luke, and Mark—for putting up with me and my obsession with writing and for loving me nevertheless.

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- 1 Thaler and Sunstein (2008), Gladwell (2005), Ariely (2008).
 - 2 Dan Lockton has [a set of papers](#) that provide an extensive review of the various domains in which intentional behavior change has been applied.
 - 3 See [recommendations to the Irish government](#), for example.
 - 4 [Kwon \(2020\)](#); [Jachimowicz \(2020\)](#)
 - 5 We've also seen what happens when incomplete *lessons* from behavioral science are divorced from the *process* of science (including rigorous testing and empirical validation)—in the furor over the UK government's initial resistance to strict social distancing based on the concept of behavioral fatigue. See [Yates \(2020\)](#) for the political debate, and [UK Behavioural Scientists \(2020\)](#) for a thoughtful response calling for proper scientific testing of the concept before using it in government policy with lives at stake.
 - 6 In *Nudge*, Thaler and Sunstein make this case, for example, and a discussion of the unintentional and intentional impact of design has long been a part of the design community (e.g., Nusca 2019, Vinh 2018).

Part I. How the Mind Works

Chapter 1. Deciding and Taking Action

On the day I got married, I was lying on the bathroom floor of the church because my back hurt to move. I'd been out of commission and in bed for almost three weeks, but now my family and my wife-to-be, Alexia, were waiting for me in the aisles. My best man Paul had to pull me up from the floor and get me out there to say my vows. My back had seized up weeks before because I hadn't been getting enough exercise.

Now I was born skinny, but that hides the fact that I've had musculoskeletal problems all my life—lower back problems, pinched nerves in my hands and neck, and so forth. I've seen many doctors over the years, and they've all said about the same thing: you'll be OK, if you just exercise regularly.

So I've long known about the importance of exercise; I don't have a problem with motivation. There's nothing more motivating or scarier than almost canceling your own wedding. I've certainly intended to exercise. But, like many other people who struggle with this, I haven't done as much as I should.

For me, and for many others, there's a gap between our sincere intention to act and what actually happens. There's something more going on in our heads and in our lives than a simple cost–benefit analysis about what we should do. Even though the benefits clearly outweigh the costs, we struggle. To change this pattern—to help ourselves and others take action when needed—we must first understand how our minds are wired.

In my research and that of many other behavioral scientists in the field, we've found that people don't always make decisions and take action in a straightforward way. People struggle to turn their intentions into action. People struggle sometimes to make good decisions—even if, at another time, they might have done fine.

We recognize this for ourselves and our own lives, but we tend to forget this when it comes to our users. We assume that if they like our products, they'll use them. If they want to do something, they'll figure out how. But they don't.

I'm not the only person who struggles with a lack of exercise. Many of your users might too. Or, they struggle with poor eating or bad sleep habits or distractions that keep them from their family and friends. Often, motivation isn't the problem: like me, they know what they should do and even want to do it. Other things get in their way. This book is about how to help your users, and all of us, change behavior when we need to.

Behavior Change...

All around us, people try to change our and each other's behavior. Negative examples are often obvious: from ads that entice us to buy stuff we don't need to apps that try to swallow our attention and time. Positive examples are there too, but perhaps not as obvious; for example, the loving parent teaching a child to share. Support programs helping addicts break free from their demons. Apps helping us track our weight and encouraging us as we diet and exercise.

In a sense, we're all in the "behavior change" business. When we're falling short of our own goals and want to make a change in our lives, that usually means our behavior must change first. Moreover, we're a social species; in order to achieve our goals, even altruistic goals of helping another person succeed, often someone must do something differently. To effect change *is* to effect behavior change.

Yet we rarely talk about it that way. In the product world, we talk about features delivered, user needs met, and so forth. Those things are all important, certainly, but none of them matters unless people adopt and apply our products (i.e., we need our users to change their behavior in a meaningful way).

Perhaps we don't talk about behavior change so directly because it's uncomfortable: we don't want to be seen as manipulative or coercive. So we end up with sanitized conversations distanced from real people changing their behavior because of our products and communications: key performance indicators for adoption and retention; objectives and key results for click-through rates and usage.

It shouldn't be that way. When we don't talk about what we're actually doing, we are both less effective at helping others when we should and more likely to try to change behavior in ways we shouldn't. This book is about designing products intentionally to change behavior—how to ethically and thoughtfully help others succeed, without, I hope, falling into trickery or manipulation.

In this book we'll have an open discussion about how to help people decide what to do and how to help them act on their intentions and personal goals. We'll talk about how to build products that influence that process and how to assemble and run a team that does so. Nothing presented here is perfect, but I hope this book can help you make better products and better serve your users.

...And Behavioral Science

One of the best toolsets to accomplish this task—intentionally designing for behavior change—comes from behavioral science. And, in addition to being useful, behavioral science is fascinating.

Behavioral science is an interdisciplinary field that combines psychology and economics, among other disciplines, to gain a more nuanced understanding of how people make decisions and translate those decisions into action. Behavioral scientists have studied a wide range of behaviors, from saving for retirement to exercising.¹ Along the way, they've found ingenious ways to help people take action when they would otherwise procrastinate or struggle to follow through.

One of the most active areas of research in behavioral science is how our environment affects our choices and behavior, and how a change in that environment can then affect those choices and behaviors. Environments can be thoughtfully and carefully designed to help us become more aware of our choices, shape our decisions for good or for ill, and spur us to take action once we've made a choice. We call that process *choice architecture*, or *behavioral design*.

Over the past decade, there has been a tremendous growth of research in the field and also of best-selling books that share its lessons, including Richard Thaler and Cass Sunstein's *Nudge*, Daniel Kahneman's *Thinking, Fast and Slow*, and Dan Ariely's *Predictably Irrational*.² They give fun introductions to the field, including anecdotes of how:

- Putting a picture of a fly in the center of a men's urinal can help reduce the mess that men make far more than exhorting them not to make a mess.
- Giving people many small bags of popcorn makes them eat less of it.³

In fact, Thaler and Kahneman have each won the Nobel Prize largely because of their work in behavioral science.

That said, we're not trying to re-create *Nudge* or *Predictably Irrational* here. This book is about *how to apply lessons from behavioral science to product development*; in particular, how to help our users do something they want to do, but struggle with. Whether that's dieting, spending time with their kids, or breaking a social media app's hold on their lives. It's about arming you with a straightforward process to design for behavior change.

Some of those lessons are what you'd expect: when designing a product, look out for unnecessary frictions or for areas where a user loses self-confidence. Build habits via repeated action in a consistent context. Some of those lessons are far less expected, and you may not even want to hear them; for example, most products, most of the time, will have no unique impact on their user's lives. For that reason, we need to test early and often,

and use rigorous tools to do so. Other lessons are simply fun and surprising; for example, make text *harder* to read if it's important that users make a careful and deliberative decision.

With that, let's dive into a primer on behavioral science!

BEHAVIORAL SCIENCE AND DESIGN

In addition to research in behavioral science, there are wonderful tools in the design community—from user-centered design to design thinking—that can help us intentionally design for behavior change. In fact, in many places where I discuss lessons from behavioral science in this book, you could readily substitute in terminology from the design community (and vice versa). People are people, and user-focused design techniques and behavioral science seek to understand them on their own terms. It's for that reason that many designers now study psychology and behavioral science in particular as part of their training and that many behavioral scientists, like myself, seek out and learn from lessons in the design world.

Throughout this book, we'll honor the existing expertise and skillset in the design community, and offer unique tools and techniques that aren't part of that discipline yet.

Behavioral science has a particular understanding of the mind that I think enriches the discussion and goes beyond current practice in design. It also has a commitment to experimental testing that is invaluable. So, throughout this book, I'll talk about many lessons and techniques that are common to the two fields or unique to design, but my focus will be on the areas that are less covered and more unique from behavioral science. As you'll see, behavioral science and design overlap and complement each other in many wonderful ways.

Behavioral Science 101: How Our Minds Are Wired

Last summer, my family and I were on vacation and having a great time. One afternoon, we decided we'd eaten out way too much and we wanted something cheaper and more familiar than another restaurant meal. So we went to a grocery store.

Now, the first thing we looked for was cereal. We found the aisle and there were far too many options to choose from. As they often do, our kids were running up and down the aisle, pulling and swinging each other around. Somehow, all of that movement makes them unable to hear us telling them to stop. It's clearly loads of fun—until they crash into something. So we had to make a quick decision.

Unfortunately, my kids and I have lots of allergies. My allergies are lethal, and my kid's allergies cause pain but thankfully not too much more. So as we're standing in the aisle trying to make a choice and keep our kids out of trouble, my wife and I were torn: we simply couldn't read all of the boxes for their ingredients.

Thankfully, we have some simple rules we know to follow. Any cereal with cartoons on the box is automatically out; those are often crammed full of sugar, and our kids have enough energy already. Second, cereals that are gluten free (which one of our sons needs) usually proclaim it proudly on the box—easy to scan for. And third, after decades of practice, I have a really useful habit: I automatically pick up food and recognize ingredients on the list that would kill me. It only takes a split second and I hardly think about it unless I see something that's a problem.

After a little while, we picked up a nice bag of corn flakes, grabbed a box of some granola-like stuff, and went on to the next aisle. No problem.

Unfortunately, we did forget to grab milk and a few other things in that aisle. We'd intended to get them, but in the moment, we missed those items on our mental checklist.

Now, when we got home, the granola stuff was actually really good. The corn flakes were terrible—in all of the hurry, we missed a key sign: dust on the bag. They'd been sitting there a long time, and everyone else clearly knew not to buy them.

In everyday life and in (true) stories like this one, we can find the core lessons of behavioral science if we know where to look. I like to start with a basic, and often overlooked, one: as human beings, we're all limited. We can't instantly know which cereal is best just by thinking about them. We have to take time and energy to sort through the options and make a decision. That time and energy is scarce—if we spend too much time on one thing, there's a cost (like our kids crashing to the shelves). Similarly, we're limited in our attention, our willpower, our ability to do math in our heads, and so on. You get the picture.

Our limitations aren't *bad*, per se; they just are facts of life. For example, I can't even imagine what it would mean to have *unlimited attention*—to be simultaneously aware of absolutely everything at once. That's just not how we're made.

Given these limitations, our minds are really good at making the best of what we have. We *economize* on our time, attention, and mental energy by using simple rules of thumb to make decisions; for example, by excluding cereals with cartoons. As researchers, we call these results of thumb *heuristics*. Another way our minds economize is by making split-second *nonconscious* judgments; for example, *nonconscious habits* are automated associations in our heads that trigger us to take a particular action when we see a specific trigger (like scanning for deadly ingredients whenever I see unknown food). Habits free up our conscious minds to think about other things.

While these economizing techniques are truly impressive, they aren't perfect. They falter in two big ways. First, we don't always make the right decision; for example, sometimes we don't pay attention to something important (dust on the bag). As researchers, we often call the results of a heuristic or other shortcut going awry a *cognitive bias*: a systematic

difference between how we'd expect people to behave in a certain circumstance and what they actually do.⁴ Second, even when we make the right choice, our inherent human limitations mean we don't always follow through on our intentions (getting the milk). We call that the *intention–action gap*.

And finally, context matters immensely. It mattered that our kids were running around; we had less of our limited attention to pay to the task (reading ingredients, remembering milk). If milk were in a different aisle, we might have seen it and remembered it. If our kids weren't running around...never mind. That wouldn't happen.

So, if I were to put decades of behavioral research into a few bullet points (please forgive me, my fellow researchers!), it would be these:

- *We're limited* beings in attention, time, willpower, etc.
- *We're of two minds*: our actions depend on both conscious thought and nonconscious reactions, like habits.
- In both cases, our minds *use shortcuts* to economize and make quick decisions because of these limitations.
- Our decisions and our behavior are *deeply affected by the context* we're in, worsening or ameliorating our biases and our intention–action gap.
- One can *cleverly and thoughtfully design a context* to improve people's decision making and lessen the intention–action gap.

Let's look at each of these points in a bit more detail.

We're Limited

Who hasn't forgotten something at some point in their lives? Heck, who hasn't forgotten something in the last hour, or the last five minutes? Forgetfulness is one of our many human frailties. Personally, the older I get, the longer that list seems to grow. There are sadly many ways in which our

minds are limited and lead us to make choices that aren't the best, including limited attention, cognitive capacity, and memories.

These limitations string together. In terms of our *attention*, there are nearly an infinite number of things we could be paying attention to at any moment. We could be paying attention to the sound of our own heartbeat, the person who is trying to speak to us, the interesting conversation someone else is having near us, or the report that's overdue and we need to complete.

Unfortunately, researchers have shown again and again that our conscious minds can really pay proper attention to only one thing at a time. Despite all of the discussion in the popular media about multitasking, multitasking is a myth.⁵ Certainly we can *switch* our attention back and forth; we can move from focusing on one thing to focusing on another—and we can do so again and again and again. But the reality is, switching focus frequently is costly; it slows us down, and it makes it harder for us to think clearly. Given that we can only focus on one thing at a time and that there are so many things that we could focus on (many of them urgent and interesting), it's no wonder that sometimes we aren't thinking about what we're doing.

Similarly, our *cognitive capacity* is limited: we simply can't hold many unrelated ideas or pieces of information in our minds at the same time. You may have heard the famous story about why phone numbers in the United States are seven digits plus an area code: researchers found that we could hold seven unrelated numbers in our heads at a time, plus or minus two.⁶ And, of course, there are so many other ways in which our cognitive capacity is limited. For one, we have a particularly difficult time dealing with probabilities and uncertain events, and with realistically predicting the likelihood of something happening in the future. We tend to over-predict rare but vivid and widely reported events like shark attacks, terrorist attacks, and lightning strikes.⁷

In addition, we can become overwhelmed or paralyzed when faced with a wide range of options, even as we consciously seek out more choices and options. Researchers call this the *paradox of choice*: our conscious minds believe that having more choices is almost always better, but when it actually comes time to make a decision and we're faced with our limited

cognitive capacity and the difficulty of the choice ahead of us, we may balk.⁸

Lastly, when it comes to our memories, they simply aren't perfect, and nothing is going to change that. And, for most of us, having a "not perfect" memory is a significant understatement. Our memories usually aren't crystal-clear videos, but a set of crib notes from which we reconstruct mental videos and pictures. We remember events that occur frequently (like eating breakfast) in a stylized format, losing the details of the individual occurrences and remembering instead a composite of that repeated experience. Additionally, in some circumstances, we remember the peak and the end of an extended experience, not a true record of its duration or intensity.⁹

What do all of these cognitive limitations mean? They are important to product people for two main reasons. First, these cognitive limitations mean that *sometimes our users don't make the best choices*, even when something is in their best interest. It's not that they're bad people; it's that they are, simply, people. They get distracted, they forget things, they get overwhelmed. We shouldn't interpret a few bad choices as a sign that they are fundamentally disinterested in doing better (or using our product); instead, it's just that their simple human frailties may be at work. We can design products to avoid overburdening users' limited faculties.¹⁰

Second, our limitations matter because our minds cleverly work around them by having two semi-independent systems in the brain and by using lots and lots of shortcuts. When developing products and communications, we should understand those shortcuts and use them to our advantage or work around them.

START WITH OUR LIMITATIONS: IT ONLY GETS BETTER FROM THERE

If you're familiar with some of the books on behavioral science for a general audience, you might have noticed that this description of behavioral science is different. Many of them start with the foolish things we do: our cognitive mistakes or mental biases. As someone who develops products, that's not a great starting place. It gives the incorrect impression that our users are obtuse, and it's not actually how many behavioral researchers see people and the mind. My goal here is to give you a deeper understanding of how decision making and behavior works and why our seemingly foolish choices arise. Your users (and you) are really smart, given our limitations.

We're of Two Minds

You can think about the brain as having two types of thinking: one is deliberative and the other is reactive; it's a useful metaphor for a complex underlying process.¹¹ Our reactive thinking (aka *intuitive*, or System 1) is blazingly fast and automatic, but we're generally not conscious of its inner workings. It uses our past experiences and a set of simple rules of thumb to almost immediately give us an intuitive evaluation of a situation—an evaluation we feel through our emotions and through sensations around our bodies like a “gut feeling.”¹² It's generally quite effective in familiar situations, where our past experiences are relevant, and does less well in unfamiliar situations.

Our deliberative thinking (aka *conscious*, or System 2) is slow, focused, self-aware, and what most of us consider “thinking.” We can rationally analyze our way through unfamiliar situations and handle complex problems with System 2. Unfortunately, System 2 is woefully limited in how much information it can handle at a time—like struggling to hold more than seven unrelated numbers in short-term memory at once! It thus relies on System 1 for much of the real work of thinking. These two systems can

work independently of each other, in parallel, and can disagree with each other—like when we’re troubled by the sense that, despite our careful thinking, “something is just wrong” with a decision we’ve made.¹³

What this means is that we’re often not “thinking” when we act. At least, we’re not choosing consciously. Most of our daily behavior is governed by our intuitive mode. We’re acting on habit (learned patterns of behavior), on gut instinct (blazingly fast evaluations of a situation based on our past experiences), or on simple rules of thumb (cognitive shortcuts or heuristics built into our mental machinery).¹⁴ Researchers estimate that roughly half of our daily lives are spent executing habits and other intuitive behaviors, and not consciously thinking about what we’re doing.¹⁵ Our conscious minds usually become engaged only when we’re in a novel situation, or when we intentionally direct our attention to a task.¹⁶

Unfortunately, our conscious minds believe that they are in charge all the time, even when they aren’t. In his book, *The Happiness Hypothesis* (Cambridge 2006), philosopher Jonathan Haidt builds on the Buddha’s metaphor of a rider and an elephant to explain this idea. Imagine that there is a huge elephant with a rider sitting atop it. The elephant is our immensely powerful but uncritical, intuitive self. The rider is our conscious self, trying to direct the elephant where to go. The rider thinks it’s always in charge, but it’s the elephant doing the work; if the elephant disagrees with the rider, the elephant usually wins.

To see this in action, you can read fascinating studies of people whose left and right brains have been surgically separated and can’t (physically) talk to one another. The left side makes up convincing but completely fabricated stories about what the right side is doing.¹⁷ That’s the rider standing on top of an out-of-control elephant crying out that everything is under control!¹⁸ Or, more precisely, crying out that every action that the elephant takes is absolutely what the rider wanted them to do—and *the rider actually believes it*.

Thus, we can *do* one thing and *think* about something different at the same time. We might be walking to the office, but we’re actually thinking about

all of the stuff we need to do when we get there (Figure 1-1). The rider is deeply engaged in preparing for the future tasks, and the elephant is doing the work of walking. In order for behavior change to occur, we need to work with both the rider and elephant.¹⁹



Figure 1-1. While the mind consciously thinks about what needs to be done at work, the subconscious mind keeps the body walking (habits and skills), avoids shadowy alleys (an intuitive response), and follows the sweet smell of a bakery (habit)

We Use Shortcuts, Part I: Biases and Heuristics

Both our conscious minds and our nonconscious minds rely heavily on shortcuts to make the best of our limited faculties. Our minds' myriad shortcuts (heuristics) help us sort through the range of options we face on a day-to-day basis and make rapid, reasonable decisions about what to do.

These heuristics are a mix of rules we use throughout our lives with obvious consequence:

Status quo bias

If you're faced with many options to choose from and you can't devote time and energy to think them through, or you aren't sure what to do with them, what's generally the best thing to do? Don't change anything. *We should generally assume people will stick with the status quo.* That's true whether it's a deep-seated historical status quo or one that is arbitrarily chosen and presented as the status quo: to change is to risk loss.²⁰

Descriptive norms—we're deeply affected by social signals

Another way we handle uncertainty in a decision is to look at what other people are doing and try to do the same (aka *descriptive norms*).²¹ This is one of our most basic shortcuts. For example, "People here are drinking and having a good time, so it's OK if I do as well."

Confirmation bias

We tend to seek out, notice, and remember information already in line with our existing thinking.²² For example, if someone has a strong political view, they may notice and remember news stories that support that view and forget those that don't. In a sense, this tendency allows

our minds to focus on what appears to be relevant in a sea of overwhelming information. It has a side effect, though: it leads us to ignore new information that might help us gain a truer picture of the world or try new things.

Present bias

Our limited attention also applies to *time*: we can only focus on a single moment at once. Our minds appear to use a simple shortcut: what seems to be most important? The present. We give undue attention to the present and value things we get now over future benefits, even if it puts our long-term health and welfare at risk. While formally studied in economics since the 1990s, the concept is an ancient one: the desire for instant gratification and the procrastination that comes with it.²³

Anchoring

It's often quite difficult to make a clear and thorough assessment of an answer. And so, when we don't know a number—like the probability of an event or the price of an object—we often start with an initial estimate (whether our own or one provided to us) and make adjustments up or down based on additional information and feedback. Unfortunately, these adjustments are often insufficient—and the initial anchor has a strong effect on the results.²⁴ Anchoring is one of many ways in which we make judgments that are *relative to a reference point*.

Others are interesting and seemingly narrow shortcuts that guide how we act, like these:

Availability heuristic

When things are particularly easy to remember, we believe that they are more likely to occur.²⁵ For example, if I'd recently heard news about a school shooting, I'd naturally think that it is much more common than it actually is.

IKEA effect

When we invest time and energy in something—even if our contribution is objectively quite small—we tend to value the resulting item or outcome much more.²⁶ For example, after we've assembled IKEA furniture, we often value it more than similar furniture someone else assembled (even if it's of higher quality)—our sweat equity doesn't matter in terms of market value, but it does to us.

Halo effect

If we have a good assessment about someone (or something) overall, we sometimes judge other characteristics of the person (or thing) too positively—as if they have a “halo” of skill and quality.²⁷ For example, if we like someone personally, we might overestimate their skill at dancing, even if we knew nothing about their dancing ability.

There are over a hundred of these shortcuts (heuristics) or other tendencies of the mind (biases) that researchers have identified. Unfortunately, these shortcuts can also lead us astray as we try to make good choices in our lives. For example, if you're a religious person living in a place where people don't speak about religion, descriptive norms apply a subtle (or not-so-subtle) pressure to avoid doing so yourself. Or a homeless person might look and smell dirty, and the (negative) halo effect could lead others to think negatively about them; they might see the person as less honest and less smart than they really are. While I've mentioned some negative

outcomes from our shortcuts and biases, it's important to understand that, at their root, our shortcuts are clever ways to handle the limited resources that our minds have.

Let's take a closer look at another way in which our minds economize: habits.

We Use Shortcuts, Part II: Habits

We use the term *habit* loosely in everyday speech to mean all sorts of things, but a concrete way to think about one is this: a habit is a repeated behavior that's triggered by cues in our environment. It's *automatic*—the action occurs outside of conscious control, and we may not even be aware of it happening.²⁸ Habits save our minds work; we effectively *outsource* control over our behavior to cues in the environment.²⁹ That keeps our conscious minds free for other, more important things, where conscious thought really is required.

Habits arise in one of two ways.³⁰ First, we can build habits through simple repetition: whenever you see X (a *cue*), you do Y (a *routine*). Over time, your brain builds a strong association between the cue and the routine and doesn't need to think about what to do when the cue occurs—it just acts. For example, whenever you wake up in the morning (cue), you get out of bed at the same spot (routine). Rarely do you find yourself lying in bed, awake, agonizing over which exact part of the bed you should exit by. That's how habits work—they are so common, and so deeply ingrained in our lives, that we rarely even notice them.

Sometimes there is also a third element, in addition to a cue and routine: a *reward*, something good that happens at the end of the routine. The reward pulls us forward—it gives us a reason to repeat the behavior. It might be something inherently pleasant, like good food, or the completion of a goal we've set for ourselves, like putting away all of the dishes.³¹ For example, whenever you walk by the café and smell coffee (cue), you walk into the shop, buy a double mocha espresso with cream (routine), and feel chocolate-caffeine goodness (reward). We sometimes notice the big habits—like getting coffee—but other, less obvious habits with rewards (checking our email and receiving the random reward of getting an interesting message) may not be noticed.

Once the habit forms, the reward itself doesn't directly drive our behavior; the habit is automatic and outside of conscious control. However, the mind can “remember” previous rewards in subtle ways, intuitively wanting (or

craving) them.³² In fact, the mind can continue wanting a reward that it will never receive again, and may not even enjoy when it does happen!³³ I've encountered that strange situation myself—long after I formed the habit of eating certain potato chips, I still habitually eat them even though I don't enjoy them and they actually make me sick.³⁴ This isn't to say that rewards aren't important after the habit forms—they can push us to consciously repeat the habitual action and can make them even more resistant to change.

The same characteristics that make habits hard to root out can be immensely useful. Thinking of it another way, once “good” habits are formed, they provide the most resilient and sustainable way to maintain a new behavior. Charles Duhigg, in *The Power of Habit* (Random House, 2012), gives a great example. In the early 1900s, advertising man Claude C. Hopkins moved American society from being one in which very few people brushed their teeth to a majority brushing their teeth in the span of only 10 years. He did it by helping Americans form the habit of brushing.³⁵

**Awake or asleep—FILM is
gluing acid to your teeth !**

**Pepsodent removes FILM—
helps stop tooth decay !**

Tooth decay is formed by acid that film holds against your teeth—acid formed by the action of mouth bacteria on many foods you eat. When you use Pepsodent Tooth Paste right after eating, it helps keep acid from forming. What's more, Pepsodent removes dulling stains and "bad breath" germs that collect in film.

FILM NEVER LETS UP! It's forming night and day on everyone's teeth. Don't neglect it. Always brush with film-removing Pepsodent right after eating and before retiring. No other tooth paste can duplicate Pepsodent's film-removing formula. No other tooth paste contains Irium* or Pepsodent's gentle polishing agent.

Don't let decay start in your mouth! Use Pepsodent every day—see your dentist twice a year.

YOU'LL HAVE BRIGHTER TEETH AND CLEANER BREATH when you fight tooth decay with film-removing Pepsodent!

Figure 1-2. Pepsodent advertisement from 1950, highlighting a cue to trigger the habit of tooth-brushing (courtesy of *Vintage Adventures*)

- He taught people a cue—feeling for tooth film, the somewhat slimy, off-white stuff that naturally coats our teeth (apparently, it's harmless in itself) (Figure 1-2).
- When people felt tooth film, the response was a routine—brushing their teeth (using Pepsodent, in this case).
- The reward was a minty tingle in their mouths—something they felt immediately after brushing their teeth.

Over time, the habit (feel film, brush teeth) formed, strengthened by the reward at the end. And so did a craving—wanting to feel the cool tingling sensation that Pepsodent caused in their mouths that people associated with having clean, beautiful teeth.

Stepping back from Duhigg's example, let's look again at the three pieces of a reward-driven habit:

- The *cue* tells us to act now. The cue is a clear and unambiguous signal in the environment (like the smell of coffee) or in the person's body (like hunger). BJ Fogg and Jason Hreha categorize the two ways that they work on behavior into *cue behaviors* and *cycle behaviors* based on whether the cue is something else that happens and tells you it's time to act (brushing teeth after eating breakfast) or the cue occurs on a schedule, like at a specific time of day (preparing to go home at 5 p.m. on a weekday).³⁶
- The *routine* can be something simple (hear phone ring, answer it) or complex (smell coffee, turn, enter Starbucks, buy coffee, drink it), as long as the scenario in which the behavior occurs is consistent. Where conscious thought is not *required* (i.e., consistency allows repetition of a previous action without making new decisions), the behavior can be turned into a habit.
- The *reward* can occur every time—like drinking our favorite brand of coffee—or on a more complex *reward schedule*. A reward schedule is the frequency and variability with which a reward occurs each time the behavior occurs. For example, when we pull the arm or press the button on a slot machine, we are randomly rewarded: sometimes we win, sometimes we don't. Our brains *love* random rewards. In terms of timing, rewards that occur immediately after the routine are best—they help strengthen the association between cue and routine.

Researchers are actively studying exactly how rewards function, but one of the likely scenarios goes like this: when these three elements are combined,

over time the cue becomes associated with the reward.³⁷ When we see the cue, we *anticipate* the reward and it tempts us to act out the routine to get it. The process takes time, however—varying by person and situation from a few weeks to many months. And again, the desire for the reward can continue long after the reward no longer exists.³⁸

We're Deeply Affected by Context

And finally, we turn to the last big lesson: *the importance of context* on our behavior. What we do is shaped by our contextual environment in obvious ways, like when the architecture of a building focuses our attention and activity toward a central courtyard. It's also shaped in nonobvious ways by the people we talk and listen to (our *social environment*), what we see and interact with (our *physical environment*), and the habits and responses we've learned over time (our *mental environment*). These nonobvious effects can show themselves even in slight changes in wording of a question. We'll examine how the environment affects human behavior throughout this book, but let's start with one famous example:

- Suppose there's an outbreak of a rare disease, which is expected to kill six hundred people. You're in charge of crafting the government's response.

You have two options:

- Option A will result in two hundred people saved.
- Option B will result in a one-third probability that six hundred people will be saved and a two-thirds probability nobody will be saved.

Which option would you choose?

Now suppose there's another outbreak of a different disease, which is also expected to kill six hundred people. You have two options:

- Option C will result in the certain death of four hundred people.

- Option D will result in the one-third probability that nobody dies and two-thirds probability everyone dies.

Which option would you choose now?

Presented with these options, people generally prefer Option A in the first situation and Option D in the second. In Tversky and Kahneman's early studies using these situations,³⁹ 72% of people choose A (versus 28% for B), but only 22% choose C (versus 78% for D). Which, as you've probably caught on, doesn't make much sense, since for both A and C, four hundred people face certain death and two hundred will be saved. Logically, if someone prefers A, that person should also choose C. But that isn't what happens, on average.

Many researchers believe there is such a stark difference in people's choices for these two mathematically equivalent options (A and C) *because of how the choices are framed*. One is framed as a *gain* of two hundred lives and the other is framed as a *loss* of four hundred lives.⁴⁰ The text of C leads us to focus on the loss of four hundred lives (instead of the simultaneous gain of two hundred), while the text of A leads us to focus on the gain of two hundred lives (instead of the loss of four hundred). And people tend to avoid uncertain or risky options (B and D) when there is a positive or *gain* frame (A versus B) and seek risks when faced with a negative or *loss* frame (C versus D).

That's, well, *odd*. It shows how relatively minor changes in wording can lead to radically different choices. However, it is especially odd since this isn't something that people would explain themselves. If they were faced with both sets of choices, they wouldn't say, "Well, I recognize that A and C have exactly the same outcomes, but I just intuitively don't like thinking about a loss, even when I know it's a trick of the wording." Instead, the person might simply say: "knowing that I can save people is important (A), and I really don't like the thought of knowingly letting people go to certain death (C)."

Or to use the rider and elephant metaphor again, the rider thinks they're in control, but the elephant really is. Our conscious rider explains our behavior

after it's happened, without knowing the real reason. We are, as social psychologist Tim Wilson nicely puts it, “strangers to ourselves.”⁴¹ To bring this back to product development, our users take actions that they don't understand but will try to explain after the fact.

Our lack of self-knowledge also extends to what we'll do in the future. We're bad at forecasting the level of emotion we'll feel in future situations and at forecasting our own behavior in the future.⁴² For example, people can significantly overestimate the impact of future negative events on their emotions, such as a divorce or medical problem. We're not only affected by the details of our environment, we don't often recognize that our environment has affected us in the past, so we don't consider the influence when we're thinking about what we'll do in the future. In a product development context, this means that asking people what they will do or what they think will happen to them in the future is fraught with problems.

Tversky and Kahneman's study demonstrates one of the key principles of behavioral science: reference dependence. In absolute terms, outcomes of options A and C are identical. But the first frame sets the reference point as people dying and the potential to save them; the second sets the reference point as people living and the potential to let them die. Whether something is a loss or a gain depends on our reference point. And, as you can see in Tversky and Kahneman's study, that reference point is malleable: it's subject to design.

We Can Design Context

Because our environment affects our decision making and behavior, redesigning that environment can change decision making and behavior. We can thoughtfully develop product designs and communications that take this knowledge into account and help people make better decisions, use habits wisely, and follow through on their intentions to act, which is the focus of the rest of this book.

What Can Go Wrong

We've touched upon the areas in which the quirks of our minds can lead to bad outcomes a bit already. It's useful to make these areas more explicit and clearer, though, since that understanding is the foundation of making things better. We can distinguish two major branches of research, both of which are useful for our purposes. Broadly, behavioral science helps us understand quirks of *decision making* and quirks of *action*.

Quirks of Decision Making

The shortcuts that our minds use lead us to rapid and generally good decisions without evaluating all of the options thoroughly. They're necessary, letting us make the best of our limited faculties, and are generally very helpful—until they aren't.

Think about what happens when you're visiting a new town and you're walking around looking for a bite to eat:

- You might look on your phone to see which restaurant has the highest and most ratings. Or, you might peek in the window to see where other people are eating—it's never a good sign if a place is deserted, right? That's the *social proof* heuristic: if you aren't sure what to do, copy what other people are doing.
- You might have seen ads touting a particular restaurant, and when you pass by the sign, it catches your eye. If you've at least heard of it, that's good! The *availability heuristic* supports that feeling.
- You might notice a chain you've been to and liked recently and figure that what's been good recently probably will be good again. That, among other things, is a *recency bias* helping you choose.
- You might look at their prices and see that one place is offering burgers at \$10, and another at \$2. You're all for saving money, but that's just too much: there must be something wrong with the \$2 place. That's the shortcut of *price as a signal of quality*.

In each case, the shortcuts help us make quick decisions. None of them is perfect, certainly. We could find ways to make them better, but by and large, these are all reasonable choices. Most importantly, they are fast: they save us from spending hours and hours reviewing all of the pros and cons of each restaurant in the city, judging them in terms of their taste and nutrition on 50 dimensions, the time to reach each one, their ambiance and likely social environment, and so on. Shortcuts like these make decisions possible and avoid decision paralysis.

Switching contexts, let's think about investing money in the stock market. You've recently received a windfall and you're looking to invest some of it for the future. You don't have much experience in investing, so what do you do?

- You might look online to see what everyone else is talking about and investing in, using social proof. Awesome! Except that's how bubbles are made: think Bitcoin or the dot-com bubble.
- You might invest in things you've heard of, using the *availability heuristic*. Excellent! Except, again, that's how bubbles are made.
- You might look at what's performed well in the past and invest in that, using *recency bias*. The problem is that past performance doesn't predict future performance. Not a great guide.
- You might look at prices—if a stock has a really high price, it must be a good investment, right? OK, you know where that goes.

And so forth. You get the picture.

When shortcuts work well, we often don't notice them; we effortlessly and quickly come to a decision. Or, in the rare cases we do notice them, we call them *clever*. In the research community, we refer to them, in the positive sense, as fast and frugal heuristics (where heuristic is another word for shortcut).⁴³

When the same shortcuts get us into trouble, however, we call them foolish: how could I have been so stupid as to follow the crowd? Since you're

reading this book, you’ve probably come across the term *bias*, which is intimately related to these shortcuts. A bias, strictly speaking, is a tendency of thought or action. It is neither positive nor negative; it just is. Most people, including many researchers, use it explicitly in the negative sense, as in a “deviation from an objective standard, such as a normative model” of how people *should* behave.⁴⁴ A shortcut or heuristic gone awry creates a bias.

Once we call something a bias, it’s easy to jump to the logical conclusion: well, we just need to get rid of them! Remove people’s biases! It’s not that simple. It’s precisely because these shortcuts are so clever and smart that they are hard to change. If we simply did foolish things, we’d eventually learn not to do them (either in our own lives or across the history of our species). But these shortcuts are not foolish at all: they are immensely valuable. They are sometimes out of context and used at the wrong time. We can’t and shouldn’t be able to simply stop social proof or the availability heuristic. That would wreak more havoc than it would solve.

The reality is that *we can’t completely avoid* using mental shortcuts. Rather, by understanding how our conscious and nonconscious shortcuts are triggered by the details of our environment, we can learn to avoid some of the negative outcomes that result from their use. So the first challenge of behavioral science—of designing for behavior change—is to help people make better decisions, given the valuable but imperfect shortcuts we all use.

Quirks of Action

I, myself, am made entirely of flaws, stitched together with good intentions.

—Augusten Burroughs

Behavioral science also helps us understand the quirks of our behavior, above and beyond our decisions, especially why we decide to do one thing and actually do something else. This understanding starts with the same base as for decisions: that we’re limited beings with limited attention, memory, willpower, etc. Our minds still use clever shortcuts to help us

economize and avoid taxing our limited resources. But these facts make themselves felt in different ways after we've made a decision. In particular, we have errors of inaction and errors of unintentional action.

In the research literature, the *intention–action gap* is one of the major errors of inaction. We've all felt this gap in one way or another. For example, do you have a friend with a gym membership, or exercise equipment at home, that they just don't use that often? Do they really enjoy giving money to a gym? Of course not. They really intended to go to the gym when they first signed up, or to use that fancy machine when they first bought it. It's just that they didn't. The benefits of the gym are still clear. And despite all of their past failures, they keep hoping and believing that they'll get it together and go regularly. But something else gets in the way that *isn't* motivation. So they keep paying—and keep failing to go.

With the intention–action gap, the intention to act is there, but people don't follow through and act on it. It's not that people are insincere or lack motivation; the gap happens because of how our minds are wired. It illustrates one of the core lessons of behavioral science: *good intentions and the sincere desire to do something, aren't enough.*

And unintentional action? I don't mean revelry that we regret the next morning. Rather, I mean behaviors that we don't intend even while we're doing them, often because we aren't aware or thinking about them. One cause of these we've already looked at: habits. Our habits allow us to take action without thought—effortlessly riding a bike, navigating the interface of common applications, or playing sports. But naturally, they can also go awry.

Do you know someone who just can't stop eating junk food? Each night, when they get home tired and need a break, on the way to the couch, they pick up a candy bar and a bag of chips and sit down with the laptop to watch videos. An hour or so later, they take a break and notice the crumpled-up wrapper and bag and throw them away. They're still hungry and hardly noticed the snacks on their way into their mouth.

There are many other examples, like when we get hooked on cigarettes (it appears the *habit* is more powerful than the nicotine, in fact),⁴⁵ on late-night TV binging, or on incessantly checking social media apps. Habits, as learned patterns of behavior, are inherently neutral. We learn bad habits just as we learn good habits: through repetition. Our minds automate them in order to save us cognitive work. For the person eating junk food, maybe it was a particularly rough time at work, or maybe it was when they first moved to the city and didn't know where to get good groceries that set up the automated routine. Regardless of the source, once that junk food habit was set, it was hard to shake.

Just as with our decision-making shortcuts, try to imagine a world in which we didn't have habits—one where we had to carefully think through every decision, every action, as if we were a teenager first learning to drive a car. We'd be exhausted wrecks in no time. We can't not rely on habits, nor ask users of our products not to do so. Rather, as developers of behavior-changing products and communications, we need to understand them and work with (or around) them. Shortcuts gone awry, habits that people wish they didn't have, and the yawning gap between people's intentions and their actions: these are the problems that we're here to solve. These are why we design for behavior change.

WHY FOCUS ON PROBLEMS?

You may have noticed that we're focusing on the *problems* of decision making and not on how people make decisions generally. That is, we're not looking at all the times that people make *good* decisions that are appropriate and right for them. That's intentional. It's because this book is about designing for behavior change to help users do better: they don't need help when their decisions and behaviors are already on track!

While a particular company may want people to make a different decision (like to buy their product), if that isn't the decision the person would make themselves given a thorough process, we shouldn't be involved. We shouldn't try to change anything. It's only when there is a gap between what a person might otherwise want to choose and do, and the heat of the moment mistake that we should be involved. Hence my, and many behavioral scientist's, focus on errors of decision making and not on persuasion or "making people do stuff" regardless of their wishes. More on the ethics of designing for behavior change in [Chapter 4](#).

A Map of the Decision-Making Process

We've talked about the different ways in which our minds make our decisions, from careful deliberative thought to shortcuts to fully automated habits. We can think about these mental tools as being part of a spectrum, based on how much thought is involved. Unfamiliar situations (like, for most people, math puzzles) require a lot of conscious thought. Walking to your car doesn't. Similarly, high-stakes decisions like "Which job should I take?" also pull in more conscious thought than "Which bagel should I eat?" Frequently repeated, low-stakes decisions like "Which way should I hold my tooth brush this morning?" don't require much thought at all and can turn into habits.

The spectrum in **Figure 1-3** provides the *default, lowest energy* way that our minds would respond if we didn't intentionally do something differently.

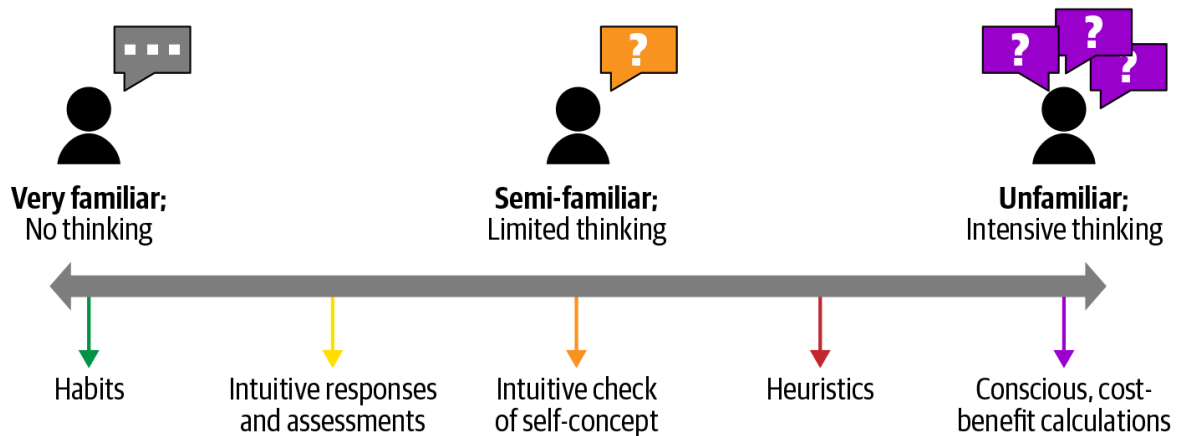


Figure 1-3. In familiar situations, our minds can use habits and intuitive responses to save work

Here are some simple examples, using a person who is thinking about going on a diet and doesn't have much past experience with diets:

Eating potato chips out of a bag

Very familiar. Very little thought. Habit.

Picking out what to get at your favorite buffet bar

Familiar. Little thought. Intuitive response or assessment.

Signing up for dieting workshops at the office

Semi-familiar. Some thought. Self-concept guides choice.

Judging whether a cheeseburger will violate your diet's calorie limit for the day

Unfamiliar. Thought required, but with easy ways to simplify.⁴⁶

Heuristic.

Making a weekly meal plan for the family based on the individual calorie and nutrient counts of hundreds of foods

Unfamiliar. Lots of attention and thought. Conscious, cost–benefit calculations.

Table 1-1 provides a bit more detail on where each of the tools on the spectrum are likely to be used.

Table 1-1. The various tools the mind uses to choose the right action

Mechanism	Where It's Most Likely to be Used
Habits	Familiar cues trigger a learned routine
Other intuitive responses	Familiar and semi-familiar situations, with a reaction based on prior experiences
Active mindset or self-concept	Ambiguous situations with a few possible interpretations
Heuristics	Situations where conscious attention is required, but the choice can be implicitly simplified
Focused, conscious calculation	Unfamiliar situations where a conscious choice is required or very important decisions we direct our attention toward

This spectrum doesn't mean that we always use habits in familiar situations, or that we only use our conscious minds in unfamiliar ones. Our conscious minds can and do take control of our behavior and focus strongly on behaviors that otherwise would be habitual. For example, I can think very carefully about how I sit in front of the computer to improve my posture; that's something I normally don't think about because it's so familiar. That takes effort, however. Remember that our conscious attention and capacity are sorely limited. We only bring in the big guns (conscious, cost-benefit calculations) when we have a good reason to do so: when something unusual catches our attention, when we really care about the outcome and try to improve our performance, and so on.

As behavior change practitioners, it's a whole lot easier to help people *take actions* that are near the “eat another potato chip in the bag” side of the spectrum, rather than the “thoughtfully plan meals” side. But it's much

harder for people to *stop actions* on the potato chip–eating side than on the meal-planning side. The next two chapters will look at both, though: how to create the good and how to stop the bad.

A Short Summary of the Ideas

Behavioral science provides a powerful set of tools to help us both understand how people make decisions and take action and to help them make better decisions or follow through on their intent to take action if they would like our help.

Here's what you need to know:

We're limited beings

We have limited attention, time, willpower, etc. For example, there are nearly an infinite number of things that your users could be paying attention to at any moment.

Our minds use shortcuts(aka heuristics)

We use them to economize and make quick decisions because of our limitations. Heuristics applied in the wrong context are one cause of *biases*: negative and unintended tendencies in behavior or decision making. Often because of these biases, there's a significant gap between people's intentions and their actions.

We're of two minds

What we decide and what we do depends on both conscious thought and nonconscious reactions, like habits. What this means is that your users are often not “thinking” when they act. At least, they're not choosing consciously.

Decision and behavior are deeply affected by context

This worsens or ameliorates our biases and our intention–action gap.

What your users do is shaped by our contextual environment in obvious ways, like when the architecture of a site directs them to a central home page or dashboard. It’s also shaped in nonobvious ways: by the people they talk and listen to (the *social environment*), by what we see and interact with (their *physical environment*), and the habits and responses they’ve learned over time (their *mental environment*).

We can cleverly and thoughtfully design a context

We do so to improve people’s decision making and lessen the intention–action gap. And that is the point of *Designing for Behavior Change* and this toolkit.

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- 1 There are hundreds, if not thousands, of papers and books one can draw from. [Benartzi and Thaler \(2004\)](#) is a good start for retirement research.
 - 2 Ariely (2008), Thaler and Sunstein (2008), Kahneman (2011)
 - 3 [Krulwich \(2009\)](#); Soman (2015)
 - 4 Soman (2015). Not all biases are directly caused by heuristics gone awry, but many can be traced back to time- or energy-saving devices in the mind. One major category that isn’t from heuristics consists of identity-preserving biases (mental quirks that make us feel better about ourselves), like overconfidence bias.
 - 5 [Hamilton \(2008\)](#)
 - 6 Miller (1956)
 - 7 For example, see Manis et al. (1993). These outcomes are actually the result of reasonable but imperfect shortcuts that our minds use to *counter* our limitations; we’ll talk about those shortcuts shortly.
 - 8 See Schwartz (2004, 2014), Iyengar (2010), and [Solman \(2014\)](#). As we should expect with all behavioral mechanisms and lessons, the paradox of choice isn’t universal or without disagreement.
 - 9 Kahneman et al. (1993)

- 10 As many designers have argued, including Krug (2006).
- 11 That is, the family of theories referred to as *dual process theory* in psychology. Dual process theories give a useful abstraction—a simplified but generally accurate way of thinking about—the vast complexity of our underlying brain processes.
- 12 Damasio et al. (1996)
- 13 There are great books about dual process theory and the workings of these two parts of our mind. Kahneman’s *Thinking, Fast and Slow* (Farrar, Straus and Giroux, 2011) and Malcolm Gladwell’s *Blink* (Back Bay Books, 2005) are two excellent places to start; I’ve created a [list of resources](#) on how the mind works (including dual process theory).
- 14 The boundaries between “habit” and other processes (intuition, etc.) are somewhat blurry; but these terms help draw out the differences among types of System 1 responses. See Wood and Neal (2007) for the distinction between habits and other automated System 1 behaviors; see Kahneman (2011) for a general discussion of System 1.
- 15 Wood (2019); Dean (2013)
- 16 I’m indebted to Neale Martin for highlighting the situations in which the conscious mind does become active. See his book *Habit* (FT Press, 2008) for a great summary of the literature on when intuitive and deliberative processes are at play.
- 17 Gazzaniga and Sperry (1967)
- 18 This isn’t to say that the rider is equivalent to the left side of the brain and the elephant to the right side. Our deliberative and intuitive thinking isn’t neatly divided in that way. Instead, this is just one of the many examples of how rationalizations occur when our deliberative mind is asked to explain what happens outside of its awareness and control. Many thanks to Sebastian Deterding for catching that unintended (and wrong!) implication of the passage.
- 19 Heath and Heath (2010)
- 20 See Samuelson and Zeckhauser (1988) for the initial work on status quo bias.
- 21 Gerber and Rogers (2009)
- 22 Watson (1960)
- 23 See Laibson (1997) for initial modeling work in economics; O’Donoghue and Rabin (2015) for a relatively recent summary.
- 24 Tversky and Kahneman (1974)
- 25 Tversky and Kahneman (1973)
- 26 Norton et al. (2011)
- 27 Nisbett and Wilson (1977)
- 28 See Bargh et al. (1996) for a discussion of the four core characteristics of automatic behaviors, such as habits: uncontrollable, unintentional, unaware, and cognitively efficient (doesn’t require cognitive effort).
- 29 Wood and Neal (2007)

- 30 There are nice summaries at [News in Health](#) and [CBS News](#).
- 31 Ouellette and Wood (1998)
- 32 There's an active debate in the field about how exactly the notion of a reward affects a person *after* the habit is formed. See Wood and Neal (2007) for a discussion.
- 33 See [Berridge et al. \(2009\)](#) on the difference between *wanting* and *liking*. The difference between wanting and liking is a possible explanation for why certain drugs instill strong craving in addicts although taking them long stopped being pleasurable.
- 34 And yes, for those of you who recall this example from the first edition of the book, it's still true today.
- 35 Duhigg's story also is an example of the complex ethics of behavior change. Hopkins accomplished something immensely beneficial for Americans and American society. He was also wildly successful in selling a commercial product in which demand was partially built on a fabricated "problem" (the fake problem of tooth film, which is harmless, rather than tooth decay, which is not).
- 36 [Fogg and Hreha \(2010\)](#)
- 37 This is one form of *motivated cueing*, in which there is a diffuse motivation applied to the context that cues the habit (Wood and Neal 2007). There is active debate in the field on how, exactly, motivation affects habits that have already formed.
- 38 Duration is covered in Lally et al. (2010), and delay in [Berridge et al. \(2009\)](#); Wood (2019).
- 39 Tversky and Kahneman (1981)
- 40 Many researchers accept this explanation, but not all. As often happens in science, there is a divergence of opinion on why framing effects like this occur. An alternative view is that people make a highly simplified analyses of the options and the two different options have two different simplified answers. See Kühberger and Tanner (2010) for one such perspective.
- 41 Wilson (2002); see Nisbett and Wilson (1997b) for an early summary.
- 42 Emotion and other examples are covered in Wilson and Gilbert (2005), and behavior in Wilson and LaFleur (1995).
- 43 Gigerenzer and Todd (1999); Gigerenzer (2004)
- 44 Soll et al. (2015), building on Baron (2012)
- 45 Wood (2019)
- 46 One such commonly used heuristic is the volume of the food—yes, how big it is. [Barbara Rolls](#), director of the Penn State Laboratory for the Study of Human Ingestive Behavior, developed a diet that leverages this heuristic to help people lose weight (see Rolls 2005).

Chapter 2. Creating Action

Americans are over 1.5 trillion dollars in debt with student loans—yet over two million students who are eligible to receive free government aid don't apply each year.¹ They don't ask for free money.²

When asked, students say that they didn't apply because they didn't know about their eligibility. However, Harvard researchers ran an experiment to see if that was really the case, and they found that a lack of information about aid eligibility doesn't affect application rates one way or another. Their research illustrates one of the core lessons in behavioral science: people don't always know why they act the way they do, and the real reasons may not be obvious.

Kristen Berman, cofounder of Irrational Labs, set to work on increasing aid applications with her team. They determined that students had to spend almost an hour to apply and undertake 20 separate actions. They believed that cognitive overload affected the students. They would mull over whether or not to apply and avoid making a decision because of its complexity. They postponed it until the next day, then the next day, and so on.

To help counter that overload (and procrastination), Irrational Labs ran an experiment in which they sent simple text messages to students. The message told them that applying for aid was part of the enrollment process (which it is, it's just a step that many students skip), and reminded them to complete it by the deadline. In other words, the “decision” was already made.

For students who hadn't applied in the previous year, that simple text message tripled the odds of applying. Extending that to the US student population, approximately 230,000 more students would apply for financial aid each year with that intervention. Hence another core lesson of behavioral science: when we understand the obstacle people face, we can help them take action.

From Problems to Solutions

Where does this understanding of the quirks of the mind leave us? It gives us the tools we need to help people make better decisions and change their behavior.

To give you a preview of where we're headed, there are two main sets of tools we'll use to help our users. The first addresses the intention–action gap. Just thinking about the action more isn't enough. Instead, we need to look at all of the pieces that come into play to create an action. There's a core lesson underlying those behavioral interventions: past a certain point, motivation isn't the sole, or even main, determinant. There are many things we might want to do or be willing to do. Which one we actually do is highly dependent on context—and thus the specific moment.

People take action (or fail to) in a specific moment. Our will and desire are certainly important—but it's not enough, especially when we're looking to design for behavior change (i.e., to do differently than the present). We need to understand what brings one action to the fore and not others.

For that, we have the CREATE framework: a *Cue*, which starts an automatic, intuitive *Reaction*, potentially bubbling up into a conscious *Evaluation* of costs and benefits, the *Ability* to act, the right *Timing* for action, and the overwhelming power of past *Experience*.

These are the prerequisites for most conscious action.

We'll use the second set of tools for poorly thought through decisions and unintentional behaviors: we'll make them more intentional by interfering with habits and slowing down rash decisions. You can think about replacing a habit or stopping a mental shortcut as CREATE in reverse: removing one or more of the key factors that leads to the negative decision or behavior. In particular, we'll avoid the *Cue*, replace the *Reaction*, rethink the *Evaluation*, or remove the *Ability*.

A Simple Model of When, and Why, We Act

From moment to moment, why do we undertake one action and not another? The six CREATE factors must align simultaneously before someone will take *conscious* action. Behavior change products help people close the intention–action gap by influencing one or more of these preconditions: Cue, Reaction, Evaluation, Ability, Timing, and Experience.

To illustrate these six factors, let's say you're sitting on the couch watching TV. There's an app on your phone you downloaded last week that helps you plan and prepare healthy meals for your family. When, and why, would you suddenly get up, find your mobile phone, and start using the app?

It's an odd question, I know. We don't often think about user behavior in this way—we usually assume that somehow our users find us, love what we're doing, and come back whenever they want to. But researchers have learned that there's more to it than that, based on how the mind makes decisions. So, imagine you're watching TV. What needs to happen for you to use the meal planning app *right now*?

Cue

The possibility of using the app needs to somehow cross your mind.

Something needs to cue you to think about it: maybe you're hungry or you see a commercial about healthy food on TV.

Reaction

Second, you'll intuitively react to the idea of using the app in a fraction of a second. Is using the app interesting? Are other people you know using it? What other options come to mind, and how do I feel about them?

Evaluation

Third, you might briefly think about it consciously, evaluating the costs and benefits. What will you get out of it? What value does the app

provide to you? Is it worth the effort of getting up and working through some meal plans?

Ability

Fourth, you'll check whether actually using the app now is feasible. Do you know where your mobile phone is? Do you have your username and password for the app? If not, you'll need to solve those logistical problems before using the app.

Timing

Fifth, you'd gauge *when* you should take the action. Is it worth doing now or after the TV show is over? Is it urgent? Is there a better time? This may occur before or after checking for the ability to act. Both have to happen, though.

Experience

Sixth, even if logically using the app is worth the effort and it makes sense to use it now, if we tried the app before (or something like it) and it made us feel inadequate or frustrated us, we'd be loath to try again. Our idiosyncratic personal experiences can overwhelm any "normal" reaction we might have.

These six mental processes—detecting a cue, reacting to it, evaluating it, checking for ability, determining if the timing is right, and interpreting it all through the lens of our past experiences—are gates that can block or facilitate action. You can think of them as "tests" that any action must pass: all must complete successfully in order for you to consciously, intentionally, engage in the action. And, they all have to come together at the same time.³ For example, if you don't have the urgency to stop watching TV and act

now, you certainly could do it later. But when “later” comes, you’ll still face these six tests. You’ll reassess whether the action is urgent at that point (or whether something else, like walking the dog, takes precedence). Or maybe the cue to act will be gone and you’ll forget about the app altogether for a while.

So, products that encourage us to take a particular action have to somehow cue us to think about the action, avoid negative intuitive reactions to it, convince our conscious minds that there’s value in the action, convince us to do it now, and ensure that we can actually take the action. That’s a lot to do! Much of this book talks about how to organize, simplify, and structure that process (and then test whether you’ve got it right).

If someone already has a habit in place and the challenge is merely to execute that habit, the process is mercifully shorter. The first two steps (Cue and Reaction) are the most important ones, and, of course, the action still needs to be feasible. Evaluation, Timing, and Experience can play a role, but a lesser one, because the conscious mind is on autopilot.

Let’s go into more detail about the six preconditions for conscious action.

DIFFERENT FRAMEWORKS FOR DIFFERENT PURPOSES

In addition to CREATE, there are many other frameworks used in the behavior change world—such as the Behavioral Insight Team’s EAST Framework, and Michie et al.’s COM-B Behaviour Change Wheel.⁴ These approaches generally draw on the same underlying literature and lessons of the mind but seek to solve a somewhat different problem. The problem I’m trying to solve here is practical product development.

EAST, for example, is a straightforward framework highlighting how people tend to take actions that are Easy, Attractive, Social, and Timely. That’s absolutely true—it’s just not a great guide for the nuances of product development. I’ve been in far too many meetings where people say, “Well, let’s just make it social, that always works!” That’s using EAST for the wrong purpose. Similarly, long-standing academic theories about behavior, like the theory of planned behavior, are very useful for predicting conscious, intentional action; they just aren’t as useful for building products that support conscious (and nonconscious) action.

In this book, I present two conceptual tools: CREATE, which helps us understand what’s required to create action, and DECIDE, which helps us discover the obstacles our users are facing and then to decide which specific techniques (out of a list of 30 or so) are appropriate for those problems. We begin looking at CREATE here and continue our discussion of it in Chapters 3 and 4, and we cover DECIDE in Chapters 5 through 15: that’s the bulk of the book.

Cue

At every moment of every day, we’re deciding what to do next. The universe of things we might do with our time is truly infinite. Our minds can’t handle that much information and they protect us from being overloaded by using a set of mental filters. For example, *inattentional*

blindness means we simply don't see things we aren't looking for when we're concentrating heavily. That's what happened in Chabris and Simons's famous studies where half of the people watching a group pass a basketball back and forth failed to see a guy in a gorilla outfit walking across the screen!⁵ Our mental filters, out of necessity, let us consider only a fraction of what's possible. A similar effect happens when we're at noisy, crowded parties—we can focus on the person talking to us, despite what otherwise would catch our attention and pull us in.⁶

In addition, *confirmation bias* shapes what we notice in our environment. When confronted with a complex environment or lots of information, concepts that we often think about and agree with bubble up to our attention. We've all seen this on social media or political discussions, where it seems like people only focus on things that support their case or political persuasion. Like with many of our mental quirks, this is a good and useful mechanism gone awry: our minds help us pay attention to what we probably want to focus on, out of a sea of overwhelming information. It also just makes us narrow-minded jerks in political conversation.

So what cues attention? We start thinking about an action for two reasons:⁷

External cues

Something in our environment can trigger us (like an email or text message) to think about it. It could be a pair of running shoes that makes us think of running or something more overt, like a friend calling us on the phone and asking why we aren't out running in the park with them.

Internal cues

Our minds can drift into thinking about the action on its own, through some unknown web of associated ideas (which may themselves have been cued externally or by an internal state like hunger).⁸

Sometimes, cues can capture our attention no matter what—like a car that’s about to hit us. Other times, we are explicitly looking for cues to act—like scanning over subject lines in our inbox or looking for notifications on our mobile phones. It’s even possible that we honestly have no idea why an action bubbles up in our minds.

LESSONS FOR BEHAVIORAL PRODUCTS

When users are just starting to undertake a new action, external cues are vital. For example, if you’re beginning to run each morning, placing your running shoes by the door is a good cue. Here are a few strategies products can use for external cueing:

- Placing the product in the user’s daily environment
- Using a slightly different cue each time to avoid being ignored
- Building strong associations with parts of a person’s existing routines

As the action becomes more familiar, products can help users build strong associations between an internal cue—like hunger or boredom—and the action.⁹

When designing for behavior change, you should also avoid, or co-opt, distracting cues that seek the users’ attention at the same time. Email inboxes are very crowded in the morning with lots of cues to act, for example.

We should also recognize that we aren’t dealing with a blank slate; our users’ attention will naturally be drawn to certain items over others because of the design (naturally) and also because of what’s already going on in their heads—which we can learn about and observe.

Reaction

Once the mind starts thinking about a potential action, there is an automatic reaction from System 1—the lightning fast, intuitive, and largely nonconscious part of the brain discussed in [Chapter 1](#) and Kahneman’s *Thinking, Fast and Slow* (FSG, 2011). In some cases, it is startling and powerful, like the desire to run the heck out of a building when you smell gas. In more common situations—like removing our running shoes or using an app—the automatic reactions are less jarring but still guide our behavior. Our conscious minds don’t really have insight into what goes on within our automatic response system.

While researchers don’t fully understand this process, we have some significant clues as to what drives our nonconscious reactions.

It’s strongly social

In many ways, we are wired to pay attention to and focus on social interactions. We intuitively assess whether something is right for us to do based on whether it’s something that other people like us seem to do. We are hesitant to take actions that our peers might disapprove of. We try to be consistent with our social commitments and our sense of identity, both of which depend on and are shaped by our interactions with others. Our social connections reach us at a level that’s deeper, less deliberative, than merely a cost–benefit analysis of expected outcomes.

It’s linked by similarity

Our minds quickly assess how we feel about unfamiliar things based on their similarity to more familiar items (aka the similarity heuristic). Sometimes those similarities express something essential—like the genre of a book or movie. But often, the distinctions are based on more cursory distinctions: shape, color, smell. This is true for fruit and for

people: it's a root cause of stereotyping, and like all mental shortcuts, it's a valuable cognitive tool that can go awry.¹⁰

It's shaped by familiarity

The more we're exposed to something, like an idea or object, the more we tend to like it (all else being equal). Researchers call this the *mere exposure effect*.¹¹ For example, advertisers rely on this principle when they buy ads to show you an image of a brand again and again—just by seeing the ad, people can come to like the brand more (again, all else being equal). More generally, our minds confound the easy-to-remember with the true; it just feels right to us when we can think about it quickly.

It's trained by experience

Our intuitive responses are the ruts cut into the earth of our mind by frequent passage. Over time, our minds learn associations; the things that we have enjoyed in the past, we learn to react positively to in the future (*operant conditioning*); even the things that are associated with good experiences in the past can make us respond positively (*classical conditioning*).¹² And even without formal conditioning, our minds learn what to expect in a familiar situation. For example, if we're thinking about walking up 10 flights of stairs, the last time we took the stairs and almost had a heart attack will color how we feel about doing it again (and this can occur *before* we consciously think about whether or not to act). Prior experience can also affect us in more immediate ways: if we've become angry, we may interpret an ambiguous situation as more hostile than if we were in a good mood to start with.¹³

Remember reference dependence, one of the lessons of behavioral science? Our experiences help set our reference point. They tell us what to expect in a situation—and thus even a very good meal at a restaurant could be viewed negatively because we expected (based on our own experience or what others had told us in the past) an *excellent* meal.¹⁴

What happens because of this reaction? First, our nonconscious mind can render a *verdict* or “gut feeling” about the action: an emotion that colors our thinking and conscious deliberation, if any. The gut feeling we get doesn’t necessarily determine our behavior. Our conscious minds can override (or ignore) what our intuitive system tells us—but it will feel wrong. And it’s hard to sustain a change in behavior if it intuitively feels wrong.

The reaction may also trigger other memories and ideas: when we start thinking about one action, we also activate memories and thoughts about other related concepts. If we’re thinking of a particular need (like hunger), our minds will search for other possible answers to the need and evaluate them as well. For example, if I’m looking at the stairs, my mind will automatically, and without my control, also activate thoughts about using the elevator or escalator.¹⁵ Our web of mental associations may lead us in an entirely different direction as well, distracting us from the original cue and task at hand.

And finally, the reaction may directly trigger an *action*. In the case of habitual behaviors, the reaction might automatically initiate the action, based on the cue. Let’s say I take the elevator every day. Once I walk in the building, I may go to the elevator and press the button without conscious thought.

LESSONS FOR BEHAVIORAL PRODUCTS

Users react to your product, and the action it supports, in the blink of an eye. You can't avoid it; it happens automatically. But from a behavior change perspective, there are particular aspects of this automatic assessment you should be paying attention to:

Trust

Your product is encouraging your users to do something. Even when they want to take the action, they will be hesitant if they don't trust the company behind that encouragement. Whether or not a user trusts the product, and company, is often an intuitive sense.

Watch where you get your product signal

If you ask people what they want to do or whether they have the motivation to use your app, you're engaging their conscious minds. But it's their intuitive minds you have to pass first, and that isn't something people articulate on surveys. Ideally, watch their behavior and don't listen to their mouths.

The first-time user experience really matters

You may be able to convince or entice someone to try out your product and action the first time. But the more your action requires repeated use, the more that you rely on intuitive reactions. And those reactions build on what they've actually experienced, the associations they've made, and the emotions they felt about your product and action.

Evaluation

After the mind is cued to think about a particular action, and assuming it hasn't been derailed by its intuitive reaction, then the action may rise to conscious awareness. This happens especially when we're facing novel situations and we don't have an automatic behavior to trigger. The conscious mind kicks in and evaluates whether the person should take the action, given the various costs and benefits.

This stage is the one that we tend to think about first when we're trying to change behavior. We try to educate people about the benefits of the action, increase their motivation with money or other rewards, and reduce the (perceived) cost of taking the action.

For example, again consider people who have the choice of taking the stairs to go up a few flights or taking the elevator (**Figure 2-1**). Let's say that their conscious mind is engaged. The common approach to encouraging people to take the stairs would be to focus on:

Highlighting benefits

Taking the stairs will get you in shape and may lengthen your life.

Minimizing costs

Taking the stairs will cost you only three more minutes, and if you go slowly, you won't sweat.

Downplaying alternatives

The elevator is slow and crowded at this time of day.



Figure 2-1. While the subconscious mind might see the stairs and think “Ugh, that’s work,” the conscious mind thinks of costs and benefits (Benefits: good exercise. Cost: just three minutes. And the elevator is crowded anyway. Done!)

There is, of course, tremendous complexity behind the deliberations we make over whether to act. How much do we really know about the costs and benefits of the action? Where did we get that information, and do we trust it? Is it worth the effort to seek out more information, or should we simply use what we have? What motivations weigh upon us most strongly at the moment we’re deciding?

These are vital questions. For now, though, let’s leave it at this: if we deem the action worth the effort *and better than the alternatives*, then we’re in business. The choice to act has been made.

Note that the “thinking” that occurs here may be extremely limited and rapid. If the action isn’t very important or is familiar, then the conscious mind may decide to go ahead without much effort. It’s when the action is

unfamiliar, or the mind decides to pay a lot of attention, that more intensive thinking occurs.

A product that promotes an important action that users “should” take (and some part of them wants to take) isn’t enough. The product must give users something they actually want, right now, more than the alternatives. Like any product, a behavior-changing product must solve a problem for its users. Otherwise, the rest of this discussion won’t help. If the conscious mind doesn’t see value that’s worth the effort, it won’t intentionally use the product or take the action. That value needn’t be purely instrumental—it can be something social, emotional, or an intrinsic enjoyment of the activity (we’ll talk more about types of motivation later)—but it needs to be there.

Remember, for habitual behaviors, this conscious awareness and evaluation usually doesn’t occur at all. But our conscious minds are happy to make up stories about why we do habitual things. Those stories are just noise and aren’t real reflections of our actions.¹⁶

LESSONS FOR BEHAVIORAL PRODUCTS

The conscious evaluation phase is what most people who are designing products naturally target—making the benefits of the application clear and removing frustrations and frictions (costs). It’s all about the conscious, quantifiable value that the product provides.

The trick when designing for behavior change is to remember that the value that matters most is the value that *the user ascribes to the product and action*, and not the value that *you* ascribe to it. If your company and your product see taking the stairs instead of the elevator as the start to massive changes in long-term health (i.e., huge benefits) but the user doesn’t see it that way, you’re misaligned.

Ability

Let’s say the choice to act has been made after weighing the costs and benefits. Is it actually *feasible* to undertake the action? If you’ve decided to

finally put aside some money for your retirement, can you actually do it, right now? The individual must be *able* to immediately take the action; the ability to act has four dimensions:¹⁷

Action Plan

The person must generally know what's required to take action. For example, they must know that setting up a retirement account requires going to a particular website, entering information provided by their employer, and so on.

Resources

The person must actually have the resources required to act. For example, they must have money available and access to a computer to go to the retirement website and set up an account.

Skills

They must have the necessary skills to act. For example, to sign up for a retirement account online, they must know how to use a computer and navigate its (too often impenetrable) user interface.

Belief in success

No one wants to feel like a failure. The person needs to feel reasonably sure that they can be successful at the action and not end up looking like an idiot. That's known as a feeling of *self-efficacy*.

If the person doesn't have a basic plan for how to take action, doesn't have the necessary resources for immediate action, or is hesitant because the action is daunting, those challenges are surmountable. But that means delay. It means that the person isn't taking the action *right now*. And that, from the perspective of behavior change, is a partial failure.

In the research community, we call these moments *decision points*:¹⁸ each time a person who has already decided to do something needs to stop to think about the next step in the action plan, gathering the resources to do it, or whether they personally have what it takes, you've created a new decision point. That decision point may pass smoothly, and after whatever frictions or obstacles are resolved, the person could take the action later. That's *if* the other preconditions for action are still in place when the person has the ability to act. However, the situation may change—other distractions could arise, the cost of action may go up, and so on.

It's because of these decision points that minor frictions can be so important to product design. For example, checking a box on a retirement form is, in itself, a trivial action. The cost is minor, especially in comparison to the benefits. And yet, we know that defaulting people in (and letting them opt out so they can express their preferences either way) tremendously boosts enrollment in retirement plans, in part because it reduces friction. The friction isn't important for the cost–benefit *evaluation*; rather, it creates a decision point at which people can become distracted, feel unsure about their mathematical ability and competence, decide they need to think about it more later, and so on.

LESSONS FOR BEHAVIORAL PRODUCTS

This step poses four possible barriers to action, which a good product must avoid. Products can readily help users by providing a clear action plan; specific plans grease the pathway to action. They can address self-doubt and anxiety about failure head-on as well by talking about other users who were successful, for example. Even minor frictions like the user not filling out another page of form fields or not having a clear understanding of the next step can create a decision point (a new point at which the individual reevaluates all of the previous steps and may get distracted or dissuaded).

Addressing deeper resource and skills gaps is trickier. With good user research, you can identify the resource constraints that particular user groups face and their current skills. You can then either accept that some users won't be served well by the product or plan around them.

Timing

You have their attention and the action is appealing and feasible. But *when* should you take the action? Why not do it a bit later? (And why not later, and later...) That's a major problem with many "beneficial" actions we want to take, like exercising, getting control of our finances, or planting a garden. We can always do them later. Even if we want to take the action, if our minds feel that there is something that's likewise desirable but more urgent, we're out of luck. We could take the action later. However, as we saw with ability barriers, if there isn't a decision that *now* is the right time to act, there's a problem. By the point the person does feel the timing is right, circumstances may have changed, and the person won't take the action for other reasons.

The decision of when to take action (i.e., its timing) can be driven both by a sense of clear and present urgency and by other, less forceful but still important factors. Urgency can come from a variety of sources:¹⁹

External urgency

In the US, we really do need to put in our taxes (or file for an extension) by April 15. Otherwise, the IRS comes after us. That's a true, external urgency; bad things happen if we deny the tax man his due.

Internal urgency

Changes in behavior may be urgent because we have a biological need that we can't ignore (hunger, thirst, etc.). However, these needs just don't apply to many actions and products. Negative mental states like boredom may provide a lesser, but still potent, urgency to act.

Similarly, we can decide that now is the right time to act (even when it doesn't feel strictly urgent) for a variety of reasons:

Specificity

Think about the two statements "I should save for retirement" versus "I should set up a retirement account on Thursday at 8 p.m., right after dinner." The latter feels more real, right? Simply by putting a specific time on an action can settle the issue of "when" to act. It also helps us *remember* to act then too!

Consistency

Another way to help us decide when to act (and to follow through on it) is to pre-commit to a specific time in the future, especially if we tell others about our commitment. That moves the action from the domain of something that we might do some time to an issue of personal consistency with our word. Our desire to be consistent with our prior

statements means that the right time to act is exactly when we said we'd act.

The decision to act at a particular time also arises, in part, from our motivation (emotional and deliberative) to take the action. An action that's really exciting and promises to be enjoyable also can feel more urgent and prompt someone to decide to act now rather than later. I've distinguished between the two concepts to analyze and address them separately, but in reality, there can be a significant blurring between them when the action is highly motivating (**Figure 2-2**).



Figure 2-2. When is eating chocolate cake urgent? (Top left: When you're hungry! Top right: When the waiter calls "Last chance for cake!" Bottom left: When it's New Year's Eve, and it's time to indulge a little. Bottom right: When you've promised yourself cake after finishing work.)

LESSONS FOR BEHAVIORAL PRODUCTS

You can think about the timing of action in terms of two factors: what the product actively does to *make* the timing ripe for action, and what it does to *align* with the times when a person is naturally inclined to take action. To make action urgent, products can use time-sensitive content like news, which is inherently timely (if you care about the content at all). NPR provides this, and so does Facebook (the latest items about your “friends”). Products can also construct urgency by creating pre-commitments or using specific dates for planned action.

Instead of making something urgent, products can wisely align themselves with events in a user’s life that already provide that urgency. The user may need to take a similar action as part of their work, for example, and the product can hook into and build on that opportunity. This is similar to the ancient Greek concept of *kairos* or the opportune time—it’s the product’s job to be there when the opportune time for action arises.²⁰

Products can use internal states like boredom to drive action, but those internal states are a double-edged sword. On one side, they can drive the particular target action, if the person thinks that the product will relieve the negative feeling. On the other side, they could drive a different action that also relieves boredom. Which do you think is more likely—your users surfing the Internet to relieve boredom, or playing with a mobile phone application that helps them plan healthy meals?

Experience

Prior experience is uniquely powerful and important when it comes to designing for behavior change. We’ve already talked about the role that it plays in forming one’s intuitive associations and reaction. It also shapes our understanding of the costs and benefits of the action, removing uncertainty and calibrating our evaluation to the actual pros and cons for us rather than the claimed or expected ones because of a company’s marketing efforts. It

helps determine whether we have the self-confidence to attempt the action. And due to confirmation bias and selective attention, it deeply shapes whether we pay attention at all.

For example, imagine two people who work at the same company and live in a stylish part of town that's a 30-minute walk to the office. One person might intuitively love the idea of walking to work because it makes them think about showing off their calves. Another person, in equally good shape and living in the same neighborhood, might intuitively hate it because they associate walking to work with growing up in poverty and not having the money to afford to drive. That's just how people are: we can look exactly the same on the surface, but our histories run through us and guide us in hidden and surprising ways.

I find it useful to call prior experience out as a separate factor in designing for behavior change to remind ourselves, as practitioners, that prior experience is fundamentally personal: it varies from user to user. It varies in ways that we don't necessarily know or understand. Even before someone uses our products, we are judged by the experiences each person has had in similar environments and with similar products. As they use our products, *history matters*. Regardless of how awesome the product is now, if we've already taught them that it's poorly built and designed, that will be difficult to overcome. Or, if people have previously tried to succeed at their behavior using your app and hadn't seen any progress, they're less likely to try again.

LESSONS FOR BEHAVIORAL PRODUCTS

Listen to your users, get to know them, and don't be surprised when two similar people behave differently in the exact same situation because of their prior experiences.

Plan for and, where possible, appropriately record, each person's history with your product—if they are the unlucky people who have faced a particular bug or unusable interface in the past, they will react differently, even when the product is fixed and perfect.

The CREATE Action Funnel

These six mental tests are prerequisites for most conscious action. They are what we need to *create action*.

Let's say you have a hundred people, all of whom are trying to better organize their email and respond to messages in a timelier manner (behavior change isn't always sexy!). All one hundred of them have set a reminder to go through their old messages and delete or respond to them on the path to a zero-message inbox. Suppose 75% of them will actually see and pay attention to the reminder despite their hectic schedules and clustered home screen (we all know that 75% is really optimistic, but let's work with that assumption). Unfortunately, some of them will quickly and intuitively close it because they don't want to deal with that annoying task. But let's say that 75% have a positive emotional reaction to the thought of cleaning their inbox (I know, wildly optimistic). The remaining people will think about it for a second, and 75% of them decide that cleaning up their inbox really is worthwhile.

Of those who have a favorable evaluation, a few of the remainder will realize they don't have enough mental energy and time to do it now and will postpone. But an impressive 75% of them still preserve. Among those who are left, an astounding 75% say to themselves, yes, absolutely, this is the most urgent and important thing I could do right now (the others let it wait because there are more pressing matters; they postpone). Finally, only 20% of people had previously failed at cleaning out their inbox and were discouraged because of that past experience or cleaned it out and it quickly became cluttered again, killing their enjoyment of it.

In the end, how many people of the one hundred who sincerely wanted to clean out their inbox actually did it? Just 18. And this is with wildly optimistic assumptions. Only a handful of the original hundred people actually respond to the cue and followed all the way through to action. That's the intention–action gap at work.

There's a nice study by Yale's James Choi and others that measured the intention–action gap in the context of saving for retirement (sadly it was

before CREATE existed): they found that only 10% of the people who declaratively committed to saving more actually did so.²¹ In other words, looking at what actually happens in practice, and the simple math of the many barriers that people need to pass to take action:

We shouldn't be surprised when sincere, motivated people do nothing.

Another way of thinking about this process is as a leaky funnel: a group of people start the process and some leak out at each step, leaving only a few of them who make it all the way (Figure 2-3). The funnel metaphor is a common one for salespeople, marketers, and product folks focused on converting potential clients into actual clients on their website.

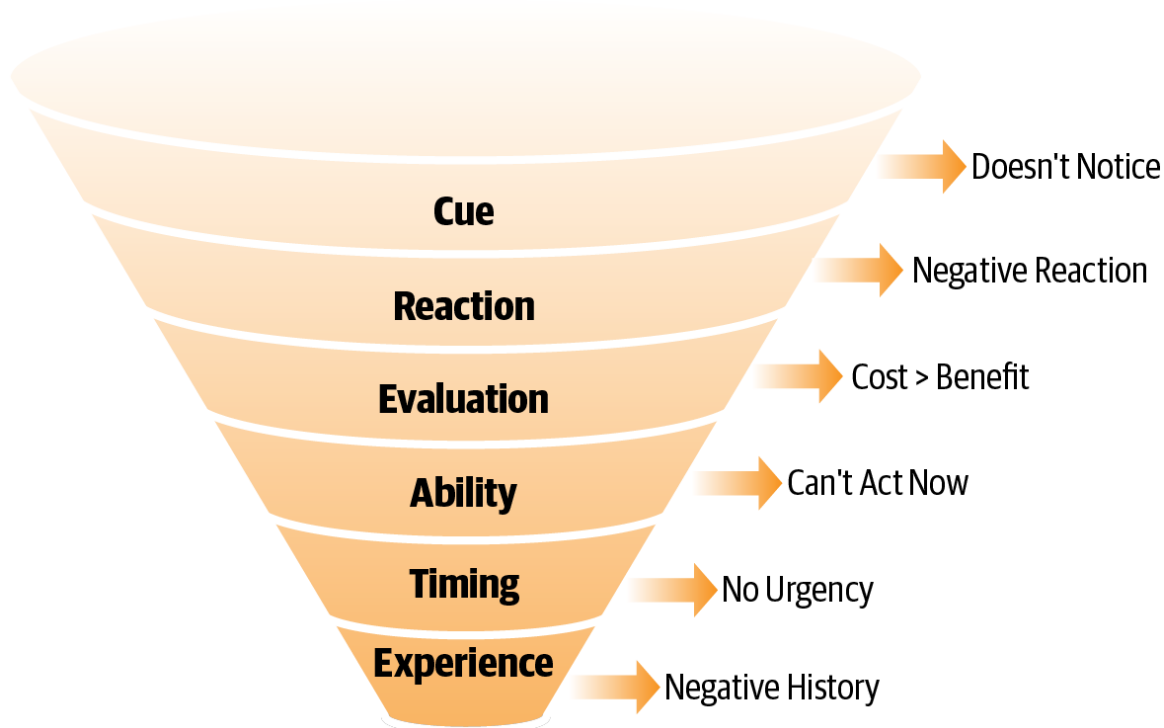


Figure 2-3. The CREATE Action Funnel: six stages that a potential action has to pass in order to be undertaken—people can drop out at every step along the way

Each section of the funnel has two leaky holes in it. On one side, people can reject the action (or the cue) because it's not valuable or urgent enough. On the other side, they can be distracted into doing something else—either

because they think of something else to do with the same effect (like surfing the Internet to relieve boredom instead of using a meal planning app) or because they are pulled into something completely different (like answering the phone).

For habitual actions, there's a simpler version of the funnel than for conscious action: Cue-Reaction-Ability. Habits effectively plug potential leaks in the conscious process, so there is very little drop-off where a conscious evaluation of the action would otherwise occur (unless the person is intentionally trying to be aware of their habitual actions and stop them) and where the assessment of urgency would occur.

Each Stage Is Relative

An important thing to remember about the funnel is that at each stage, the person only continues on if the action is more effective or *better than the alternatives*. There are always alternatives, including other cues that seek to grab us, other actions we're intuitively and consciously assessing, or other priorities that could be urgent.²²

From a product design perspective, that means you should consider not only how well the product guides the user through these stages but what else is competing for the individual's scarce time and mental resources. Removing distractions is a key part of structuring the individual's environment, which we'll return to in [Chapter 9](#).

It also means thinking about what the user is currently doing with respect to the action. Let's say the product seeks to promote dieting in order to lose weight. What is the person currently doing? Avoiding any thought about dieting? Trying something and failing? Asking friends for advice, but never acting? *Whatever the user is currently doing—that's the main behavioral competition for the product*. One shouldn't assume that products interact with a user who's a blank slate. Rather, the product needs to beat an existing behavior and do so at each stage of the CREATE Action Funnel. Each stage is relative to the alternatives.

The Stages Can Interact with One Another

I've presented a nice, neat model with six stages of processing. At a high level, that's correct. The details are much more complex, of course. One issue we haven't talked about much is how these different processes interact with one another.

While all six factors must be in place at some level for conscious actions, weakness in one area can be counterbalanced by strength in another area. For example, things that are really easy to do (like grabbing a bottle of olive oil out of the cabinet instead of an unhealthy hydrogenated oil blend) don't need to have significant conscious benefits (it might make you a little healthier) or positive intuitive feeling. This is one of the lessons that BJ Fogg incorporates in his Behavior Model²³—in economic terms, the factors are partial substitutes for one another.

In terms of the ordering, the funnel is a useful way to remember the activities the mind performs, but it isn't a perfect representation of sequence of processes. The first two steps generally come before conscious awareness, as shown, but sometimes intuitive reactions can occur after (or as part of) conscious deliberation.²⁴ For the next three, there's some evidence that the evaluation of an idea (e.g., value, timing, and ability) is pursued simultaneously by different parts of the mind,²⁵ and there can be some interaction between them. These complexities don't affect the central lesson, though: intentional actions need to pass all five stages, and there is often a significant drop-off at each step.

MORE EFFORT WON'T BUY YOU MUCH

One research lesson BJ Fogg built into his Behavior Model is that making the action easier or making the user more motivated won't buy you as much as you might think. His model has three factors: motivation, ability, and trigger. He defines motivation as pleasure/pain, hope/fear, acceptance/rejection (which has elements of an emotional reaction and a conscious evaluation), and ability as the lack of costs, roughly; he includes both intuitive and deliberative elements in both. The trigger part of his model corresponds to the cue in the model presented here (Figure 2-4).

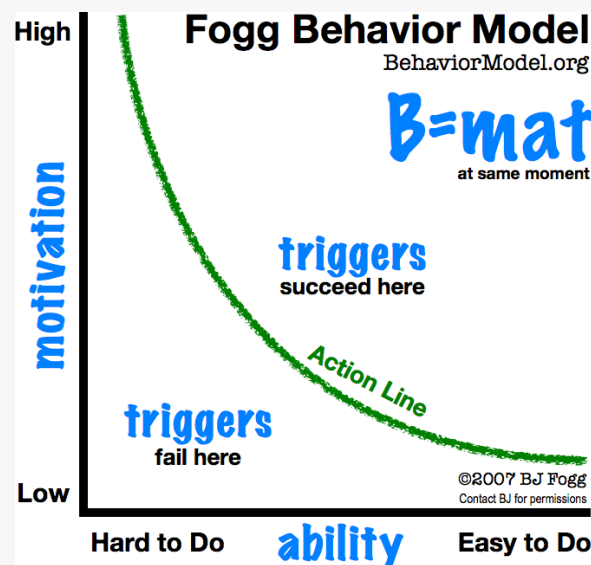


Figure 2-4. *Fogg's Behavior Model* showing the diminishing marginal returns that happen with extra motivation or increased ability to act

Fogg argues that in order for an intentional action to occur, you need all three elements. You can encourage an action by increasing a person's ability to act (decreasing costs) or increasing motivation.

In each case, the boost it provides to the person decreases as the action becomes easier and more motivating (i.e., when the action is very difficult, a bit of help to make it easier is very powerful). When the action is already easy, making it even easier isn't going to change behavior as much. In economic terms, this is known as diminishing marginal returns. It's a good practical lesson for product designers.

The Funnel Repeats Each Time the Person Acts and Each Time Is Different

People don't stay in the funnel over time. They drop out somewhere or they take the action—either way, the moment passes and they're out. Each time people think about taking the action, the process repeats: a cue leads them to think about it, they react intuitively, and so on. Thus, repeated actions require multiple passes through the CREATE Action Funnel. However, the funnel is subtly different each time. This is especially true when the person is deciding whether to take the action a second (or third, etc.) time.

Let's say you've gone to the gym for the first time. Here are some of the things that change from the first time you planned on going to the second time you're thinking about going:

Your relationship to the action has changed

You now know how the gym operates, where the equipment is, and so on. So your cost to use it decreases. But you also know more clearly whether you like going to the gym or not. So your intuitive reaction and conscious evaluation change too.

You've changed

If you did well your first time exercising, you have more confidence (increasing the perceived feasibility); if you weren't able to do the exercises you had hoped, you have less confidence.

Your environment may have changed

You may set reminders for yourself to go back to the gym (creating cues) or set expectations among your family that you will continue

going (creating urgency and increasing benefits). You may have friends at the gym who expect you to return.

WHAT'S STOPPING YOUR USERS?

Here's another way of thinking about the CREATE Action Funnel and the six stages that a potential behavior passes through: what *blocks* your users from taking action? What are the cognitive and practical barriers they face?²⁶

Problems with cues

The user forgets to act or has limited attention. Nothing in the environment reminds the user to act.

Problems with the intuitive reaction

The user doesn't trust the product or the company behind it. The action is unfamiliar and feels foreign.

Problems with the conscious evaluation

The user just isn't very motivated to act. The costs of taking the action are too high.

Problems of ability

The user doesn't know how to actually do it or doesn't have what they need to act. The user fears failure.

Problems of insufficient urgency

The user procrastinates and puts off the action until another day, which never comes. Or, other urgent issues block the user from the action.

Problems of prior experience

Your product reminds the user of a badly designed, difficult-to-use one they tried in the past. Even out of the gate, your product is negatively judged.

Whether you look at it from the perspective of what's required for action (the CREATE Action Funnel) or from the perspective of barriers to immediate action, the same factors are required for individuals to successfully act.

A Short Summary of the Ideas

Here's what you need to know. For someone to take a conscious action, six things must happen immediately beforehand:

1. The person responds to a *Cue* that starts their thinking about the action.
2. Their intuitive mind automatically *Reacts* at an intuitive level to the idea.
3. Their conscious mind *Evaluates* the idea, especially in terms of costs and benefits.
4. They check whether they have the *Ability* to act—if they know what to do, have what they need, and believe they can succeed.
5. They determine whether the *Timing* is right for action—especially whether or not the action is urgent.
6. They aren't turned off by a prior negative *Experience*—that overwhelms the otherwise clear benefits.

These six events can be visualized as a funnel, like a conversion funnel in ecommerce websites. If the person passes all six stages, then they will act.

A quick way to remember the preconditions for action is the acronym CREATE: Cue, Reaction, Evaluation, Ability, Timing, Experience. At each step, people drop off because they fail to see the cue, don't consider the action worthwhile, or don't find it urgent. Each step of the way, they can also become distracted and diverted into taking other actions.

If the action requires conscious thought (System 2), our minds engage in all six steps. If it doesn't require conscious thought (System 1 only), then part of the process is short-circuited, and the cue, reaction, and ability are most relevant (CRA).

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- 1 New York Federal Reserve (2019); Kantrowitz (2018)
 - 2 This case study derives from phone interview and subsequent email exchange with Kristen Berman of Irrational Labs.
 - 3 I'm thankful to BJ Fogg for stressing that behavioral prerequisites must occur at the same time. It's something he talks about in the Fogg Behavior Model (Fogg 2009a) and that sets his work apart from other models of behavior and intentional action—which too often focus on the raw materials of behavior (resources, motivation, etc.) but not the *timing* required for action.
 - 4 Service et al. (2014); Michie et al. (2011)
 - 5 Chabris and Simons (2009)
 - 6 Aka the *cocktail party effect*. My thanks to Peter Hovard for suggesting its inclusion here.
 - 7 For lack of a better term, I'm using *thinking* to refer to preconscious sensory processing and reactions and, later on, conscious thought.
 - 8 I'm indebted to Nir Eyal for reminding me of the importance of internal cues and showing me how products can move from relying on external cues to internal cues over time.
 - 9 Eyal (2014)
 - 10 Pomeroy (2013)
 - 11 Zajonc (1968)
 - 12 For summaries and links to the broad body of research, see Wikipedia's articles on [operant conditioning](#) and [classical conditioning](#).
 - 13 See Litvak et al. (2010) for a summary.
 - 14 See contrast effects in psychology; for example, see [Cash et al. \(1983\)](#) for an early study.
 - 15 Many thanks to Keri Kettle and Remi Trudel for their feedback on an early draft of this chapter, and for bringing up the intuitive needs assessment and search for alternatives.

- 16 See Dean (2013) for a great overview.
- 17 In this general concept of *ability*, I'm combining disparate elements from the self-efficacy literature (Bandura 1977), work on goals and implementation intentions (Gollwitzer 1999), and “weak” rational choice models of resource constraints, like the Civic Volunteerism Model (Verba et al. 1995). You may be familiar with the term “ability” from the Fogg Behavior Model. I’m using it in a different way here—as the perceived and actual capability of the individual to take the action. Fogg uses the term for how “easy” or “simple” the action is; i.e., the lack of costs (Fogg 2009a).
- 18 See Soman (2015) for a nice discussion of this in a business context.
- 19 Beshears and Milkman (2013). Also, these factors serve as cues as well as urgency. They grab our attention and provide urgency to act on it now. Hat tip to (h/t) Paul Adams.
- 20 Many thanks to BJ Fogg for introducing me to the concept of Kairos.
- 21 Choi et al. (2002)
- 22 This concept of ever-present competition is found in social marketing (e.g., Grier and Bryant 2005) but is considered in few other behavior change perspectives.
- 23 Fogg (2009a)
- 24 There can be an interplay between the deliberative and intuitive minds, as our conscious attention shifts, potentially intervening in an automatic process, or relinquishing control back to automatic processes. See Wood and Neal (2007) for a discussion of some of these scenarios.
- 25 Brass and Haggard (2008)
- 26 Many thanks to John Beshears and Katy Milkman for presenting the idea of basic cognitive barriers to action (procrastination, forgetfulness, and a lack of motivation) when they came to speak at the Action Design Meetup, April 2013 (Beshears and Milkman 2013).

Chapter 3. Stopping Negative Actions

Breaking: “FBI Agent Suspected in Hillary Email Leaks Found Dead of Apparent Murder-Suicide”

It’s on NPR.¹ It’s been talked about all over the internet. It’s also completely fake.

Such news stories are often really interesting: they excite our emotions, confirm our suspicions, and help us see that other people like us are out there who “get it.” In other words, they build on nondeliberative “fast” thinking from System 1, deploy confirmation bias, and leverage social proof and other behavioral techniques. It’s no wonder that people trust these fake news stories, forward them to others, and start to integrate them into their view of the world.

What can be done about it?² Researchers Sander van der Linden and Jon Roozenbeek developed a psychological vaccine called Get Bad News, building on a body of research known as cognitive inoculation theory. Get Bad News is an online game in which players try to create a fake news website and build a base of loyal followers by using the tactics of actual fake-news purveyors. By experiencing fake news tactics in a controlled environment where participants can see how they work without being duped by them, people can become inoculated against the real thing. In van der Linden and Roozenbeek’s research, they tested the platform with 15,000 participants and found that people could better spot (and resist) fake news because of it.³

Qatar’s B4Development Foundation and Nudge Lebanon partnered with van der Linden and Roozenbeek to apply the same approach to radicalization to help people who might be recruited into terrorist or other extremist organizations recognize the tricks recruiters use and resist them.

Together, they developed the game Radicalise, in which the participant plays the chief recruitment officer for a fictitious extremist organization. Players use social media to hook their audience to get them interested in the organization and its narrow interpretation of the world and to start to hate others who oppose them. Individuals have been randomly assigned into a treatment (using the game) and control group (not using it), and the team tested how well participants could identify the manipulateness of sample WhatsApp posts. While the project is ongoing, it appears that the treatment group is better able to identify manipulative messages and to identify people who might be susceptible to recruitment.

Radicalise, and the fake news research that it builds on, is a promising example of how behavioral techniques can be used to hinder bad decisions and actions in the future.

Sometimes, users of our products can't succeed at their goals because they make a series of bad choices that they wouldn't otherwise want to make, or act out previously learned bad habits. In both areas, lessons from behavioral science could help, starting with an understanding that the same environment facilitates desirable and undesirable choices alike. For example, we're far more likely to overindulge when alcohol is present and available than when we need to go search for it. Similarly, we need more self-control and are more likely to falter when others around us are doing so.⁴

Often, people think that to resist temptations, they need an iron will. However, one of the most effective ways to help block rash decisions or actions is to intentionally design the environment to hinder them. Researchers call this *situational self-control*.⁵ For example, we can help someone be less subject to peer pressure to overdrink by relocating the person away from binge drinkers or surrounding the person with more responsible drinkers.

The process is effectively CREATE in reverse; you can think about it in four steps:

1. *Identify how CREATE supports the negative behavior.* What Cues it? What causes the person to have a positive Reaction and Evaluation? What makes the person Able to act immediately and prioritize it as Timely over other things?
2. Come up with strategies *to change the environment to create obstacles*—adding friction, removing cues, etc.—and to lock them in over time.
3. *Double-check what type of behavior it is.* If it is habitual (nonconscious, without intention), there are extra techniques you can use (discussed in “**Changing Existing Habits**”). If it is an effortful conscious choice, then special attention should be paid to supporting the Evaluation.
4. *Set up a feedback loop* to check in and see if you are being successful at stopping the behavior. We need feedback loops to stop behaviors just as we do when we’re starting them because we’re just not that good at seeing averages and trends in our behavior over time.

Step 2 in particular deserves extra attention. How do we *create* obstacles? We’ve spent our time thus far talking about how to remove them. But the same logic can work to add them.

Using CREATE to Add Obstacles to Action

Let’s look at each stage of the action funnel to see how to interfere with problematic actions. Consider someone who checks their phone frequently and finds themselves ignoring their family and friends. The behavior might be an automatic habit, or it might not, depending on how long it is continued in the same context—we’ll start by looking at the nonautomatic version of phone checking:

1. Cue

Just as the secret to getting attention is to put the item in the line of sight, to avoid attention, *get it out of sight*. For my wife and me, one way we've tried to avoid checking our phones is to remove them from our room so we have fewer distractions from spending time with each other.

2. *Reaction, especially our social reaction*

How can we use our innate social sense to help us avoid bad behaviors? One way is to intentionally surround ourselves (at the moment of temptation or otherwise) with peers who *don't* act that way or actively disapprove. Another way is to avoid being around friends who *do* engage in it and trigger us to do so as well. In the context of phone checking, that means seeking out people who have gone cold-turkey on their phones, or intentionally spending time with that annoying person who always gives us a hard time when we're using our phone in public.

3. *Evaluation*

Think about the consequences of the action and see how to make those consequences more prominent, or make the benefits of not doing it more vivid and real. It doesn't have to be the most important aspects of the costs and benefits (in the long term)—but rather something you can focus on and change your framing of it in the moment. For example, add a timer on the phone that shows how long it's been since you've unlocked it. You could also significantly increase the cost of action by *making it more difficult*. On my phone I have the bad practice of checking the news many times a day. To make it more difficult, I

canceled my online news subscriptions and removed the apps from the phone.

4. *Ability*

We can also add *small frictions* that cause pause without fundamentally changing the big picture costs and benefits of an action. I moved some of my most distracting apps (those I felt I couldn't simply delete) to a folder labeled "distractions." It only takes a second to overcome, but it slows me down, giving me more opportunity to think about what I'm doing. I set a simple password on Amazon Video for the same reason.

5. *Timing*

How can we remove a sense of urgency from the parts of our lives that distract us from our goals? I'll admit I struggle with this one greatly, and I don't have an answer that has worked for me. I try to increase the urgency of things that matter to me—like setting a deadline for myself to write this book—to crowd out other things. That helps, but not as much as I'd like. Many people have found mindfulness practice helpful to remove the false urgency from life's many distractions.

6. *Experience*

If someone wants to hinder a negative behavior (and that's the only type of behavior we want to help stop), their prior experience is generally pretty negative.

Changing Existing Habits

Sometimes helping people take action requires intentionally stopping a habit. For example, at some point, improving fitness through exercise means not just exercising more but also sitting less. And that means overcoming an existing habit. Unfortunately, it can be extraordinarily difficult to stop habits head-on. Brain damage, surgery, and even Alzheimer's disease and dementia sometimes fail to stop habits, even as other cognitive functions are severely impaired.⁶ BJ Fogg, for example, argues that stopping existing habits is the hardest behavioral change task to undertake.⁷

Why are habits so difficult to change? First, remember that habits are automatic and not conscious. Our conscious minds, the part that would seek to remove them, are only vaguely aware of their execution;⁸ we often don't notice them when they occur, and we don't remember doing them afterward. Across dozens of studies on behavior change interventions, researchers have found that the conscious mind's sincere, concerted intention to change behavior *has little relationship to actual behavior change*.⁹

Second, it's because habits never truly go away—once a habit is formed (i.e., the brain is rewired to associate the stimulus and response), it doesn't normally un-form. It can remain latent or unused, but under the right circumstances, that circuitry in the brain can be activated and cause the habitual behavior to reappear.¹⁰

Another way to think of habit cessation is this: if stopping bad habits were easy, we wouldn't need so many darned books on everything from stopping smoking to dieting.¹¹ Nevertheless, we can draw lessons from the literature on habit formation and change, which can save product teams needless pain and suffering. There are four main options that product teams can take to handle an existing habit:

Attention

Avoid the cue.

Reaction

Replace the routine by hijacking the reaction.

Evaluation

Cleverly use consciousness to interfere, including using mindfulness to avoid acting on the cue.

Ability

Crowd out the old habit with new behavior.

In each case, the person doesn't engage in a direct confrontation to simply suppress the habit. That takes constant willpower, which is finite and often unsustainable.

Attention: Avoid the Cue

The cue signals the brain to engage in the problematic behavior; one way to stop a habit is to avoid the cue. For example, in addiction counseling, counselors advise addicts to change their environment so that they don't encounter the things that remind them to act. If you always stop for a drink when you see the bar on the way home, then change your route home so you don't see the bar anymore.¹²

Designing a *product* to help people avoid cues is especially tricky. First of all, most cues for bad habits are, by definition, outside of the behavior change product. People use the product in order to change the habit—the product didn't cause the bad habit. So, the product must help the person avoid the cues themselves; it must provide guidance and instruction. And the individual must first know what the cues are—and be able to successfully avoid them.

Second, because the routine is outside of the product, the application usually won't know if the person has engaged in the behavior. It's up to the user to report falling off the wagon—which is doubly difficult. External monitoring systems are required—like the breathalyzers that alcoholics

install in their cars to avoid driving drunk. Much more is required in the case of chemical addictions like alcoholism, but we can learn from these efforts as we design products to stop less intractable habits.

While this route is clearly challenging, there are products that have successfully done it. One example is **CovenantEyes**—software that helps people who are struggling with sexual addiction or want to avoid the temptation before a habit is formed (see **Figure 3-1**). It helps users avoid cues (by filtering out sites with explicit content) and/or automatically monitors web usage to inform accountability partners of when the person does access pornography.

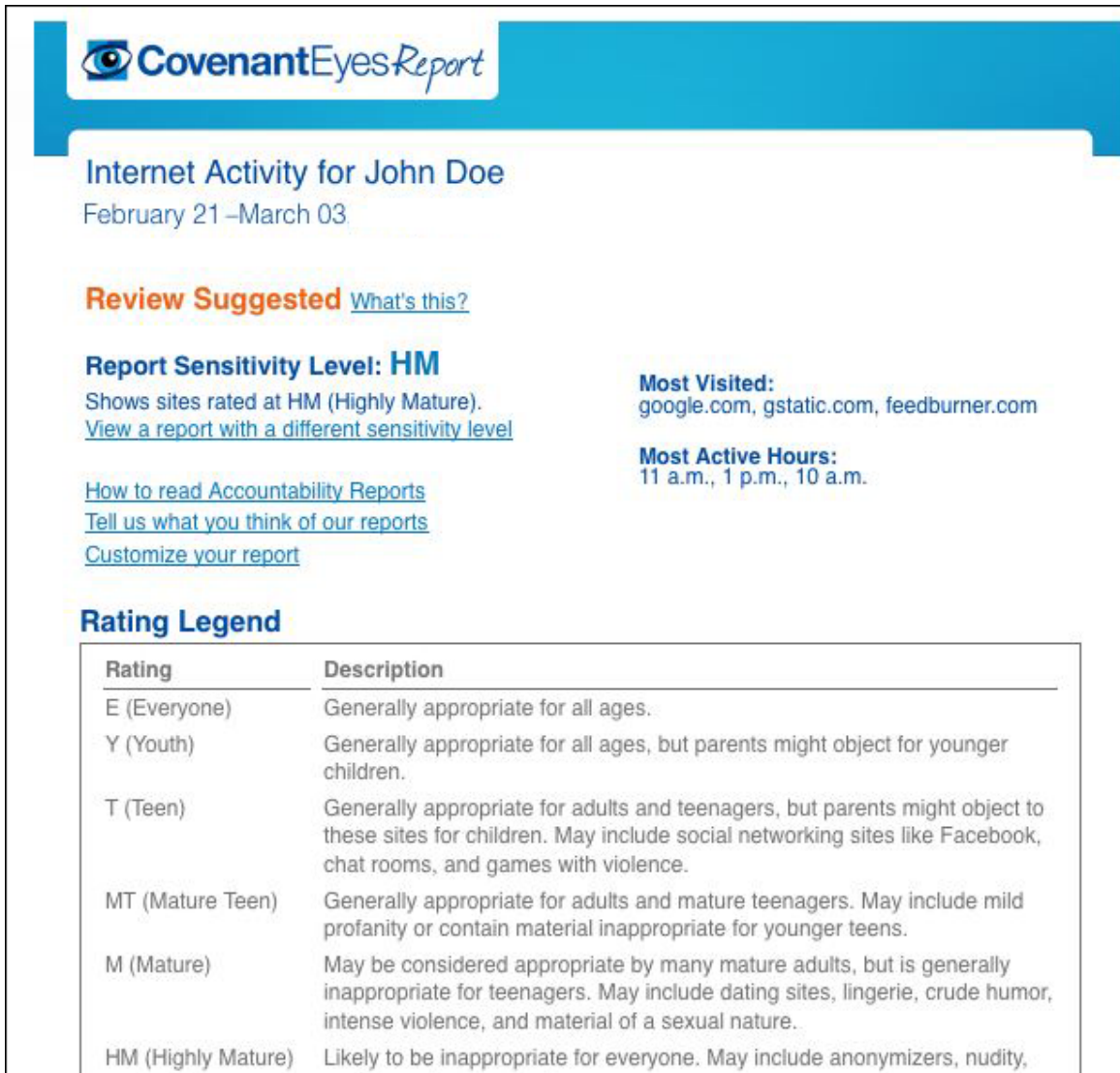


Figure 3-1. CovenantEyes, an application to stop the habit of viewing sexual material online, via filtering and automatic monitoring

Reaction: Replace the routine by hijacking the reaction

The other strategy that products can use to change a bad habit is to transition an existing cue and reward to a different (more beneficial) behavior. In *The Power of Habit* (Random House, 2012), Duhigg describes two elements that are needed: routine replacement and a real belief that the habit can change.

Routine replacement works by hijacking the cue and the reward and inserting a different routine between them. He uses the example of taking a snack break when you're not really hungry. The cue may be that you're

having a down moment at work or watching a commercial on TV. The reward would be the relief of (momentary) boredom and the pleasant crunching sensation of the snack. To hijack this process, one needs to:

1. Identify the trigger and the reward (if appropriate).
2. When the trigger occurs, consciously engage in a different routine that provides a similar reward (like doing a crossword puzzle during commercials).
3. Continue that conscious switching of routines until the new habit is instilled.

The process of consciously replacing routines is also known as *competing response training*. It is used in the treatment of people with Tourette's syndrome (involuntary tics) and has shown dramatic results in experimental testing.¹³

For especially difficult habits, like smoking and drinking, swapping in a new routine isn't enough, though. The new reward is never quite like the old one. Swapping can handle everyday behavior, but when times are tough, people can be immensely tempted to "fall off the wagon." Something else is needed to get through those dark times and back to the day-to-day humdrum that they *can* handle. That something else can be faith that the hard times will pass. It can be a religious faith, a personal faith in themselves, or a faith in others that pulls them through. Either way, it's an internal narrative that things will get better.

How does routine replacement work in practice? One of two ways. First, you can ensure that the product itself is present at the moment when the cue normally occurs. At that moment, it would remind or entice the user to do the new routine instead of the old one. After the routine is done, it would reward the user—or encourage them to reward themselves.

The other route is trickier and is needed when the product *isn't* present when the user encounters the cue. As with avoiding the cue, the product must advise and prepare the individual for the moment of temptation and find some way of tracking what action the person took. **Changetech** has an

intensive program of support and tracking that accomplishes this, with more than four hundred points of contact with individuals during their smoking cessation program. And its method has shown positive results in randomized control trials.¹⁴

An example of in-the-moment hijacking of habits that we're all familiar with is shopping in brick-and-mortar stores with a smartphone:

1. *Cue*

See a camera, computer, or something else you like.

2. *Old routine*

Pick it up, go to the cash register, buy it.

3. *New routine*

Look it up on the phone, compare price (usually lower), and buy it.

4. *Reward*

Feel great about saving money, imagine yourself using the cool camera, and so on.

This habit hijack is killing brick-and-mortar stores, of course.

Evaluation: Use conscious interference

Our big brains are really good at blocking our own autopilot; properly deployed, they can interfere with habits in progress *without requiring direct willpower to overcome the action*. Thinking = bad, for a habit at least. In sports, masters of their game sometimes “choke” because they consciously cut into a process that normally runs on autopilot, and this happens in any field of mastery.¹⁵ To interfere with a habit, think about it. Look especially for what triggers it, then closely examine the routine that's normally automatic—just by thinking about it (consciously), we can interfere with its smooth execution.

Products that do this should be present at the time of action and can grab the user's conscious attention to their behavior. The Prius is well known for functioning this way. The car's consumption monitor provides ongoing, immediate feedback about the car's gasoline consumption. This in-the-moment feedback can break people out of their existing driving habits by making them consciously aware of what's going on, causing them to use less gasoline, aka *the Prius Effect*.

In order for this approach to work, like all habit intervention (and habit formation) approaches, it must be voluntary. If someone doesn't care about mileage or finds the car's consumption monitor annoying, they won't listen to it. It starts with the conscious choice to act.

Evaluation: Increase attention with mindfulness

Another, subtle way to overcome bad habits is by employing mindfulness. Mindfulness is a concept used in Buddhism to refer to awareness of the present moment and its experiences without judging or trying to control them. It's a mental state of openness and acceptance of events and sensations as they occur. Mindfulness-based therapies are increasingly popular in the treatment of mental conditions such as acute stress, anxiety, and depression. Similar to mindfulness meditation in Buddhism, these therapies entail an intentional focus on the present moment without interference or judgment.¹⁶

By bringing into conscious awareness the cues that trigger habitual behavior, it's possible to be aware of the trigger without acting on it. As someone who personally has practiced this approach, it's fascinating and nonintuitive. You don't fight the habit; instead, by noticing the trigger (and the urge to respond), you sap the habit of its power. You don't need to respond to the habit when you're being mindful.

Mindfulness has been shown to be powerful in a variety of contexts; for example, in limiting undesired but habitual binge drinking.¹⁷ A number of apps such as **Headspace** and **Calm** support mindfulness to reduce stress or increase focus, though do not target habit change in particular.

Ability: Increase the power of other behaviors

Another way of approaching habit change is to crowd out the old habit with new behaviors. In this method, you *focus on doing more of what you want instead of less of what you don't want*. The consequence is that you don't have the time or energy to do the prior action anymore—the relative ability decreases.

For example, think about someone who is in bad physical shape, spends lots of time watching TV, and has bad eating habits. The person starts to go to the gym to exercise more (creating a new habit). As they go to the gym, they meet new people and enroll in exercise and cooking classes with them. Slowly, the amount of time available to watch TV decreases. The person simply isn't at home as much, which leads them to avoid the old cues to watch TV. Also, because of the cooking class and new ways of eating and cooking, they simply don't have the hunger and opportunity to use their old eating habits, which are slowly being replaced.

Naturally there are multiple forces at work in their life, such as changing self-identity and changing social norms. However, as the structure of their daily life changes, the old habits fade away—not through a direct assault but because other things are taking up their time and satisfying their hunger pangs. This only works if they get far enough down the path of habit change—and don't quit going to the gym soon after signing up, as so many people do. The initial choice to push ahead before the habit is formed is a conscious one.

Rushed Choices and Regrettable Action

If your users face an important decision that they're likely to rush or not properly think through, what obstacles can interfere with that choice and hasty action? Rushed choices are different than intention–action gaps since the cue is already there (they are making the decision) and the rush often comes from the person deciding based on their intuitive reaction instead of a careful deliberation or evaluation.

Behavioral researchers Soll, Milkman, and Payne put out a guide to debiasing a few years ago that provides some excellent pointers on how to do this.¹⁸ The core lesson: you either change the person or the environment. To change the person you'd:

1. Educate them beforehand, so they'll have the information they need for the decision ready when the time comes. For example, teaching people about the risks of assault at campus parties with heavy drinking.
2. Provide an effective rule of thumb, so they can still make a quick decision—but it will be based on a rule that gets them to a similar place as a long thought through deliberation. For example, the rule that you should never take on a mortgage payment over 28% of your income.
3. Teach them to use more formal decision-making aids, like a checklist for the many factors a pilot needs to consider before landing a plane.

To change the environment you'd:

1. Slow people down—especially with friction. If people are making a poor intuitive choice, add friction to switch them from System 1 thinking to System 2 thinking (i.e., change the *Ability* to act immediately).
2. Lessen the consequences of their biases. For example, a default contribution rate on a retirement plan both makes it easier to take the action of contributing (as we've discussed earlier) and makes it easier to make the good choice about how much to save—as long as the default is appropriate for the person.

In financial services, disclosure rules are meant to slow people down and give them time to think through a big commitment—like a mortgage. Unfortunately, people eventually learn to skip over these slow-down devices. For that reason, stronger slow-downs are sometimes required. In

Denmark there is a 48-hour *cool-off* period before someone can receive a payday loan to reduce the incidence of impulsive borrowing.¹⁹ There's **active debate in the United States** on requiring a 24-hour waiting period before purchasing a gun.

One of the ways to avoid rushed choices and regrettable action is to try to change a person's time perspective. As already mentioned, we can't focus on all points in time at once, and we give undue focus to the present and near-present costs and benefits of an action. That makes perfect sense when we're facing a tiger; less so when deciding whether or not to have another drink or watch another episode. As Seinfeld nicely captured it: when we're deciding on that nighttime drink or episode, we're Night Guy, thinking only about what's happening now, and not the suffering Morning Guy will endure because of our choices.²⁰ And so, making Night Guy (or whenever the present-tense mind is focused on) connect with the future can help avoid bad choices.

Finally, another technique is to bring attention to otherwise ignored aspects of a decision. In an experiment conducted at the Financial Conduct Authority of the UK, researchers found that a pop-up message of warning successfully turned participants' attention to an important detail of the purchase: the fees they were paying.²¹ And, in one of my favorite experiments of all time, Dan Egan at Betterment tested an in-the-moment intervention to avoid a hasty investment sale. When people intended to sell an investment, he popped up a box reminding them of the tax consequences of the sale, slowing them down and having them look at the decision from a different perspective.²² If there is one thing that people hate more than losing money on an investment, it's paying taxes!

WHERE TO LEARN MORE ABOUT IMPROVING COMPLEX DECISIONS

Even when people are thinking carefully, they don't always make good choices and may want help. They may be overwhelmed with the complexity of the choice, unable to do the necessary math in their heads, etc. In other words, the Evaluation phase can be a problem. These types of challenges aren't the focus here (since we're focusing specifically on *behavior change*, not how to do mental math), but at least I can point you in the right direction. Two good places to start:

- Russo and Shoemaker's *Winning Decisions* (2002)
- Hammond et al.'s *Smart Choices* (2002)

Judgment and decision making is the umbrella term of this area of research.

A Short Summary of the Ideas

Users aren't a blank slate. For example, in order to spend more time with their families, they often will need to spend *less* time with their phones. If they want to eat healthier, *not* eating a tub of ice cream at a time would help. And so, often when designing for behavior change, we need to think about how to stop or hinder existing (negative) actions, as well as starting new ones.

Here's what you need to know:

- Self-control often isn't enough on its own—if people want to stop a negative behavior and struggle, exhorting them to push harder is likely to be ineffective, condescending, or both.
- Instead, we can help people use what's known as *situational self-control*; just as we can shape an environment to encourage action, we can shape an environment to slow down rash decisions and

interfere with undesirable habits and behaviors. We control the situation (environment) in order to control behavior.

- The process uses the CREATE framework in reverse:

Identify how CREATE supports the negative behavior.

What Cues it? What causes the person to have a positive Reaction and Evaluation? Etc.

Find ways to change the environment to create obstacles

Add friction, remove cues, etc. For example, add an automated lock on the phone after a certain amount of screen-time.

Set up a feedback loop

To check in and see if you are being successful at stopping the behavior, and adjust accordingly.

- If the behavior is habitual, here are specific techniques to focus on:

Avoid the cue altogether

For example, if seeing the bar triggers stopping and getting a drink, avoid seeing the bar.

Hijack the cue and trigger a different behavior

Build up a new (positive) habit that uses the same cue. For example, see the bar: call spouse and talk about your day.

Deploy intentional mindfulness

Acknowledge and be aware of the trigger without overtly exerting willpower.

While the focus of this chapter and this book is on behavior, there is also literature on improving the mental process of decision making: to be more careful, for example, in selecting mortgages or jobs. See Soll et al. (2015) for a short summary of that research, but major approaches include:

- *Educate* beforehand so people will have the information they need for the decision ready when the time comes. For example, provide detailed training on the mortgage process.
- *Provide a rule of thumb* so they can still make a quick decision. For example, the rule that you should never take on a mortgage payment over 28% of your income.
- Use a *formal decision-making aid*. That is, train them to use a mortgage evaluation tool (instead of training them to learn how to evaluate it themselves).
- *Slow people down*—especially with friction. Time-delayed disclosures (HUD-1 forms) are an example in the U.S. context.
- *Lessen the consequences* of their biases. For example, by using regulations to limit the rates and fees that lenders can charge borrowers, especially uninformed or first-time borrowers.

1 This statement is true but misleading. It is on NPR; it's just in [an article that critiques fake news](#).

2 This case study derives from a phone interview and subsequent email exchange with Fadi Makki and Nabil Saleh, both of B4Development and Nudge Lebanon.

3 [Roozenbeek and van der Linden \(2019\)](#)

4 Hofmann et al. (2012)

5 Duckworth et al. (2016)

6 See, for example, [Eldridge et al. \(2002\)](#).

7 [Fogg \(2009b\)](#)

8 Dean (2013)

9 Webb and Sheeran (2006)

- 10 And, in cases of chemical addiction, there are added layers of difficulty that make defeating addiction beyond the scope of this book. For example, drugs can cause brain's receptors for key neurotransmitters to change, requiring additional levels of stimulation to attain the same experiences that were had before the drug was used. While many of these techniques are also used for addiction, I don't try to cover the extensive research on addiction.
- 11 They're difficult to change on their own, and, of course, there are also other factors associated with the habit that make it more difficult to change—like peer pressure, chemical addiction, etc.
- 12 Changing circumstances is used widely beyond addiction counseling as well. See Wood et al. (2005) for research on this method.
- 13 Piacentini et al. (2010); Dean (2013)
- 14 Many thanks to Sebastian Deterding for mentioning this example. For more information, see [Brendryen and Kraft \(2008\)](#).
- 15 [Baumeister \(1984\)](#); Gallwey (1997)
- 16 Hofman et al. (2010); Shapiro et al. (2006)
- 17 [Chatzisarantis and Hagger \(2007\)](#)
- 18 Soll et al. (2015)
- 19 H/t Paul Adams. Payday loan companies know this would deter people, so they renamed and retargeted their products to get around the law ([Toft 2017](#)).
- 20 See Seinfeld's discussion with Jay Leno about [Night Guy](#).
- 21 Hayes et al. (2018)
- 22 See [Egan \(2017\)](#), under "Tax Impact Preview" and prior work on tax avoidance, Sussman and Olivola (2011).

Chapter 4. Ethics of Behavioral Science

Researchers at Princeton developed an automated tool for searching websites for dark patterns: “user interface design choices that benefit an online service by coercing, steering, or deceiving users into making unintended and potentially harmful decisions.” In analyzing 11,000 websites, they found 1,841 dark patterns. They even found 22 third-party companies who offer “dark patterns as a turnkey solution”; in other words, digital manipulation as a service.¹

*The term dark pattern was coined by UX specialist **Harry Brignull**, who categorizes 11 different types, from confirmshaming (guilting the user into opting in) to privacy Zuckering (you can probably guess). He hosts a “Wall of Shame” of companies clearly trying to trick their users and demonstrates how Amazon makes it nearly impossible that someone will discover how to cancel their account, which Brignull nicely calls a “roach motel”: you can enter, but you can never leave.*

Sadly, cases of such deceptive techniques aren’t hard to find in practice. A recent New York Times exposé, for example, detailed how the company thredUP generated fake users to make it look like other people had recently purchased a product and saved money to encourage real customers to make a purchase themselves.²

There’s a rightful backlash against the application of psychology and behavioral techniques in the design of products and marketing campaigns. Products that seek to manipulate users—to make them buy, get them addicted to our products, or change something deep and personal about their lives like emotions—have started to gain the rightful negative scrutiny they deserve.

In April 2019, Senators Mark Warner of Virginia and Deb Fischer of Nebraska introduced legislation they called the Deceptive Experiences to Online Users Reduction (DETOUR) to make it illegal for large online services to:³

- “Design, modify, or manipulate a user interface with the purpose or substantial effect of obscuring, subverting, or impairing user autonomy, decision-making, or choice to obtain consent or user data”
- “Subdivide or segment” users into groups for “the purposes of behavioral or psychological experiments” without informed consent
- Operate without an independent review boards for the approval of behavioral or psychological experiments

The effort by Senators Warner and Fischer is clearly targeted at social media, search, and ecommerce companies, which have been some of the worst offenders in terms of data privacy and tricking individuals into giving their consent to data usage. But the work on dark patterns, and sadly, the daily experience of anyone with an email inbox or mobile phone, shows that the deception and abuse don’t stop there. And we’re kidding ourselves if we think we’re not part of it.

Thus far in this book, we’ve talked about how to help our users succeed at their own goals; this chapter takes a different angle and seeks to accomplish four things:

- To show the extent of unethical manipulation of users
- To think about where things have gone wrong
- To show that each of us is as likely to be unethical as anyone else given the right circumstances
- To look at ways to clean up our act

Digital Tools, Especially, Seek to Manipulate Their Users

The Princeton study nicely quantifies how common dark patterns can be—but their analysis focused specifically on shopping sites in 2019. Is this a widespread problem? There don't appear to be other large-scale quantitative studies (yet), but many government and watchdog groups have analyzed the practices of major digital companies and found them to be, nearly across the board, manipulative. For example, in a 2018 report, the Norwegian Consumer Council analyzed how Facebook, Google, and Windows discouraged people to exercise their privacy rights. Google is under fire for obtaining consent for location tracking through deception. ProPublica shows how Intuit has tricked people into paying for tax filing—even when it is free. Apple has even changed its App Store guidelines to hinder apps from tricking people into subscriptions.⁴

And while these issues have gained prominence recently, they clearly occurred before. Do you remember when people trusted Facebook, or at least didn't think it was evil? One of the initial chinks in their armor came with an experiment they ran to manipulate user emotions, reported by the *New York Times*, *Wall Street Journal*, and others.⁵ The *Times* story started with “To Facebook, we are all lab rats,” and went downhill from there:

Facebook revealed that it had manipulated the news feeds of over half a million randomly selected users to change the number of positive and negative posts they saw. It was part of a psychological study to examine how emotions can be spread on social media...

I wonder if Facebook KILLED anyone with their emotion manipulation stunt. At their scale and with depressed people out there, it's possible,” the privacy activist Lauren Weinstein wrote in a Twitter post.

This study was part of a collaboration with academic researchers and was widely condemned—even if it was misunderstood and blown out of proportion.⁶ After that study came Cambridge Analytica, and many more high-profile stories of broken trust with Facebook and other major

companies. LinkedIn paid millions of dollars in a class action lawsuit because of its use of trickery to use people's contact lists.⁷

I'd challenge the reader to think of three major digital companies that *don't* try to trick you into giving consent to using your data, sign you up for things you don't want, or encourage you to binge on their products despite your better judgment. As the *Financial Times* nicely summarized, for many companies "manipulation is the digital business model."⁸ And despite the negative attention recently, the unfortunate revelations continue; for example, Flo Health's app reported to Facebook about its users' intention to get pregnant and the timing of their periods, without informing users.⁹

But it's also important to note that it's not a problem of "those big companies over there." In the broader product development, marketing, design, and behavioral science communities, we brag quite publicly about the ways in which we can manipulate users into doing what we want. At numerous behavioral marketing conferences, for example, speakers talk about their special expertise at changing user behavior by understanding the psychological drivers of customers and using that to drive a desired outcome. They often throw around some behavioral techniques like peer comparisons and claim a tremendous rate of success for their clients.

Similarly, marketing and design companies tout their ability to change user purchase behavior, without any concern or discussion about whether the products being sold are appropriate or wanted by the end user. One example (among many) from the field is that of System 1 Group, a marketing agency named after a core psychological (and behavioral) model of decision making, which advertises on their website how they use "behavioral and marketing science to help brands and marketers achieve profitable growth." As written in their promotional book, *System1 Unlocking Profitable Growth*,¹⁰ "Designing for System 1 (i.e., avoiding conscious thought among customers) can also boost the profitability of promotions and point-of-sale materials. To increase rate of sale, shopper marketing must help people make quicker, easier, more confident decisions."

System 1 isn't, I believe, a particularly egregious example. I personally know some of the people there and at similar companies in the field; they are reasonable people who are trying to catch the wave of interest in psychology (and behavioral science in particular) to do marketing and advertising more effectively for their clients. That wave of interest—and manipulative techniques that accompany it—wasn't something they created.

While it's easy to find examples of digital companies (or companies in the digital age that advertise online) doing questionable things, this is not a new problem. One of the great pioneer researchers in the field, Robert Cialdini, learned about the strategies and tricks of persuasion by doing field research with used car salesmen and other in-person persuaders,¹¹ and many analyses have been written about the physical and psychological designs of casinos.¹² What's different is that behavioral science is both documenting and explicitly contributing to these efforts—especially in digital products.

Researchers and other authors like me have actively spread these techniques. In our community, we write books on topics such as how to:¹³

- Make games and apps “irresistible” so you can “get users...and keep them”
- Build habit-forming products based on “how products influence our behavior”
- Apply behavioral research to adjust packaging, pricing, and other factors to create consumption
- “Harness the power of applied behavior psychology and behavior economics to break through these nonconscious filters and drive purchase behaviors”

I've known some of these authors over the years, and they aren't bad people; they are sharing techniques that can help product designers make better products—ones that people enjoy using and want to use. They seek to develop relevant and engaging marketing campaigns that are tailored to their audience's interests. But empirically, the techniques we talk about are used in many other ways as well, which aren't so beneficial.¹⁴

My own writing—including the first edition of this book—clearly falls into this category as well. We may want to help users, but we shouldn't be blind to what has actually happened.

Where Things Have Gone Wrong: Four Types of Behavior Change

In the **Preface** and throughout this book, we've talked about two different types of behavioral products:

- Behavior change is the core value of the product for users
- Behavior change is required for users to extract the value they want from the product effectively

In the first case, products explicitly seek to help users change something about their lives, like exercise bands, sleep habit apps, and mindfulness apps. In the second case, products use behavioral techniques so the user can be more effective at using the product itself; these types of applications and products vary from making app menus manageable to helping users customize their displays and focus on what they care about.

Both of them have something clearly in common: they seek to help the user do something they already wanted to do. That's the focus of this book: voluntary and transparent change. There's another type of behavior change that hasn't been our focus thus far, but now must be:

Behavior change is about helping the company achieve something, and the user doesn't know about it or doesn't want it.

From what I've seen in the industry, that's the most common type of all, and it's time to call a spade a spade. Our industry uses consumer psychology, behavioral science, and whatever other techniques it can to push people into doings they don't fully realize, and wouldn't want to do if they were fully aware.¹⁵

Facebook's emotion study? That was about Facebook, not about helping its users. Marketing campaigns to use psychology to push a product (regardless of what the product is and who the audience is)? That's clearly about helping the business's profits, without thinking about the user and their goals and needs. If we refer to the issues that make people uncomfortable (effective persuasion or coercion, often hidden from users), that study rings alarm bells on all accounts.

To be fair, there are many cases where unethical use of behavioral techniques isn't *intentional*, however, where by accident or through competitive pressure, firms have adopted an approach that tricks their users without setting out to do so. Teaser rates are an example: where competitive pressures and historical practice in the credit card industry lead to unsustainably low opening interest rates. Such teaser rates are only viable for a company because they are replaced with much higher rates later (as sophisticated buyers know, but the unsophisticated fall for), or user behavior triggers punishingly high rates that make up the difference in profits. In theory, credit card companies would be better off if they all ditched teasers and used more transparent pricing, but if any single company did so without the others, they would lose market share. Manipulative techniques don't always signal malice; still, unintended but obvious manipulation is our responsibility just as intentional manipulation is.¹⁶

So, is the solution simply "don't do that"? If only it were that easy. Instead, we have bad actors that poison the water for everyone else, products that seek to be addictive, and problems of incentives in our industry that lead us back to problematic uses.

Poisoning the Water

Applied behavioral science has a reputation problem; there's no easy way for users to distinguish between "good" and "bad" actors—between companies that are using behavioral science to help them and those that are using it to hurt them. And, especially when companies overhype how effective behavior science is at changing behavior (as marketers often do),

people can assume that behavioral techniques are inherently coercive—that is, able to make people do things they don’t want. What else should people expect from titles like *The Persuasion Code: How Neuromarketing Can Help You Persuade Anyone, Anywhere, Anytime* and *How to Get People to Do Stuff: Master the Art and Science of Persuasion and Motivation*?

That’s based on hype, though, and not the real state of the research. The main issue of impact in the research community is over techniques that don’t replicate (i.e., that don’t seem to have a real effect at all), that aren’t clear whether they generalize (i.e., all effects are context specific, and we don’t fully understand in which contexts a particular technique helps or not), or that backfire (i.e., something that has a positive effect to help people in one context but actually makes things worse in another context). That’s why throughout this book we talk about the importance of experimentation: behavioral science has an incredible set of tools, but they aren’t magic wands. The hype in the industry makes it seem like our tools are magic and that makes us all look bad.

Beyond telling thoughtful companies to simply *stop* using behavioral science in ways that users won’t approve of, we have a problem of how to stop the bad actors who don’t want to stop, or at least differentiate everyone else.

Addictive Products

In addition to products that go against what users want in the short term, there’s another highly problematic category of uses:

Behavior change helps the user do something that they currently want to do, but we know they are likely to regret in the future.

What’s an example of this? Any product that seeks to *addict* or *hook* its users, without their users asking for it, from mobile games to social media. We can see the backlash and discomfort in the field, from Ian Bogost’s labeling of cell phones (BlackBerry, at that time) as the “Cigarette of this Century” to more recent stories in the *New York Times*, the *Washington*

Post, and on NPR.¹⁷ These products can harm users directly (like cigarettes harm people's lungs) or indirectly by dominating what's known as the *attention economy*:¹⁸ eating into their users' time and attention to the point that they crowd out meaningful interactions with friends, family members, and others.

Now, the term *addict* is used loosely in the field and often doesn't refer to what a medical doctor would call addiction. There's also healthy debate on whether technological products are actually as addictive as some researchers say.¹⁹

From the perspective of those of us who design for behavior change, the point remains, though: if a product or a product designer *tries* to addict its users (even if it can't achieve it in the medical sense), there's probably something wrong. The language in the field of designers seeking to hook people is disturbing—to the extreme of Mixpanel's congratulatory report called "Addiction."²⁰ Digital products seek to hook people on something they want right now (even if they may not have wanted it before the advertising campaign or sense of FOMO took root) but hurts them in the long run. Naturally, the individual user makes a series of choices that lead to that bad outcome. As a field, we should take responsibility for our actions; that doesn't imply that others shouldn't take responsibility for theirs as well.

So we could simply say, "Don't addict people." And indeed, some brave voices in the field do, like Matt Wallaert, chief behavioral officer at Clover Health.²¹ But even more than products that explicitly go against their users' wishes, this is a hard one to tackle. There's an easy path to self-justification, and the business incentives are huge: products that really could hook their users would be immensely profitable.

That brings us to the question of *incentives*. Simply put, would a company avoid designing for behavior change if that means hurting their business? When we would objectively see an action as ethically dubious, would the product managers, designers, and researchers working on the project see it as such *in the moment of action*?²² It appears that, in many cases, the

answer to these questions is no. To understand and address these challenges, let's take a short detour from the ethics of behavioral science into the behavioral science of ethics.

The Behavioral Science of Ethics

There is significant research literature on how ethical behavior is influenced by our environment, both explicitly in behavioral science and in the older social psychology research tradition.

Researchers find that our environment impacts not only everyday behavior but also *moral* behavior. There is a long history of work showing how factors in our environment shape whether we act responsibly or not; for example:²³

- When people hear the cries of someone having an epileptic seizure in another room, the more people who hear, the less likely that anyone responds.
- People are more likely to help others when the person gives a meaningless reason for requesting help, instead of simply asking for help without a reason.
- People are more likely to cheat on a test when they can't be caught, when they see others cheat, and when they can rationalize it as helping someone else.

My personal favorite is the story of the seminary students.²⁴ In that study, researchers had seminary students do an activity, at the end of which they had to go to another building (not knowing that the travel to the other building was, in fact, the key part of the study itself). The researchers varied the degree of urgency with which the students were asked to move and varied the activity the students undertook before traveling to the other building. In one version of the pretravel activity, the students prepared to discuss seminary jobs; in another, they prepared to discuss the story of the Good Samaritan. The requests to travel had one of three levels of urgency.

In each case, the seminary students passed by a man slumped in an alleyway. He moaned and coughed, and the researchers had observers record whether the seminary students would stop and help the individual.

The level of urgency mattered—the more urgently the student was supposed to reach the other building, the less likely they stopped. The pretravel activity (thinking about the Good Samaritan story) did not. The ineffectiveness of thinking about the Good Samaritan story makes the research more dramatic and interesting, but the truly important finding was how simply being asked to hurry changed the behavior of presumably moral and helpful people. In particular:

- In the least-urgent situation, 63% of the people helped the man slumped in an alley.
- In the medium-urgency situation, 45% stopped and helped.
- In the highest-urgency situation, 10% did.

A temporary and, in the grand scheme of things, largely unimportant detail (whether the person was asked to hurry or not) had a massive effect on the person's behavior. To put it bluntly, these things *shouldn't* matter—not if we're good and thoughtful people, right? But yet they do. And as much as we might condemn the students in this famous study, I'm sure we can all remember similar times in our lives as well—when we had something on our minds and didn't take the opportunity to help someone in need.

The research on moral behavior and how it's shaped by our environment ranges from the comical to the truly troubling. It raises serious concerns: how can people be ethical in one situation but unethical in another situation that is only slightly different?

It also raises questions about what it means to be a good or moral person. Gil Hamms comes to this conclusion about avoiding the evil within: “we should seek out situations in which we will be good and shun those in which we won't,” or as Kwame Appiah puts it:

A compassionate person can be helped by this research, by using it to provide a “perceptual correction” on how we see the world, and using them to reinforce the good in our behavior and avoid the bad.²⁵

We'll Follow the Money Too

What does this literature tell us? The first lesson from the literature is that good intentions aren't enough; people's environments affect ethical behavior just as they affect other areas. And, “people” includes us (heck, most of us aren't as ethical as seminary students to begin with). Thinking otherwise requires a potent combination of arrogance and self-deception.

What environment are we in? Our environment, in which we apply behavioral science, by and large, is not directed to help individuals flourish and prosper. Many companies sincerely would want their users to be happy and successful, but their first priority in applying behavioral science is to increase profits, either for themselves directly or by serving as consultants and providers to other firms. The most money is to be made either using behavioral science to increase the profitability of products (regardless of the user's interests and needs) or hooking users on products that look interesting in the short run but can cause significant downsides in the long run.

It's neither a new discovery nor necessarily a negative thing that companies want to increase their profits; both good and bad come from our system. There is a problem, however, when those of us in the product, design, and research communities ignore the fact that we are affected by our environment. We should expect ourselves to follow the money just like anyone else and to use behavioral science in unethical or questionable ways. We shouldn't be naïve about how our *behavior* will diverge from our *intentions*.

The second lesson is much more hopeful, though. Despite our impressive ability to self-deceive and the many ways in which our environment can nudge us to act unethically, we can also design our environment to

encourage ethical behavior—that is, to turn our intentions to act ethically into action.

A Path Forward: Using Behavioral Science on Ourselves

How might a company or individual change their environment to support ethical uses of behavioral science? We can find many such techniques once we start to think about the problem as a behavioral one; in particular, as a gap between our intentions and our subsequent actions.

Assess Intention

As with any intention–action gap, the first question we should answer is whether we intend and prioritize helping users succeed. In other words, is the company actually concerned with the ethical behavioral science as we’ve defined it here; that is, does the company want to help the end user change behavior in a transparent and voluntary way?

This isn’t a glib question, nor one where the opposing side is evil or full of bad people. Many companies find their true north in accomplishing something that’s never been done before or in providing stable jobs for their employees. Similarly, most consulting companies are first and foremost concerned with providing value to their clients and not in judging what that means for the end user. These aren’t inherently bad companies; they are just companies for which the rest of this section won’t be relevant. Behavior change, even within our own companies, should be *voluntary* and *transparent*.

Assess Behavioral Barriers

Your company might already be applying behavioral science, and you might have a sense of where trouble could be in the future (or present). If the challenge is one of not taking a particular ethical action, debug it with the CREATE Action Funnel we’ve used throughout this book. If the challenge

is one of existing and problematic habits, look to the cues of those habits and disrupt them. Above all, check the core incentives. Despite all of the nuance that behavioral science can provide about how people's decisions are shaped by social cues and other factors, often the simplest economic reason is the best place to start: we do what we're paid to do. If your company hasn't started applying behavioral science, but you're concerned about where things might go in the future, again, basic incentives (not intentions) are often the best place to start.

The specific barrier or challenge matters: there isn't a magic wand here any more than there is in any other part of behavior change work. That being said, we can point to some techniques that might help, depending on the particular behavioral barriers you face in your company.

Remind Ourselves with an Ethics Checklist

A simple way to keep ethical uses of applied behavioral science front and center is to remind ourselves, such as with a humble checklist. What do you consider important in a project? Write it out. Condense it into a few questions to evaluate each project. That's something we've done on my team at Morningstar. Then, with that checklist or set of questions, print it out, post it prominently, and if possible, set up a process where other people outside of the team review it.

As with other behaviors, often we simply fail to take action in the way we desire because we get distracted by other things and lose focus or forget; a checklist helps fix that.

Several groups in the field have drawn up ethical guidelines, from the Behavioral Scientist's Ethics Checklist by Jon Jachimowicz and colleagues to the Dutch Authority for Financial Markets' Principles Regarding Choice Architecture.²⁶

Here are some rules that I find useful for this purpose:

- *Don't try to addict people to your product.* This should be obvious, but clearly needs to be reiterated.

- *Don't harm your users.* The phrase I use with the team is to always keep our work “neutral to good” either explicitly helping or doing something that users don't mind and won't cause harm. It can be difficult to know for sure that you're helping the user, but if even your own team doesn't think it will help or if people wonder if it might be harming users, that's a big warning sign.
- *Be transparent: tell users what you're doing.* Directly telling the user what you're doing shouldn't cause a problem and is a good, simple check on excess. A related technique is to imagine that your work becomes front page news—would your users be upset? Would your company survive? This technique is useful, but it's hypothetical. Even better is to tell them up front.²⁷
- *Make sure the action is voluntary.* The user should be able to decide whether or not to participate in the product or service. For example, an app that's required at work to monitor employee productivity isn't optional; the job may be “optional,” but the app isn't.
- *Ask yourself* whether you'd want someone else to encourage you to use the product. Is this product really designed to help you? Would you encourage your child or parents to use it?
- *Ask others,* especially strangers, if *they* would trust the application.

Create a Review Body

Checklists are great, but not very valuable if you don't use them or you get into the habit of marking off all the questions by rote. Having an external review body—external to your team, or even external to your company—can help here. In the academic community, the Institutional Review Board (IRB) serves this function, with an independent board that reviews the ethical considerations of any research study.

Most private companies do not work with the IRB or other external body. But they can create an internal one without much difficulty. However,

remember the research in self-deception: the more closely aligned the review body is to the people being reviewed (the more the reviewers see the reviewed as friends or colleagues to support), the less valuable it is. In other words, it helps to have strangers on the review body. Or, if strangers can't be found, try adding a few jerks!

Remove the Fudge Factor

One of the key lessons in the self-deception research of Ariely, Mazar, and others is that self-deception relies on a “fudge factor”: the capacity to bend the rules a bit and still see ourselves as honest people.²⁸ That fudge factor is largest in ambiguous situations (where it isn't clear whether you really are bending the rules or not) and in situations where it's easy to rationalize the rule-bending (when you're helping others or when you see others bend the rules—as previously mentioned).

To limit self-deception, we should try to limit the fudge factor, especially ambiguity and rationalization. To remove ambiguity, we can make the rules crystal clear with a plain-language internal policy, or we can create feedback processes to frequently check whether we've strayed from our internal rules. To remove rationalization, we can intentionally set an ethical reference group that is the best of the best; we can ensure that senior leaders set the tone not only that unethical behavior won't be tolerated, but that it won't help the company or other employees in the long run.

Raise the Stakes: Use Social Power to Change Incentives

Another technique we can use is to intentionally raise the stakes against straying from an ethical path by telling others about our commitments. In other words, don't just set a checklist: tell your customers, your employees, and your friends and family, that you've made a particular commitment. Tell them the rules you'll follow for designing products and applying behavioral science.

If your company has a reputation for honesty (which ideally yours does), this means using that reputation to help keep yourself in line; it's at risk if you stray. As part of this technique, welcome attention—being transparent about how you're applying behavioral science is good in its own right, but it also helps raise the stakes to not stray. If it fits your company culture, you can also be a bit preachy: call out the abuses of other groups in the field. In addition to perhaps helping clean up our field, it has another effect: because people really dislike hypocrites, this approach makes it very risky to stray.

Remember the Fundamental Attribution Bias

Working on behavior change in a company adds the extra layer of complexity: within the company there can be a thoughtful range of opinions and priorities. They may not like your approach to improving ethical behavior—thinking it unnecessary or actually counterproductive. And when they don't like it, it's simply easy to see them as naïve, deceived, or unethical themselves. In other words, that there's something wrong with them. I find that, as a behavioral scientist, it's valuable to start with the assumption that we're all wrong; that the other person seeks to do good, but is just as imperfect as I am.²⁹ It's a small step to try to counter the fundamental attribution bias: to assume that other people's "bad behavior" is because they are bad people, while ours we excuse away.

Use Legal and Economic Incentives as Well

Behavioral science provides a great set of tools to close the gap between intention and action. Sometimes, though, there really are bad actors who have no intention to doing right by their customers, their employees, or others.³⁰ And in these situations, we shouldn't be afraid of more traditional techniques to regulate abusive practices (legal penalties and economic incentives). Legal approaches can include the proposed DETOUR Act, which, at the time of writing this, is imperfect but could be revised and restructured to provide thoughtful legal oversight and penalties where they are lacking. Economic incentives might include taxes on the use and transfer of personal data (making some deceptive practices less lucrative).

Why Designing for Behavior Change Is Especially Sensitive

Thus far, we've talked about abuses in the field and how to potentially counter them. However, are these abuses particular to behavioral science? I'd argue that they really aren't. We shouldn't design products that addict people—whether we use behavioral science or not. We shouldn't trick users into buying things they don't want, nor into giving uninformed “permission” to hawk their data.

Behavioral science offers a set of ideas for design (what to change) and measurement (how to know if it worked). Where the idea for a design comes from shouldn't matter as much as whether the targeted change (the ends) and how you do it (the means) are themselves ethical. In other words, the simplest answer to the question “When is it ethical to use behavioral science to try to change user behavior?” is this: “in the same situations where it's ethical use non-behavioral techniques to try to change user behavior.” Unethical efforts shouldn't be tolerated either way, and ethical designs should be fine with or without a layer of behavioral science.

While theoretically correct, that answer isn't terribly helpful. There is something different about behavioral science in product design that makes people uncomfortable. We can think about and understand why. From the perspective of a user, four factors are likely at work:

Persuasion

It's inherently unsettling to think that any product is trying to “make” us do something.

Effectiveness

It's especially unsettling when is a technique appears to be universally effective; that is, that it compels us to act or do something and we have no control over it.

Transparency

It's worse when the technique is hidden; we never know it's happening or know only after the fact (and thus feel tricked).

Attention

Behavioral science has the word *behavioral* in it and talks explicitly about trying to change behavior. That draws our attention to it, whereas in another case, we might not know about it.

The first three factors—persuasion, effectiveness, and transparency—aren't really issues of behavioral science at all. We should always be concerned about them; doing something against someone's will (effective coercion), especially without their knowledge (i.e., lacking transparency), should raise hard questions.

In terms of attention, though, behavioral science is special; people pay more attention to, and thus are more unsettled by, sketchy uses of behavioral science. In the design and research community, we should embrace that critical attention. Rather than try to dismiss it as behavioral science being treated unfairly, let's use this attention to have an honest conversation about persuasion, effectiveness, and transparency.

If there's something that we're doing that relies on people not paying attention, that's a pretty clear sign we shouldn't be doing it. And yes, there are absolutely such cases—because of behavioral techniques or not—people are rightfully upset once they become aware of how products were designed and function. In other words, let's apply behavioral science to ourselves to welcome the attention and scrutiny and special attention our field gets to raise the stakes to unethical conduct. Because sadly, we need it.

BEHAVIORAL DATA AND USER DATA OVERALL

One could easily take the words “behavioral science” in this chapter and substitute them with “data about users,” and the challenges would be nearly the same. In recent years, we’ve seen the same breaches of trust with user data (in fact, with some of the same companies): with near daily revelations of companies selling our data, not handling it carefully, and so on. Similar issues about user control over their outcomes and transparency of what’s going on arise with data handling as with applying behavioral science.

There’s an active debate over appropriate handling of user data—and frankly one that is better developed and more thoughtfully articulated than the discussion over using behavioral techniques with users. In Europe with GDPR (General Data Protection Regulation), and likely soon in California, Washington, and other states in the US, governments are setting standards for the appropriate use of user data.

In the case of GDPR, the new rules center around a few simple principles of transparency and control: users of data should be transparent, and control should remain in the hands of the person that data is about. While certainly not perfect, these guidelines provide a nice template. They change the incentives of technology companies and other data gatherers so that significant fines and penalties are at risk if companies don’t comply. Some companies will always try to skirt the rules, but these rules set a clear baseline. They’ve helped draw attention to the issue among consumers and data scientists, among others.

In the behavioral community, however, there aren’t any such government guidelines yet. The proposed law by Senators Warner and Fischer would set such guidelines, but in the absence of it, we should look to set these guidelines ourselves.

A Short Summary of the Ideas

Here's what you need to know:

- While there are always gray areas, ethical behavior change is not a subjective, squishy thing. There are manipulative, shady practices in our industry, and they are rightfully being called out by journalists and regulators. We need to clean up our act.
- Other disciplines also have manipulative practices—Cialdini, for example, learned from used cars salesmen—but designing behavior change is drawing particular scrutiny because we do so intentionally and at massive scale. We should welcome that scrutiny; obscurity is never a solid ethical defense.
- We do need guidelines for our work. For example:
 - Don't try to addict people to your product.
 - Only apply behavioral techniques where the individual will benefit.
 - Tell users what you're doing.
 - Make sure the action is optional.
 - Ask yourself and others if they'd want to use the product.
- We're all human though, and guidelines aren't enough. We will fudge things and stretch the rules just like anyone else. Applying behavioral science on ourselves means:

Fix the incentives

If you're paid to drive sales, you're going to drive sales. If there aren't clear goals or incentives to ensure clients will benefit from the product or service, then it's too easy to fall into murky territory.

Draw bright lines

Ensure that whatever guidelines you set are straightforward and clear so any reasonable observer can talk whether they are being violated or not.

Set up independent review

Is there a third party, not on your team, who reviews the applications of behavioral science in your work?

Support regulation

Yes, I said it. While bills like DETOUR are flawed, some regulation is coming, and it's better that we make it thoughtful and effective. Like it or not, the best way to align incentives, draw bright lines, support independent review, etc., is to hold our organizations legally liable for not doing so. Regulation and penalties force attention to the issue.

Avoiding coercion doesn't mean that you encourage users to do anything they want to do. Your company will have, and must have, a stance on the behaviors it wants to encourage. "Dieting" and "eating everything you want" aren't two equally valid options for weight loss. One works (sometimes) and the other doesn't. You can talk about and be up front about that stance. If you're helping people diet, don't be ashamed about it. Do it, and do it proudly, but be transparent and make participation optional.

Many types of products, even those that are explicitly coercive, can be good and useful. The ankle bracelets used for home detentions probably fall into that category. On net, society is better off because of their use. But that's a different type of product than we aim to develop here, and they deserve scrutiny and thought. Here, my goal is to spur ideas about products that enable voluntary behavior change so that we are clear about what we're doing and the means we're using to affect user behavior.

Talking about product ethics may be an unusual topic for a book aimed at practitioners. But we can't outsource ethics. We should feel proud of our work. Part of that means double-checking that the product is truly voluntary, is up front about the behaviors it tries to change, and seeks to make a beneficial change for its users.

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- 1 Mathur et al. (2019)
 - 2 Valentino-DeVries (2019)
 - 3 These quotes come from Reuters (2019) and GovTrack (2019).
 - 4 Norwegian Consumer Council: Forbrukerrådet (2018); Google: Meyer (2018); Intuit: Elliot and Waldron (2019); Apple: Lanaria (2019)
 - 5 Goel (2014); Albergotti (2014)
 - 6 Kramer et al. (2014); My thanks to Ethan Pew for pointing out that the effect size was much smaller than was represented in the media.
 - 7 Roberts (2015)
 - 8 Murgia (2019)
 - 9 Schechner and Secada (2019); h/t Anne-Marie Léger
 - 10 Kearon et al. (2017)
 - 11 Cialdini (2008)
 - 12 See Schüll (2014), for example.
 - 13 These quotes come from the Amazon book descriptions for Lewis (2014), Eyal (2014), Alba (2011), and Leach (2018), respectively, as of June 2019.
 - 14 Nir Eyal's book provides perhaps the clearest example. As one author puts it in describing his book: "the well-known book by user experience expert Nir Eyal was a hit because it showed developers exactly how to create addictions. Yet readers often forget that Eyal gave ethical guidelines for using this 'superpower,'" Gabriel (2016).
 - 15 In addition to Brignull's Dark Patterns site mentioned earlier, an examination of how (usually unintentional) bad design can harm users can be found in *Tragic Design* by Savard Shariat (2019); h/t Anne-Marie Léger.
 - 16 See Gabaix and Laibson's (2005) shrouded attributes model; h/t Paul Adams.
 - 17 Bogost (2012); see Alter (2018) for a book-length analysis of addictive products and their repercussions.
 - 18 Thanks to Florent Buisson for the suggestion of including the attention economy here.
 - 19 Gonzalez (2018)

- 20 This report has been removed from Mixpanel’s website, but was up when I last accessed it in June 2019, tauting the benefits of addicting your users.
- 21 Wallaert (2019)
- 22 Or hide behind either intentional ignorance or a cynical take on the mantra that “no design is neutral” and therefore all designs are permissible?
- 23 See Appiah (2008) for a summary. These examples come from Latané and Darley (1970), Langer et al. (1978), and Ariely (2013); the last study also provides a nice summary of how self-deception operates in daily life, what exacerbates it (ambiguity, cheating to help others, seeing others cheat), and what minimizes it (clear feedback on what dishonesty is, supervision/surveillance).
- 24 Darley and Batson (1977)
- 25 Appiah (2008)
- 26 Jachimowicz et al. (2017), Dutch Authority for the Financial Markets (2019); h/t Julián Arango
- 27 Telling people what you’re doing doesn’t necessarily mean to have a big banner saying, “Look here, we’re testing the impact of this button color on speed of response.” That creates an annoying experience for the user and an unrealistic environment for measuring the impact of an intervention (see [Chapter 13](#) on experiments). Rather, it means being clear that you’re running tests and what you’re trying to accomplish overall; h/t Fumi Honda for raising the issue.
- 28 See Ariely (2013) for a summary.
- 29 Aka, Hanlon’s Razor: “Never attribute to malice that which can be adequately explained by stupidity”; h/t Paul Adams. See [Wikipedia](#) for a brief history of this aphorism.
- 30 Thanks to Clay Delk for stressing the need for tools to handle *intentional* bad actors. While there are variety of frameworks out there for organizing the tools for regulating behavior and society, my favorite framework comes from Lawrence Lessig’s *Code* (Basic Books, 1999): law, market, architecture, norms. Behavioral science tends to focus on architecture and norms; we should never forget the power of the law and market.

Part II. A Blueprint for Behavior Change

Chapter 5. A Summary of the Process

The Applied Behavioral Science team at Walmart and Sam's Club understands how the mind works and that there is often a gap between mind and action. For example, even customers who are interested in their products may not return to the store in the future. They also know that theory isn't enough: to put these ideas into practice, they need a step-by-step and repeatable process.

Min Gong, head of Applied Behavioral Science at Walmart and Sam's Club, refers to their process as "the 4-D's":

- Define the business case and problem.*
- Diagnose the status quo and opportunities for change.*
- Design and test the proposed solution to the problem.*
- Decide on whether to scale up and implement the solution more broadly.*

For example, the team was recently asked to help drive in-store trips and membership renewal at Sam's Club. First, they worked with their business partners to more tightly define the problem and constraints. Together, they found that what was really needed was a marketing strategy that was cost-effective, helped address behavioral biases among customers, and supported membership renewal.

The team studied current engagement and renewal decisions among Sam's Club members to diagnose gaps between current and desired behaviors, and identified strategies to build habits that led to tangible business impact. They designed 20 different RCTs (randomized control trials, or A/B tests) across six rounds of testing based on this understanding. As Min says about

their efforts, “Often we spend 70% of the time iterating on the solution; it’s not something you should expect to get right immediately.”

The winning idea was a progressive-incentive program to reward members as they returned to the store over time and it supported long-term relationship and habit building. Given its demonstrated impact, and low cost, they’ve decided to implement it across Sam’s Clubs. Their clear, repeatable process allows the team to focus on translating insights into practical, field-tested interventions.

Understanding Isn’t Enough: We Need Process

Thus far, we’ve covered the easy part: how the mind works. We’ve also started learning how to *intervene* in the process of decision making and action to help people do better. There’s a problem, though. These interventions are unlikely to work. Or, more accurately, they are unlikely to work in a given situation without forethought, adaptation, and learning in the field. That’s because *all* new products and features, whether informed by behavioral science or not, are unlikely to have a meaningful effect on their users’ lives without tuning them to a particular environment and target audience.

Behavioral science helps us understand how our environments profoundly shape our decisions and our behavior. It shouldn’t come as a surprise that a technique that was tested in one setting (like a research laboratory) doesn’t affect people in the same way in a real life. To be effective at designing for behavior change, we need more than to understand the mind; we need a process that helps us find the right intervention, the right technique, for a specific audience and situation.

What does this process look like? I like to think about it as six steps, which we can remember with the acronym *DECIDE*: that’s how we decide on the right behavior-changing interventions in our products and communications.¹

First, *Define* the problem. Who's the audience, and what is the outcome we're trying to drive? Second, *Explore* the context. Gather qualitative and quantitative data about the audience and their environment. If possible, reimagine the action to make it more feasible and more palatable for the user before we build anything.

From there, *Craft* an intervention—a behavior-changing feature of the product or communication. We craft both the conceptual design (figuring out what the product should do) and the interface design (figuring out how the product should look). As we prepare to *Implement* the intervention, we consider both the ethical implications and how to instrument the product to track outcomes.

Finally, test the new design in the field to *Determine* its impact: did it move the needle or did it flop? Based on that assessment, *Evaluate* what to do next. Is it good enough? Since nothing is perfect the first time, we'll often need to iteratively refine it.

In shorthand:

1. *Define* the problem
2. *Explore* the context
3. *Craft* the intervention
4. *Implement* within the product
5. *Determine* the impact
6. *Evaluate* what to do next

TALKING ABOUT PRODUCTS

The process of designing for behavior change is used in many situations: developing new products or features, refining existing ones, or developing marketing and communications. For ease of reading, I generally refer to all of these as developing a “product.” Where the differences between working with an entire product versus a feature or a communication are important, I call that out in the text. Otherwise, product is used in the general sense of something we build that can change behavior.

It’s important to emphasize that the process is inherently iterative. That’s because *human behavior is complicated*, and thus, stuff is hard! If you could simply wave a magic wand and other people would act differently, we wouldn’t need a detailed process for designing for behavior change (and would be very disturbing). Instead, there’s a cyclical process of learning about our users and their needs and our efforts to address them if they miss the mark. The most overlooked yet most essential part of the process isn’t great ideas and nifty behavioral science tricks; it’s careful measurement of where our efforts go awry and having both the willingness and the necessary tools to learn from those mistakes. You can see this illustrated in [Figure 5-1](#).

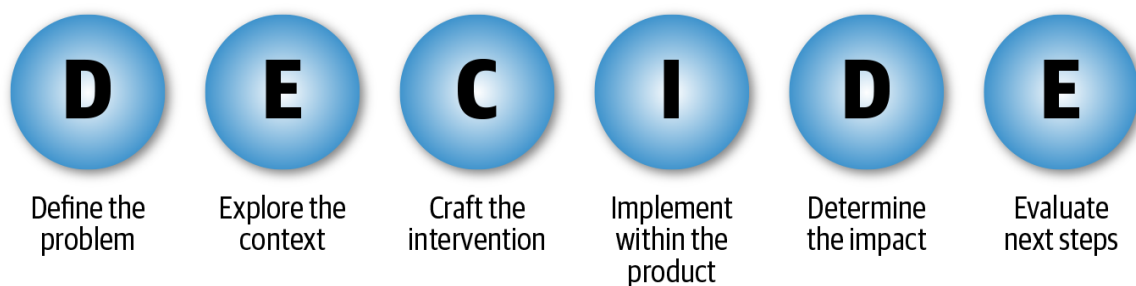


Figure 5-1. DECIDE: A six-stage process for applying behavioral science to intentionally change user behavior

The first part of the book laid the foundation: understanding how the mind makes decisions. DECIDE takes that foundation and builds real, impactful products on top of it. However, let’s not overclaim its uniqueness.

The Process Is a Common One

The field has now matured to a state in which a range of approaches are available for applying behavioral science to products and communications. When *Designing for Behavior Change* was first written, groups like the Behavioural Insights Team in the UK and ideas42 in the US were starting to do this work, but their underlying design processes weren't widespread nor fully solidified. There were (and still are) lots of books about biases and nudges; *Designing for Behavior Change* was one of the first public manuals on the *process* of how to do this stuff.

That's changed. ideas42 has a published process;² Dilip Soman's book *The Last Mile* (Rotman-UTP Publishing, 2017) outlines how to “engineer for behavior change”; Clover Health's Matt Wallaert recently published a book on his Intervention Design Process (*Start at the End*, Portfolio, 2019). And to be frank, they all look pretty much the same. I'd like to think that's because everyone else copied *Designing for Behavior Change*'s process, but that's not true. Rather, we all copied common sense and the scientific method.

At a high level, we all settled on some similar ideas. That's because, in retrospect, they are simply what's needed to do this work effectively. We need a base of knowledge about human behavior. In Chapters 1 through 3 I cover the foundation you need before designing for behavior change; others, like ideas42 and Soman, similarly have it as a precondition. We need to study the details of a particular situation and the problem at hand (“Define & Diagnose” in ideas42's terms; a “Behavioral Diagnosis” in Irrational Lab's terms). We need to come up with a proposed solution and implement it (Soman's “Select Nudges and Levers”). We then need to see if it works, and iterate if not. Got it. As a field, we now have a blueprint for behavioral design—the details of what we call each step differ, as do some of the particular substeps along the way, but at a high level, it's a shared blueprint. And indeed, in many ways it is a shared blueprint with the design community as well; many similar design and problem-solving frameworks exist. Originally launched in 2004, the Design Council's (2019) Double

Diamond is one prominent example; it's a nonlinear approach to the design process, and problem-solving overall.

So if you don't choose to use my process, I won't be at all offended. If you're using any other well-baked and thorough process, you're following a similar path. What really matters are the details: how you actually do it in practice, what tools you have to carefully diagnose behavioral obstacles, and how you select among interventions. To use a common term in design, it's the *artifacts* you generate and the specific process you use to generate them that matter. That's what we turn to next. I hope the detailed tools and artifacts that the rest of the book covers will be valuable for your work to make designing for behavior change real in your organization.³

The Details Do Matter

To make things concrete, **Figure 5-2** shows how the six DECIDE stages of designing for behavior change can be applied, and their specific deliverables (or artifacts). It shows you the specific outputs that the team will generate within each stage of the process.

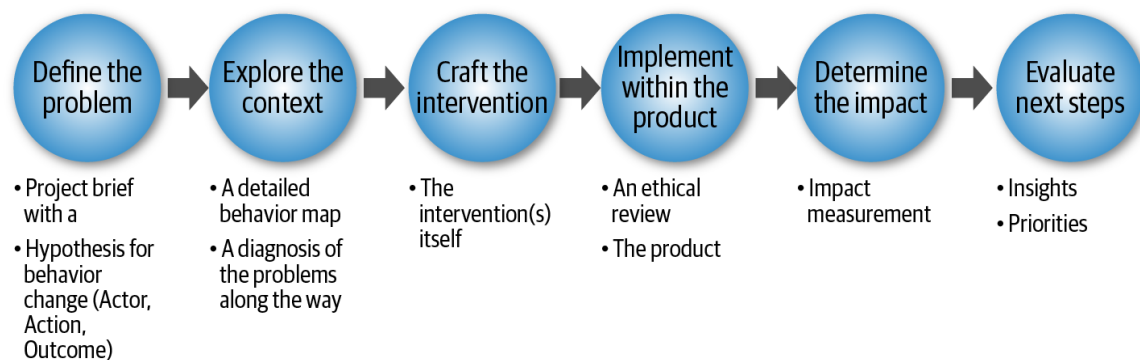


Figure 5-2. The outputs of Designing for Behavior Change at each stage of the process

Let's assume that a company (or NGO, government body, or individual entrepreneur; I use "company" as a convenient shorthand for any individual and organization making behavior change products) is developing a new product.

To start things off, the company gains an understanding of how we make decisions and how our cognitive mechanisms can support (or hinder)

behavior change. The two main topics to cover are the prerequisites for action, which are summarized in the CREATE Action Funnel. From there, we move to DECIDE. Here is how the DECIDE process works, and the outputs the team will generate at each stage:

1. *Define*

With that knowledge in hand, it clarifies what, specifically, the company wants to accomplish with the product and for whom: the *outcome*. Perhaps the company seeks a world full of (newly) healthy people. The company then identifies the particular group it wants to make healthier (let's say, office workers) and the action it generally *thinks* will help (let's say, walking more); that's the *actor* and *action*. If the company is working with an existing program, then the action may be where there appears to be a current bottleneck (like office workers with an existing walking program, where people don't show up regularly).

2. *Explore*

With these goals in hand, it's time to get out in the field and see if those goals are, in fact, realistic and wise. How does the target audience think about the action? What alternative actions could we take, and how do we evaluate them? Exploration is about gathering the *data* we need to better *diagnose* the behavioral obstacles users face, and inform a thoughtful design.

3. *Craft*

The company crafts an *intervention* or series of interventions that will help overcome the obstacle, and writes it up as a behavioral hypothesis, a story about how the user will interact with the product. The interventions

are incrementally built up by changing the *action* itself, the *environment*, and the *user's preparation* to act.

4. *Implement*

Next, it's time to actually build *the product*. We take a step back and perform a final *ethical review* of the proposed product. If everything checks out, we proceed to development. Some engineering compromises and trade-offs naturally occur, and the team reviews them as well for their behavioral impact.

5. *Determine impact*

Once a version of the product is ready for field testing, the team starts to gather quantitative and qualitative data about user behavior to form an initial *impact assessment* of how the product is doing.

6. *Evaluate next steps*

A careful, structured analysis of that data leads to *insight and ideas* for improving the product. That could lead the team members to revise their underlying concept of whom they are helping and how, and generate a new behavioral map and interventions accordingly. The process continues inward until the desired level of impact is achieved. With each revision, the team makes *changes and measurements* of how those changes impacted user behavior.

If the company already has an existing product and wants to refine it, the process is similar but often is shorter and more focused. It too starts with a problem, an existing challenge or bottleneck in the product. But, unlike the more speculative work of new product development, companies will likely have empirical data on the actual behavioral patterns of users—to help

identify obstacles and their causes. There will be strong in-house views about what needs to be done, which is both a blessing and a curse. That makes analysis easier, but also makes it harder to see alternative interpretations (i.e., confirmation bias sets in). For that reason, we intentionally set aside time to reimagine the “obvious” solution in the exploration phase. We may discover new avenues and opportunities, in which case the rest of the process becomes more like new product development work.

Since We’re Human Too: Practical Guidelines and Worksheets

Earlier, I mentioned that *understanding* how to change behavior isn’t enough because our interventions require adaptation and targeting to a specific audience and situation. That’s certainly true, and that’s one reason we need a *process* for designing for behavior change. But there’s another one too: we’re just as human as anyone else. We’re unlikely to put the ideas from the previous chapters into practice. A collection of stories about our biases may entertain us, it may even inspire us to design products differently, but that’s not the same thing as effectively changing *our* behavior.

Instead, we should apply the lens of behavior science on ourselves as product managers, designers, and researchers to think about and overcome *our own* obstacles to designing for behavior change. One of the best ways to do that is to have a written process to follow, with checklists and worksheets, so we don’t need to rely on our own overtaxed System 2 to figure things out in the middle of a project. At the end of most of the subsequent chapters, I offer exercises from worksheets that you can use. These are collected in a complete start-to-finish **workbook**, giving you practical tools to design for behavior change in your work.

I think about it like this: Chapters **1–3** are what most people want and expect from a book about designing for behavior change. The rest of the book is what I sincerely believe is actually needed to do it effectively. So

please stick with me. We’re going to talk about processes, checklists, and experimental design, not because they are exciting or popular; instead, because they’ll help you succeed.

Putting It into Practice

Now that we have a good understanding of the theory behind applied behavioral science from **Part I**, it’s time to put these ideas into practice. Applied behavioral science has settled on a certain blueprint for behavioral design; many teams have independently come up with a similar process on how to be successful. This chapter introduces our take on that shared process: DECIDE. As we dig into each step in the process in what’s to come in **Part II**, you’ll find a “Putting It into Practice” section at the end of each chapter. It includes both a summary of the key lessons and a set of exercises you can use to apply these techniques yourself.

Here’s what you need to do. While different teams use different names, we can summarize this process as DECIDE:

1. *Define* the problem—identify the people you’re working with (actor), what you and they are trying to accomplish (outcome), and how you plan to drive that outcome (action).
2. *Explore* the context—learn more about the context in which your users live and act, and align your initial plan for action with what is realistic and helpful to the user.
3. *Craft* the intervention—a new screen, feature, product, communication, etc., that helps someone overcome the obstacles they face.
4. *Implement* within the product—build the new intervention into your product, feature, or communication, and include metrics and behavioral tracking as a core part of the product itself.
5. *Determine* the impact—by gauging their reaction and checking whether it is having the effect you’re looking for.

6. *Evaluate* what to do next—by learning how to further improve its impact and judge whether those additional iterations are warranted.

The DECIDE process is fundamentally about problem-solving: because applied behavioral science at its best is about helping your users overcome obstacles to the choices and the actions they want in their lives. It's a process shared with user-centered design and many other communities.

Where designing for behavior change differs from a general problem-solving process is in the specific tools you use along the way; for example, with a particular process for identifying behavioral obstacles—which we'll call a *behavioral map*. With an understanding of the many ways one can misinterpret changes in user behavior and thus the tools for rigorous objective measurement. With a set of specific behavioral techniques that researchers have discovered over the years—from temptation bundling to implementation intentions. It's these specific tools that'll be our focus for most of the remainder of the book.

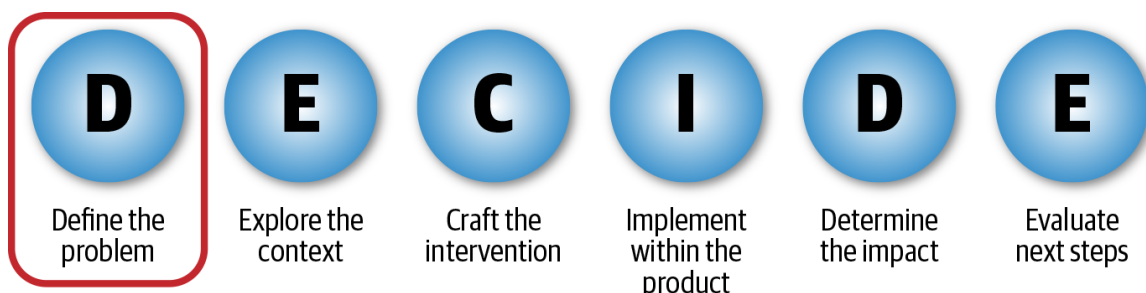
Workbook Exercises

Check out the worksheets at the end of each chapter in the rest of **Part II** for tools to help you apply the techniques from that chapter.

You can download **the full workbook**, which collects all of the worksheets in one place for you to use.

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- 1 For those of you who read the first edition of this book, you'll notice that this structure looks different. I took three phases from the first edition (discovery-design-refine) and broke them into two sections each—to better outline the specific steps you take. The core idea is quite similar, but the structure is different and I've added new details and tips along the way.
 - 2 They adapt it to different domains of application, but the core is similar. See **Darling (2017)** for an example. Irrational Labs has a toolkit by Dan Ariely, Jason Hreha, and Kristen Berman, *Hacking Human Nature for Good* (2014).
 - 3 There are a range of similarities between the detailed process and that of other authors. I'll cite them along the way. We've each tried to solve the same fundamental problem. I think this book brings together a more holistic and detailed toolset than most, but if you disagree, that's perfectly fine. In the end, what matters is what helps you and your team do good by your users. Use what works for you!

Chapter 6. Defining the Problem



If you currently save for the future, what's your goal? I know it sounds strange, but if I were to ask you in a few hours, the answer might be different. That's because we tend to respond to questions like this with what comes easily to mind, either because it's something we think about frequently or we were recently exposed to.

If I asked you right after you'd gone through an expensive neighborhood, you might answer that you wanted to save up to afford a nice house and lifestyle. If you'd just spoken to your father who was recovering from an illness in the hospital, you might answer that you wanted to have what you needed to pay for medical expenses in the future. But in reality, both are true, and it would be hard to tell which was more important.

At Morningstar, my team thought this might be a problem among everyday Americans—we'd found research documenting it in particular circumstances, such as MBA students not providing stable answers about their goals for a summer internship and major executives not giving stable answers on the goals of their company.¹ If so, it would represent a real challenge to how we encourage people to save and invest, since knowing why people save is key to helping them commit to, and follow through on, their savings goals over time.

We decided to assess the problem among two populations: a nationally representative sample of working Americans over 18, and a subsample specifically of investors. We wanted to determine whether goal instability was in fact a problem worth solving, and if so, test a potential action that would help: using a master list of goals to prompt people to think more broadly.

With an outcome in mind (reducing goal instability), an action (master list), and two target populations, we could design a specific intervention: an online tool to help people analyze their own goals. Indeed, we found that it both seemed to be a major problem and that we could address it. In a randomized control trial, testing different ways of asking people about their goals, we found that when presented with our tool, nearly three-quarters of the participants reevaluated and changed one of the top three goals they'd initially reported, and an amazing 24% changed their top-priority goal.

Overall, the participants' new goals were longer-term, and more specific, than their initial responses. We published the results and implemented the concept in a software tool for our clients. This was possible because we started with a clear understanding of the problem, the outcome we sought to drive, and a specific behavioral action that a behavioral actor would undertake.²

When Product Teams Don't Have a Clear Problem Definition

I fear that we've all been there: six months into a two-month project, with no end in sight. At the beginning, someone thought everything was obvious and charged ahead. Of course we know what we're doing, let's get on with building it! Six months later, you look back and wonder where all the time has gone and why the product is still off course.

In my experience, the root cause of many bad designs—when designing for behavior change or otherwise—comes from a lack of clarity from the start. It comes from a researcher or product manager who had a great idea about

how to fix something and rushed to implement it without thinking through what was really needed. It comes from a product team taking orders from an executive to “just build this” and never really getting the opportunity to learn *why* that was being built or questioning whether it was the right way to do it. And so you end up with a mess—or a misaligned product. Or, as Yogi Berra once put it, “If you don’t know where you are going, you might wind up somewhere else.”³

The problem isn’t unique to designing for behavior change, of course, and there are good tools for problem definition throughout the business world and in the design community. Here, we’re going to take a particularly behavioral approach to this age-old problem. For us, Defining the problem centers on:

The target outcome

What is the product supposed to accomplish? What will be different about the real world when the product is successful? For example, users should have less back and neck pain, with 50% fewer doctor’s visits and physical therapy sessions over the next six months.

The target actor

Who do we envision using the product? Who will do something differently in their lives and thus accomplish the target outcome? For example, sedentary white-collar office workers who don’t exercise regularly.

The target action

How will the actor do it (to the best of our knowledge)? What behavior will the person actually undertake (or stop taking)? For example, users should go to the gym twice a week, for 30 minutes each time.

AN ONGOING EXAMPLE: FLASH

In this chapter, and many of the following ones, we'll show how designing for behavior change works with a running example: an exercise app. Imagine that you work at a B2B company that offers wellness programs to employees of your clients. You already have a number of programs in place, but now your company wants to develop a new product to help people become more physically fit through exercise—especially by going to the gym. You can think of it as Fitbit-style app for a B2B context. The tentative name is *Flash* (to evoke images of rapid, positive change in the user's life).

Your company has seen a suite of wearable computing devices fail in the market and has decided to try something simpler (and less expensive to develop): an app on the employee's mobile phone. In particular, this is the problem definition you're given:

Target outcome

Have less back and neck pain, with 50% fewer doctor's visits and physical therapy sessions over the next six months.

Target actor

Sedentary white-collar workers (employees of client companies), who don't exercise regularly

Target action

Users should go to the gym twice a week for at least 30 minutes at a time.

We'll flesh out this example over time, Exploring the context, Crafting the intervention, etc.—going through the full DECIDE process. As you'll see, this is a *difficult* problem and one that is difficult to build a successful app around. We'll use the example to show the challenges of

designing for behavior change, as well as the process. Along the way, we'll also use other examples to show a range of situations in which these techniques can be applied, and just to keep things more interesting.

Throughout the book, we'll use those three terms, outcome, actor, and action, to Define the problem we're looking to solve. We'll use them first to express the initial *intentions* of the company—the intended target audience, behavior, and outcome. The real world isn't so simple, however. As we learn more about the realities of our users and their situations, we often discover that our intentions aren't realistic. And so, we'll evaluate and refine these three ideas until key stakeholders are on the same page, and potential problems have been identified and resolved. We'll build a clearer and more impactful problem definition over time, centered on these three concepts.

Here's what we'll cover in this chapter, to clearly define the (behavioral) problem:

1. Clarify the overall behavioral vision of the product.
2. Identify who we are trying to serve—the users.
3. Identify the user outcomes sought.
4. Document our (initial) target action.
5. Define success and failure according to that action.

Your company may already have completed some of the initial stages of the problem definition process. If so, you should feel free to jump ahead to the part that's relevant. However, you may find it useful to quickly scan the earlier parts to see if there's anything you missed. At the end of this chapter, I introduce a worksheet that helps with problem definition (“**Worksheet: The Behavioral Project Brief**”). If you can't complete the worksheet, check back for the relevant section on how to handle it.

Start with the Product's Vision

Inspiration for a product or a new feature can come anywhere, from a client request to inspiration that strikes while taking a shower. Either way, you start the problem definition by simply getting the vision of the product down on paper. The vision can be general and somewhat vague—that's fine. It allows us to start the conversation about the more specific, concrete impacts that the product should have—the target outcome.

As an example, let's start with a product vision rooted in the mission of an organization. For example, **the Sunlight Foundation**, a nonprofit that works toward government transparency, gives its mission statement:

Our Mission: The Sunlight Foundation is a national, nonpartisan, nonprofit organization that uses civic technologies, open data, policy analysis and journalism to make our government and politics more accountable and transparent to all.

The vision for each of its products clearly follows from this organizational mission. For example, its Web Integrity Project “monitor(s) changes to government websites, holding our government accountable by revealing shifts in public information and access to Web resources.”

For a company developing health and wellness programs, the product vision might be to “help people take control of their weight and health.”

Nail Down the Target Outcome

After you've recorded your general vision for the product, ask, What should be different about the world when the product is successful? What's the specific, concrete change that should occur because of the product? What could a third party see, hear, or touch? What meaningful thing in the real world *changes* because you've done your job?

The answer to these questions is the product's (or feature's) desired *outcome*. It's the tangible thing that the company *seeks to accomplish* with the product (remember, I use “company” as shorthand for corporation,

nonprofit, or government agency). You might call it a goal, but I like the word *outcome* rather than *goal* because it feels more concrete—it focuses attention on something changing in the world.

Write down that desired outcome (or outcomes).

Clarify the Outcome

Next, let's refine your target outcome with some probing questions. We'll use the example of an environmental cleanup program, in which we move from a vague outcome (decreasing pollution) to one that is clear and more precise. Specifically, we ask:

Which type?

Does the product ultimately seek to change something about the environment (e.g., clean water) or about people?

Where?

What is the geographic scope of the impact (e.g., Chesapeake Bay)?

What?

What is the actual change to the environment or person (e.g., decrease nitrogen pollution)?

When?

At what point should the product have an impact? We're looking for the order of magnitude. Unlike the others, this doesn't need to be precise at this point—"in a few months" or "in five years" is fine.

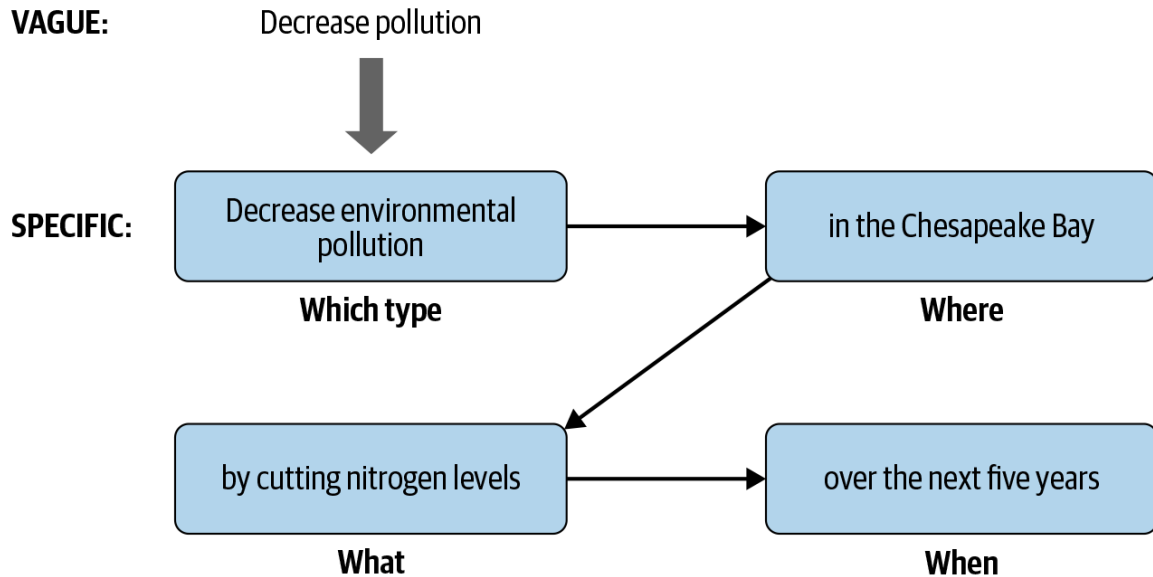


Figure 6-1. How to turn a vague outcome into a specific, measurable one

Write down the answers to these questions with a simple, clear statement that summarizes them. For example, “This product should cause a decrease in nitrogen levels, an environmental pollutant, in the Chesapeake Bay over the next five years.” The new, clearer outcome statement is displayed in **Figure 6-1**.

Based on that statement of the desired outcome, define a metric that you can use to gauge whether the product is successful or not—nitrogen levels in the water or an employee’s weight, for example. You don’t need to settle on the exact measurement yet—but if you can’t define one at all, then the outcome isn’t concrete enough.

Here’s an example of clear outcomes that are readily measurable and ones that aren’t:

Clear, measurable outcomes

Employees with a BMI over 25 will lose 10 pounds. Teenagers in San Jose won’t smoke.

Unclear, difficult to measure outcomes

Users will gain experience with exercise. Users will understand the dangers of smoking.

Having a clear outcome in mind doesn't mean that you're omniscient or that you're locking your company into a strict path for the next few years. Instead, it's a solid point of reference. As problems arise, you can settle disagreements of fact (does this design work or not?) by measuring against the outcome. And, you can settle disagreements of vision (is this the right goal for the product or not?) by redefining the target outcome when needed—*as long as the new outcome is also clearly defined.*

That and nothing else

One of my favorite techniques to refine a team's desired outcome is to ask, "If you got exactly what you asked for (the outcome you've identified thus far), and absolutely nothing else, would you be satisfied?" For example, imagine a team that's working to reduce teen pregnancies and had developed an educational program. They decide that their target outcome is that teens are aware of how pregnancy could disrupt their schooling and future career path. If the teens were educated but didn't care (and did nothing differently), would the team consider that a success? Certainly not. The team would then refine the target outcome to be concrete: they want to decrease teen pregnancy. Education is simply one of the tactics that they *believe* will lead to that outcome.

Vanity metrics—metrics that make a company feel good but don't give an accurate sense of whether the product and company are on the right track—fail on these criteria. For example, consider page views—the stereotypical vanity metric. Let's say a company had *zero* revenue from its flagship consumer product but had lots of page views on the product's website. That usually wouldn't be considered a success. And, vice versa: if the product brought in lots of revenue, but for some reason it had few page views, that *would* be considered a success.

Avoid states of mind

The teen pregnancy example illustrates a common problem that many companies face as they define the product's target outcome: they want to talk about something within their user's heads. Education. Confidence. Skill at doing something. There are two reasons why states of mind are problematic.

First, they are difficult to measure in a consistent and unambiguous way. States of mind can be measured with surveys, but the results of those surveys are highly dependent on how questions are framing, the question order, when and how they are administered, and more. *Highly dependent* means open to debate, misunderstanding, and argument. That's exactly what we're trying to avoid.

Perhaps more importantly, though, states of mind probably *aren't what the company actually wants*. Consider an NGO that trains low-income people in developing countries to be entrepreneurs; do they want the people to know what it takes to be an entrepreneur, or do they actually want the people to start new, successful businesses? If the clients knew how to be entrepreneurs but never actually started businesses, would the NGO consider the program a success? Probably not.

Instead, we want the outcome to be something observable, outside of the individual's head, absolutely unambiguous (to avoid arguments within the company about whether something is successful or not), and easy to measure (so you can gauge success quickly). *The target outcome will define the success (or failure) of the product*. Avoiding states of mind as the outcome doesn't mean that states of mind aren't important; a particular perspective (like wanting to have a bright future without having a baby as a teenager) may be absolutely necessary.⁴ However, it's just not enough.

BEING EFFECTIVE SOMETIMES MEANS BEING CONTROVERSIAL

Often, we focus on states of mind because they are noncontroversial. Everyone can support education; fewer people are comfortable talking about what the education is supposed to do—change people’s behavior, which then changes something in the real world.

Remember, we’re not trying to be noncontroversial. We’re trying to be *effective* at actually changing behavior and helping our users. That takes a measure of self-reflection and honesty—what is your real goal?

Prioritize and combine

If you’re stuck with more than one outcome, that’s fine. Multiple outcomes need extra work, though. First, they need to be organized. If there’s a clear, top-priority outcome, excellent. If not, get the stakeholders together and see if there is a majority opinion on what’s most important. Or, take the list of desired outcomes and ask for each one: would the product still be “successful” if this did not occur? Drop them, one by one.

If winnowing down the list isn’t feasible, there is another, more challenging route. Create an aggregate outcome that combines the contenders for top priority. To do this, you need to get very specific and define a formula that combines them in a way that can get everyone on the same page. This formula is a formal definition of success for the product.

For example, say the two *highest priority* outcomes are “employees will have lower blood pressure” and “employees will lose 10 pounds” A definition of success that combines them both would be “the success of the product is defined as one point for every decrease in average blood pressure across the target population and two points for every decrease in weight, in pounds.”

Unfortunately, most companies just don’t have enough information to understand the complex impacts of the product across related behaviors before the product has been built. Personally, I’d avoid making up a

formula that combines multiple outcomes and settle on one outcome as the top priority.

Avoid stating how the product (might) do it

You may have noticed that there's an important question I didn't suggest asking at this point: how?

For now, try to avoid delving into how the product works its magic (i.e., what action it encourages users to take to make the outcome happen). We'll get to that shortly, after we're armed with additional information.

We focus on outcomes instead of actions for three reasons: first, because there may be many ways to accomplish the action, and the "best" one may not be what comes to mind easily. Second, the link between any behavior and the outcome is uncertain, so we keep our eyes on what really matters: the outcome. And finally, because of "compensatory behaviors." Human behavior is a complex web of checks and balances, not something you just change with a clear effect. A well-known example is moral licensing: sometimes when people exercise, they feel so good about what they've done, they go out and eat unhealthy food. They nullify any benefit that the exercise might have had on their weight.⁵

Why go through the trouble?

Why do we need to define the desired outcomes so carefully and clearly? We want to *pull problems into the present* and resolve them. If the team has a mess of conflicting objectives, or something that can't be measured, then there are problems lurking in the future. The team is likely to argue over what the product should look like as each member tries to meet conflicting, unstated objectives. The head of the company may think the product isn't successful, but the engineers do. The grant funding agency (for NGOs) might cut off funding because it had implicit assumptions that aren't met about what the product would do.

A clear statement of measurable outcomes, which key stakeholders sign on to at the beginning of the project, can resolve many of these problems early.

And, just as with any product development process, finding and resolving problems early on is much cheaper than trying to fix things later.

In addition, a clear outcome statement is essential for future revisions to the product to improve its impact. It forms the basis for measuring the product's success, finding problem areas, and gauging whether proposed changes to the product are worth their salt.

What if no one can agree on the product's intended outcome?

One possible result of this process is that there simply isn't a clearly defined outcome for the product to achieve. Either the stakeholders fundamentally don't agree or the product was poorly conceived and doesn't have real-world outcomes. In that case, the product shouldn't proceed in its current form. In the spirit of failing fast, that's a really good outcome—and should be embraced, painful as it is.

That doesn't mean the team needs to have *consensus* about what the ideal product should do; few companies operate that way. But everyone should know what to expect and sign on to the goal once the decision has been made about the product's intended outcomes. If there are still deep divisions in the team, then problems lie ahead. The team should move on to other interesting products, or change its membership, rather than argue for months over a product that's ultimately doomed.

Define the Metric to Measure Outcomes

Next, you define your outcome *metric*. The metric is what tells you, as unambiguously as possible, whether the target outcome has occurred, and at what level. The outcome metric should flow directly from the target outcome itself. It's how you determine if the outcome is there or not. You should define and write down a formula, even a dirt simple one, which says how the outcome is measured. Here are some easy ones:

Company income

Money received from client companies over the course of a month

User weight

Body weight without shoes, measured in the morning after breakfast

And here's one that is a bit more complex:⁶

Neighborhood connectivity

Number of times users attend social gatherings with their neighbors over the course of a month

Ideally, the metric should be:⁷

Accurate

It actually measures the outcome you want to measure.

Reliable

If you measure the exact same thing more than once, you'll get the exact same result.

Rapid

You can quickly tell what the value of the metric is. Rapidness encourages repeated measurement and makes it easier to see if a *change* in the product was effective.

Responsive

The metric should quickly reflect changes in user behavior. If you have to wait a month until you can measure the impact of a change (even if it only takes a minute to measure it; i.e., even if it is rapid), that's 29 days wasted that you could have been learning and improving the product.

Sensitive

You can tell when small changes in the outcome and behavior have occurred. For the developers among you: floating-point values are great; Booleans are not.

Cheap

Measuring the outcome *multiple times* shouldn't be costly for the organization or it will shy away from measuring the impact of individual changes to the product and have difficulty improving it.⁸

Quite a lot there, right? Yes, but that doesn't mean you need to obsess over the perfect metric. What we're really looking for is a quick check of sufficiency. Treat this like a checklist—for a given outcome metric, ask: is it specific enough that there won't be too much disagreement when it's measured? Is it reliable enough that it won't fool the team into thinking the product is working when it actually isn't?

Working with Company-Centric Goals

Thus far, we've talked about a product development process that's focused primarily on the user and what the product can do for them. But I've found that companies sometimes take two very different approaches to behavior change. They can either:

- Focus on how the product will *benefit the user*, which in turn helps the company's bottom line
- Focus on how the product will *benefit the company*, by way of providing value to the user

This difference is in how companies think about the *value* of changing behavior; it isn't related to the behavior itself. The target behavior could be inside the product or outside the product, socially important or trivial; that doesn't matter as much.⁹

In the first case, which we'll call a *user-centric approach*, the company might have a vision of improving financial wellness (like Acorns or Mint) and need to figure out what actions users can reasonably take that will best make that happen. In the second case, a *company-centric approach*, a company might have a purely self-interested goal, like increasing customer renewals, but needs to provide real value to the user in order for that to succeed. The second approach includes red-blooded capitalists as well as NGOs that need to make their case to their funders. There's nothing wrong with that approach—as long as the companies build products that people like. But it does mean that the process is different.

In the user-centric approach, the process of Defining the behavioral problem is this:

Product Vision (for the User) → User Outcome → Actor → Action

In the company-centric approach, we add another step:

Product Vision (for the Company) → Company Objectives → User Outcome → Actor → Action

The product vision is *why the product is being developed*, at a high level; and the company objectives are *what the company seeks to achieve*, for itself, by building that product.

Since the previous section highlighted examples of the user-centric approach, let's now walk through the company-centric process (you can safely skip this if you're using a user-centric approach).

State the vision

As before, the Definition process starts by writing down the high-level vision the company has for the product. That vision, however, should answer how the product will generally benefit the company. For example, this product should:

- Expand the company's appeal into new markets.
- Increase revenue.

- Demonstrate the expertise and capacity of the organization to take on new projects, to support new grant funding.
- Increase public awareness and interest (or brand prestige) of the organization.

State the company's objectives

With the company's vision for the product in mind, translate that vision into one or more specific, measurable objectives that benefit the company. For example, ask:

- How will the product be judged a success or failure at fulfilling the company's vision? How will success be measured?
- What would a third party observe about the company that's different because of the product? Increased retention of customers? Increased upsell? More referrals of new customers?

The company's objective might be to win 35% of the wellness program market among technology companies in the next year. Or it might be to win at least one million dollars of additional grant funding for the next year. Write down that initial company objective.

Based on that initial statement of the company's objectives, fine-tune it by asking who, what, when, and where the outcome occurs (as we did in the previous section).

Define the user outcomes

Building your business or establishing your expertise to funders is great, but your users probably don't care. Sorry. You need to deliver something of value to them. Without that value, you can't meet your business objectives.

So, putting aside the financial (or other self-interested) goals of the product for a moment, what does the product mean for users? We want to define the measurable changes in the world that are caused by the product *that the user would care about*.

Here are some questions to help draw out those outcomes:

- What does the product deliver? What's its core value proposition as the user would see it and measure it?
- After users have used the product, what's different about the world?
- What tangible thing would a user look at (or hear or see), sometime after using the product, and say, "Hey, I want to use that product again"? (The product itself doesn't count, sorry.)
- How would you know that your users have gained the maximum value from your product?
- What do the users *do* because of the company's increased brand awareness?

For example, consider the company objective of winning 35% of the wellness program market among technology companies in the next year. The specific user outcome might be what we identified above for our *Flash* app: the product helps the user decrease doctor and physical therapy visits by 50% over six months. Or the product might help the user drop two waist sizes.

In the previous section, we discussed a set of rules and tips for clarifying target outcomes. All of them apply for company-centric goals as well: avoid states of mind, make sure the outcome is measurable, find disagreements early, and don't obsess about *how* the product will achieve these outcomes yet.

A Quick Checklist

In summary, this is what makes a good outcome, both for the user-centric and for company-centric goals:

- The outcome is *what will be different* in the real world when the product is successful.

- The outcome should be something *tangible*, and not something in the user's head. Often the user's head is just a proxy for what the company really cares about—a tangible outcome that occurs because the user's knowledge or emotions have changed. For example, lower BMI, or pounds lost, instead of knowledge about the importance of exercise and maintaining one's weight.
- The outcome should be something *unambiguously measurable*. For example, say your product is supposed to “decrease government corruption.” What is corruption, exactly, and how is it measured?
- The outcome should be able to *signal success*. If you can say, “Well, if X didn't happen, the product would still be a success,” then X is not the outcome we're looking for.
- The outcome should be able to *signal failure*. You must be able to think of a plausible case in which the outcome would indicate that the product is failing.

Who Takes Action?

Usually the company's business goals and market research specify who a product is supposed to serve: the people who buy the product (private company) or the people who they are tasked with helping (NGOs and government agencies). Here, we need to dig a bit deeper. We want to know not just who the product will serve, but who is it that will take action? Whose behavior are we seeking to change?

Often the person being served and the person taking action are the same: user = actor. But that's not always the case. The user may influence another person, who then does the real work. This is common in situations like B2B sales (where the user is a different person than the buyer) and with advocacy websites that try to influence policymakers to change regulations (like on the cost of gasoline), which then change behavior in society to drive outcomes (like lower greenhouse gas emissions).¹⁰ For the sake of

simplicity, though, and to use language people are familiar with, I'll assume they are the same for this discussion.

Write down the target actor as specifically as possible: age, gender, location, number of people, and so on. It can also help to write down who is *not* being targeted; for example, people without smartphones, the wealthy, or expats.

In all likelihood, some target actors aren't a good fit for the product, and you'd waste your time trying to target them. True. For now, we want to know the *potential* actors you should investigate further. You may end up targeting only a subset of them.

If the company has no idea who it is trying to serve and who needs to take action, then it's back to the basics. A behavioral product, like any product, must serve a user need. It can only help people change their behavior to the extent that they care about the product at all. To get a handle on the unmet user need, you need a traditional market research or (non-behavioral) product discovery process. That's beyond the scope of this book; from here on, I'll assume you have a sense of the target user already.

Document Your Initial Idea of the Action

Thus far, we've intentionally not talked about how your actors would accomplish the outcome. That's to help us separate and clearly think about the outcome, without assuming a particular way to accomplish it. But, of course, we have ideas about what action the person should take. We need to record that idea. It won't necessarily be correct—but by writing it out, we'll better be able to evaluate and refine it.

What defines an action? I think about it as two blanks you want to fill out in this sentence:

Our product will help the user [start/stop] doing [describe action].

For most companies, especially those with existing products on the shelf, the universe of possible user actions is tightly constrained by the company's

business model, product strategy, and internal culture. At HelloWallet, for example, we looked for actions that would be appropriate for a wide range of users and weren't being well covered by existing products, that fit our company mission. So job-hunting tools were out, as were mortgage finders and pure lead-gen tools.

Next, ask specifically, "How does the action cause the target outcome?" The action should directly and clearly cause the target outcome. If the current action doesn't do that, is there another, later action that *does* cause (or is a missing piece that's required to cause) the outcome? If so, focus on that one instead. We want to target a behavior that directly supports the outcome.

For example, let's assume we want to drive community volunteerism. The action "Go to a seminar about the importance of community engagement" *might* lead people to get more involved in their communities. The direct link between the action (go to seminar) and outcome (community members volunteer) is a bit tenuous though. What if the person doesn't pay attention? What if they've been forced by their spouse to attend such events in the past? A better, more immediate action is "Volunteer at a local soup kitchen." That's the real point of the seminar and has a direct link to the outcome we care about. A seminar might be a useful tactic along the way, but it's only a tactic to support action—it's not the target action itself.

Clarify the Action

As we did for the company's target outcome, we're looking for a concrete, specific definition. A physical, measurable action someone will take. Avoid actions that just affect the person's mental states (reading educational material) and dig deeper into what the person does with their new education that causes them to *do* something differently and achieve the outcome.

For example:

Outcome sought

People don't get lung disease.

Vague action

Users avoid cigarettes. (Does that mean they cut down on smoking? Go cold turkey?)

Action that's too far removed from the outcome

Users attend a seminar about the dangers of smoking (OK, but do we care if they attend? Or that they actually stop smoking?).

Clear action

Users don't buy cigarettes at all.

You'll notice that the action doesn't specify how exactly the product will help the user not buy cigarettes. It could be by helping them avoid stores where cigarettes are sold, or by decreasing the desire to smoke with nicotine patches.

Like the target outcome, the target action isn't written in stone. It should be clearly defined so that it can be built into the product and clearly measured. The definition, and measurement, will help with fine-tuning the product. They'll also help with revisiting and revising the target action itself, if the need arises.

A Metric for Action

As we did with the outcome, we want to translate the target action into a specific metric that will assess whether people are taking action. The action metric tells you whether (and to what degree) the user is taking the target action, the action that is supposed to drive the desired outcome. If the desired outcome is a specific level of weight loss and the action is exercise, a sample metric would be "How much is the user exercising, and how often?" A good action metric must pass the same tests as the outcome metric: accurate, reliable, rapid, responsive, etc.

The action metric covers *what* is measured, *how* it is measured, and for *how long*. For example, one way to define purchase behavior for a particular product is: money spent (not flexible future commitments) through product sales and subscriptions over a 30-day period.

It also needs to be specific, because if the value changes over time, specificity helps us determine if it changed because of your product. If the definition is unclear, a change in the data may be caused by a change in how people interpret or measure it.

Here are two examples of action metrics:

Action: “the user exercises”

Bad metric

User exercises = how much the user reports exercising each day. This is a bad metric because (a) the user may not know how hard they are exercising without the help of a timer, heart rate monitor, or other tracker and (b) the user may stretch the truth.

Good metric

User exercises = how long and how hard a heart rate monitor automatically tracks the user exercising each day.

Action: “the user studies a new language”

Bad metric

User studies = an expert evaluation of language proficiency on a lengthy written exam. This metric is problematic because it focuses on the intended *outcome* rather than on the *activity* that

we assume, rightly or wrongly, leads to that outcome. It also takes a long time to measure and can't be measured frequently (without annoying the users!).

Good metric

User studies = time spent within the application, or number of exercises completed with a minimal level of accuracy.

Clearly there are trade-offs when creating action metrics. The most accurate metric may take too long to gather, or the cheapest metric may not be reliable. Again, there's no need to obsess over it—we're looking for an action metric that is sensitive enough to quickly show when there's a problem and accurate enough to not mislead the team.

Look for the Minimum Viable Action

The *minimum viable action* (MVA) is the shortest, simplest version of the target action that users absolutely must take to so that you can test whether your product idea (and its assumed impact on behavior) works.¹¹ It builds on the Lean start-up concept of the minimum viable product: the smallest set of features that allows the product to be deployed and tested in the field.

How do you find the MVA? Look over the action you've proposed. I think of the minimum viable action as something that you arrive at by cutting back on what naturally came to mind the first time. Cut back from the obvious until only the necessary remains:

1. *Cut repeated actions down to the first action.* If you have a repeated action, can you start by supporting a one-time action? One-time actions are easier for users to perform and engineering teams to build than repeated ones, all else being equal. And they still provide valuable insight into whether the software works at supporting the behavior. For example, if you want to help people

lose weight by getting them to run two miles twice a week, see if they will run or jog *at all* before trying to change their regular routine.

2. *Cut big actions down to simpler ones.* Even if the target outcome won't be achieved initially, can you cut the action down into something smaller and shorter but still fundamentally the same basic task? The goal is to test the core premise, as with a minimum viable product (i.e., instead of asking people to run frequently, can they start with a group jog?)
3. *Drop steps in the sequence altogether.* Can you identify the high-risk, most uncertain aspects of the action (i.e., having people feel comfortable running alone in a new place) and either remove them altogether from the target action (e.g., run at work with colleagues) or field test them before developing the lower-risk aspects? Similarly, are there steps that are just nice to have and can be removed?

I find that people don't naturally think about the smallest possible action to change behavior. We like to think big. That's fine. It's useful, good, and most natural (i.e., easiest) to express that big vision first. That provides a blueprint that the team can go back to and draw upon as the product develops.

However, once those big behavior change thoughts are up on a board, and we see all of the pain we're thinking of putting our downtrodden users through, then reality should hit—the more work that users have to do, the less they are likely to do it (with some important exceptions I'll cover later). Hence, the MVA.

To demonstrate the idea, let's use an example of helping users learn Spanish. Here are actions that the team might target:

- Complete an online training course.
- Visit Spain for a few weeks to be immersed in the language.

- Label each item in the household with their Spanish names.

With simpler MVAs, you can test the core assumptions of the approach and its impact more quickly:

- Complete a single module of an online training program.
- Get a Spanish-language conversation partner who is committed to only speaking Spanish with the user.
- Label a few items in daily use with their Spanish names.

OK, we have a rough idea of the potential action that users might take. The next chapter will dig deeper into exactly who the users are and whether that action is realistic for them to take.

A Hypothesis for Behavior Change

You now have all the things you need to determine what success and failure would mean for your product *before you build it*. You know who the product is supposed to serve. You know what real-world outcome that action is supposed to cause. You have an initial sense of the action you want to drive in order to accomplish that outcome. You don't have all of the details, of course, but that's OK. At this stage, you have the rough outlines you need.

Write down a sentence that says what the product is supposed to be doing, and for whom. For example:

By helping overly sedentary white-collar workers (actor) to go to the gym (action), our product will cause them to have less back and neck pain and go to the doctor and physical therapist 50% less often (outcome).

As other authors have thoughtfully noted, we can think about this as our *hypothesis for behavior change*.¹² The general format is:

By helping the [actor] [start/stop] doing [describe action], we will accomplish [outcome].

It's a nice way to structure the outcome-action-actor, and it reminds us that everything we plan to do is just a plan. It's full of assumptions about the real world, and calling it a hypothesis (which it is) helps us remember that it's something we need to test to see if, in fact, we're right and we'll have the effect we hope.

Now, from your user research about what's feasible for users to accomplish and from your market research about what will differentiate and sell the product, you should be able to add in specifics about the proposed product and its impact on your business. For example:

By helping 25 to 35-year-old white-collar employees of tech companies in urban areas (actor) to go to the gym twice a week for at least 30 minutes (action) the product will cause them to decrease their doctor and physical therapy visits for back or neck pain to 50% less than they otherwise would have had over the following six months (outcome). When successful, it should double our current revenue (company objective).

In this statement, the team is saying: if this happens, the product will be a success; if not, it will be a (complete or partial) failure. Later on, we'll translate this statement into a set of metrics to gauge whether the product actually succeeded or failed at its goals.

Our goal isn't to create a false sense of security by thinking we can forecast the future and how the product will actually play out. There are lots of assumptions built into this definition of outcome, actor, and action. We want to draw out those assumptions, to get something that can be explicitly tested and then revised as lessons are learned. Once they're on paper, we can review and improve them—through pre-mortems, thinking-hat reviews, and such.¹³ And, perhaps most importantly, it helps us fail fast—to make sure that the key stakeholders in the company are on the same page before building the product. If they aren't, now's the time to fix it.

Defining success and failure ahead of time doesn't mean that we can't change the goalposts now and then. As we learn more about the market, product, and the company's other opportunities, our understanding of

what’s “good enough” will change. But make sure the team knows about the change and understands the rationale. No one likes a moving target—especially when that means making the job harder for the team midstream and without explanation (raising the standard) or it seems like accepting failure (lowering the standard).

Examples from Various Domains

Desired outcomes and target actions can be a bit abstract, especially given the tremendous range of possible products that can affect user behavior. So let’s look at some concrete examples (Tables 6-1 and 6-2). Since the approach is somewhat different for user-centric versus company-centric products, I’ve broken them up into two different tables. I’ve started both tables off with our sample product—the Flash running app—to show how the analysis would proceed from each perspective.

Table 6-1. User-centric examples

	Example 1	Example 2	Example 3
Product	Flash, an exercise app	Financial wellness app (Acorns)	Cigarette Cessation
Vision	Help people take control of their health	Provide broad access to financial guidance	Help smokers quit and avoid cancer
Outcome	Less back and neck pain (fewer doctor and physical therapy visits)	Americans have sufficient emergency savings	Cigarette smokers stop smoking
Actor	White-collar technology company employees	Gig economy workers in San Francisco	Long-term smokers who want to quit but have failed with nicotine patches
Action	Go to the gym twice a week for at least 30 minutes	Users automatically transfer money to savings account each month	Smokers switch from cigarettes to vaporizers and decrease total nicotine intake by 50%

Table 6-2. Company-centric examples

	Example 1	Example 2	Example 3
Product	Flash, an exercise app	Grocery store website	Breathalyzer
Vision	Expand into adjacent wellness markets	Expansion of business into upscale grocery shoppers	Make company's breathalyzers standard in cars across the country
Company objectives	Increase revenue from corporate wellness program clients	Double number of upscale buyers	Attain 25% market share in three newly entered states
User outcome/user need	Less back and neck pain (fewer doctor and physical therapy visits)	Learn how to cook healthy meals	Prevent accidents and tickets from drunk driving
Actor	White-collar technology company employees	Upper-income Denverites commuting from the suburbs	Drivers in the targeted states, who have had a suspended license, a ticket, and/or an accident from a DUI in the past
Action	Go to the gym twice a week, for at least 30 minutes	Users take free cooking classes on grocery website	Stop default driving home behavior 75% of the time after heavy drinking, using company's breathalyzer before starting the car

Reminder: Action != Outcome

Despite our best efforts, there may be a big leap between the outcome and the action. Will transferring small amounts of money into a savings account (when the user can choose to transfer it back or spend it) really solve the broad lack of emergency savings in the United States? That's a tough question to ask. The purpose of this chapter was to provide a clear direction for the product development process, uncover hidden assumptions, and, sometimes, determine that a pivot is needed, and a different approach or behavior should be targeted.

Sometimes the outcome that's important is, in fact, an action that the user takes. The first example could easily have had “users exercise” as the outcome, as well as the action. This occurs when the action itself has an unambiguous, real-world outcome (like exercise). However, I argue that companies should avoid equating the two in most cases. It's an easy way to hide assumptions about why the action is important and thus to potentially choose the wrong action.

Putting It into Practice

This chapter started off the DECIDE process by Defining the problem; in particular, with the outcome we're seeking to achieve, for whom, and our initial idea of how (an action). Here we summarize the key lessons and show you how to use the exercise in the workbook.

Here's what you need to do:

- Define the real-world outcome that the product should have. Avoid states of mind—focus on measurable outcomes that define the success or failure of the product.
- Translate company-specific goals (e.g., increased profit) into real-world outcomes that the user actually cares about.

How you'll know there's trouble:

- The company can't agree on the intended outcome of the product.
- The company knows what it wants but doesn't offer something that users care about.

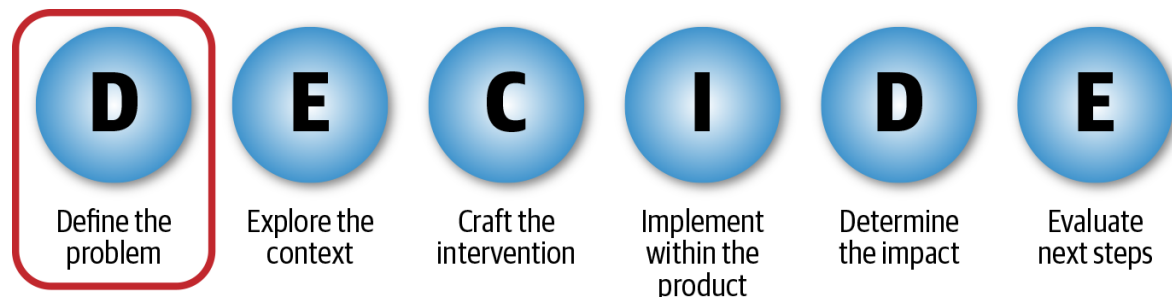
Deliverables:

- The behavioral project brief: a clearly defined outcome, a clearly defined target population (actor), and an initial idea for the action the actor will take. You can bring all of these together into a *hypothesis for behavior change*.

Learning about a technique often isn't enough on its own. Perhaps the biggest obstacle readers faced with the first edition of *Designing for Behavior Change* was that they were unclear on how exactly to act on it, and that kept them from getting the most out of the book.

For Defining the problem and each of the other steps in DECIDE, we have exercises your team can go through to implement the process. For all of the exercises, we'll use a consistent example throughout: the Flash app introduced in the sidebar at the beginning of this chapter. In this example, your company provides wellness software to employers, who then provide it to their employees. You're working on a mobile wellness app to help people exercise—it'll be a new product for your company, and you're just starting on the project.

Worksheet: The Behavioral Project Brief



Goal: Understand and articulate the aims of your team's new product—the outcome, action, and actor it should target.

Project: Flash app

☒ New product, feature, or communication?

☐ Change to existing product, feature, or communication?

Vision: Briefly describe *why* you want to change behavior and how this product fits in. Flash will help employees take control of their health.

Outcome: What do you hope to achieve with the product? Consider both the company's objective, as well as the real world, measurable change users will see and value. Then, drill down and define a rough metric your team can use to evaluate the product and an idea of what success looks like in numbers.

Company objective: Increase revenue from corporate wellness program clients.

Real-world outcome: Less pain (back, neck, etc.)

Performance metric: Doctor and physical therapy visits

Definition of success: 50% decrease in doctor/PT visits

Actor: Who is the specific user (or person involved in the product) who causes the outcome?

Who is the actor? Sedentary white-collar workers

Action: What does the actor do/stop doing to accomplish the outcome? This an initial idea; we'll refine it later.

What is the action? Go to the gym twice per week

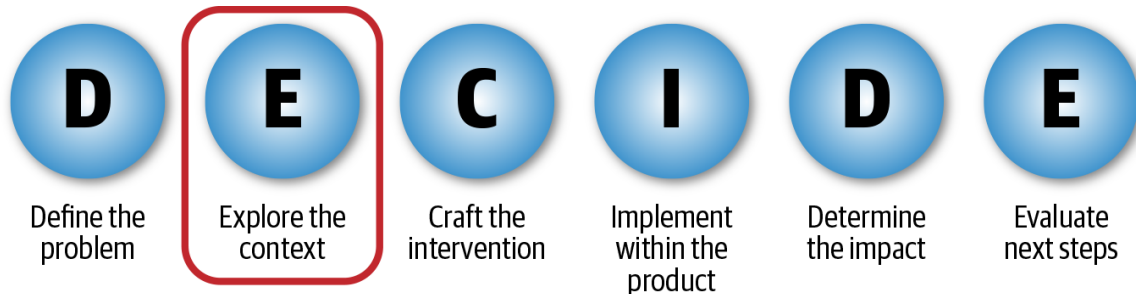
A Hypothesis for Behavior Change: Alternatively, you can write out this information as an explicit hypothesis to remind the team that nothing is for certain, and you'll need to test that hypothesis in practice through the product:

By helping [actor] sedentary white-collar workers ☒ Start ☐ Stop doing [action] go to the gym twice per week, starting with 30 minutes per session, we will accomplish [outcome] decreased pain and a 50% decrease in doctor or PT visits over six months..

-
- 1 Bond, Carlson, and Keeney (2008); Bond, Carlson, and Keeney (2010)
 - 2 Detailed results from this study can be found in Sin, Murphy, and Lamas (2018).
 - 3 Yogi Berra was a professional baseball player and coach, known for his (mal)aphorisms. For example, see "*Yogi Berra Dies at 90*" in the *LA Times*.
 - 4 My thanks to Brian Merlob for pointing this out.
 - 5 Again, h/t to Brian Merlob for this one.
 - 6 Originally inspired by the (now-closed) community connections start-up, Neighborsations.
 - 7 There are a variety of perspectives on what makes a good metric but no generally accepted and applied definition. These are characteristics that I've found to be important.
 - 8 It may take an up-front investment (that isn't cheap) to make recurring measurement cheap. We want to set up data collection that will be cheap and easy to check whenever there is a change to the application. Survey data, for example, is often "cheap" to measure the first time, but the cost usually remains the same with each iteration (and survey data is plagued with biases, as we discuss in "*Figure Out How to Measure the Outcome and Action by Hook or by Crook (Not by Survey)*"). Ideally, we want automatically gathered administrative data—collected from the original source without human intervention or extra costs. Asking people what they spend money on is a survey. Their actual credit card transactions are administrative data.

- 9 In other words, this is a different distinction than the one I made in the [Preface](#) about behavior change affecting a behavior *within a product* or *outside of it*. In either case, a company could start with the benefit for the user or the benefit for the company.
- 10 My thanks to the folks at ForumOne for pointing this out.
- 11 In the context of habits, BJ Fogg has a similar idea in his new book, *Tiny Habits* (Fogg 2020).
- 12 My thanks to Rajesh Nerlikar for first introducing me to the idea. Matt Wallaert has a similar idea in his book, *Start at the End* (Wallaert, 2019).
- 13 On pre-mortems, see Klein (2007)—h/t Paul Adams. On thinking hats, see De Bono (2006). There are also many assumption-mapping techniques to draw out (and question) assumptions in a decision or process. As discussed in [Booth \(2019\)](#), Shopify has a fun one called the *assumption-slam*—h/t Anne-Marie Léger.

Chapter 7. Exploring the Context



Clover Health’s Behavioral Science Team shows how we can combine qualitative and quantitative research to Explore the context of our users’ lives. From survey data, the team knew that some members reported difficulty getting the care they needed. So, the team tried to understand the underlying causes.¹

After talking with clients and checking the data, they found that people in lower income neighborhoods especially reported not getting quality care. The team hypothesized: perhaps lower income neighborhoods attracted lower quality doctors, making it essentially an issue of access?

However, the data showed this simply wasn’t true: low-income neighborhoods had high-quality doctors and enough to serve Clover’s members there. Members simply weren’t finding their way to those doctors. But why?

In interviewing members, they learned that the problem wasn’t access but learned behavior. Because many of these members had faced a lifetime of medical racism, they simply no longer believed that good doctors existed—and so they didn’t look for them. In a self-fulfilling prophecy, the very

inequity they'd faced in the past led them to face continued inequity in the present. They took whichever doctor they found. If they happened to get a good doctor, they received good care. But if they ended up with a bad doctor, they simply ascribed it to an unjust system and continued to see them.

With this understanding of the context of people's lives, the behavioral science team developed a number of interventions, from providing an annotated map of good doctors in the area to calling members and introducing them to better doctors in their neighborhoods.² Overall, more than 80% of targeted members were redirected to high-quality, equitable care.

Each of us has different routines, experiences, and ways that we respond to the world. To design for behavior change, we need to discover the right action for your users based on this complex terrain of routines, experiences, and responses. The action must be effective at helping them achieve their goals. And, at the same time, you must balance user needs against the needs of the company building the product—to generate revenue and to cost-effectively deploy design and engineering resources.

In **Chapter 6**, we clarified what the company sought to accomplish with the product (the *target outcome*, like people having less back and neck pain) and identified a potential action that users could take to make that happen (such as going to the gym or going on a diet). These steps helped us identify what we want to accomplish. If only the world would oblige and make people's behavior conform to our plan. Naturally, it doesn't.

Now it's time to confront our assumptions and goals with real users. We'll revisit the outcome, actor, and action and evaluate them according to company and user needs. Along the way, we'll also gather vital information about the target users for designing the product itself. The end result is a refined problem definition, especially of the *action* we want to target and a *diagnosis* that captures why we believe that action isn't currently occurring.

Here's how we'll do it:

1. Get to know your users and how they feel about the target outcome and action.
2. Generate a list of other possible actions they could take.
3. Evaluate the list of possible actions and select the best one.
4. Express that big action as a series of micro-steps.
5. Diagnose why the action isn't occurring currently.

What Do You Know About Your Users?

Clover Health's example of underserved patients and medical racism illustrates the importance of learning about your users—their real lives and circumstances, their desires and interests, and especially how they relate to the behavior you're seeking to change. We need to gather data, both qualitative and quantitative, to better target our interventions and find realistic solutions for our users' lives.

How Do They Behave in Daily Life?

First the team should seek to understand where users are starting from. In particular, what are they currently doing?

I'll use an example from my past life as a microtargeter (someone who analyzes data about large numbers of individuals to identify people who are likely to take action and what will appeal to them). One of my company's clients was an advocacy organization, which I'll call ActMore so they don't feel obliged to sue me. ActMore is an environmental NGO and wanted to help its constituents become more deeply involved in ActMore's community of like-minded environmentalists. They had a large number of people who had signed up for their email list and newsletter and had said they wanted to do more but hadn't yet really gotten involved. First ActMore needed the basics—information about age, location, income level, race, gender, political interests, and so on. My company used data that the

organization already had to fill in most of the basics. We could use that to target the appeal and provide guidance on the website.

That type of analysis is all standard stuff and is well covered in any book on product development or assessing market opportunities (as well as in political advocacy books, in ActMore's case). We then started on the more interesting part, though, focusing on behavior. We wanted to understand how strong the members' interest would be in a specific event the organization was putting together—a rally on Earth Day. We were looking specifically for divisions within the member base: groups of people that would respond differently to appeals to join the rally. Each group would get its own personalized appeal that made sense given its background and level of experience. The product we were shaping was an outreach campaign and associated website (and, as I remember, there was a phone-calling component as well). So, we started digging into the data available about the members to understand who the product would need to serve.

With ActMore and other organizations I've worked with, I've found the following questions most relevant to learn about the user base from a behavior-changing perspective:

Prior experience with the action

Do the users have experience taking the target actions? (For ActMore—have they been to other, similar rallies?) How do they think about the action? Are there strong emotional associations, or is this fresh ground? It's much easier to increase an action than to start a new one. Existing *habits* around the goals are especially important.

Prior experience with similar products and channels

If the product employs email and a website, do some users have regular access to a computer (and know how to use it) and others don't?

Relationship with the company or organization

To put it bluntly: do users trust you? You'll have a harder time making your case, and need a different set of appeals, for the people who don't trust you versus those who already know you and love you.

Existing motivation

Why would users want to achieve the outcome, completely separate from what the product offers them? In other words, what can the company build upon so it doesn't need to do all of the work itself? One especially powerful aspect of motivation is social motivation (positive or negative). What will users' friends and family think if/when they take each of the actions on the list? Will they face a community of support, ridicule, or simple apathy?

Physical, psychological, or economic impediments to action

This isn't as common, but sometimes arises. Are there groups of users for whom the action is especially difficult? For example, users who are homebound or don't have the money to travel to the rally (we faced this with ActMore, in fact).

These five things make up the behavioral profile of the users. To gather this information, you can use the standard tools of market research and product development—look for existing quantitative data on user demographics, deploy field surveys, and conduct qualitative research with users in focus groups and one-on-one interviews.³ If at all possible, *include some direct observation in the field*—see how people actually act (and not merely what they say they do). If you're just exploring the idea for a product informally or if no direct contact is possible with the user base (if business or privacy restrictions make it infeasible), then talk to people who have had contact with the target users and glean what you can from them.

This approach obviously builds on existing tools and techniques. The innovation here is in adding questions directly focused on the target action, not how people feel about the product or their “user needs” per se, but their actual experiences, motivations, and problems vis-à-vis the company’s target outcome and target action.

As you observe your users, you may witness or think up completely new ideas about what behaviors to change. For example, for the Flash exercise app, you may have thought about having users go to the gym, but after observing them, you realize that eating healthier food (enabled by shopping at a different grocery store) could be much more effective at controlling weight and decreasing back pain. Add that idea to the list and evaluate it along with the others. You may also realize that one (or more!) of the ideas just doesn’t make sense. But if you know at this point without a doubt that one of your early ideas isn’t feasible, save yourself time and cross it off the list.

Along the way, the team might also identify particular terms and concepts that resonate with the users; that’s not the focus now, but it’s still useful—put them aside to inform the UX design later on.

BUILD ON WHAT YOU KNOW

Getting to know your users is important in product development and in design regardless of whether or not you're applying a behavioral science lens. And indeed, many of the techniques that you should use are common across these communities: direct observation, journey maps, personas, and so forth.

Here, I present versions of these techniques that I find are particularly relevant for applied behavioral work. However, you may have learned these techniques somewhat differently—no problem. Build on what you know. As we discussed in **Chapter 5**, user-centric problem-solving isn't something unique to behavioral science at all; it's a shared process in which we are each learning from each other. Take the parts you find useful, and apply them to the foundation you already have.⁴

How Do They Behave in the Application?

If your company or organization already has an existing application or product, great. It can be used to understand the diverse groups of users you have and how they will respond to the new product or feature being considered. If you don't already have a product serving these users, you can safely skip this step.

In my personal example, we couldn't work with ActMore's previous products to learn about users, so I'll stretch my example a bit. Assume that ActMore already had a mobile and web platform for facilitating political action called "ActMore Now!" In studying an existing product like ActMore Now!, start with standard questions from the user testing arena: how do users feel about the application, what features are they lacking, what different types of people use the product, who is most active in the system and why, etc. Then add new questions focused on behavior:

Prior experience with the action

What features of the existing application are similar to the targeted action? Have the users built up existing habits that can be leveraged for the new targeted action?

Prior experience with the product

Which features were unsuccessful? What did those failures reveal about the characteristics of the users (particularly low attention space, impatience, or lack of background knowledge about a topic)?

Relationship with the company or organization

Are they showing that they trust you with their current usage of the application?

Existing motivation

What motivations or interests are underlying the most successful features of the application? How does using the current application interact with users' daily lives, and especially their social lives? Are there communities built around using the application?

As you can see, the questions are very similar to those asked when analyzing users outside the context of the product: motivation, prior experience, and trust. But the answers are all the more valuable as a guide because they relate to user behavior in the context of the product itself.

Practically, this process means watching, interviewing, and/or surveying existing users to understand their views of the application, their frustrations, and their joys. Make sure to include some direct observation of people as they go about their lives and use the application.⁵ It also means analyzing existing usage patterns within the application to see what parts of the application have been successful at catching users' attention. Especially important is measuring the behaviors of users on tasks related to the

application's top-level goal and analyzing how the population has responded to existing interventions.

Behavioral Personas

Next, you can use the information you've gathered to identify broad groups of (potential) users within your target population. I know the term is a bit loaded, but I like to generate formal user personas—short descriptions of archetypal users with a simple background statement about a sample user's life. (You can accomplish the same thing in your own way, without the life-stories part, as long as you get a clear idea of the groups of users within the population.) Since designing for behavior change entails changing nitty-gritty details of people's lives, it's valuable to keep in mind vivid, realistic, specific personas—not an amorphous vague concept of “our users.”

Unlike traditional user personas, these personas are all about behavior: groups of users who are likely to interact with the application differently and who are likely *to respond differently to behavioral interventions*. Each persona should have information about the topics discussed; **Table 7-1** gives one way to organize them. The examples are inspired by my work at HelloWallet and Morningstar, where the target action is saving money for emergencies among a population that hasn't used online savings tools before.

Table 7-1. Two behavioral personas on saving money for emergencies

	Frugals	Spendthrifts
Experience with similar actions	Always keeps emergency savings.	No real experience saving for emergencies.
Experience with similar products	Hasn't needed to use online products —saves by default.	Only passing familiarity with online saving tools, and they seemed boring.
Relationship to the company	None.	None.
Existing motivations around emergency saving	Saving for emergencies is clearly important, and something this group already does. So why listen to advice? Key motivations and uncertainties include: are they saving enough? What else and when should they save for other things?	Saving for the future is far away and not a motivation on its own. However, this group wants to be able to continue to live a fun lifestyle. Saving for future <i>fun</i> (especially when short on cash and they don't want to look boring) is a possible motivation.
Hard barriers to action	N/A	Doesn't have excess money to save currently
Sample bio	Jane is 33, married, and fears falling into poverty like her parents did when her father lost his job at the auto factory.	John is 28, single, lives with friends, and spends everything he's got on good food and good times.

Here are three approaches to synthesizing behavioral personas. First, you can use the four questions (experience with similar actions, experience with similar products, etc.) to spur ideas for personas in an open-ended way. For example, is there a group of users that is clearly more experienced taking this action than others? What are they like? Who would be an archetypal example of that group?

Second, you can use the four questions in a more formal way to generate a fixed set of options to explore. For example, take the four questions and consider them each as a simple yes or no question. Look at each possible combination of answers to the questions (there are 16 of them). On a first pass, you can eliminate most of the resulting options as not relevant to your

population (e.g., physical impediments may not be relevant to your product; that cuts the options down in half immediately). After that quick pass through the options, you can ask whether you really have any users who fit those criteria and what they are like as a group. Each remaining option gives you a persona.

Third, when you have historical data available about your users and their behavior, you can build them using statistical or machine learning techniques, looking for which users' characteristics best segment the population in terms of outcomes. A decision tree (random forest, etc.) or a simple regression can accomplish this. We use this approach at Morningstar.

My preference is the data-driven approach. When historical data isn't available, however, the more formal second method, odd as it may appear, completely covers the range of possible users. It makes you think about each "type" and decide whether or not it's relevant. Ideally, your personas should be *exhaustive and mutually exclusive*; that is, every person in the target population *best* fits into one and only one persona. You also can do this by drawing a simple box that represents your entire user population (Figure 7-1). Each line marks off another segment of the population, until you've covered everyone.

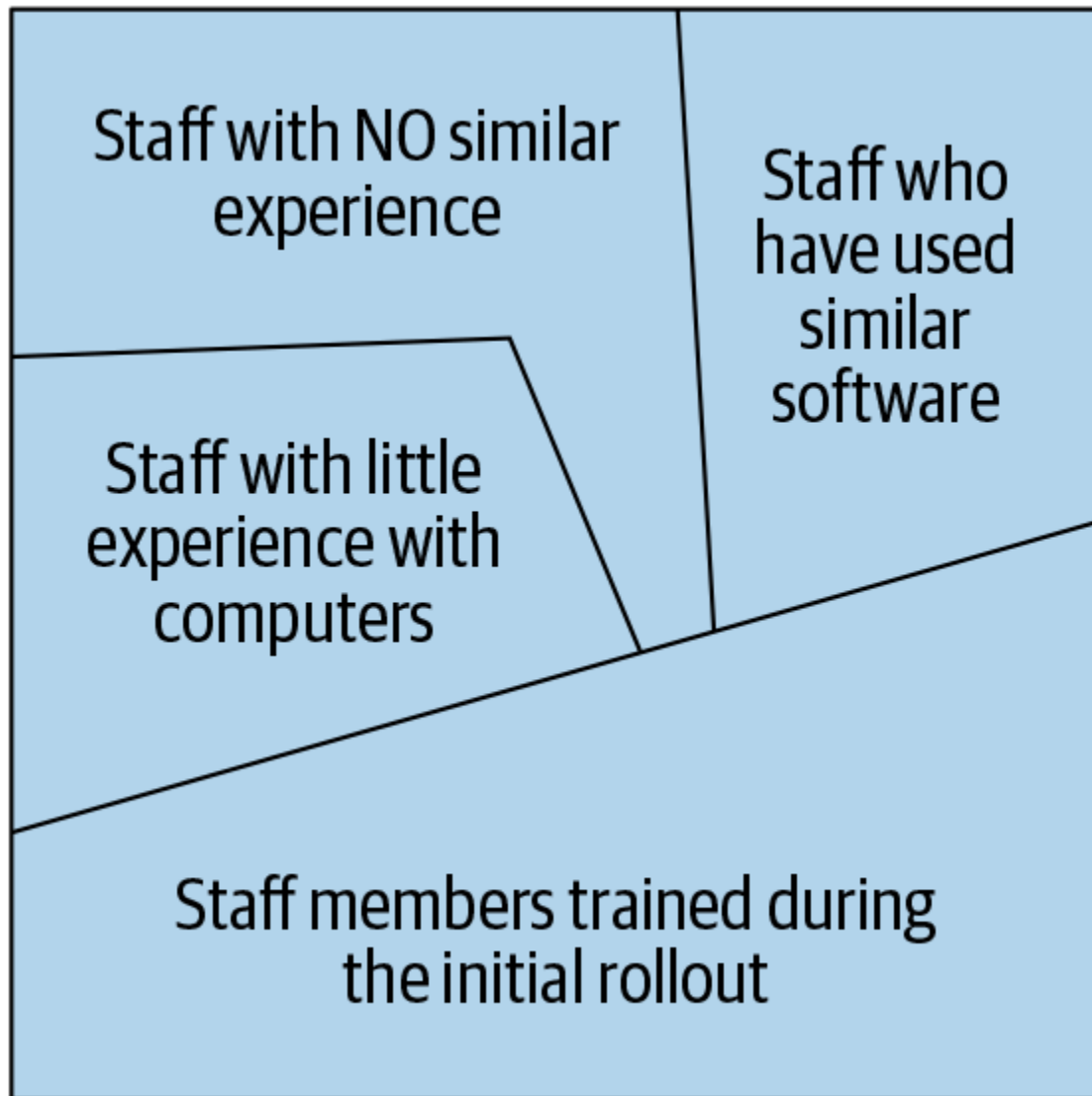


Figure 7-1. A sample population breakdown to generate personas (this is from a start-up that encouraged people to use a new software package their employer purchased for them)

For each persona, mark it off as a portion of the box, with a rough estimate of the group's size. If personas overlap—or one is part of another—that's OK. Look at each overlapping piece as a separate group of people. As you're running out of ideas, ask, "Who isn't in one of these groups? What are they like?" Label each group and see if some of them are redundant from a behavioral perspective (i.e., you expect them to respond to the product's appeal to change behavior in the same way). Then, use those informal groups to come up with more detailed personas.

As we study our users, to understand (behaviorally) diverse segments within the population, we also start to understand the context in which they act. What does it really mean for someone to sign up for our service? It seems like a simple straightforward thing: just sign up! But what's simple to us isn't simple to them. We discover that, and the real obstacles our users face, by digging into the micro-behaviors on the path to action.

The Behavioral Map: What Micro-Behaviors Lead to Action?

Barack Obama was in the midst of the struggle over the Affordable Care Act (aka Obamacare). His team wanted to mobilize supporters to call into radio programs in support of the legislation. In the last election cycle, they had built an impressive set of online tools to onboard potential supporters and get them involved in the campaign—from calling potential voters to making campaign contributions themselves.

But calling into a radio show? That was a particularly nasty challenge, and something that most Americans aren't familiar with doing, especially about a piece of new, complex legislation.⁶

How did the campaign do it? **Figure 7-2** gives a screenshot from the campaign's online mobilization platform. They intelligently structured the action into something volunteers could reasonably do. They broke the action down into three manageable chunks. They automated parts of the process, such as figuring out what number to call. They simplified and provided defaults for other parts of the process, with a script of discussion points to mention during the call. They gave clear instructions. They provided positive encouragement.





ON THE AIR

The fate of health reform has been a focus of debate in living rooms and offices, on TV and online -- and on talk radio. And since millions of folks turn to talk radio as a trusted source of news and opinions, we need to make sure OFA supporters are calling in with a pro-reform message.

IT'S TIME TO GET ON THE AIR

- 1 LISTEN, THEN CALL THE SHOW**
Dial the number listed below
- 2 ADD YOUR VOICE**
Use the discussion points below
- 3 REPORT YOUR CALL**
Tell us how it went

1

LISTEN, THEN CALL

On the Media



LISTEN NOW WITH


CALL NOW (646) 829-4074

Can't get through? GIVE ME ANOTHER SHOW

CALLING TIPS

- Be polite, respectful, and clear. Remember, you represent Organizing for America.
- How radio stations will connect you will depend on the show. Some radio shows may connect you right away, but most will take your name and basic info and put you on hold. You may or may not be able to hear the broadcast on the show while you're waiting. You may hear the radio host say something like, "Hello, we have your name on the line with us."
- Some hosts may challenge your views. Stay calm and firm. Sharing a personal story about how health reform affects you and your family is a great way to show the importance and urgency of health reform.
- If you can't get through, don't worry! If the show you call is busy or not accepting calls at the moment you call in, simply click "Give me another show" to find another.

DISCUSSION POINTS

These points are only to provide extra information and suggestions. Your personal story will make the most compelling message.

- For most Americans, their health care plan covers too little and costs too much. Far too many people delay or even skip the care they need because they simply can't afford it.
- The plan the President laid out includes the largest health care tax cut for middle class families in history and makes coverage more affordable for tens of millions of families and small business owners and expands coverage to over 31 million Americans who are currently uninsured.
- This plan will give millions of Americans new choices in health insurance by making coverage more affordable, ending the denial of coverage for pre-existing conditions, putting power in the hands of consumers instead of insurance companies and providing one of the largest tax cuts in history while also reducing our national deficit.
- Reform couldn't be more urgent -- just this month consumers in California were told their premiums could go up as much as 39 percent.
- Too many in Washington are now saying that we should

2

ADD YOUR VOICE



Use the discussion points on the right in your conversation -- but remember that the most important part of your call is your own story about why you support reform.

3

REPORT YOUR CALL

Your Email

Feedback for us:

Was your call on the air?

Select One
▼

SUBMIT YOUR REPORT

Figure 7-2. Image from barackobama.com; snapshot taken by The Political Guide in February 2010

Building the Behavioral Map

You know what you want to do (help volunteers call a radio show on behalf of your cause). You know something about your users (they are interested volunteers, but most have never called a radio program as part of an advocacy campaign). Now what?

Well, in order to call into a radio program, the volunteer will need to:

1. Find a quiet time and place with a radio and a phone.
2. Identify the radio program.
3. Listen to the radio program for an appropriate time to call.
4. Get the number to call.
5. Work up the gumption to actually call.
6. Call the program.
7. Convince the person screening calls that the volunteer has something interesting (and not crazy) to say.
8. Say something intelligible on the radio show itself.
9. Tell the volunteer HQ that the call was made so other volunteers can spread out their efforts to other shows.

That's a heck of a lot to do. Imagine if your product simply told users to find a radio program and call them about this issue. Each person would have to plan out the long list of things required, find the confidence to try this new strange thing, and not get distracted by other concerns along the way. They would also need to do some serious prior planning—planning ahead to find the program, find time in the day to call with having access to a radio, thinking about what to say, and so on.

Good luck. In fact, only a fraction of Americans has *ever* called into a talk show, and an even smaller subset has called in a premeditated, advocacy-oriented way.⁷

To use another example, for someone to take up running as exercise, there's a lot more required than simply walking out the front door and starting to

run. Some of the prior steps include (a) getting running shoes, (b) identifying a route and reasonable distance, (c) finding the time to do it, (d) remembering to do it, (e) making sure you haven't eaten heavily beforehand, and more. Again, a heck of a lot to do.

That's why we have products that can help people take action and make unwieldy tasks like these feasible. The process starts by writing out the obvious steps a user would normally take to complete the action. Make it detailed. List each physical and mental piece of work that's required, like the radio program example. When we're seeking to *stop* a behavior, the behavioral map outlines all of the micro-steps that the person takes leading up to engaging in the behavior. Each of those steps provides an opportunity for intervention—to help the person change. It's the same idea—break the big action into micro-behaviors.

Now that you have a basic list of steps, let's flesh it out into a full behavioral map.

Write or Draw It Out, and Add Behavioral Detail

The behavioral map is a depiction of the individual steps users take from whatever they are doing now, all the way through using the product and completing the target action (or for stopping an action, what they do that leads up to the action to be stopped). Some of those steps will occur within the product, and some require behavior that is completely outside of it. The map examines, at each step of the way, what's going on with users and *why* they would continue to the next step.

For those in the UX world, this should sound familiar, and intentionally so. You can express the behavioral map with a variety of design tools. I'm partial to customer experience maps like the one depicted in [Figure 7-3](#). They not only can include the stages of the individual experience, but they can also draw out the “customer types” (similar to our personas), areas of frustration and delight, and user emotions along the way. Related tools include a touch-point inventory and map, empathy maps, and journey maps.⁸

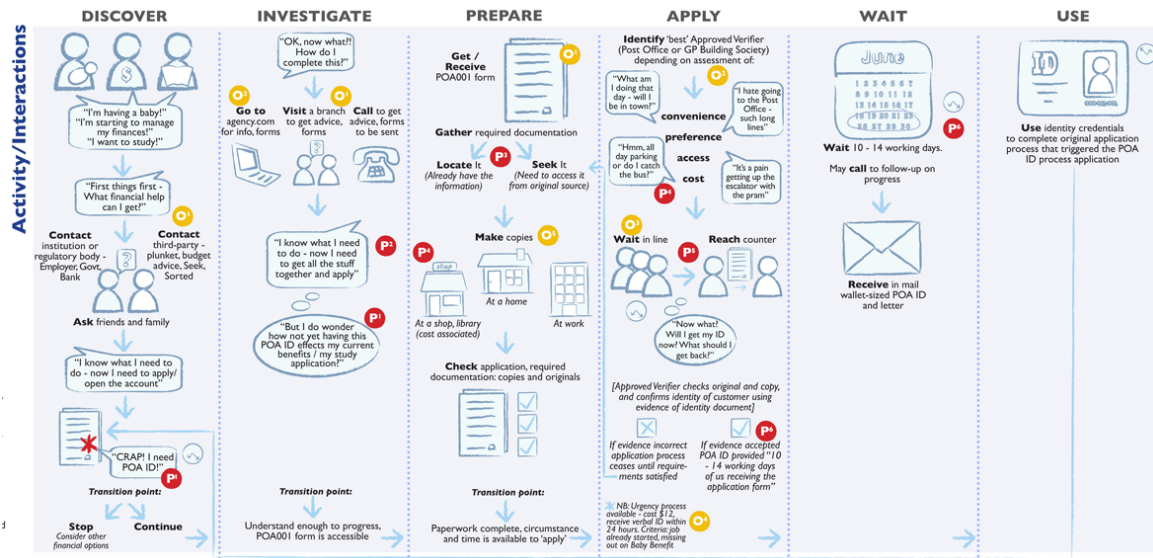


Figure 7-3. Part of a customer experience map from *Mel Edwards at desonance*

I don't have a graphic design background, so I've used much simpler tools to accomplish the same thing:

- The humble flowchart with notes in a sidebar
- A written narrative describing the user's experience and mental state at each step
- A hierarchical outline where each top-level point is a step in the user journey and under each point is a description of what's happening with the user during that step

No matter which tool you use to express it, developing the initial behavioral map is a bit different than a normal customer experience map process (if you're familiar with that process; if not, don't worry). Namely, make sure you cover the following points:

1. Write/draw out the rough sequence of steps in the real world—not just in your product—that a user must take to complete the action. (That's what we did in the previous section, listing the nine steps that person would have to take to call into a radio program.)
 - a. To be successful, you'll usually need to plan for how the product indirectly affects those other steps.⁹

2. Label each of the steps as follows:

- a. Something the user must do *within the product*
- b. Something *the product should do* in response to the user
- c. All of the other things that need to be accomplished “in the real world” (outside of the product) on the behavioral map

3. Look for missing steps, especially for new users.

- a. Take the perspective of a completely new user—one who has never interacted with your product. Are there additional steps required in the beginning (e.g., registration)?

4. Look for one-time steps.

- a. Take the perspective of an experienced user—one that has often interacted with your product. Are there steps that can be skipped for experienced folks?

For example, in the case of the Obama campaign, with volunteers calling in to radio programs, some of the steps clearly had to occur outside of the product, such as “Find a quiet time and place with a radio and a phone” and “Listen to the radio program for an appropriate time to call.” Most of the rest could be done within the app. Repeat-callers might need to do steps 2 or 4 (“Identify the radio program” and “Get the number to call”).

That’s the initial behavioral map. Nothing too fancy. Now let’s figure out where to focus our attention.

THERE'S RARELY A BIG CHOICE

Applied behavioral science means looking at people's behavior in a different light. Instead of assuming that people make clear choices and then act accordingly in one decisive action, we look at the many small actions people take and how those come together to have a broader effect—intended or unintended. And, even when people are being very intentional and thoughtful, we look at how a single big choice can be derailed at numerous places along the way. That's why we need a behavioral map: to find the small steps that are hidden within the seemingly simple big choice.

New Products or Features Versus Existing Ones

The behavioral map provides the series of micro-behaviors that lead up to the target action. For an existing product, the map is *descriptive*, like the volunteer radio program system's, which describes what people currently need to do. We use the data we've gathered from the existing product and about people's real lives interacting with the product to carefully trace out the micro-behaviors.

For new products (and features), the behavioral map is necessarily more speculative. It helps us make that hypothetical future user interaction more real by tracing through in detail what will really be required to use the product or feature.

In both cases, we usually find that the reality—what people do (or will do)—is far more complicated than we think. And that there are numerous steps along the way where people could get stuck. As you'll see, the behavioral map will help diagnose behavioral challenges—whether existing challenges in existing products, or challenges down the road, waiting for us in the products and features we plan to build, if we don't think them through and eliminate them now.

The Behavioral Map for Stopping Behaviors

What does it really take for someone to quit smoking? Often smoking cessation is presented as a command, such as, “Just stop! Don’t light up (or vape)!” But the reality is more complex. There’s a social and physical path that leads up to each instance of smoking or vaping. There are the colleagues who also smoke. There’s the box of cigarettes or Juul cartridge easily at hand. And of course, there’s the internal dependency and strength of habit. As with starting an action, stopping an action isn’t simply about that moment. Rather, it’s a series of moments and micro-behaviors that lead up to a fateful choice or behavior.

Just as these micro-behaviors present potential frictions to beneficial action, they also present opportunities to hinder negative ones. When creating the behavioral map to stop an action, pay close attention to the micro-behaviors that lead up to the final act and look for *vulnerable* or easily disrupted ones.¹⁰ It may be far easier to block one of the micro-behaviors that lead up to smoking than to stop the urge to smoke in the moment when there is a cigarette in front of you.

For example, if the cigarette weren’t there cuing the person, or the colleagues at work weren’t asking the person to hang out, the fateful decision of “smoke or not smoke” wouldn’t even occur. The behavioral map helps us find those creative opportunities for intervention rather than taking on the hardest point in the process (usually the final step).

Is There a Better Action for Them to Take?

You can think about the behavioral map as helping us understand where to intervene—which micro-behavior leading up to the action really needs our attention. The behavioral map can also show us just how hard it would be to intervene at all: How hard it will be to change behavior.

Sometimes the wisest choice to make isn’t to plow through, but rather to rethink the plan altogether. To see if there is a better action to target—one that can meet the goals outlined in [Chapter 6](#), but in a more efficient and effective way. Let’s look at how to thoughtfully reevaluate our core assumption about the action a user should take.

Techniques for Generating Ideas

How can you figure out the actions that people could take? What types of actions would make the target outcome happen?

There are lots of brainstorming and creative thinking techniques out there. I cut my teeth reading Edward de Bono, who popularized the term *lateral thinking*,¹¹ but you can use whatever works for you. Either way, don't stop until you have at least five different actions.

If you don't know where to start, here are some approaches that can help:

- What does someone do right before the outcome occurs?
- What's unique about the company? What user actions are easier to facilitate because of those unique aspects of the company (specialized skills, a special relationship with the users, etc.)?
- What do users already do that's similar?
- Why aren't people making the outcome happen?
- Why would users want to make the outcome happen? What action is most natural for them to take if they are motivated?
- Observe your users in practice. People find creative ways to change their own behavior all the time. Watch them for inspiration on what the product can do.
- Draw from a list of random words (yes, I really mean random—this is a technique from Edward de Bono). How is the word related to the outcome? How would a person act based on that word, in support of the outcome?

If possible, start small. Go with the bite-sized, easy things that the person could do to accomplish the outcome. That will make it faster to test and can be expanded upon later if needed. Look for the existing skills and habits of the users wherever possible and build on them. Try some crazy ideas. At this stage, don't self-censor and limit actions that seem impossible.

The action doesn't need to be something that the user would do while using the product itself—dieting is a great example. People don't (usually) eat while they are logged in to a dieting app on their computer or phone. Dieting occurs when people make choices about what to eat and how much. Many dieting applications are designed to help inform and prepare individuals so that when they *are* making food and eating choices, they avoid temptation and make better choices for themselves. There's a danger there, though—the further removed the product is from the action itself, the less likely it will be that it causes people to take action.

As you think of actions, there's necessarily a leap that occurs between the action and the outcome—the assumption that the action will actually work and produce the outcome the user and company seeks. We'll draw out that assumption and judge how risky it is a bit later. For now, I suggest focusing on coming up with some cool new ideas, even if some of them are uncertain.

The Obvious Is Our Enemy

These techniques for generating alternative actions are useful when you've hit a dead end; when you've explored the context of your users' lives, mapped out the micro-behaviors that would lead up to action, and figured out that it's just too hard. Then you go back to the drawing board.

Personally, I try to use these techniques from the beginning, not just when we hit a dead end. When I'm first approached with a behavior change idea or problem and asked how to support a target action, I try to put on the brakes. I try—I certainly don't always succeed—to question the action itself that we're looking to support.

For any long-standing and difficult problem, there's usually an “obvious” action for their users to take. The challenge is that for long-standing and difficult problems, the *obvious is our enemy*. Once we start thinking about the problem, a solution will often pop into our heads. It will intuitively feel right and we'll want to move on it. It's a cognitive mirage. Our minds have the nasty habit of conflating the ease of thought with truth;

that's our availability heuristic at work. But if a problem is difficult, what comes to mind quickly has probably already been tried *and failed*.

For example, let's say you've defined your target outcome as helping users put more money into savings. The obvious answer is to set a budget and spend less on something. It's obvious but it's also a really hard action for most users to undertake (at least, head-on). Other, less obvious actions might work better (e.g., automatically deducting the money from your paycheck so it's never in your checking account to tempt you). Similarly, in the case of retirement, when people don't save for retirement, the obvious answer is financial literacy. If people just knew about retirement, they would save. Unfortunately, the facts just don't show that.¹²

And so, when you're faced with a difficult problem, I encourage you to look for the nonobvious. Write down the obvious answer and then force yourself to come up with five other, unrelated solution. Use random word association. Use lateral thinking. Use whatever get you beyond your intuitive sense of a solution that *feels right*.

That said, we don't always need the big guns. For easy problems the obvious is obvious because it's actually the right approach to take. You just need to take that obvious path and follow through. In the first edition of the book, I placed this section front and center—pushing everyone to think of alternative actions as part of the initial problem definition phase. It's now here and optional to save you time and energy when this level of effort just isn't needed.

Select the Ideal Target Action

Once you've generated a list of alternative actions, what do you do with it? Combine them, and narrow down the list.

Before getting too fancy, first remove actions that are directly blocked by known impediments, especially if similar actions were tried but weren't successful in the past. Next, take each action and score it along the following criteria, ranking it *low*, *middle*, or *high*. To make the process more

concrete, imagine that the target outcome is to help users learn a new language:¹³

Impact (on outcome)

How effectively would it achieve the outcome?

In other words, assume that every user does the action, without reservation—how much would it help? When learning a new language, the action of repeating one word is not very effective and gets a “low”; practicing some sentences might be “middle”; immersing yourself in a foreign country gets “high.”

Motivation (for user)

What motivation do users have to perform the action?

Draw upon the data about the users’ existing motivations and their social interactions around the product. Users may be really excited to travel to a foreign country or make new friends by practicing their language skills in person (those two options get a “high” rating). They may have little interest (and negative associations) with rote memorization (that’s a “low”).

Ease (for user)

How similar is this action to things that the users already do in their daily lives (including their interaction with the existing product)?

Immersion in a foreign country to learn a language would (usually) be “low.” Repeating words or practicing sentences could be medium or high, depending on what you’ve learned about your users. Existing habits always get a “high” rating. Actions that require users to *stop* existing habits always get a “low” rating (see [Chapter 3](#)). This process may require subject matter experts to gauge the likely impact of the action on users.

Cost (for company)

How easy would it be for the company to implement a solution around the action?

True language immersion would be “low” for online products. Live conversation with native speakers could be a “medium,” depending on the existing resources and capabilities of the company. Providing scripts for users to repeat would be easy. This rating may require a lead engineer to assess potential resource costs.

With these ratings in hand, look for obvious outliers. If there’s a standout winner, great. If not, remove any hands down losers. If this doesn’t narrow down the list enough, make a judgment call about what’s most important and feasible for the company, given their business strategy. If resources are tight, then the cost of implementation naturally becomes rather important!

Behavioral economics can be useful here—certain behaviors and ways of thinking are inherently more difficult for (most) users, and we covered some of the high-level lessons in **Chapter 1**. For example, behaviors that require extensive mental calculations are difficult and often avoided. Actions that focus on long-term gains over short-term losses are also contrary to much of our cognitive machinery (losses are more painful than gains are good, short-term gains are valued more highly than long-term ones).

Beyond that, there’s not more guidance to give here, unfortunately. Keep narrowing down the list until one top choice remains or there are two neck-and-neck options that can be tested in practice.

With the new action in hand, update the problem definition. Write out the new action and fill in these details: whether it requires starting or stopping, who does it, and the micro-behaviors that lead up to it.

Updating the Behavioral Personas

If and when the team decides to change the target action, we must revisit the behavioral personas and see if they need updating. In other words, quickly look over each of the five questions (experience with similar

actions, experience with similar products, etc.) and generate additional personas as needed for our new user action. Often, the resulting personas may be the same across various actions, but be prepared for them to be different. Remember that unlike normal UX personas, again, behavioral personas are *relative to and defined by a particular target action*: since the goal is to create a set of personas that are likely to respond differently to the product's attempts to change behavior.

Diagnosing the Problem with CREATE

Now that we know what micro-behaviors individuals need to take in order to succeed, we can look more tactically at the changes we want to make to the environment. For that, we return to the CREATE Action Funnel from Chapters 2 and 3 both to help people take positive action and to hinder negative ones.

BEHAVIOR IS ALWAYS CONTEXTUAL

Action always occurs in a given moment. While preparation can occur over time, any *specific action* only occurs in the moment. While this may seem obvious, it's a surprisingly difficult lesson to internalize and understand the ramifications of. It means that even if someone thinks about the action (or product) often, it doesn't matter if they don't think about it in the specific moment that matters. Similarly, if people are generally motivated to act, but demotivated for whatever reason at a particular point in time that matters—no action.

Diagnosing Why People Don't Start

I've worked with many individuals and companies over the years to write out their behavioral maps. Often, when participants have mapped out a behavior they want to support, they say two things:

- Wow, there are lots of steps! What we're asking our users to do is just much more complicated than we thought!
- I know what do to!

Simply by exploring the problem in a more structured, focused way, the solution might become clear and straightforward. When it's not, we can use the CREATE funnel to help us *diagnose* the core behavioral challenge (and then work to fix it).

To see how this works, let's continue the previous example of helping political volunteers call a radio program. We outlined nine distinct steps, from finding a quiet time to make the call to telling the volunteer coordinator afterward. With this behavioral map in hand and our data about our users and their situation, we ask, "Where are the problems?"

- For existing products, where are people dropping off? At which stage are they stopping? Naturally, some people will falter at different stages, but we're looking for where the largest group gets stuck.
- For new products, the analysis is more hypothetical, but similar. Given what you understand about your users, where are they likely to struggle in this series of micro-behaviors?

For existing products and features, this analysis is based on the empirical data you've gathered about your users. **Figure 7-4** is an example from *Improving Employee Benefits*, a book I wrote for Human Resources professionals on behavior change in the context of retirement plans, health and wellness programs, etc.

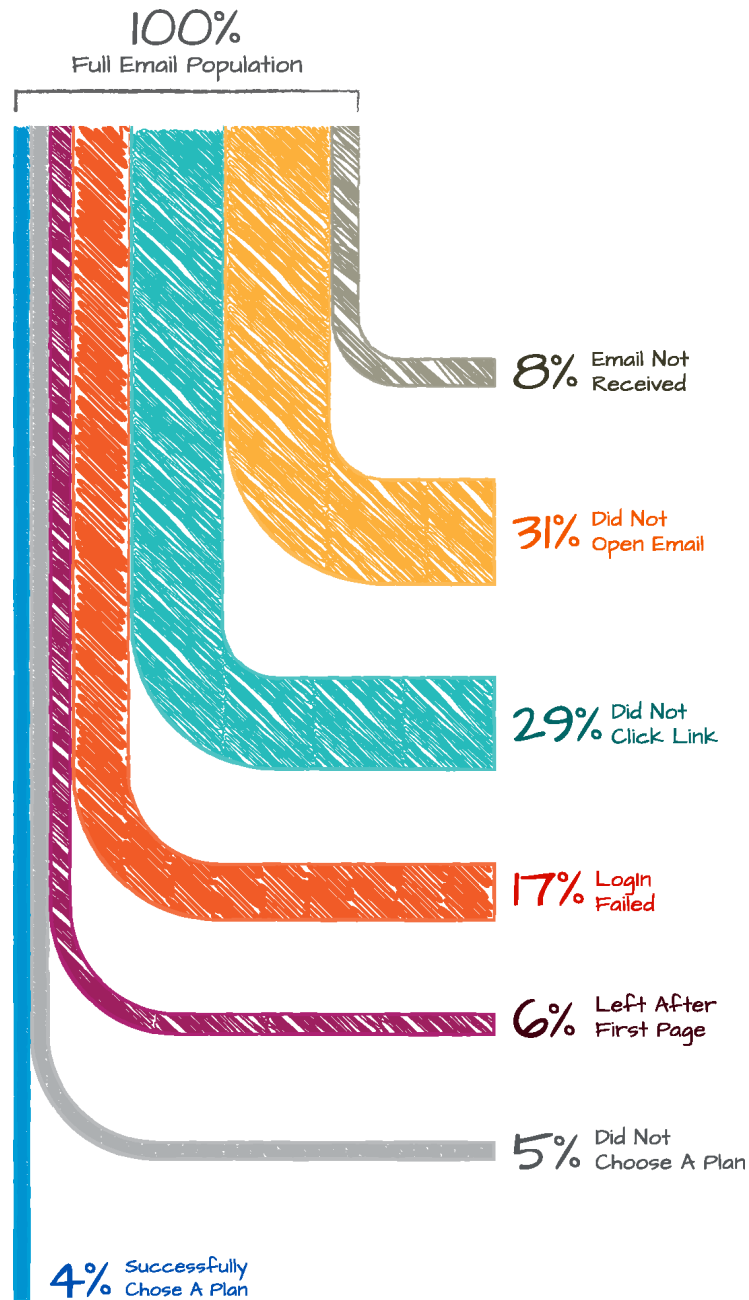


Figure 7-4. An easy way to visualize where employees are dropping off, with a conversion funnel

Once we've identified a problematic step, we use CREATE. During that specific step, and in that specific moment of potential action, is there a cue in place to take the action? (And is it sufficient?) What emotional or intuitive reaction does the person have toward that action? Is the action motivating, and do the benefits outweigh the costs? And so on. Whenever the answer is no, that's a behavioral obstacle. That's where we'll focus our attention in the design process.

In the case of the radio-program example, an obvious problematic step would be “actually making the call”—that is, overcoming the fear and uncertainty (an emotional reaction) of undertaking this new and unusual action.

Diagnosis is thus a three-part process:

- First, we identify the micro-behavior that stops people (or for new products, are likely to stop people). That’s our behavioral map.
- Second, we check which micro-behavior seems to be a problem. Where are people dropping off (or likely to drop off)?
- Third, we use the CREATE Action Funnel to determine the likely behavioral cause.

With these three in hand (a map, a point—or points—in the map that’s a problem, and a cause for that problem), we have our diagnosis. We have what we need to start designing a solution.

That’s for *starting* an action. Let’s look at the similar, but slightly different, case of *stopping* an action.

Diagnosing Why People Don’t Stop

The good news about stopping an action is *diagnosing* the problem is easier—even if solving it isn’t. Each of the CREATE factors must be in place for the person to take the (negative) action. Diagnosis entails outlining what those factors currently are. What’s the cue that triggers people to take the action? What’s the intuitive or emotional reaction they have? And so on.

When you’re helping the user stop an action, it is essential to know if the action is habitual; that is, whether it is something the person consciously thinks about or not. That’s because, as we talked about in [Chapter 2](#), habits work differently than conscious behaviors. In a habit, the Evaluation, Timing, and Experience steps are largely skipped (or they are hardwired internally). A Cue triggers a Reaction, assuming the person has the Ability

to take action. We focus on the C-R-A part of CREATE, rather than the whole step.

Here's an example of diagnosing the point of intervention with a common habit: binge watching online TV. What are the microactions that lead up to binge watching?

- Picking up the computer or phone.
- Logging into the Netflix app or site.
- Selecting a movie or series.
- Watching it.
- Letting autoplay load up the next episode.
- Continue watching it, etc.

At each step, there are either habitual elements (C-R-A) or conscious choices (CREATE). Picking up the phone is often habitual. What's the Cue? It's the sight of the phone (external cue) or a moment of boredom (internal cue).

The diagnosis for a behavior you want to stop entails:

- Identifying the micro-behaviors that lead up to action—the behavioral map.
- At each micro-behavior, determine if it's habitual or conscious (often if the final action is habitual, so will the lead up to it, but not necessarily).
- Use CREATE for conscious actions and C-R-A for habitual actions to map out the current enabling factors for each micro-behavior.

The behavioral map, our understanding of the nature of the behavior, and list of enabling CR(E)A(TE) factors is just what we need for designing a solution.

To give you a preview: we've all heard the advice to avoid distractions by leaving one's phone somewhere you can't see or hear it. That's an example of intervening at the first microaction (and, specifically, intervening in the Cue stage of the CREATE Action Funnel). Let's say that isn't possible, and you can't stop the person from having the computer or phone handy. But you can make changes to how the person logs in (changing the password to something that isn't memorable!) or to the autoplay process (changing the settings so autoplay doesn't occur).

The behavioral problem with autoplay is that it is cheating: it takes the burden of work from the individual and places it with the software. We can counter that by undoing that clever design and bringing the burden of work back to the individual. In other words, turn off autoplay. When *hindering* an action, we'll use this behavioral map and list of enabling factors to look for a micro-behavior that is both essential to the subsequent process and feasible to change. Then we use CR(E)A(TE) to identify where to *create* a behavioral obstacle.

Putting It into Practice

This chapter is all about checking your assumptions about your users and their situation so you can refine our vision what they'll do and diagnose the specific behavioral obstacle(s) they face.

Here's what you need to do:

- Research and document the characteristics of your users, especially around prior experience with the action, prior experience with the product, existing motivations to act, their relationship with the company (trust), and barriers to action.
- Generate behavioral personas—groups of users that you expect will respond differently to your product's attempts to change behavior.

- Develop a behavioral map showing the sequence of steps the actor takes to go from their current state to taking action.
- Rate alternative actions for users to take in terms of their effectiveness, cost, motivation, and feasibility for the user.
- Select the ideal target action, based on these criteria.

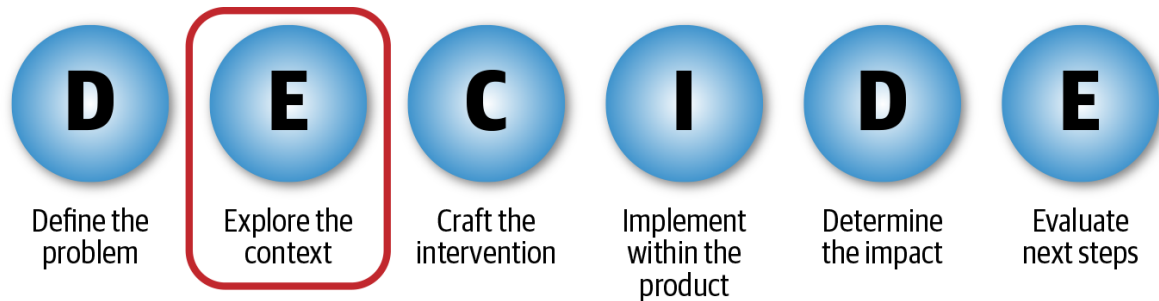
How you'll know there's trouble:

- It looks like all of the users are alike—they usually aren't. You probably haven't dug deeply enough into their existing experiences and behaviors.
- When rating potential actions, all of the actions are rated similarly, or all of them are too expensive to the company or infeasible for the user to be realistic (sorry, go back and think up more user actions).

Deliverables:

- Detailed observations about your users.
- A set of user personas, indicating the main groups of users (or potential users) of your application and their characteristics.
- A behavioral map outlining the microactions the person takes.
- On that behavioral map, the particular obstacle(s) you believe your users face or will face—a diagnosis.
- A potentially updated project brief, with the target outcome, actor, and action.

Worksheet: The Behavioral Map



Describe each micro-behavior the user needs to take to move from inaction to action. Then ask whether the six CREATE preconditions for action are in place at each step. Check existing preconditions off the list and describe them briefly, for reference. Where one is missing, think about how you can restructure the action, change the environment, or educate the user to help move him or her through the process.

What Is the User's Initial State? Sedentary, does not normally exercise

What Does the User Do First? Opens email inviting him or her to download the app

- ☒ Cue: Email from employer
- ☒ Reaction: Neutral, receives various emails from employer
- ☒ Evaluation: Low cost to open, generally important
- ☒ Ability: Easy
- ☒ Timing: Emails from the company should be opened quickly
- ☒ Experience: Neutral

What Does the User Do Next? Installs the app using employee ID and unique password

- ☒ C: Call to action in email
- ☒ R: Ug. Another app
- ☒ E: Could help me get in shape
- ☒ A: Doesn't have employee id handy
- ☒ T: Sitting at home anyway, nothing pressing
- ☒ E: Neutral

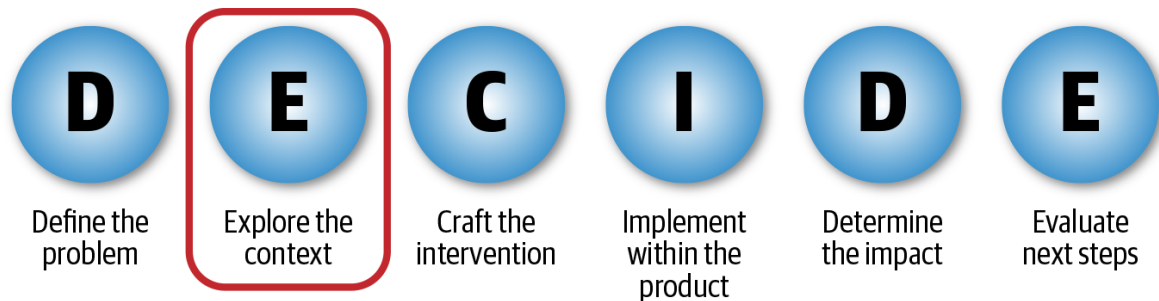
What Does the User Do Next? Goes to a class at the gym!

...and so on for each micro action. Look for the first major obstacle: that's your behavioral diagnosis. There may be subsequent problems, but if they

don't get past here, they aren't relevant.

In the case of our exercise app, we find two problems early on in the process: the user's negative emotional reaction to downloading another app, and the inability to log in to create an account.

Worksheet: Refining the Actor and Action



For some products, especially existing ones, the actor and action are obvious and not in doubt. In that case, skip this exercise. However, as you Explore the context, you may learn more about your users and their situation and realize that your initial assumptions were incorrect. If you think the product might appeal to or help another group, or that your team is offering the same tired solutions that haven't worked in the past, this worksheet can help you revisit and refine them.

Action: Brainstorm four very different actions that people can take to achieve your target outcome because of your product or communication. When thinking through possible actions, keep these points in mind:

- The obstacles people currently face to achieving the outcome
- What needs to happen right before the outcome
- How your company is uniquely positioned to help people achieve the outcome
- What people who currently achieve the action are doing

Action 1: Solo run twice per week, starting with two miles.

Action 2: Write down exercise goals.

Action 3: Get a personal trainer at the gym.

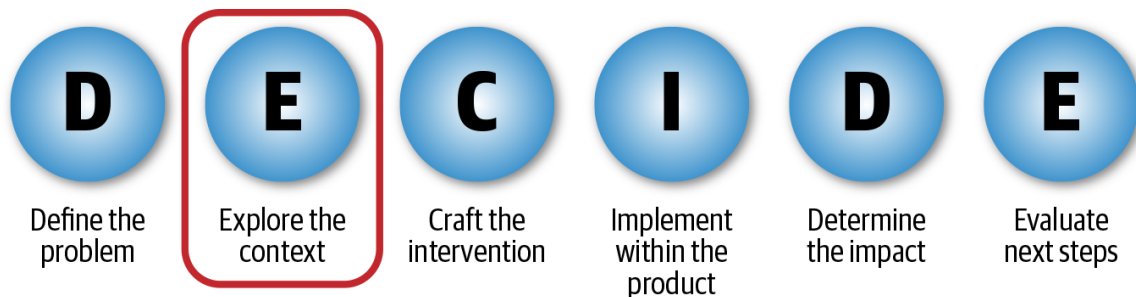
Action 4: Participate in an in-person workplace fitness program.

For the sake of argument, let's assume that for our Flash exercise app, the company has locked in on the action (going to the gym) and the target user (white-collar relatively sedentary employees), and we'll continue with those

targets throughout the rest of the process. In the [workbook](#), you'll find an additional exercise you can use to brainstorm different types of actors and evaluate among possible actors and actions..

- 1 This case study is based on an interview and subsequent email exchange with Matt Wallaert.
- 2 They assessed doctor quality via a combination of self-reported patient satisfaction, patient outcomes, cost, and availability.
- 3 Since we were doing microtargeting, the end result of this process was a set of machine learning models of the propensity of ActMore's members to respond to different product features, which we then field tested before rolling out the product for real. We used quantitative data from the organization and from third-party providers. But the core concept is the same in a less quantitative-data-heavy environment. Who are the users, and how will they respond differently to appeals to change their behavior?
- 4 My thanks to Darrin Henein for highlighting our shared base of techniques.
- 5 Thanks to Jim Burke for highlighting the importance of direct observation.
- 6 This tactic isn't unique to the Obama campaign and was somewhat controversial for trying to generate a sense of (extraordinary) support on the ground for the legislation. But it's a darn good example of helping people voluntarily take an action they wouldn't otherwise take.
- 7 There doesn't appear to be much data on the topic, but even when radio was more popular, a 1993 Times Mirror Center for the People and the Press (predecessor of the Pew Research Center) survey estimated the percent of Americans ever calling into a talk show at any point in their lives at 11%, and roughly half of them had actually been on air. See Pease and Dennis (1995).
- 8 See Kolko (2011) for various examples of journey maps, touchpoints, and concept maps. Xplane developed the empathy map; take a look at [an interactive example](#). You can use tools like the [Touchpoint Dashboard](#), but a whiteboard or some sticky notes work well too.
- 9 The reason I start with the real world is that for many behaviors (e.g., controlling finances, exercising, getting politically active, decreasing energy usage), the product only interacts with part of the story, but all of it determines whether users will actually change their behavior.
- 10 My thanks to Clay Delk for highlighting this point.
- 11 De Bono (1973)
- 12 [Fernandes, Neydermeyer, and Lynch \(2014\)](#)
- 13 This technique of rating potential actions is inspired by BJ Fogg's method *Priority Mapping*, in which he rates behaviors on ease of implementation and effectiveness. Also, there's certainly a science to teaching people new languages, and I won't go into those methods here; this is just a stereotyped example.

Chapter 8. Understanding Our Efforts: A Brief Story About a Fish



Imagine that you're walking along a beach. You see a fish stranded in the sand, flopping around a few feet away from the water. Let's further imagine that you're feeling more helpful than hungry. What do you do?

Do you walk up to the fish and yell, "What's wrong with you? Don't know how important it is to be in the water? If you just understood the value of water like I do, you'd get into the water!"

Or do you calmly and thoughtfully try to teach the fish, "Let me explain to you how your gills work. Your gills extract dissolved oxygen in the water and put it in your bloodstream, where the rest of your body can use it. If you aren't in the water, you can't get dissolved oxygen, and you'll soon die."

No. That's obviously foolish. The fish isn't lying in the sand because it lacks motivation, nor because it lacks understanding. Maybe it doesn't know the details, but it "knows" the most important part: it needs to get back into the water.

Instead, there are really four things you can do in that moment.

First, you could pick up the fish, hold it in your hands, and walk it over to the water where you drop it in. All the fish needs to do is stay put and not flop around too much in your hands—it's not trivial, but it's a whole lot easier for the fish than to get back into the water on its own.

Second, there might even be a way you could take care of everything. You'd somehow put the fish to sleep, transport it to the water, and then wake it up again. The fish wouldn't have to do anything at all (it wouldn't even need to resist the urge to flop around).

But if you aren't in the mood to get your hands slimy holding the fish, you could dig a channel of water in the sand, leading from the fish, down to the water. You could then take some water and pour it over the fish. That would make it easier for the fish to do what it already wants to do: follow the water back to safety. That's our third option.

And, fourth, if you had had the luxury of spending time beforehand to train the fish how to flop better, that training might have been useful. But when the problem is acute, you usually don't have that luxury.

Either way, yelling, educating, or relying on the fish to rescue itself aren't great ideas.

When designing for behavior change, our first instinct is to yell: to try to motivate someone to do something. Or, we try to ram facts and figures down their throats, thinking that if they just understood the problem as well as we did, they'd act. Neither is very helpful. Our users (who aren't anything like fish, but at least the story get the point across) are the ones with the problem; they may not know all the details of the situation, but they usually understand the basics already.¹

Instead, where we should start is by taking a hard look at the action itself, the thing that people are struggling with. Is there an easier way to accomplish the same end? Something that is more natural for the person to accomplish?

We should (1) *change the action* to make it easier. The most extreme version of changing the action is to completely take the burden of the work

onto ourselves, or (2) *do it for them* so the person doesn't have to do anything at all. Alternatively, we might look to (3) *change the environment* to make it more likely for the person to take action. Finally, if we have the time and ability, we might (4) *change the person* to help prepare them for the moment of action (see **Figure 8-1**).



Figure 8-1. The four approaches to behavior change: do it for them, change the person, change the action, or change the environment

Each of these options shapes the context of action. Each is designed to increase the likelihood of a good outcome; they are the four main ways we intervene in a situation to engender behavior change.

The purpose of the design process is to *craft a context* that facilitates (or hinders) action. Crafting a better context rarely entails yelling at a fish.

Let's look briefly at the extreme case of changing the action: doing it for them. That's when we *cheat* at designing for behavior change and do whatever is needed on their behalf. It's great when we can take this magic solution, and we should always look for it, but we shouldn't count on it being available. Afterward, the next few chapters look at the details of changing the action, environment, or person.

Do It for Them When You Can

While you can make an action rewarding, easy, familiar, socially acceptable, or any of the other things we talked about in **Chapter 2**, the activity still involves work for the user. Ideally, the company could find ways to shift the user's burden onto the product by identifying clever ways to make active participation unnecessary beyond informed consent. That's what I call cheating—substituting the user's nasty problem with a much simpler one: deciding whether they want to let something else (the product) take the action for them. As you'll see, this strategy is only available in certain cases, but when it is feasible, it is immensely powerful.

Exactly how a company can “cheat” depends on whether the target action is undertaken once or infrequently (like buying running shoes) or repeatedly (like going running each morning). I'll talk about each of these two situations in turn.

Strategies to Cheat at One-Time Actions

To cheat at a one-time action, either you can take care of the action yourself directly, by automating it, or you can make it a side effect of another action. Let's look at both in more detail.

Automate it

To automate an action, the company first finds a way to take the action on the user's behalf. Then, usually the company combines it with another work-saving technique: defaults. They give the user a choice about whether the product should take the action on their behalf, where the default answer is “yes.” The user can say “no” if they so choose.

Most automation is invisible—you don't even think about it as automation; it just happens. In fact, we're not used to seeing the automation that's all around us, so we rarely think of it as a solution. To that point, the most common reaction I get to proposing automation is, “That's great, but there's no way that will work here. You can't automate this behavior.” Well, maybe. Here are some examples to show how automation works in real life:

Behavior change sought: have users save for the future

Two of the greatest success stories in recent history in helping users save money are 401(k) autoenrollment and autoescalation.² (For non-US readers, 401(k)s are retirement savings plans provided by an employer to employees.) Under autoenrollment, individuals who are eligible to participate in their company's 401(k) plan are defaulted into contributing to the plan but are given the option to not contribute if they wish. Contributions are then automatically routed to their 401(k). Similarly, autoescalation increases the contribution rate over time, but the individual can opt out at any time.

The initial action users take is often quite minimal; for example, signing their name on a package of new-employee documents. Afterward, contributions to the 401(k) plan are automatically deducted from their paychecks and placed into their retirement account on their behalf. Instead of requiring that an individual choose to contribute to the retirement plan each month (or choose to find the HR representative with the necessary paperwork required to enroll in the plan), this process effectively removes the work required by the user.

401(k) autoenrollment is a powerful example of increasing savings, but it also can skirt the line between *voluntary* behavior change and trickery. Some employers strive to inform employees about their retirement plans and default contributions. In other cases, the employees don't know about their accounts until they leave their job and get a check—which they quickly spend on non-retirement needs, since they weren't informed and invested in the process in the first place.

Impact of automation and defaults

It's significant in this case: autoenrolled, defaulted-in plans have nearly *twice* the participation of non-defaulted plans.³

Behavior change sought: have users take high-quality photos rather than crappy ones

High-end camera manufacturers have a problem: many consumers want lots of features, but those same features make the camera sensitive to user mistakes and result in bad pictures.

Good cameras have a simple solution that help people take quality pictures, but still provide power options (and a premium price): the cameras have powerful built-in features that can handle much of the work for the user. These features (like autofocus or red-eye reduction) are defaulted on, making the camera dirt simple to use and providing a good picture in most scenarios. In addition, the cameras have all of the fancy bells and whistles that make the product more attractive and expensive than a bargain-basement camera.

Similar combinations of automation and defaults are common in computer software (“Would you like the standard install or the scary customized one?”)—the options are there, but the software makers have provided intelligent defaults so most people don’t have to worry about them and install the software without getting themselves in trouble.

Impact of the automation and defaults

Apparently, cameras still can’t help us take *interesting* pictures. More seriously, though: do any mass market cameras *exist* anymore that don’t have intelligent defaults for things like contrast, white balance, and F-stop?

Make it incidental

If the action can’t simply be defaulted away, there’s another clever option—have the action come along for the ride with something else that users are going to do anyway. In other words, don’t have them think about doing the action at all. Make the action happen automatically when the user does something else—something that is inherently more interesting or engaging—but leave the option for the user to decline the action if they so choose. Here are two examples:

Behavior change sought: improve people's intake of vital vitamins and minerals

OK, before I go into the solution, what's the most effective way to improve the amount of vitamins and minerals people get? Convince them of the benefits? Pay them to eat well? Run a public campaign with celebrities endorsing vital minerals? How about this: put it in the food people already eat—with their consent, and without removing other food options. For example, put iodine in salt.

Iodine deficiency is the leading preventable cause of mental retardation.⁴ It causes stunted growth, infant mortality, lower IQ, goiter (big lumps in the neck), and more.⁵ Two billion people suffer from insufficient iodine. Iodine costs virtually nothing to produce and add to salt.

The story of iodized salt also shows that defaulting can't be allowed to turn into coercion, either practically or ethically. At various times, people around the world have objected to iodine being added to their salt without their consent, causing these iodization campaigns to fail (and iodine deficiency to continue). When there is no way to opt out (non-iodized salt) and no consent, it's not "defaulting"—it's just an ethically problematic mandate. There must be consent among the population *first*.

Impact of making iodine incidental

In many of the places where iodized salt has been used (with consent), the problem of iodine deficiency simply ceases to exist; that's the ideal outcome of any behavior change strategy. In the United States, iodine deficiency is rarely an issue anymore, except where it hasn't been made incidental.

Behavior change sought: have people (voluntarily) contribute money to savings

One solution in this case is a savings lottery, aka prize-linked savings accounts. A prize-linked savings account is like a lottery in that people can “buy” multiple tickets.⁶ Each ticket is a contribution to their savings account. Like any lottery, there’s a jackpot—a big pool of money that one (or more) winners get. Unlike a normal lottery, the participant doesn’t lose the cost of their ticket; it’s just deposited into their savings account.⁷ There’s a significant upside, but little downside, to participating.

For users who enjoy playing lotteries, the *savings* part is incidental: they would spend money on the lottery anyway;⁸ the fact that they don’t lose the ticket money is a nice added bonus, but not the primary reason for them to participate.

Impact of making it incidental

Prized-linked savings programs have been highly popular around the world for centuries, starting in Britain.⁹ They have recently gained traction in the US through the tireless work of Commonwealth, a Massachusetts-based NGO.^{10, 11}

And there are many more examples that we rarely think about in our daily lives. If you want your toddler to take a pill, you crush the pill up and put it in some juice they like. The toddler doesn’t care or know (and if they don’t know, they can’t complain) about the pill; it’s incidental. The juice is what matters.

Even when behavior change is a *side effect* of the product, the same ethical rules should apply as when the user directly takes action.¹² Is the side effect something that is truly beneficial for the user? Would they be surprised or upset if they learned about the side effect? Or, even better, have you told your users that it’s occurring?

Strategies to Cheat at Repeated Actions

You can also use these two approaches, defaults and making the action incidental, with repeated actions. For example, with prize-linked savings programs, the savings lottery can be repeated each month to encourage sustained savings contributions. Each time the person acts, savings are incidental. Similarly, each time the person uses an application, they can encounter the same (configurable) defaults.

In addition to these two approaches, another becomes possible with repeated actions: you can turn the repeated action into a one-time action by *automating the act of repetition*.

Automate the act of repetition

Taking an action repeatedly is inherently more difficult than taking that action once, even when the person learns how to do it better over time. So, why not turn a repeated action into a one-time action?

In this scenario, the individual takes a one-time action to set up or accept the automated process, and then the rest is handled without their intervention by the product itself. The principle is simple and is very similar to defaulting a one-time action: use behind-the-scenes magic to shift the work from the user to the product.

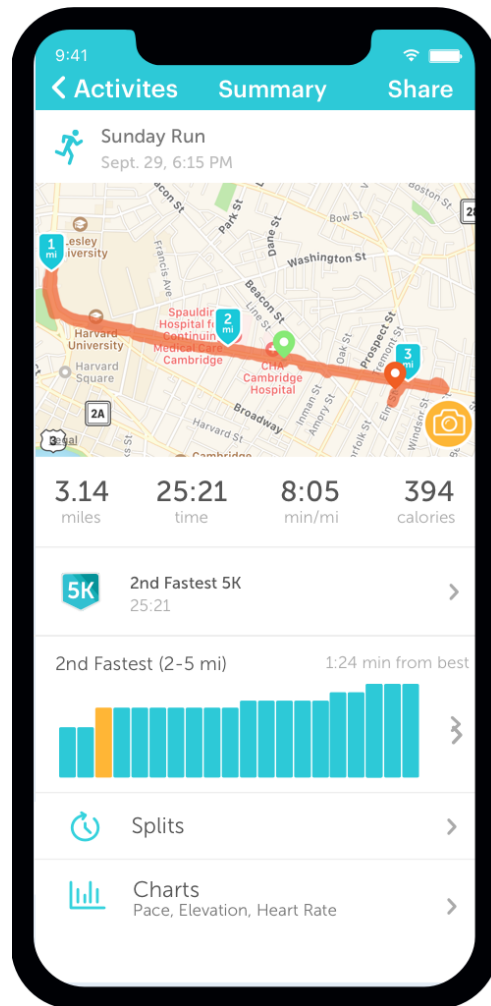


Figure 8-2. Runkeeper, an app with automated exercise tracking (image by Runkeeper)

Some great examples of automating repeated behaviors in the health space are exercise trackers that people carry with them throughout the day. These include bands from Garmin and Fitbit and apps (e.g., Runkeeper) that use GPS or a phone's accelerometer to accomplish this without a separate device (see **Figure 8-2**). These apps and devices automatically log and compare exercise against a user's target. They've successfully taken something that is annoying but beneficial (logging exercise in a journal, comparing it to one's daily goals) and made the work magically disappear. Once *exercise tracking* was automated away, companies could focus on more interesting (and user-beneficial) target actions—like helping users *exercise more*.

Another example of automating behavior comes from the personal finance space, with software that automatically categorizes transactions and tracks spending—such as Acorns, Mint, and numerous bank websites. In the “old” days (i.e., the 1980s), if you wanted to know how much you had in your checking account, you had to track your spending and balance your checkbook (remember checks?). When ATMs became popular in the 1990s, you also had to track your cash withdrawals. If you had a credit card, it would send you a monthly statement, but before that arrived, you were out of luck.

With personal finance applications, tracking expenses can happen automatically. Each individual transaction is automatically logged, categorized, and, where relevant, compared against a goal or budget item. As with many other forms of automation, once the action is automated for the user, the product team is then free to focus on more interesting and difficult-to-change behaviors, like helping users stay within their budget. But that wouldn’t be feasible for most users if they are wasting their time tracking their spending first.

The most powerful combination of all is to combine automation with defaulting—automation makes it a one-time action, and defaulting makes it little more than an acceptance of that automation. I didn’t go into detail about this earlier, but 401(k) auto-enrollment is such an example—the savings contributions are automatically deducted, and the default is to enroll in the program.

But Isn’t Cheating, Well, Cheating?

Before I move on to other behavior change strategies, I’d like to confront an implicit assumption that I’ve seen in many do-good products—that doing good requires making our users work hard. If we, as people designing for behavior change, want to help people take an action, we should be pushing people to climb that hill! We know it’s hard, they know it’s hard, and that’s what makes it worthwhile, right?

Well, no.

If the goal is to make people healthier, the action is consistent with that goal (say, by making the food that people already eat magically become healthier but taste and cost the same), and automation doesn't have nasty side effects, does it inherently matter if the user doesn't have to work hard for it? I just can hear my own inner do-gooder say, "Well, that misses the point—we want people to make wise choices, learn about the wonders of nutrition, be grateful for all the energy we put in to help them," etc. Sometimes, when we design for behavior change, we secretly want more than to simply *help* our users: we also want them to be a certain type of *virtuous person*. We want them to learn all the important things we know, we want them to care about the things we care about, we want them to exert real effort toward their goal—to demonstrate their determination and commitment. However, these desires are about the behavioral designer, not about the user and what helps the user succeed.¹³

That's why it's vital to be clear about the *end goal* of the product. For example, teaching people what we know about healthy eating is a laudable goal. But do we really only want to teach people? Or, do we teach in order to help people change their eating habits, which then makes them healthier in the long term? If we could jump ahead, solve this *very particular problem* and move on to something else, wouldn't that be a good thing? Maybe making food healthier helps with the goal of decreasing vitamin deficiencies, but it doesn't solve the issue of cardiovascular disease. Great—once the food solution is in place, then you can devote your energies to the next problem: helping people decrease cardiovascular disease.

Any product will have multiple aims. But there should be one clear thing that you gauge its success against—a final outcome or goal (that one thing can be a composite of multiple smaller things). We covered how to identify and fine-tune the product's goal in Chapters 6 and 7; let's assume you've already done that. When you're clear about what exactly is being sought, go for it, even if it feels like cheating because it doesn't make people suffer. There are no martyrs in beneficial behavior change. The point of making work magically disappear is that you can move on and help your users with other, more intractable problems.

There's good behavioral science behind this point too. In short, our self-conceptions are constantly adapting based on our own behavior. We often forget or ignore the reasons why we do things and develop a story of who we are based on what we observe about our own behavior.¹⁴ For example, if we successfully contribute money to a retirement plan, even if we were defaulted into it, we suddenly feel that that's something we can do—we're savers! The pride that people feel at saving money through automatic enrollment is real and should not be discounted. That self-conception as a saver then has knock-on effects for other related behaviors—we're prepped for future action.

A classic study in this field is by Freedman and Fraser,¹⁵ in which the researchers started by asking homeowners to put a small sticker in their window encouraging safe driving. Weeks later, this randomly selected group was far more likely to accept a large lawn sign about safe driving than other homeowners; a whopping 76% of them accepted the large sign, compared to 17% who hadn't been asked to show the small sticker. In other studies, homeowners were also more likely to accept other non-driving-related lawn signs. The homeowners started to see themselves as people active in their community, which had broad effects on their behavior.

There are cases when this doesn't work, of course. When people don't know the action occurred at all, then the self-conception doesn't change—but in that case it isn't a voluntary behavior change at all; it's behind-the-scenes trickery.

Cheating at the Action Funnel

Remember the CREATE Action Funnel—**Figure 2-3**? It's difficult for a user to pass all of the way through the funnel from the initial cue to a conscious choice to act with sufficient urgency. The *cheating* strategy takes the funnel and changes its meaning. With a conscious choice to take a hard action, success occurs when the user passes through the funnel. When the product cheats, success occurs when the user agrees to the action occurring but *doesn't* pass through the funnel to stop it from occurring.

When You Can't Do It for Them, You CREATE

When you have the ability to perform magic, you should: magically make the user's behavioral challenge go away (while giving the user the option to say no). Sadly, magic isn't as common as we'd like. And so, the next two chapters walk you through the details of crafting an intervention—how you specifically change the action, environment, and person to help facilitate (or hinder) action. The solutions—interventions—are intimately tied to the behavioral diagnosis from [Chapter 7](#). Solutions, like behavioral problems, employ the CREATE Action Funnel.

As you move into developing a solution, here are some additional lessons from behavioral science about the philosophy or approach to designing for behavior change:

- Remember to look beyond the user's motivations
- Remember to look beyond the screen

Look Beyond Motivation

As we saw in the opening story, the fish had all the motivation it needed to get back in the water—its problems lay elsewhere. That part is meant to encourage us to look beyond the user's motivation when designing for behavior change, and it warrants a bit more attention. That's because one of the first places that designers and product managers look to change behavior is to increase motivation. In my experience, this is the tactic that most of us think of when we're asked how we can encourage people to do X. The problem is, extra motivation is often not what's really needed, or at least, not on its own.

Here's a real-life example: the financial rewards of investing in a retirement plan when your employer contributes money (i.e., a matching contribution) are tremendous. We're talking about free money that grows for decades, tax free, with the magic of compound interest. Employees need to personally contribute some of their own money to get the match, but that's something they know they need to do anyway in order to enjoy their retirement (and

not live in their kids' basement when they're 70). But up to 50% of people won't do it unless something more is done to encourage behavior change¹⁶—like automating the process (automatic enrollment) or forcing people to make a choice (a strong trigger to act).

To make this point another way, think about this: what is the most effective way to earn gads of money? That is, how can you receive the most financial reward? For most of us, we simply have *no idea*, and we don't waste our time looking for that “optimal-money-producing” answer. We've picked the best career path among those presented to us through a mix of what we're aware of, what seems feasible, the diverse set of motivations and options that fit our personal stories, and prior experiences. Money, as with any other type of motivation, is only part of the story.

Here are some challenges that arise when focusing too much on motivation:

- When a product is helping a person take an action they already want to do, the person by definition already has some motivation. For most “good” behaviors, like exercising, everyone has already told that person how important it is—another voice in the choir isn't going to add much. (Though sometimes the motivation is too far in the future, as mentioned earlier.)
- There are *always* competing motivations to do other things. Understanding which exact action we take, why, and when, is the kicker. The answer is often not “the most motivating one.”

This isn't to say that making people more motivated isn't important—it is. It's just not enough. Tactics to increase motivation (like highlighting the user's existing reasons to act, or experimenting to find the right motivator for your users) work best when carefully executed along with other parts of the CREATE Action Funnel from **Chapter 2**. More motivation improves the chances that a person will successfully pass the intuitive reaction and conscious evaluation components of the funnel. But don't forget that there's other work to be done as well!

Let's say you offer a badge of recognition when users complete an action, which is a tactic to increase motivation. That's useful but needs to be combined with the rest of the CREATE Action Funnel. For example, for the users to benefit from that extra motivation and eventually take action, they first need to be aware of the reward. In other words, they need to be *cued* to think about it. The target action must also *out-compete* (be relatively more motivating, more urgent, etc.) other potential actions the user could take. Those two tactics—cueing and blocking the competition—are coming next.

The Value and Limitations of Educating Your Users

Educating users in a behavior change context is all about giving them the information they need and then hoping that when the time comes to act, they will make a well-informed choice.

Focusing on information is a noncontroversial and common approach to behavior change. We think that everyone would believe like we do if only they had access to the same information and training. This has been an approach taken by many of my otherwise favorite NGOs and government agencies.

Unfortunately, information doesn't equalize us and doesn't make us behave the same. As we saw in [Chapter 1](#), conscious information might have nothing to do with action at all—the action might be habitual or based on intuitive reactions. Or, when it does have an impact, it might be filtered through all of the rest of our experiences and information.

Providing users with information can be immensely powerful. But we should be thoughtful about how and when it's applied. Education efforts falter when:

- The action that people will take isn't consciously thought through at all—it's habitual or otherwise automatic.
- People are overwhelmed with too much information.
- The information comes too long before (or after) the decision needs to be made. We rapidly forget unconnected, unused facts.

An example of an education-only approach that clearly didn't work is mortgage disclosures in the United States. Mortgage lenders are required to provide reams of documentation on everything from how the loan works to the fact that most older homes have lead paint in them. This is important information—and rarely read. It's too much to take in, not structured to demand attention, and isn't clearly actionable. The only "action" the mortgagee can take by the time they get all of the disclosure documents is to walk away, with no house, and no certainty about where the escrow money goes.

To better understand when education can be effective, let's use a common example: educating people about the importance of saving for retirement with financial literacy seminars. In the 1990s, there was a significant increase in the use of retirement planning seminars as employers shifted from pension plans to 401(k) plans that individuals directly contributed to and managed. These seminars, and other financial literacy programs in high schools and beyond, have been the subject of quite a lot of controversy, with numerous researchers questioning their impact.¹⁷

Consider three different approaches that a retirement education product could take. It could educate people about *why* an action is important, *how* the action works and the raw data needed for a good decision, or *what to do* to take the action:

Why

By and large, we already know that saving for retirement is important. No one wants to die in poverty. However, for other beneficial actions, users of the product may not honestly know the importance of the action; for example, I didn't know that skin cancer screening was important for (relatively) young people until recently.

How

We don't understand the inner workings of 401(k) plans and score poorly on financial literacy questions about basic topics like compound interest.¹⁸ And there's evidence that this knowledge helps us make good financial choices.¹⁹ But if the information is delivered too far in advance from a moment of decision, as many financial literacy programs are, we simply forget it.

What to do

We don't know what to do when confronted with dozens of 401(k) options—in fact, we often take a naïve strategy of putting an equal amount into each fund we're faced with. A simple heuristic for easy diversification—use a stock market index fund or a target date fund—can simplify and shape that decision to users' benefit.

Which type(s) of education is best depends on the particular situation. With voluntary behavior change, we assume that the user already has some motivation to act. Information about *how* a system works can be fascinating for those already in the know, but overwhelming and too removed from the actual decision for those who aren't. Logistical information (*what* to do) can provide clear actionable guidance and increase the user's ability to act immediately, a key component in the CREATE Action Funnel.

Reach Out of the Screen

Part of the philosophy of designing for behavior change is that we're not limited to actions within our products. Or, to come back to the story of the fish: even if your product isn't on the beach at that moment, you can still help change the context of action via the type of action being asked, the environment it's done in, and the preparation of the user.

The context consists of two things:

The product itself

For software applications, the primary environment enveloping the user is the product—particularly, the web pages where the user takes action or the tiny progress screen on the Fitbit One tracker.

The rest of the user’s local environment

While using your application, the user is also embedded in a physical environment (trying to use your application while on the subway, with a spotty Internet connection) and a social environment outside of the application (a set of expectations from friends on the “right” behaviors).

You have the most control over steps that occur within the product itself, of course. But you can also use the product to *reach out of screen and touch the user’s daily life*. For example, guidelines given within the product, and even the marketing materials that surround the product, define the action for the user. They shape how the users think about the action and what it is they try to accomplish in their daily lives outside of the product. The basic definition of what behavior is being changed (and how) can and should be tailored to fit the user base.

Putting It into Practice

It’s easy to get excited about our products and how they can solve our users’ problem. We assume our users will be just as excited about our solution as we are; we just need to help them understand how absolutely awesome it is. But when they don’t immediately “get it,” we want to yell at them. We want to scream about the benefits. We want to beat them over the head about *their* problem and how we’re solving it for them.

If you’ve been in product development or marketing for a while, you’ve likely been in meetings where everyone is convinced there’s something

basically wrong with the user, and we just need to educate/motivate them, and they'll use our solution. It's because of those painful meetings that I developed the Story of the Fish. To remind ourselves, in a cutesy story, that usually there isn't anything wrong with our users. Rather, that there's something wrong with the *context*. *Designing for Behavior Change* is about changing that context, usually in one of four ways: ²⁰

- Do it for them by magically taking away all of the burden of work from the user
- Structure the action to make it feasible (or, in reverse, more difficult) for the user
- Construct the environment to support (or block) the action
- Prepare the user to take (or resist) the action

The first one is really a special case: great when you can find it, but not something you should count on.

Here's what you need to do:

- Ask whether you can fully automate the action (i.e., remove the need for the person to do anything at all).
- Look in particular for defaults (automatically setting reasonable options where the user has to make a choice), making it incidental (bundling a beneficial side effect with something else the user does, like enriching bread with vitamins), or automating repetition after the user has made an initial choice (like automatic deposits to savings, once the user sets up the transfer).
- Don't force your users to suffer and work through the action because "it's good for them." If you can help them achieve the outcome, do it—and save the noble suffering for more difficult problems.

How you'll know there's trouble:

- The user can't stop you from taking action on their behalf, or can't do so in an easy and obvious way. That's not designing for behavior change, that's simply coercion.

Deliverables:

- Where possible, a magic solution—you do it for them.

Exercise: Review the Map

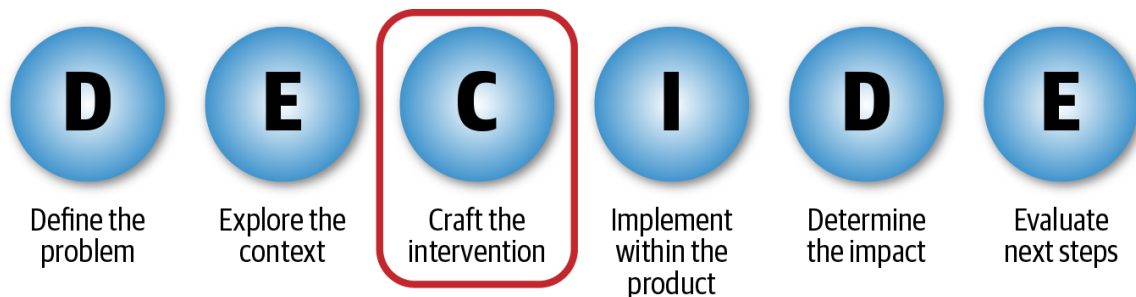
Look over your behavioral map from “[Worksheet: The Behavioral Map](#)” and the obstacle you diagnosed as the most likely problem. Is it possible to simply remove that obstacle altogether—to skip that step or to take the burden of work from the user and unto your product? If so, excellent—look for the next behavioral obstacle in the map, if any, and try to do the same. If there aren't additional obstacles, skip the next few chapters and go straight to implementing the solution ([Chapter 12](#)).

In the case of our exercise app, one of the obstacles we identified is that the user doesn't have their employee ID on hand when they receive the invitation email: an Ability obstacle. We can skip that step by embedding it in the invitation email itself. Unfortunately, there's still another obstacle remaining: a negative Reaction to the idea of downloading and using another app. So we move on to crafting an intervention to overcome it.

-
- 1 If they think there's a problem to be solved or don't think it's important despite being aware of the problem, we shouldn't be involved in the first place—that's persuasion or coercion.
 - 2 For many Americans, the behavior change isn't what our policymakers and companies intended—it's become a short-term savings vehicle. But the impact on savings is still amazing. See Fellowes (2013) for a discussion of the downsides of autoenrollment and autoescalation.
 - 3 Nessmith et al. (2007)
 - 4 [McNeil \(2006\)](#). Though, to be fair, I found that citation from [Wikipedia](#).
 - 5 [American Thyroid Association \(2020\)](#)
 - 6 Legally, prize-linked savings is usually structured as a sweepstakes in the US; no need to get into the legal definitions here, though.

- 7 Tufano (2008)
- 8 Filiz-Ozbay et al. (2013)
- 9 Murphy (2005)
- 10 One could also point to state lotteries as examples of making a behavior, contributing money to schools, incidental. In California alone, they have funneled \$24 billion to “education funding” since 1985 (Strauss 2012). But state lotteries are also a great example of how different the mind is from a conscious budgetary process. In our minds, school contributions are a side effect of our lottery purchases. No extra work. In reality, state budgeters consciously know that a dollar is a dollar and move the money around from one budget category (schools) to another. Changing *lottery participant* behavior doesn’t mean that you’re also changing the behavior of accountants!
- 11 Another great example of making savings behavior incidental comes from the IDEO/Bank of America “Keep the Change” program. The program rounds up purchases made on debt cards to the nearest dollar and takes the difference between original cost and the rounded version and automatically deposits it into the person’s savings account. The person does nothing differently—savings are incidental.
- 12 My thanks to Peter Hovard for highlighting the ethical concerns with behavior change as a side effect.
- 13 This image of a virtuous person is modeled after ourselves (the behavioral designers), of course. When we try to make our users more like us, it’s like a form of self-affirmation. And sadly, yes, I’ve seen this in my own and many others’ work.
- 14 Wilson (2011). The related tendency to create consistent (non-dissonant) stories about our own behavior has been in everything from North Korean gulags to the “foot in the door” technique of successive commitments in sales. See Cialdini (2008).
- 15 Freedman and Fraser (1966)
- 16 Nessmith et al. (2007)
- 17 Bayer et al. (2009); Mandell and Klein (2009); Lyons et al. (2006); Fernandes et al. (2014)
- 18 Lusardi and Mitchell (2007)
- 19 Hilgert et al. (2003)
- 20 This process of changing the context of action has similarities to Sebastian Deterding’s descriptions of game design (Deterding 2010) and differentiates it from a traditional UX process, where only the product (tool) itself is designed.

Chapter 9. Crafting the Intervention: Cue, Reaction, Evaluation



One of the earliest and most powerful demonstrations of behavioral science in government came from the UK's Behavioural Insights Team. They communicated with people who had tax debts and shared with them the fact that most people do in fact pay their taxes. That encouraged people who hadn't paid to do so themselves. In other words, they used descriptive norms. The results were clear and powerful: a 5.1 percentage point increase in the payment of liabilities within 23 days.¹

Since then tax compliance, not surprisingly, has been a popular area for applied behavioral science among governments around the world. These efforts are often very low cost and have a long history of success.

One such effort comes from the small, underdeveloped country of Kosovo, where the World Bank's Mind, Behavior, and Development Unit (eMBed) and the German Development Agency GIZ helped the country's tax authority design, implement and evaluate three experimental trials to improve tax reporting and collection.² In the US, many Americans grumble

about our taxes and how the government doesn't need them; in places like Kosovo, low tax revenue means the government struggles to provide basic services to their citizens.

The team tested messages meant to encourage residents to report their tax liabilities—using SMS, email, and physical mail. Within the communications, they employed:

- A social norms approach: “7 out of 10 firms submit their declaration on time. Don't wait, be part of the MAJORITY!”*
- An appeal to citizenship: “Not paying your taxes places an unfair burden on your fellow Kosovo citizens. Please don't be an irresponsible citizen.”*
- A focus on benefits: “Did you know that your VAT contribution is invested in your city?”*
- And reframing non-payment as an active (intentional) choice, instead of inaction: “If you do not declare now, we will consider it to be your active choice...”*

Overall, they were able to successfully increase tax reporting; the physical letters, for example, increased reporting by 73% among all intended recipients (companies and individuals), with a tremendous 431% increase among individual tax payers who successfully received the letter.³ Like many behavioral studies, tax compliance studies show how low-cost, straightforward marketing and communications can deliver outsized results.

We've arrived at the fun part: crafting the intervention itself. What should our products and communications *actually* do to facilitate or hinder action? We defined the problem in **Chapter 6**. We explored the context and diagnosed the behavioral problem in **Chapter 7**. We quickly checked whether there was a way to magically make the problem go away and do everything on the user's behalf in **Chapter 8**. Let's say there's no magic solution, and it's time to solve the problem directly.

Thankfully, solutions follow directly from our diagnosis of the problem. Sometimes they are easy and straightforward as well. If your diagnosis shows that your users don't know about your new feature (the Cue is lacking), well, the solution is obvious: show them the new feature. When it's not so obvious what to do, we can draw upon the many research studies conducted in behavioral science. These studies can be grouped into ways to encourage beneficial action and ways to hinder negative actions. We'll start with encouraging beneficial ones.

As you saw in [Chapter 7](#), we diagnose why users don't take action in terms of one or more behavioral obstacles: a missing Cue, a negative emotional Reaction, and so on. We use the CREATE Action Funnel. For each CREATE obstacle, behavioral scientists have developed a set of interventions to help overcome that obstacle.⁴

Without further ado, let's introduce the interventions.⁵

[Table 9-1](#) offers two dozen tactics you can use to facilitate action, organized by the part of the CREATE Action Funnel that they affect most strongly. The following sections describe each of these cognitive mechanisms and how you can deploy them to the user's advantage. Many of the tactics listed here have been briefly mentioned earlier in the book, when we first introduced how the mind works. In those cases, we'll focus on how that tactic can be employed in practice. The goal of this section is to provide a quick reference to each of the major tactics you can use to craft your interventions all in one place.

Table 9-1. Tactics to support action

Component	To Do This	Try This
Cue	Create a cue	Tell the user what the action is
		Relabel something as a cue
		Use reminders
	Increase power of cue	Make it clear where to act
		Remove distractions
	Target a cue	Go where the attention is
Reaction	Elicit positive feeling	Align with people's time
		Narrate the past
	Increase social motivation	Associate with the positive
		Deploy social proof
	Increase trust	Use peer comparisons
		Display strong authority
		Be authentic and personal
		Make it professional and beautiful
Evaluation	Economics 101	Make sure the incentives are right
	Highlight and support existing motivations	Leverage existing motivations
		Avoid direct payments
	Increase motivation	Test out different types of motivators
		Leverage loss aversion
		Use Commitment Contracts
	Support conscious decision making	Pull future motivations into the present
		Use competition
		Make sure it's understandable
		Avoid cognitive overhead
Ability	Remove Friction	Avoid choice overload
	Remove Friction	Remove unnecessary decision points
		Default everything
		Elicit implementation intentions

Component	To Do This	Try This
	Increase sense of feasibility (self-efficacy)	Deploy (positive) peer comparisons
		Help them know they'll succeed
	Remove physical barriers	Look for physical barriers
Timing	Increase urgency	Frame text to avoid temporal myopia
	Increase urgency	Remind of prior commitment to act
		Make commitments to friends
		Make a reward scarce
Experience	Break free of the past	Use Fresh Starts
	Break free of the past	Use Story Editing
		Use slow-down techniques
	Avoid the past	Make it intentionally unfamiliar
	Keep up with changing experiences	Check back in with users

So let's look at each of them in turn. Afterward, we'll return to the other behavioral challenge: hindering negative action.

Cueing the User to Act

The sight of the overgrown grass prompts you to mow the lawn. A TV commercial for steak reminds you that you're hungry. For many behaviors, the motivation is often present, but it's in the background. Something needs to cue you to think about it *now* rather than later: that cue is the first step in the CREATE Action Funnel we talked about in [Chapter 2](#).

Cues, wisely placed, are essential for behavior change. This is true for nonconscious habits—a cue in the environment starts a habitual routine—and for conscious decisions to act.

Ask Them

One simple way to cue people to act is just to ask them. Yes, it's obvious. Yes, it's simple. And yet, we forget to do it—because our products are so

awesome and we assume people are already thinking about using them.

I know, you can't imagine that anyone would make such an obvious mistake. But we all do all the time. Do you include a link to your website at the bottom of your emails? If so, do you actually ask people to look at your site, or do you hope it's obvious? Do you post your Twitter handle on your messages or blog posts, hoping people will follow you? Readers could figure out the action we intend (view website, follow on Twitter), sure. But the more mental leaps that are required between what we see (Twitter name) and the action, the less likely is it that the action will cross our minds before we're distracted by something else.

Dustin Curtis ran a set of experiments on how he presented his Twitter handle to readers of his blog.⁶ He started with a simple informative statement: "I'm on Twitter," in which "Twitter" was a link to his page. 4.7% of readers clicked. Then, he did the obvious—which apparently isn't so obvious to the rest of us—he told people what the action was. "Follow me on Twitter." Boom—7.31% of users clicked. And, even clearer: "You should follow me on Twitter here"—12.81% of users clicked. There are multiple effects at work in the last statement (a personal request, specificity, etc.), but the effect of requesting the action is undeniable. One lesson is simple: *directly, and unabashedly, ask people to take action.*

If you do it nicely and don't ask too often, asking rarely leads to less action than not asking. Asking for action within a software product has three distinct effects:

Cueing (attention)

Not only are people busy, but their attention is extraordinarily limited.

Dean Karlan (among others) shows that increasing mere attention to an issue is a key factor in driving behavior—especially if the person already has the motivation to act.⁷

Obligation

It's uncomfortable to say no to a reasonable request. If the company (and especially, a particular *person* who the product personifies) can be seen as a friendly, anthropomorphized presence, then this can help spur action.

Immediacy/urgency

Most “good” actions, like saving money, exercising more, or smoking less, are things that a person can do at any time, and therefore can be put off. Asking people to do it *now* (with some reason for the urgency) helps people get over the “I’ll do it later” hump.

It doesn’t take much to ask users to act. Emails. Text messages. Big honkin’ Act Now buttons. These are obvious and effective ways to trigger action.⁸ Don’t waste time on complex psychological approaches to help people to act if you haven’t already tried the obvious ones.

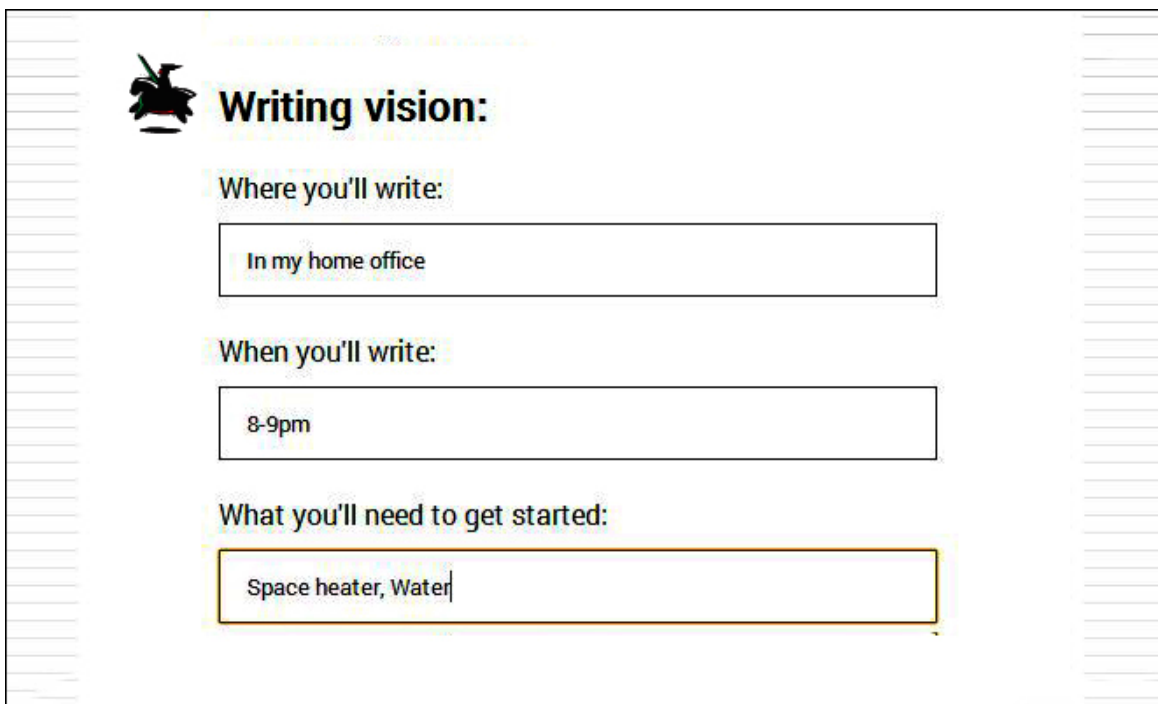
Relabel Something as a Cue

Another way to cue action is to help users reinterpret an existing feature of their environment as a cue. Let them specify something that they see or hear normally in their lives—like the morning show on their favorite radio station. Then have them associate an action with that cue (e.g., “Once the morning show finishes, go running” or, “On Thursday, when I exit from the metro, I’ll go buy my running shoes”).

Simple if/then rules like this have been used for thousands of years—and your product can help people use them by building an association between something they’ll see, and something they want to do. More recently, researchers have experimentally established the impact of *implementation intentions*, in which people make specific plans for action in the future.⁹ Implementation intentions are a way to tell the mind to do X whenever Y happens. They pull the burden of thinking from the future to the present,

allowing the person to invest time in setting up the plan to act now, and simply executing it automatically when the environment cues action in the future.

Here's a personal example of how setting up a concrete plan establishes a cue to act in the future. To write this book, I used a simple online program by Anna Tulchinskaya that encourages writers to write regularly. **Figure 9-1** shows what I filled out when I first signed up. I set a plan to write every day, at a certain time of day, in a certain place. So when I saw the clock, that became my cue for action.



The image shows a screenshot of a web form titled "Writing vision:" with a small icon of a person writing. The form has three sections, each with a label and a text input field. The first section is "Where you'll write:" with the entry "In my home office". The second section is "When you'll write:" with the entry "8-9pm". The third section is "What you'll need to get started:" with the entry "Space heater, Water". The form is set against a background of horizontal lines, suggesting a notepad or lined paper.

Writing vision:

Where you'll write:

In my home office

When you'll write:

8-9pm

What you'll need to get started:

Space heater, Water

Figure 9-1. My plan to write each night

Make It Clear Where to Act

We scan, we don't read. Don't expect users to read lots of text on your page. The two-second rule is a good test—if you don't get the gist in a two-second glance at the page, you risk losing the reader's attention. Krug's *Don't Make Me Think* (New Riders, 2006) gives a great overview and practical examples, and Johnson's *Designing with the Mind in Mind*

(Morgan Kaufmann, 2010) talks about the visual perception system and related psychology.

Some of the key things that we quickly recognize are the ways in which we can interact with a page (*affordances* per Norman's classic *The Design of Everyday Things*, Basic Books, 1988)—what looks like it is clickable, doable, or can otherwise get you off this page and quickly on to the next one. The lesson is simple: make buttons look like buttons, and make anywhere else people are expected to take action clearly a place where they *can* take action.

Remove Distractions: Knock Out the Competition

There's a flip side to encouraging a behavior that hasn't received nearly as much attention in the behavior change world. Namely, each distinct type of behavior is in competition with (almost) every other type¹⁰—competing to grab the user's very limited attention (i.e., competing triggers), to claim the user's time (i.e., competing to be easier and faster), and to be the most motivating. One can draw out these competing or blocking factors with a series of questions:

- What in the environment already has the user's attention and thus *crowds out awareness of your action*?
- Similarly, is the environment crowded with other actions that are already *easy or simple* to take?
- What in the environment *demotivates* the individual or, more subtly, *motivates the individual to do other things* and thus crowds out the target action?

When faced with serious competition, here are three strategies to counteract it.

First, if the competing factors are within the application, the product team needs to make hard choices and potentially decrease the attention/motivation/ease provided for other behaviors. Often you really

don't need to change all of them—just focus on pulling the users' attention to one thing at a time. If you have their attention at the time they are doing what they need to do, it doesn't matter (as much) that the application motivates them to do other things at other times. One straightforward way to minimize competing attention-getters is to simplify: remove other calls to action, remove distracting text, and take out anything else that isn't essential from the page. Put those other actions in another part of the application that is clearly, conceptually different from the current one.

Since users are scanning a screen and trying to save work, they'll all too likely just to click on the first thing that looks clickable. So make a single, clear call to action if your goal is to get the person to keep moving through the screen. Remove extra links and buttons, or place them in a distinctly lower level in the screen's hierarchy.

Second, you can use competing factors to your advantage. If the user is really engaged in something else, look for a clever way to connect it to your target action. Wherever the user's attention already is, that's the best place to be. That's why many applications are built for Facebook—that's where users are already putting their attention.

Third, there's the brute-force approach—shout louder for attention, be more motivating, and make using the product easier than breathing. I don't recommend this. If the user is doing something (else) there's (a) probably a good reason, and (b) it takes more than a slightly better behavior to overcome an entrenched one. There are real costs to switching behaviors; for example, we've already discussed the challenges of changing habits. However, if you can't directly dampen the other actions or find a clever way to use them to your advantage, this may be your only option. Over time, you can build up competing habits and experiences within the application that crowd out those other actions. Or, you can go back to the drawing board and find a different target action that doesn't compete so strongly with other existing behaviors.

Go Where the Attention Is

Where's the easiest place to get someone's attention? Where their attention already is. That's the logic behind marketing swag at conferences.

Pharmaceutical salespeople are well known (and rightfully critiqued) for giving away tons of “free” pens, clipboards, stickers, and such to doctors, all with their logo on them, so that when the doctors are prescribing medicine, they'll be reminded of the particular company's products.

The same logic underlies many wearables for a far more laudable purpose. If someone wants to exercise but keeps forgetting, what do you do? You could devote a significant ad budget to signage, video testimonials from athletes, etc. Or, you could simply give them an exercise band that doubles as a watch (or, increasingly, a watch that doubles as an exercise band). They wear it on their wrist because it performs a key function and, in the process, are frequently cued to think about exercise.

If you want to get someone's attention to a recurring activity—like taking time out of their day to meditate—you could try having them install an app on their phone that triggers a reminder message, since their attention is often on their phone. Or, you could give them a calendar invite that sets up a recurring appointment—since again, their attention is often on their calendar.

Align with When People Have Spare Time

In my research, I've found that the single most powerful factor in whether someone pays attention to your cue is when you cue them; that is, whether or not you align with when they have bandwidth to pay attention.

Over the years, I've run over a hundred studies of time-of-day and day-of-week effects on different populations and regularly see changes in response by three times within the exact same population and with the exact same content. **Figure 9-2** illustrates results from one set of those tests. This particular study entailed emailing employees at a large manufacturing company in the United States—we found that by far the best time to contact them was during the start of the workday on Tuesday.

Before you set all of your marketing campaigns and product launches to occur on Tuesday morning, there's an important caveat. Each group of people, and indeed each person, has a different *structure of attention*: a different rhythm to their days and weeks. People working the night shift will be able to pay attention to your cues at different times than those on the day shift. In addition to the manufacturing population in **Figure 9-2**, we ran a similar set of experiments with other groups—including a large minimum wage population in the service sector. Many of them worked two jobs, and weekdays were absolutely terrible times to reach them. Instead, it turned out that Sunday evenings and holidays were by far the best time to gain their attention—on the order of a two to three times improvement.

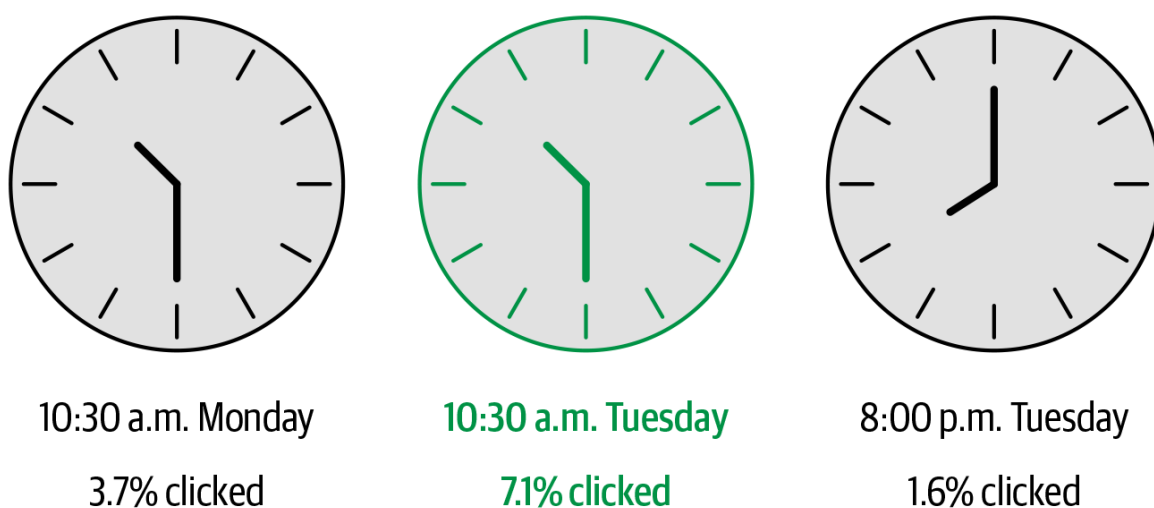


Figure 9-2. Aligning when you contact people with when they have attention to spare can have a tremendous effect on response rates

So when you try to interact with people is tremendously important. But there's no simple rule: it comes from your understanding of your population and their structure of attention.

Use Reminders

We all forget things in our daily lives—even things that are important to us. Yet we often don't consider simple forgetfulness as a cause of inaction among our users.

Researchers indeed find that people don't follow through on an action they want to take simply because they failed to remember it.¹¹ Reminders don't need to be fancy or complicated; an email, text message, call, in-app message, etc. can be enough. Don't assume that because the action is important people will remember. We all have busy lives—and so do your users. In some of my own research studies using email to reach diverse populations, I've generally found that two follow-up reminders lead to a roughly 50% increase in response relative to the first communication. It's not an iron-clad rule, of course, and it helps when those reminders occur at different times—in case you didn't align with people's spare time.

Bonus Tactic: Blinking Text

Blinking text is a really great cue. It never fails to catch our attention. And it's also darned annoying, because it catches our attention and won't let go. If possible, use blinking *scrolling* text. Right across the top of the screen.

OK, please don't do it. Seriously.

The Intuitive Reaction

Once a cue catches the user's attention, the mind reacts—often in the blink of an eye. Regardless of the overall merits of the action (and product), that reaction can cause the user to shut down. Here are some techniques to address that problem.

Narrate the Past to Support Future Action

Our self-narrative is how we label ourselves and how we describe our behavior in the past; products can help people see themselves differently. The goal, from a behavior change perspective, is to *help people see themselves as someone for whom the action is a natural, normal extension of who they are*.

In other words, if you want to help people begin exercising (like Fitbit's app does), help them see themselves as people who have already been

exercising in small ways and just need to do more (e.g., first-time Fitbit users may be surprised to find out how far they normally walk each day).¹² An easy way to support this process is by merely asking people about things they've done in the past that are related. And congratulate them for the work they've already accomplished.

Essential to a supportive self-narrative is the belief that one can actually succeed in the action (i.e., users need to feel that the action is under their control and that they have the skills and resources to—potentially—make it happen). That's the sense of self-efficacy discussed in “Ability”.¹³ Reminding people about their prior successes at related tasks can help build that sense of self-efficacy; so can the “small wins” and positive feedback described in the last two chapters.¹⁴

Bring Success Top of Mind

Similar to renarrating the past to shape someone's self-conception, a related technique is to redirect someone's current attention to prior successes. We all each have frames of reference with which we interpret and respond to the world. Those frames of reference are selectively activated, based on our (very) recent experiences.

If you're asking someone to commit to running once a week, get the person thinking about previous times they've run first (as long as that experience was positive)! When you ask them to commit to running, the benefits of running will be clearer and more salient in their minds.

Associate with the Positive and the Familiar

Chapters 1 and 2 talked about how, for many of our choices in life, we intuitively know whether taking an action feels right or not to us. A big part of that is our prior *associations*—our learned experience that buying a fancy pair of shoes is going to make us feel great when we walk out of the store with them, at least for a few days.

Products can build these associations to help a person change behavior. In Chapter 8, we talked about changing the *action* itself so that it leverages

prior experiences. Here, the *product* can be changed so that it helps users make the mental connection between the action they want to take and their prior experiences. I call this a *behavioral bridge*, because it helps the user cross from one type of behavior to another by making it less “new” and difficult. The bridge connects past experience with future actions.

Here’s an example: Jive Voice is a conference-calling application that allows people to switch from using a dial-in number and long PIN code to using a simple link in a URL. Dial-in numbers and PINs are frequently misplaced and annoying to enter. When a user clicks the URL, Jive Voice calls them and patches them into the conference line. The challenge is that using a URL is new and strange. In the product, the company highlights the new and unique aspects (ease of use, etc.) but is also careful to leave a behavioral bridge in place—a comforting bit of information about how users can treat it like a normal conference call if they need to, since the underlying technology is a conference line with a dial-in number and access code.

Deploy Social Proof

If we see that other people are taking an action, we’re more likely to feel that the action is valuable and worthwhile. It’s a quick gut check—if that person does it, it must be OK, right? This is one of the major ways that our minds save work and quickly make decisions in uncertain situations.

Using social proof is a key tactic in sales and persuasion with a long research tradition behind it.¹⁵ You can convey the fact that other people are taking the same action by using people’s faces or short testimonials. Different genres of social proof include user or expert testimony testimonials (often on a product page), celebrity endorsements (in movies or TV ads),¹⁶ or online reviewers (think Amazon). In addition to having as long research tradition, it is used and abused extensively in marketing campaigns, from paid testimonials by fake experts to comments supporting a product by people who look like everyday users but are actually trained actors.¹⁷ For more information on this topic, see [Chapter 1](#).

NO MAGIC WANDS

Throughout this book, I provide the tools you need to find the behavioral processes and product features that work in your particular context, and verify their impact with your specific set of users.

What I can't do is give you the secret behavioral tricks that will always change user behavior in predictable ways. That's because such magical formulas simply don't exist (run away from people who tell you they do!). All behavior change interventions interact with an individual's desires, prior experiences, personality, and knowledge to produce their unique impact on that person. There is just too much variation across people for any approach to always work.

Most of the approaches and lessons that I talk about here have been tested either in a researcher's laboratory or in a specific product setting. In most cases, I've also observed these techniques in practice in my own work or through the dozens of companies I've interviewed and learned from. Unfortunately though, there are very few studies out there that apply and rigorously test theories of behavior change in ways that can be generalized to lots of other products. That's something we strive for on my team—but even then, it's difficult to make the case that what works for us, in helping people take control of their finances, is going to work the same way for someone else's dieting software.

We are all still in the early stages of learning how to use products to help people change their behavior. So in Chapters 12–14, I provide some guidelines on how to test specific interventions in your product, and help you move the field forward at the same time. I encourage you to contribute your findings to the broader community so we can all learn and develop our skills together.

Use Peer Comparisons

Being told about, and compared to, the actions of our peers can be immensely powerful. It's a specific form of social influence, like social proof. Our behavior frequently conforms to what we believe our peers do (i.e., *descriptive norms*), compounded by the usually false belief that our peers are watching our behavior and judging it (*spotlight effect*). This effect has been shown in everything from energy usage to voting.¹⁸

For behavioral products, the implications of peer comparisons are tremendous. Social norms are an incredibly powerful part of our microenvironments and can encourage (or discourage) action. The same is true within the context of each individual screen that the user interacts with.

To use this technique, compare the user's performance to a reference group that they care about (their friends, colleagues in a similar job), and try to ensure that the reference group you choose is doing *better* than the user. A note of warning, however: peer comparisons encourage people to move toward the norm (the average for the reference group). So if you tell them they are already going better than most people, they may just relax and don't work so hard. That negative effect can be counteracted with an explicit social approval (Great job!) for exceeding the norm.¹⁹ Figure 9-3 shows one of our studies at HelloWallet that encouraged people to save.

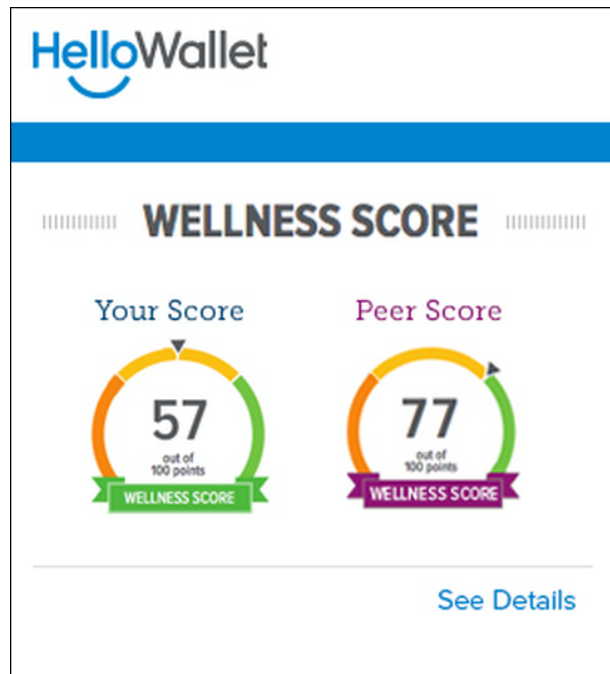


Figure 9-3. A peer comparison I helped develop to encourage people to save

Display Strong Authority on the Subject

People are more likely to trust those who they see as an authority on the subject. If you're telling your users that they have to do action X in order to build up to their goal of Y (and it's true), then speak with authority. Don't write wishy-washy text. Make sure your credentials can be seen without beating your users over the head with them.

There are great studies on how people wearing suits, or with professional titles, are simply assumed to be more credible and trustworthy. The use of (perceived) authority is also a favorite tactic in sales and persuasion. See Cialdini's discussion of the underlying research (2008).

Be Authentic and Personal

People pay more attention to personal appeals to act than to impersonal ones. If you receive a letter with a handwritten envelope, how likely are you to open it? How about one with a standard machine-printed address? The reasons are manifold, but we are more likely to ignore machine-generated, impersonal appeals than tailored, personal ones.²⁰ We have an almost

automatic response of “this is spam” for any email or letter from impersonal sources.

Here’s a great example of using personalization and authenticity to cut through the noise and get people’s attention. In Oregon, there’s a lottery for free healthcare for people who can’t afford it. But some of the people who sign up for and end up winning the lottery don’t open the letters notifying them that they’ve won. And so, they miss their chance at free healthcare.

ideas42, the leading behavioral economics consultancy in the United States, devised a simple outreach campaign to the winners of the Oregon healthcare lottery. They notify people that they’ve won with a postcard featuring the smiling faces of the people at Providence Health in Oregon that will help them sign up for their healthcare. The recipient’s name and address are handwritten on the postcard. **Figure 9-4** shows a sample.

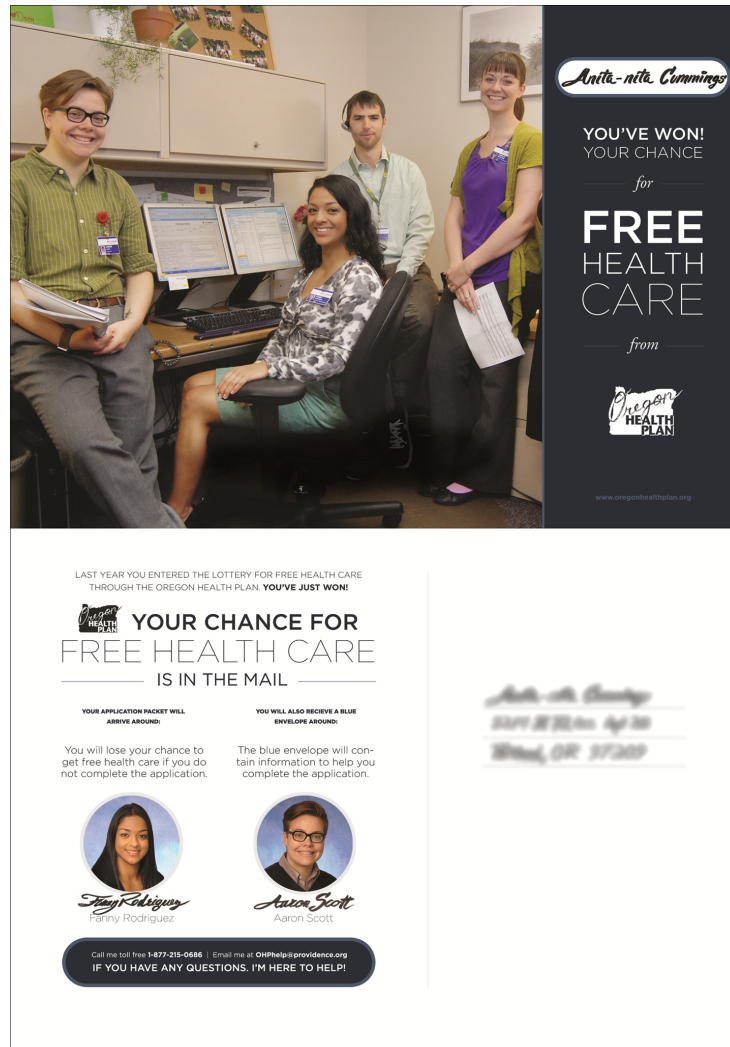


Figure 9-4. A postcard developed by ideas42 to help winners of the Oregon healthcare lottery get past their automatic rejection of form letters and read enough to see that they've won free healthcare

Remember, we've been conditioned to reject impersonal and computer-generated appeals. Most of us have an intuitive reaction against them. To avoid that intuitive reaction, our products need to do something different, something that's good practice anyway: be authentic and personal.

Make the Site Professional and Beautiful

And finally, let's not forget the basics. We rarely consider unprofessional-looking websites and apps to be credible.²¹ If you're trying to help someone take an action, don't make them have an intuitive reaction of distrust. That's unnecessary friction on the action. Like it or not, we assume that scammers

make bad websites and apps. We even find it easier to *use* clean, well-designed products.²² Even if your app is created to help people, it has to look good.

People really do judge a book by its cover. You could try to argue with the shallowness of people's intuitive reactions all day long. Or, you could just design a nicer cover. After they open the book—i.e., start using your app—then they can discover how beautiful it is on the inside, too.

The Conscious Evaluation

A person's conscious evaluation is similar to the stereotypical view most people have of decision making: do the benefits outweigh the costs? What are the alternatives, and how does this action stack up against them? While similar, it certainly isn't a perfect cost-benefit analysis: because people are often distracted, they may not carefully think about all of a particular action's benefits (or costs!). They may have limited information. And even when they do think carefully and have the necessary information, they may be overly focused on the present (present bias) or miscalculate costs or benefits (e.g., exponential growth bias).

Because of these imperfections, we should ask once again: what are ethical ways to design for behavior change? The core costs and benefits of an action are obviously very important; if the costs outweigh the benefit, the user shouldn't do it. Changing the core incentives to make it more valuable for the person, decreasing cost or increasing the benefits, seems appropriate and ethical. And there's no particular magic or mystery here, so we'll only touch upon that briefly.

While there are certainly gray areas, hiding the costs of an action sounds and likely is manipulative and dishonest. Highlighting existing (but unattended to or miscalculated) benefits seems generally permissible—though each case of designing for behavior change should be still reviewed, as noted in [Chapter 4](#).

Thus, in this section, we'll focus on these three options: increasing benefits, decreasing costs, or highlighting existing benefits. In terms of the costs, we'll look at substantive changes, and not minor ones. Minor costs are surprisingly important—many of the early examples in behavioral science involved tweaking form fields, defaults, and such. Small changes can lead to outsized effects. However, these effects are not (primarily) because of the conscious calculation that people make—an extra form field shouldn't affect a conscious cost-benefit analysis at all. Rather, they cause us to pause and face a decision point (a concept we introduced in “Ability” and will show in more detail as we craft interventions to increase their Ability in the next chapter).²³

That said, let's start with the most basic of all: incentives.

Make Sure the Incentives Are Right

Behavioral economics, and with it the broader field of behavioral science, in large part arose as an extension to or even a correction to traditional economics. Traditional economics focuses on a person's preferences and the optimal route to fulfill those preferences. In many economic models, this boils down to the simple observation, “People want to be paid, so pay them and they'll do stuff. Pay them more, and they'll do more.” Behavioral economics shows that this isn't always the case, and people have many motivations above and beyond receiving something in return.²⁴ People are motivated by altruism, by a sense of self-esteem, etc. That's all true, but we should never lose sight of this simple fact: people really are motivated by getting stuff, especially by getting money.

If the behavioral obstacle that a user faces is one of Evaluation—they don't see the benefits outweighing the cost—then we should fix that first. Make sure it really is in the users' narrowly defined interest to take the action. If it isn't, (a) it's likely very difficult to overcome that basic problem of incentives, or (b) it's likely going to involve trickery.

So if your product isn't very good and doesn't benefit your user enough to justify the cost—from their perspective, not yours—designing for behavior

change isn't going to help you. Fix the product first. It must solve a user need in a way that the user feels is worth the cost. That may require lowering your price. It may mean building a better product. Either way, it's not something you can avoid. And yes, this is actually something I've had to say many times when people ask me how to promote a product that their target audience doesn't want. Yep.

OK, let's assume you've covered Economics 101. Now for the more interesting stuff.

Leverage Existing Motivations Before Adding New Ones

Does your product need to add a new motivation for users to act, highlight existing ones, or both? First, understand what currently motivates users to act. Use the information you learned about your users in Chapters 6 and 7, about why they want to take the action. Maybe their doctor has told them to exercise more; maybe they really enjoy running but can't seem to fit it into their schedule. Since we're often distracted and thinking about other things, simply reminding people of their existing motivation at the moment of action can be powerful. *And it's really cheap to remind people of what they already care about; it's much more costly to add a new motivation.*

If you're not sure what currently motivates your users, you can do some simple field tests—check how important particular motivations are versus other things in the person's life. A good way to gather that information is to present a series of trade-offs—ask which of two things the person wants more (e.g., as motivations for exercise: “living five years longer” versus “going on a date next month”). It's less ideal to simply ask people, “How important is this to you?” because we often don't have a real baseline against which to answer that question, and it engages a different part of our minds than the one that usually makes the actual decision to act.

Another reason the existing motivations are important has to do with extrinsic versus intrinsic motivation.²⁵ *Intrinsic motivation* comes from the inherent enjoyment of the activity itself, without considering any external pressure or reward. *Extrinsic motivation* is the desire to achieve a particular

outcome, such as receiving a reward for it (like money or winning a competition).

Your users can have preexisting intrinsic *and* extrinsic motivations, and your product can leverage both to drive behavior. But when the product *adds* a new motivation to act, the source of that motivation is almost always, by definition, outside of the user and outcome-oriented, or extrinsic. For example, people using the Fitbit One often have both a preexisting intrinsic motivation and a new extrinsic motivation: an inherent enjoyment from exercising and using one's muscles, as well as the desire to reach a particular goal and be congratulated for it by the product.

Intrinsic motivations can keep people going when the product isn't directly involved in their lives. New extrinsic motivations, provided by the product, can't do that. They are effective only when the product is directly involved: when they stop, so do the users. If your product adds extrinsic motivations, it can also crowd out people's existing intrinsic motivations—meaning they lose the joy of doing something for its own sake if they start being paid to do it.²⁶

However, that doesn't mean new extrinsic motivations are always a bad thing; they just have to be used judiciously:

When the person doesn't have a strong existing motivation for a particular step in the sequence of actions

For example, someone really wants to get healthy, but doesn't see how regular blood pressure checks are important. A little boost can help.

For one-time actions where crowding out intrinsic motivation is irrelevant

For example, someone really wants to exercise but has no motivation to go buy gym clothes. An incentive can get them past that barrier and closer to their goal.

To help users transition from extrinsic to intrinsic motivation—to get people started as they find the joy of the activity itself

For example, conversation clubs can use a small incentive (free dinner) to get together people who are learning a new language for the first time. While they are there, they experience the intrinsic joys of being immersed in the language, which pulls them forward for future learning.²⁷

Avoid Direct Payments

In line with our discussion about leveraging existing motivations before adding new ones, you could just pay people to click on your button. But I don't recommend it. If you need to pay people to do something that's supposed to be a voluntary behavior change, you're probably not connecting that small action with the reason they want to change their behavior in the first place.

There is extensive evidence that financial incentives induce individuals to undertake behaviors that they would not undertake.²⁸ People are motivated by money. No great surprise, right? However, when a person is already inclined to take the action, financial incentives can backfire by decreasing preexisting internal (intrinsic) motivations; the individual is more likely to stop the behavior after the incentive is removed.²⁹ Similarly, other social motivations are crowded out when we start thinking about our behavior in terms of being paid to act.³⁰ Direct payments are less likely to cause problems with one-off behaviors, like signing up for the gym. But they can undermine long-term intrinsic motivation, like actually going to the gym over time!

So, bringing together the various points thus far about the user's conscious Evaluation: you do want the basic incentives to be aligned. That is, it should be in people's interest to take the action. But if you find yourself adding additional payment on top of that, to make the action "more motivating," it may not have actually been in their interest in the first place (incentives were off) or you're not connecting with and leveraging the existing intrinsic motivation the person has (and risk crowding them out).

Leverage Loss Aversion

People respond much more strongly to losses than to gains—they are “averse” to losses. In fact, in many scenarios people will be willing to forfeit twice as much money to keep an item that they already have (and have no other personal attachment to) than they are willing to pay to purchase an otherwise identical item. There’s a detailed literature on special cases of loss aversion, but that general rule holds true in many cases: *losses are roughly twice as motivating as gains.*³¹

Loss aversion is a very powerful tool to help people change their behavior. By selectively framing the presentation of a desired action as *avoiding loss rather than gaining benefits*, the application can trigger a strong gut reaction to act. For example, it can be much more persuasive to tell someone they’ll lose sexual potency unless they get in shape, rather than telling them they’ll gain more attractive abs.³²

When leveraging loss aversion, though, remember that your users can just stop using your product to avoid loss and the negative emotions that come with it. The product must be seen as worthwhile and enjoyable overall—loss aversion should be used only on the margins.

Use Commitment Contracts and Commitment Devices

Loss aversion, when used too often or people actually have to experience the loss (instead of merely the prospect of it), has an obvious downside: you can end up *punishing your users*. If you give people a consistently bad experience, in most cases, they will stop using your product and do something else with their time. If you could hypothetically force people to endure your punishment, that might be effective. But you can’t, and the user has the option to ignore or avoid you.

Not punishing your users doesn’t mean completely avoiding the *threat* of properly selected punishments. One powerful type of threat is a *commitment contract*, in which people pre-commit to taking an action, and they forfeit something they care about if they fail to follow through.³³ For example, [stickK.com](https://stickk.com) employs commitment contracts to generate creative, personal

punishments, like automatically donating money to an NGO you hate if you fail to lose weight. Importantly, their punishments are self-imposed and self-calibrated; people choose their own punishment. We react much more negatively to externally imposed punishments than we do to self-imposed ones.

Overall, the trick is to carefully use the threat of punishment (and ideally, a self-imposed one) to motivate action without actually punishing people and driving them away.

Another, related technique is to use a *commitment device*, in which people lock in their choice to not act in a certain way in the future. It is like a commitment contract, but it's more extreme: the future action is closed off, instead of threatening a punishment. Taking disulfiram before a night of potential drinking is one such device; it makes the person who doesn't want to drink feel sick if they cross the line and do.³⁴

Test Different Types of Motivators

As humans, we don't lack for things that could motivate us. Money. Food. Control. Esteem. Researchers have tried to make sense of our motivations for decades,³⁵ from Maslow's hierarchy of needs (we address deficiencies in a successive set of needs, from basic comfort to self-actualization) to von Neumann and Morgenstern's expected utility theory (we should do what provides us the most benefit). I won't try to argue which motivations are most important for all of humanity but rather will make an observation. The most important form of motivation is the one *that's actually compelling for your users, given their life circumstances*. Identifying that motivation is part of getting to know your users and what resonates with them.³⁶ It may also entail experimentation—trying out a cash payment or public acclaim or providing a sense of mastery. Three big areas you can explore with your product are:

- Quasi-monetary rewards like points redeemable for cash (while being wary of crowding out other motivations and of needing cash because the product is fundamentally misaligned with user needs)

- Progress and achievement rewards (including badges and other gamification techniques)
- Social motivations, like status or esteem of peers
- Intrinsic benefits like exploring something new (the product can accent the intrinsic rewards that users already receive)

Also, try varying the motivation over time—we become satiated in any single area, at least in the short term and start looking for new rewards. That’s obvious with food (if you’re no longer hungry, more food just isn’t that motivating), but it also applies to other forms of reward (if you’ve won a competition against your friends 10 times in a row, winning again isn’t that interesting).

Use Competition

To call out one of the existing social motivations people have: competition, judiciously used, can be quite powerful. We all have a natural competitive side—though it’s much stronger in some people than others. Usually, you’d build a competition into the overall product, but it can be deployed at a page level too. For example, imagine a page that has people match Spanish words to their English meanings to help the users learn Spanish. The page could include a counter of how many correct answers the individual has versus others on the page at the same time.

Pull Future Motivations into the Present

We like stuff now, rather than later. We’re far more motivated by current goods and experiences than in future ones, even after accounting for inflation, uncertainty, and so on. This *temporal myopia* (focusing on the present even to our own detriment), aka *present bias*, is deeply ingrained and something that too many behavioral change programs forget.

For most people, most of the time, “a few years from now” doesn’t exist. It’s not real, and whatever happens *then* isn’t motivating *now*.

And that presents a serious problem. Let's say we sincerely want to slim down our weight to avoid heart disease, or we may really think that saving for retirement is important. But if the threat of heart disease or the need for retirement money is still many years off, it just isn't real to us.³⁷ Daniel Goldstein refers to this as the struggle between the present and future self.³⁸ We have noble long-term goals but are tempted to do other things in the present.

How can we make that future motivation affect our near-term behavior? We can use moments of strength (when we are actually thinking about the future) to lock in that motivation. Commitment devices—described in an earlier section—are one option. An extreme version of them, called the Ulysses contract, was described in Homer's *Odyssey*: Ulysses had the crew members on his ship tie him to the mast so that he was physically unable to respond to the alluring call of the mythical (and deadly) sirens. In a Ulysses contract, people make binding commitments that restrict what they can do in the future.

Another method is to try to bring the future into our current awareness. For example, researchers have used photo imaging techniques to help people visualize what they will look like in the future and act according to their future self's motivations.³⁹

Dan Ariely tells a personal story about how he turned a long-term motivation into something meaningful and useful in the present with “reward substitution.”⁴⁰ He needed to take a highly unpleasant, painful medication for over a year that had a long-term benefit (beating a disease). But that long-term benefit wasn't enough to overcome the temptation to stop taking the medication. So, he linked taking the medication to something that he enjoyed in the near-term—in this case, watching movies. He'd only watch movies right before taking the medication, effectively substituting one motivation (beating the disease) for another one (enjoying the movie).

If these don't work, we can forget about the long-term motivation altogether and simply look for a completely different motivator that isn't far off in the

future. For example, instead of talking about the long-term health benefits of getting in shape, highlight the immediate benefits it will have on someone's love life.

Each of these is a technique to make the action motivating now, when it otherwise would be far in the future. Just remember: when we ask people to just think about what a wonderful retirement they'll have in 20 years or all the things they'll be able to do after they lose three hundred pounds, we're asking them to do something that's deeply foreign to how our brains are wired.⁴¹

A Few Notes on Decision Making

In this chapter, and indeed in most of the book, we focus on facilitating (or hindering) action. As we briefly talked about in Chapters 1 and 3, there's another body of work in behavioral science around how to make better decisions: how to help people slow down and make the choice they'd make if they really thought it through. Here are some techniques that can help there, which are similar to the Evaluation stage of the CREATE funnel.

Avoid Cognitive Overhead

One way to think about the mental cost of your target action is *cognitive overhead*, or “how many logical connections or jumps your brain has to make in order to understand or contextualize the thing you're looking at.”⁴² Figuring out what to do shouldn't be guesswork for the user. That may mean making the action slightly more *difficult* to undertake in order for it to be easy to understand.⁴³

David Lieb gives a great example of product that is physically easy to use but still is costly to the user because of cognitive overhead.⁴⁴ Here's his hypothetical user thinking through a QR code, “So it's a barcode? No? It's a website? OK. But I open websites with my web browser, not my camera. So I take a picture of it? No, I take a picture of it with an app? Which app?”⁴⁵ Forcing your users to think about what to do should be reserved for cases

where their input is important and will shape their outcomes; don't force your users to expend energy because the product is confusing.

Make it straightforward and clear what the user needs to do each time the user has to make a logical leap from, "Oh, if I do this, then this will probably happen, but I'm not sure," that's costly. It takes time and energy away from the task at hand.

Make Sure Instructions Are Understandable

This one is relatively straightforward. Look at the behavioral map and specifically micro-behaviors where the user is told what to do next. Write down how those parts would be described to a prospective user in at most two sentences. Thinking about the behavioral personas identified in [Chapter 7](#), would those users understand the description? As needed, run it by some sample users.

Avoid Choice Overload

A growing body of work demonstrates the difficulties individuals face when confronted with too many choices. Despite the common wisdom that "more choices are better," two problems arise. First, people may refuse to make any decision at all. Second, people may regret the choices they made in an impossible search for the optimal choice.⁴⁶

For example, an often-cited study by Iyengar and Lepper placed two different displays of jam in a grocery store: one with 24 jams, and one with 6.⁴⁷ The 24-jam display attracted 60% of customers, but only 3% of those shoppers ended up buying any of them. The 6-jam display attracted 40% of customers, but 30% of them bought one. Subsequent studies have also shown that satisfaction with one's choice, *whatever it is*, decreases with the number of options one had to choose among.

There is an obvious implication here when constructing individual pages in an app—avoid situations in which the user has to choose among a large number of options (if you want the user to make a choice and be happy with it). There is also a less obvious lesson: be wary of users (and fellow

employees) who say they would really like more options. The person is probably telling the truth, at least from the perspective of their conscious deliberative self, but that doesn't mean providing more options is the right thing to do.

Slow Them Down

Avoiding overhead, ensuring clarity, and avoiding choice overload all seek to decrease the effort that the conscious mind needs to exert—to help people focus on what matters in a decision and not put it off because of its complexity. What if people aren't making a conscious evaluation at all? Here, we look to the techniques in the broader judgment and decision-making literature, in particular, intentionally adding friction to the process so that it is difficult to act on an intuitive reaction. You can try to require a waiting period before making a decision, make the problem intentionally more onerous, or even make the text more difficult to read. Check back in [“Rushed Choices and Regrettable Action”](#) for more information on this topic.

Putting It into Practice

There are a great variety of approaches to choose from when crafting interventions to support action. In this chapter, we reviewed those for the first three behavioral obstacles in CREATE: Cue, Reaction, and Evaluation. Let's take a look at the crib notes.

Here's what you need to do:

- Behavioral solutions often follow directly from a clear diagnosis of the problem. If you don't have your users' attention to a new feature, well, get their attention. If they dislike how your product looks, change it. Spending enough time on the diagnosis can make crafting the intervention really straightforward.
- When the solution isn't obvious, however, we have many techniques to draw upon. These include removing competition for

the user's attention, social proof, loss aversion, and commitment contracts.

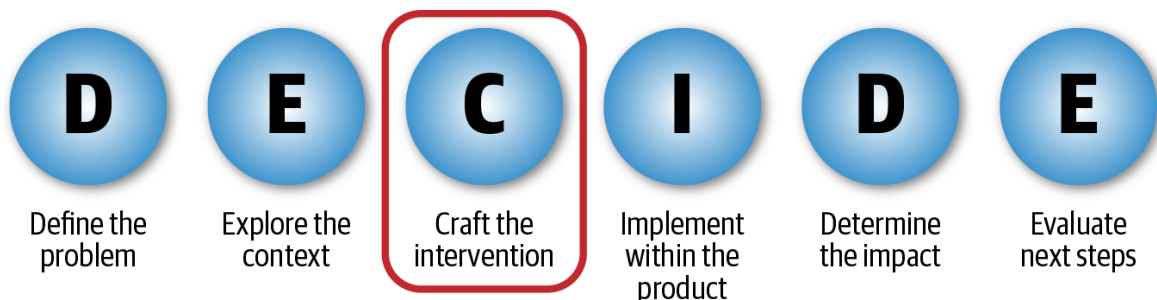
How you'll know there's trouble:

- When you're not clear which obstacle users face (go back to **Chapter 7** to diagnose it).
- When increasing the users' motivation seems like the obvious and only solution (go back to **Chapter 8**).

Deliverables:

- One or more intervention to try with your users, to see if it helps them take action and overcome their obstacles.

Worksheet: Evaluating Multiple Interventions with CREATE



When your team is evaluating alternative interventions for a particular micro-behavior or step on the behavioral map, you can quickly assess the strengths and weaknesses (from a behavioral perspective) of each using a checklist like this:

Condition	Current state for step: Installs app	Intervention	
		1: Tout benefits of app	Intervention 2: Social proof
Cue to think about taking action	☒	☒	☒
Emotional Reaction	☐	☐	☒ Sees testimonials from others about using it, thinks it might be OK to install. (If majority are using it, you could also use descriptive norms here).
Conscious Evaluation of costs and benefits	☒	☒ Increases motivation, but that's not the primary problem	☒
Ability to act (resources, logistics, self- efficacy)	☒ (After removing the need for an employee ID in Chapter 8)	☒	☒
Timing and urgency to act	☒	☒	☒
Prior Experience taking the action	☒	☒	☒

- 1 For a summary of this and other international studies in tax compliance, see Hallsworth et al. (2017). For more information on the BIT's work in this area, and their many other groundbreaking projects, see Halpern (2015).
- 2 My thanks to Abby Dalton at eMBED for suggesting this study, among the many the World Bank has implemented. This writeup is based on their report, "Promoting Tax Compliance in

Kosovo with Behavioral Insights” (Hernandez et al. 2019), and subsequent email exchanges with Abby Dalton.

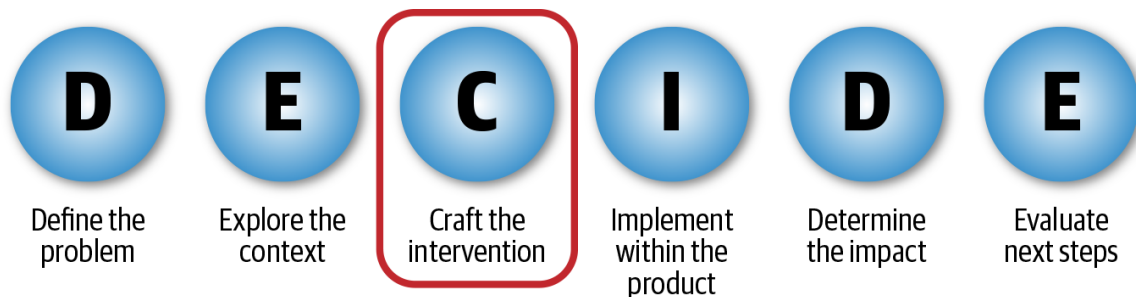
- 3 That is, on the treated population of individual taxpayers. The prior figure, 73%, was for the intent-to-treat population of both individual and corporate taxpayers. There was a significant problem of nondelivery because of bad contact information.
- 4 CREATE is my framework for organizing the myriad behavioral findings out there; the original researchers did not use this framework—in the behavioral literature, there often isn’t any discussion of organizing principles like this. Instead, each paper studies each behavioral mechanism on its own. Dan Lockton provides a good (and unfortunately rare) example of systematically organizing these tactics—he discusses them as eight “lenses” for thinking about behavior change (2013).
- 5 This presentation in table form is inspired by a conversation with Nir Eyal and ideas42’s Behavioral Map.
- 6 Curtis (2009)
- 7 Karlan et al. (2011)
- 8 By effective, I mean they engender more action than not using them. This technique is really obvious, but there are actually experimental studies that show that they work. See guessthetest.com for some examples of optimizing these simple calls to action.
- 9 Gollwitzer (1999)
- 10 We talked about this briefly in [Chapter 2](#)—that at each stage of the CREATE Action Funnel, the action must be *relatively* better than the other potential actions the person is thinking about undertaking.
- 11 Guynn et al. (1998)
- 12 [Clear \(2012\)](#) expands on this concept further in his Layers of Behavior Change model. He describes three layers of progressively increasing power over behavior: appearance, performance, and identity.
- 13 [Bandura \(1977\)](#)
- 14 On the other extreme, generating a supportive self-narrative might require overcoming *learned helplessness* (Maier and Seligman 1976). If people have failed repeatedly and believe they had no control over the outcome, they can simply stop trying. For example, a student who has repeatedly failed at math despite hard work may shut down and think they simply aren’t smart enough to handle it. Learned helplessness is difficult to overcome; products have to find creative ways to reinterpret past events and have users develop other ways of explaining future ones. Show that the person does have control over their future and that the causes of past failures don’t apply to the present situation.
- 15 Cialdini (2008)
- 16 And sometimes of celebrities [making fools of themselves doing so](#).
- 17 A few of the [many examples](#).
- 18 Energy usage: Cialdini et al. (1991); voting: Gerber and Rogers (2009)

- 19 Schultz et al. (2007)
- 20 See, for example, Garner (2005); Noar et al. (2007).
- 21 Fogg et al. (2001)
- 22 See Anderson (2011) for examples.
- 23 Thank you to Emiliano Díaz Del Valle at IMEC for raising questions about the ethical basis of designing for behavior change when the person does not feel motivated to act.
- 24 And, to be clear: traditional economics does not solely focus on monetary incentives nor that preferences are limited to money. It's rather that many economic arguments and models take this simple form.
- 25 Deci and Ryan (1985); Ryan and Deci (2000)
- 26 Deci et al. (1999). While there are various forms of extrinsic motivation, there is always an element of external control; we feel intrinsically motivating things are things we *want to do*, and extrinsically motivating things are things that we *need to do*, even if it is to get a reward that we want and choose. When a “want to” is turned into a “need to” by adding extrinsic motivation, it's hard to go back to feeling that it's something we want to do. The destructive sense of external control is lessened when the outcome we seek (the extrinsic motivation) is aligned with our other goals and desires. Such integrated motivations are less likely to undermine intrinsic motivations.
- 27 An activity can move from relying on an extrinsic motivation to an intrinsic one over time in stages. For example, consider a kid who plays the piano under the watchful eye of a parent. Over time, the kid can internalize the parent's wishes and hear their parent's nagging voice in their head (an *introjected motivation*; Ryan and Deci (2000), with thanks to Sebastian Deterding). Later, the kid might learn to really enjoy playing the piano—making it an intrinsic motivation.
- 28 Jenkins et al. (1998)
- 29 Gneezy et al. (2011)
- 30 Ariely (2009)
- 31 Kahneman and Tversky (1984)
- 32 Kolotkin et al. (2006)
- 33 They leverage loss aversion, the cognitive quirk in which we work much harder to retain the things we own (or otherwise feel to already belong to us) rather than to earn something of equivalent value.
- 34 See Rogers et al. (2014) for a summary of commitment devices in health, for example.
- 35 Millennia, really. For example, Plato saw desires coming from three parts of the soul (Blackson 2020).
- 36 Understanding your users' landscape of motivations also allows for clever techniques like temptation bundling (Milkman et al. 2013)—in which you make something people really like, such as reading *The Hunger Games*, conditional on something people like but aren't as keenly

motivated by, such as exercising at the gym. That doesn't mean you can hold the things that people love hostage to something they hate. Instead, the researchers focused on intentional, voluntary bundling—allowing people the option to get the book and exercise at the same time.

- 37 In economic terms, we “discount,” or place less value on, things that are in the future. The further in the future they are, the less we value them.
- 38 Goldstein (2011)
- 39 Hershfield et al. (2011)
- 40 Ariely (2009)
- 41 See, for example, Laibson (1997), Kirby (1997).
- 42 Demaree (2011)
- 43 Lieb (2013)
- 44 Ibid.
- 45 Ibid.
- 46 See Iyengar (2010); Schwartz (2004)
- 47 Iyengar and Lepper (2000)

Chapter 10. Crafting the Intervention: Ability, Timing, Experience



My wife has a Fitbit—a small exercise tracker that hooks onto her clothing, displays progress on a screen, and sends over detailed information to a computer or smartphone.

The Fitbit does many things right to help encourage exercise. It automates two very annoying (and therefore action-inhibiting) parts of the exercise process: it automates the process of tracking how much exercise the person has had, and it automates uploading that information onto a computer or phone. Those are examples by shifting the burden of work from the user to the product (aka “cheating”).

The device also uses a number of other behavioral techniques to help people exercise. For example:

- *It reminds people to exercise. It gives random Chatter messages on the screen; I still smile as I remember when I first saw the message Walk Me; that is, it provides a (funny) cue.*

- *It provides immediate and meaningful feedback. Shortly after my wife got it, I remember her looking at the screen and seeing she'd walked something like 9,945 steps. She just started running around the room, to break the 10,000-step threshold. That is, it creates urgency (Timing) by establishing a near-term goal—even if it's arbitrary. Even though the benefits of exercise are long term and abstract, a 10,000-step goal is immediate and real in the present.*

In the previous chapter, we covered interventions to use when encouraging an action that's blocked by a Cue, Reaction, or Evaluation. Here, we'll cover the second half of CREATE: Ability, Timing, and Experience.

The User's Ability to Act

Ability is most obviously the physical means to do something: having shoes to go running, having healthy food to eat. From the perspective of behavioral obstacles, we can also think about it as the means to act without further thought and without fear of failure. Every time your user stops to think about what to do next, there is an opportunity to be distracted. Each micro-behavior in the behavioral map can become an obstacle simply because it requires an extra iota of thought, effort, and confidence. Your user might really want to study a new language and was about to download the next lesson, but during the moment it took to look up the website and download it...the phone rang. Your user might really want to apply for a new job, but that line about needing to present in front of others terrifies him.

In this section, we'll look at each of these types of Ability: the physical ability to act for each micro-behavior, the sense of self-confidence it takes to proceed, and the mental ability to follow through from step to step without stopping to think and needing to concentrate on what to do next.

Remove Friction and Channel Factors

Small frictions play an outsized role in behavioral science; much of the initial work in the field looked at how simple form fields and minor hassles blocked people from following through.

We've talked earlier about the importance of automation: when you can take the burden of work from a user (e.g., manually making a 401(k) transfer), it's more likely to get done! Automation is often combined with a default: the user, by default, transfers work to the product unless they opt out. Both of these techniques, however, are powerful on their own. Let's look at setting defaults, separate from the issue of automation.

An early (and startling) example of this comes from the realm of organ donations. Organ donations are an ethically important subject. We literally have the potential to save someone's life. There are huge variations in participation in organ donation programs across countries, with many countries either having 98%–99% of the population agreeing to donate their organs upon death, while in others, only 0%–10% plan to do so. Even neighboring countries with similar histories and cultures—like Germany and Austria—show these variations. Germany has a 12% rate; Austria has a 99% rate.

The reason for these differences isn't because of a deep-seated ethical or religious understanding of organ donation. It's likely because Austria defaults people *into* their organ donation program and lets them readily exit if they choose. Germany defaults people *out* and lets them readily enter if they choose. It seems that what matters is the simple act of checking (or unchecking) a box on a form. That's the incredible power of a small friction (merely checking or unchecking a box) and the default presented to people.¹

What can we learn from this? Obviously, set *appropriate defaults*. But more generally, look for ways to *remove these small frictions*.

Remove unnecessary decision points

Removing the need for users to do extra work is a high-level behavior change strategy (cheat), and it should be used within particular interactions

as well. If you don't need to ask a question of the user, don't. If you can save the user from scrolling down the page, excellent. That's just another small but frictionful activity that the user needs to take on their path to action. Removing these frictions can slightly decrease the cost of action, all else constant, but most importantly, removing such frictions removes intermediate decision points and opportunities to be distracted. If the person has chosen to do something, let them go ahead and do it—the more times you stop them, the more often they can be derailed.

That isn't to say that users can't do work. There may be really important information below the fold, and the user really does need to read or act on it. However, if there is a choice between accomplishing the same task with or without additional form fields and user work, choose the route with less work.

Set appropriate defaults

Even if there isn't a big choice you can default in your product—like organ donation—look for the small choices as well. For example, keep in mind the individual input fields within an application. Assume that many users will stick with whatever default value you give them. This occurs because people are in a hurry and don't fully read the questions posed to them, because they are unsure of what the question means or because they simply do not have a strongly held preference. Thus, defaults matter not only because they create decision points (and hence distraction, etc.), they also change outcomes for the person.

Default values can be immensely useful (a) where the default response can move the individual closer to action, (b) where power users can fine-tune their responses, and (c) where everyone else can breeze past the defaulted values. However, default values should be used only where nonresponse is acceptable; it shouldn't be used where essential information is gathered. And, since users will make up fake answers (or simply disengage) when forced to answer questions that they can't really answer, it is better to altogether remove questions that users don't have answers to and can't be

defaulted. When default values are provided, the answers should be interpreted as one-part truth and one-part nonresponse.

For example, let's say your application asks users if they have kids. If there's special advice that's only relevant for people with kids, then default the answer to "no kids." Let those users who do have kids, and are paying enough attention, indicate it to receive the special content.

Elicit Implementation Intentions

As you may recall, implementation intentions are specific plans that people make on how to act in the future.² They are a form of behavioral automation, telling the mind to do X whenever Y happens. The person does the work of thinking through what needs to be done *now*, and then when the action is actually needed, there's no need to think and no logistical barrier to action—the person just executes the action. Implementation intentions should include the event that triggers action, the context for that action, and the physical things the person should do. For example: "On Friday at work, if my supervisor yells at me about the project, I'll leave the room and take a short break rather than yelling back."

You can encourage the user to create a future action plan (implementation intentions) wherever the user is committing to take some future action, *especially* when that action is outside of the application. Making a specific, concrete plan of attack can help the person follow through with the action, even when the product isn't there to remind them.

For behavioral products, deploying implementation intentions can mean adding text boxes where the user describes how they'll take the action. The key is to make people think consciously about the concrete actions, and, if possible, visualize undertaking those actions. The challenge is that implementation intentions are a friction; they slow people down and make them do additional work.³ As we talked about in the previous section, that's not such a good idea on its own. Rather, you can think about it like this: if an unimpeded path (removing friction) doesn't get people over an obstacle on their own, you need to prepare them for that obstacle (with

implementation intentions or other such techniques) to get over it when they reach it.

Peer Comparisons Can Help Here Too

We talked about peer comparisons in the context of the person's emotional reaction: knowing that other people are spending less money on electricity than you, or voting more often than you, can trigger a strong intuitive response. In addition, peer comparisons can have an effect on our sense of ability. I think about peer comparisons as having both a “yes you can” (Ability) dimension, as well as the “yes you should” (Reaction) side.

If we think a task isn't achievable, we have better things to do with our time. On your pages, make sure not only that the person *can* do what's needed but that they know they can do it, too. One way to accomplish that is through the peer comparisons described earlier; show the user that other people are successfully taking the action. Then they know, yes, it's probably something they can do as well.

Remember, however, that peer comparisons are complex. If the peer group is too far ahead of the individual, that can be demotivating (I'll never catch up), and if the peer group is behind the individual, that can also be demotivating (ach, I can just relax a bit—I'm doing better than everyone else already).

The Other Side of the Wall: Knowing You'll Succeed

In the last week, how many times have you tried something that you were pretty sure you would fail at? Something important, where other people would know that you failed and likely judge you about it? I'm guessing not many times at all. That's because our minds prune these options from the tree; we don't really think through how to do improbable actions (beyond daydreaming—here I'm talking about intentional action).

If you think you aren't going to be able to do something, you're less likely to even try. The underlying research comes again from Bandura's concept

of self-efficacy. The belief that you yourself can be effective at the task with similar lessons comes from research on goal completion.

Helping your users know that they'll succeed can be as complicated as an in-depth training program and building up their expertise and confidence for a hard action. It can also be as simple as reframing the action to make it feel more familiar and feasible. **Figure 10-1** shows one such simple experiment I reported on with John Balz in a previous paper.⁴



Figure 10-1. Simple changes in wording to help people know what they're getting into (and whether they'll be able to complete it)⁵

Look for “Real” Obstacles

The obstacles we've talked about thus far under Ability are in some sense psychological—from a strictly rational, cost–benefit perspective, they shouldn't be obstacles. But people do face real obstacles of course to using a particular product or service, like not having their password or an internet connection to use your app. It's always obvious in hindsight, but I've certainly fallen prey to missing these obstacles myself. In one study I ran a while back, we emailed a call to action to our users, using email addresses they didn't have passwords for...and sat wondering why no one responded. So this one is a simple reminder: look at your usage data and do qualitative research with your users to make sure you didn't miss simple stuff while you searched for more fancy behavioral obstacles.

Getting the Timing Right

Ideally, the action is inherently time-sensitive: people need to take the action immediately because of some existing, external rationale—like taxes

on April 15. However, when that's not possible, there are a few other tactics that you can use.

Frame Text to Avoid Temporal Myopia

We're wired to value the present far more than the future—that's our temporal myopia. We talked about that earlier, when we looked at ways to motivate the user with immediate rather than future rewards. Well, what if you're stuck, and the basic structure of the application and its core motivation are already fixed? You can still avoid the curse of temporal myopia by crafting the descriptions you provide to the user.

When designing for behavior change, this means being very careful about the framing of time. Look for ways to frame benefits in terms of immediate or near-term gains; in the exercise example, reference how the person will feel and look better *almost immediately*. The opposite is true for *pain and effort*: effort that occurs sometime in the future is much easier to commit to than effort right now. So if the pain and effort need to be discussed at all, put it in the future as much as possible. Benartzi and Thaler do this beautifully with their Save More Tomorrow plan—people commit now to saving (i.e., *pain*) at a future date.⁶

Remind of a Prior Commitment to Act

We don't like to be inconsistent with our past behavior. It's very uncomfortable, and we have a tendency to either act according to our prior beliefs or change our beliefs so that they are in line with our actions.⁷ One way to achieve this is to have the user impose urgency on themselves—promise to take the action at a specific time, then come back to them and remind them at that point. In addition to their other reasons to act, that will spur them to follow through, to avoid feeling inconsistent.

Make Commitments to Friends

Another way to create urgency to act is to make specific promises to do so to your friends. Social accountability is a powerful force—we don't want to

let our friends down or lose esteem in their eyes. The next sidebar illustrates that point with a personal story from my friend Justin Thorp.

OUR FRIENDS HOLD US ACCOUNTABLE

I've always been a big guy. Back in the fall of 2009, I was clocking in around 280 and was getting fed up with being winded when I ran up a few flights of stairs. I knew it was time to do something different.

So I decided to get into running. That was a daunting task for me. I could barely run down the block. Being a nerd, my first thought was... what's better than exercising? It's exercising with technology. So I perused the app store and got Runkeeper—an app that used my phone's GPS to track how far, how fast, and where I ran. I was instantly hooked. It allowed me to see my progress throughout my running journey.

After a while, I noticed little Facebook and Twitter share buttons at the bottom of the Runkeeper report. With the press of the button, I could share my runs with my friends. I was like “what the hell” and hit the button not really thinking much of it.

A few days passed and all of a sudden my friends and coworkers started noticing my runs. They were commenting on my Facebook posts. They were cheering me on via social media. When I wouldn't run, my boss would ask, “Justin, why didn't you go running today?”

When I got up in the morning and didn't want to go running, I'd hear the voices of my friends and supporters in my head. I didn't want to let them down. They believed in me and believed that I could do it.

And I did. I lost 50 lbs. I ran the Cherry Blossom 10 Mile Race. I gained a ton of confidence. And I still run regularly. It's become a great way for me to get exercise and clear my head.

—Justin @thorpus

Our friends have a wide range of effects on our behavior, as we've talked about previously under the power of social proof and descriptive norms.

But telling our friends what we're doing has a particular power to push us to act when we say we will. It's not just the action by which they judge us but whether we kept our word overall—and that includes timing.

Of course, it all depends on who we look to for support and accountability. If we turn to people who really don't care about us or who don't value the activity we're trying to undertake, then their disinterest can sap us of motivation. Products can mitigate this by explicitly asking people to identify friends and colleagues who will support them or by matching up the person with other users who are seeking to change the same behavior or have experience providing support (i.e., products can construct a local network of peers who *will* push us to succeed). [Coach.me](#) does something akin to that.

Make a Reward Scarce

You can make a reward for the action scarce (“the names of the first hundred people losing one hundred pounds will be featured on our website”) or artificially time-sensitive (“act in the next five minutes and you'll get another 10 points”). This is another favorite sales and marketing tactic.⁸ It's best for one-off actions and not repeated behavior. If you try to repeatedly push for a behavior with scarcity people will stop believing you. Also, you run the risk of desensitizing the person to normal scenarios that aren't artificially scarce or time-sensitive.

This technique has been abused in the field, however, by creating scarcity for the benefit of the company that is disingenuous and hurts the customer. [thredUP](#), which we talked about in [Chapter 4](#) on ethics, is one such example; hotel booking sites like [Expedia](#), [Hotels.com](#), and [Booking.com](#) are additional negative examples that have rightly been called out by regulators.⁹

Handling Prior Experience

People's prior experiences shape their reactions in ways that can be difficult to foresee and even comprehend. Our intuitive reactions, knowledge about the costs and benefits of an action, and our sense of self-efficacy are all guided by the associations and information we've built up over time. The previous sections on obstacles arising from Cue, Reaction, Evaluation, etc., all speak to common challenges that people face because of how our minds are wired, or even because people generally have some similar experiences in life. This one, Experience, is the wildcard. It's a reminder that no matter what general lessons we glean in the research community about behavioral obstacles people face, and the tools to help overcome them, all are dependent on the particular experiences of an individual.

For example, loss aversion is certainly powerful in general. However, someone who was raised with the constant threat of loss may be especially sensitive to it or may have learned to intuitively reject it. People who actively practice Buddhism may be less responsive to most of the techniques discussed under the Evaluation phase if they are able to release the hold of material desires on their lives more effectively than the rest of us.

On a more day-to-day level, people who have seen disingenuous and manipulative ads for bad products that employ social proof (expert testimonials and the like) may reject any appeal that uses that technique. And finally, people may rightfully distrust anything you say if they had a bad experience with a prior version of your product or service, which had hyped up, inaccurate claims about its benefits.

So, what can we do when someone's prior experience creates an obstacle to something they would otherwise want to do? While there is less research in this area, here are a few ideas and approaches.

Use Fresh Starts

Fresh starts are special times in our lives when we feel a new opportunity to change something about ourselves.¹⁰ Research by Hengchen Dai, Katherine Milkman, and Jason Riis at the University of Pennsylvania about fresh

starts finds this: *people are disproportionately likely to make major life commitments during times of transition.*¹¹ For example, these scientists looked at how people are more likely to make commitments like exercising more or eating better over New Year's (New Year's resolutions), birthdays, and marital changes.

The behavioral logic is this: when we've struggled to do something in the past, fresh starts give us a reason to hope things will be different this time around. We mentally separate out our experiences from before the fresh start and label them as irrelevant or outdated ("That was *last year!*"). The time after the fresh start has a newness free of our historical baggage that lets us try something different or recommit ourselves to a prior goal we've failed to achieve.

If there's something your users tried to do in the past but struggled with, like exercising regularly, then these special fresh start moments can help them reset the clock, to have renewed vigor and a sense of hope that they otherwise would not have, given their prior experience. New Year's resolutions are the stereotypical example, but these fresh starts can center on other events as well—moving house or changing job, for example. Many religious traditions have “fresh start” periods as well, such as Lent; websites like [FaithGateway](#) send out emails and change their web design during Lent to highlight the special time for users to recommit to and re-invigorate their spiritual path.

A Fresh Start can make the action and context feel special and allows people to put their past experiences in a separate historical category that doesn't doom them to repeating those mistakes again: *the future can be different, if you make it so.*

Use Story Editing

In “[Reaction](#)” we talked about Tim Wilson's research on the self-narrative: the story we tell ourselves about who we are, based on our past experiences and our understanding of our future path. There, we focused on bringing together related past experiences—especially successes—to make a new

action feel more familiar and natural. Wilson also discusses a related technique: story editing (helping people “edit” their self-narrative to reinterpret negative past experiences).

One of the best studies I’ve ever seen on behavior change was one Wilson’s on story editing. He and his coauthor, Gilbert, took a group of first-year college students who were struggling—they weren’t doing well in school and were worried about their future—and randomly assigned the students into one of two groups: one group received a short, 30-minute intervention; the other received nothing special.¹²

Wilson was concerned that the students saw themselves as failures. His intervention entailed giving the students information about potential *interpretations* of their bad performance in school:

*We gave them some facts and some testimonials from other students that suggested that their problems might have a different cause...namely, that it’s hard to learn the ropes in college at first, but that people do better as the college years go on, when they learn to adjust and to study differently than they did in high school...*¹³

The randomly selected group that reinterpreted their bad grades got better grades in the future. They got better grades *all the way to their final year in college*; they also were less likely to drop out of college. While the study did not track their full academic performance over time, we can posit that the effects were not immediate. Rather, it appears that students would have slowly changed how they saw themselves and gradually changed the amount of effort they put into their studying after this initial push.

A 30-minute intervention that changed performance for years? Impressive.

Wilson is a leading proponent of the idea of story editing more broadly; like the students in his experiment, we can reinterpret what’s happened to us in the past by changing the story we tell ourselves about it—our self-narratives.¹⁴ That reinterpretation then affects our future behavior. When we change our behavior, we also change the experiences we’ll have in the future, making them marginally more likely to support our self-narratives.

And with each new experience, our internal story of who we are changes a bit more, spurring a new cycle of behavior change.

For the students, it would have worked like this: Wilson helped half of them interpret their performance differently. Those who saw themselves as going through a temporary tough spot (and not as failures) would be slightly more likely to work harder and perform better on the next test. They would then look back at that (improved) performance and reinforce their understanding of themselves as students who *could* study and could overcome the challenges of first-year life. They would then work even harder on the next test, perform better, and so forth. With time, the internal stories, or self-narratives, of the two groups diverged, thanks to a small push from the initial intervention.

We interpret and reinterpret our experiences every day of our lives and thus shape our self-narratives and our future behavior. These cycles of interpretation and behavior can clearly support beneficial changes, like studying more. They can also lead to negative ones, like when someone feels like a failure and doesn't put in effort to try to change that. It depends on how we use our past experiences and whether we see ourselves in control of the outcomes of our lives.

Use Techniques to Support Better Decisions

If a person has a strongly negative emotional reaction to an action, based on their prior experience, or similarly obsesses about a single facet of an action's costs, it may help to treat it as a decision-making problem, rather than a problem of action. We covered this body of literature in **Chapter 3** in the context of helping people make more careful conscious decisions.¹⁵ Here are some of the key points:

- Slow thinking allows for more careful thinking.
- Add friction to slow people down by adding cognitive overhead and adding the number of steps required for action.
- Direct attention to important but ignored facets of the issue.

Make It Intentionally Unfamiliar

While I haven't seen a research study specifically test this idea in the field, another technique comes to mind that is inspired by existing work. If prior experience with a familiar (same or similar looking) product or communication causes a negative reaction that blocks action, you could intentionally change the look and feel to no longer trigger that reaction. This is appropriate only when someone faces a reaction they themselves would want not to have—i.e., in calmer moments, that they would want to take the action.

It's a technique that has clearly been used by many an unscrupulous company as well. When my wife and I were on a vacation in the Caribbean, we came across a travel service that looked wonderful. It offered great discounts on vacations in the future, at what seemed like a reasonable (but not unbelievably low) price. I checked out the company online, and everything looked OK. Only a few weeks later did we find out that the company had repeatedly changed names—whenever bad reviews and lawsuits caught up with it, it simply changed names and marketing campaigns. Same company, same (bad) service, but new skin. They kept bringing in customers by making their brand intentionally unfamiliar. You've probably come across companies on Amazon that have done the same—they change the name of their product or of their company—to avoid people's prior bad experiences with them and their negative reviews.

Thankfully, we can envision more beneficial uses as well. Think of someone who has struggled with weight loss in the past and doesn't think they'll ever succeed. Intentionally creating a service that looks and feels different than a standard offering in the market (that didn't work out for them), like a meal-delivery service or a meal planner, could help them give it another try. Even though the service itself is the same, the environment around the individual may have changed, and they could be successful now—if only they could get past their prior negative experiences. Which raises the final point for this section—remember that people change.

Check In Again: You're Not Interacting with the Same Person

Not only are people's experiences different from one person to the next, but every day, your users are changing and adapting—in both universal ways (aging) and idiosyncratic ways (getting married, having children, etc.). The person you emailed six months ago is different from the person you're communicating with now. Through your user research, you can *check back in with them* and gain insight into how your user base is changing over time. If the company has had a retirement workshop in the time since the last retirement communication, build on that, especially if you can segment the communication to target those who attended.

Putting It into Practice

In this chapter, we walked through interventions you can use for the second set of behavioral obstacles in CREATE: Ability, Timing, and Experience. Let's take a look at the crib notes.

Here's what you need to do:

- Ideally, once a user has made the decision to act, they can flow from one micro-behavior to another on the path to the final action. Unfortunately, large and small Ability barriers get in their way, in which they are lacking physical resources (having a password), lacking a sense of self-confidence, or needing extra time and thought to proceed.
- To remove physical barriers, the solution is usually pretty obvious: if we're careful to watch for each micro-behavior along the path and notice them. To remove barriers of self-confidence, we look to peer comparisons, and removing uncertainty about what's ahead. To remove frictionful pauses (decision points) we use defaults and simplify interactions.

- People are naturally focused on immediate tasks and needs—so activities that benefit us in the long term are easily lost. To counter that, we reframe how we talk about the benefits, create current scarcity, or focus people’s attention on action now through personal and social commitments.
- Negative prior experiences can cause them to lose sight of the broader benefits of an action, and each person’s experiences are idiosyncratic. To help people move past those negative experiences, we can use the concept of fresh starts (birthdays, major life events, and such) to reset the clock, story editing to reimagine what those experiences portend for the future. We can also avoid intuitive associations and invoke deliberative System 2 thinking by *adding* friction and slowing the person down. Or, we could simply avoid those prior experiences by making the action look and feel like something unfamiliar.

How you’ll know there’s trouble:

- When the product is simply hard to use—behavioral techniques can help smooth minor frictions and challenges; it can’t fix a broken product.
- When we’re using behavioral techniques to cover for prior failings of the product: we’re trying to avoid prior bad experiences people have had with our own products and convince them they should spend more money on it, etc., when it hasn’t really improved.

Deliverables:

- One or more interventions to try with your users to see if it helps them take action and overcome their obstacles.

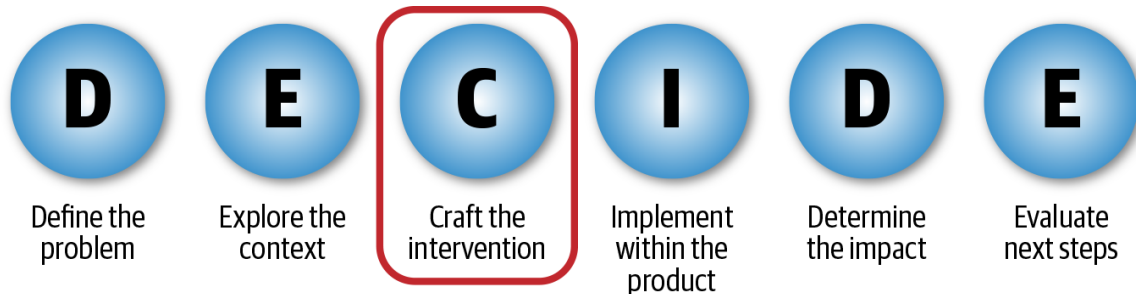
Exercise

You’ll continue to use “**Worksheet: Evaluating Multiple Interventions with CREATE**” for obstacles related to Ability, Timing, and Experience. In

addition, the table of suggested interventions to support action is reproduced in the workbook for your ease of use.

-
- 1 Johnson and Goldstein (2003). Technically speaking, this analysis shows the marginal impact of a default, given automation that is already in place—since no one can remove their own organs after death, with or without a default. But the point is the same: defaults can be logically separated from automation and have powerful marginal effects.
 - 2 Gollwitzer (1999)
 - 3 Thanks to Paul Adams for the tip.
 - 4 Balz and Wendel (2014)
 - 5 Wendel and Balz (2014)
 - 6 Benartzi and Thaler (2004)
 - 7 See Festinger (1957)
 - 8 Cialdini (2008); Alba (2011)
 - 9 See Monaghan (2019) for a story on how the UK’s Competition and Markets Authority has clamped down on these techniques. Once more, I am indebted to Paul Adams for the tip!
 - 10 This section draws from Wendel (2019).
 - 11 Dai et al. (2014)
 - 12 We discussed this study briefly in Chapter 1, as an introduction to the idea of *self-concepts* or *self-narratives*.
 - 13 Gilbert and Wilson (2011)
 - 14 Wilson (2011)
 - 15 See Soll et al. (2015) in particular.

Chapter 11. Crafting the Intervention: Advanced Topics



New Moms is a small nonprofit in Chicago serving young mothers at risk of homelessness with a range of programs—from low-income housing to doulas and parent education. Their job training program, however, faced many challenges: mothers were interested, but few people actually made it into the program. In marketing terms, their conversion funnel faced steep drop-offs at each stage of the process.

*Unlike most small nonprofits, however, New Moms has long studied behavioral and brain science. “We’re big fans of ideas42 here,”¹ says CEO Laura Zumdahl. Under the direction of Dana Emanuel, their Director of Learning and Innovation, they partnered with the **behavioral science team at MDRC** to identify obstacles facing young mothers in the program. And, as is often the case with any program or product, there were many. Here’s what they found:*

- *Their marketing materials talked about the long-term benefits of the job training program, in technical terms (who doesn’t get excited about “job skills”?). In their behavioral analysis, they realized that the young mothers were strongly present biased and*

rightfully focused on near-term goals like feeding their kids. So the team changed the materials to “message to the mothers’ motivations,” as Dana described it. In other words, money for their family and a flexible schedule. It’s a lesson that many “good for you in the long run” products would benefit from learning too...

- *In order to sign up for the program, the mothers had to fill out multiple pages of forms and meet multiple requirements. Most of it was simply unnecessary, they realized, and created unnecessary barriers to entry (i.e., hassle factors). So they simplified the sign-up process considerably.*
- *Once new mothers were in the program, the organization asked them to set quarterly to six-month to one-year goals. Many of the mothers had never worked, and long-term job goals were simply abstract ideas. Those goals didn’t help them, and they got discouraged. So New Moms worked with them to set a series of more immediate and manageable one-week and daily goals, which ladder up to a bigger transformation. These small wins helped the mothers see their progress and keep them on track over time.*

From small nonprofits to international technology companies, the problems our users face are remarkably similar because they arise from how our minds are wired and how we develop products and services without understanding that mental machinery. And, as with many products and services, New Moms found that there wasn’t a single barrier, but rather many that mothers faced over a multi-step process. Their thoughtful analysis uncovered what obstacles were at work in each case and how to overcome them to better serve their users.

The behavioral solutions we discussed in Chapters 9 and 10 are about point-in-time interventions to support action: ways to facilitate action at a particular point in a behavioral map. In this chapter, we look at three extensions: working with multi-step interventions over time, building habits, and crafting interventions to hinder negative action.

Multi-Step Interventions

For some products—and many communications—the behavioral map is straightforward and simple, or there is a single point of failure that needs attention. Here we take a step back and look at cases where the process is more complex and you need to address the behavioral map’s series of micro-behaviors as a whole to make the process more feasible for users overall. In particular, we’ll look at how to simplify the map, how to make it easier, how to provide feedback along the way, and how to build habits.

Combine Where Possible

Each step in the sequence should represent the *largest* possible chunk of the work that is still understandable and feasible. Look for ways to combine multiple steps into one. *Largest?* Yes. There is a tension between breaking the action into steps to make each one more manageable and having so many steps that it overwhelms the user. There’s no hard-and-fast rule, but keep this in mind especially when you see any sequence that’s more than a dozen individual steps!

Again, Cheat If You Can

See how to shift the burden of work at each step from the user to the product—that is, use the “cheating” strategy within each of the individual steps the user has to take. Look over the behavioral map. Is it possible to automate away the whole action? If not the whole action, is it possible to cheat at any of the individual steps along the way?

With the radio program from **Chapter 7** (in which Obama’s campaign sought to encourage people to call into a radio program to voice their support), complete automation wasn’t feasible. But certain parts of it could easily be automated or defaulted. For example, the platform could:

- Automatically match the user up with a call-in radio program and provide the necessary phone number to call, saving the user the need to research relevant radio shows.

- Make finding a radio unnecessary (incidental to using the software), by streaming the call-in program through the app itself.
- Provide intelligent defaults for what to say while on the air (e.g., a simple script that the user can edit and personalize as needed).

By simplifying the process, automating, defaulting, or making steps incidental, you remove unnecessary work for the user. In terms of the CREATE Action Funnel from **Chapter 2**, that means decreasing the costs of action (part of the mind’s conscious evaluation) and increasing the basic physical ability to take action. This simplification process allows you to focus attention on more intractable behaviors or more exciting (to the user) parts of the application. Another way to accomplish this is to build habits over time—something we’ll return to shortly.

Provide “Small Wins”

In addition to being *easy*, each step should be *meaningful* enough that the user can feel a sense of accomplishment afterward. It’s up to the product to help users feel that accomplishment (often by presenting it as progress toward the target action), but the step itself needs to support it. In the research literature, this is called “small wins”—if, after each small step, people feel they’ve done something and they’re closer to finishing their goal, they are then more likely to continue.

For example, in my prior work at HelloWallet, we had a tricky problem when we designed our guidance—how much should you encourage people to commit to save each month? Authors such as Jean Chatzky tell their readers to save about \$10 a day.² That is simple and clean, but either way too easy or too hard for many users. We have users who struggle to put aside \$10 each week, let alone each day; we also have users to whom \$10 is a rounding error, and a laughable goal. So, we constructed a meaningful step on the path to greater savings that provided small wins, regardless of their financial means—we calculated the target amount as a percentage of income (and then rounded to the nearest clean, simple number that people would remember).

For an action to provide a small win, it needs another characteristic—it must be clear to the user that the step has actually been completed. In other words, there must be a clear definition of success or failure and feedback about whether that has occurred. Weight loss applications are a great example—they set a specific, unambiguous weight target, and encourage people to change their weight. An example of an unclear target would be a step that tells users to cut down on smoking. It doesn't allow users to know if they are succeeding; if users have to wonder whether they've done it right or not, they are distracted, and you lose them.

Generate a Feedback Loop

Multi-step, and especially quickly repeated, actions provide another opportunity for behavior change: by enabling users to adjust course over time, to better meet their goals.

The current crop of wearable computing products, like the Apple Watch, Fitbit Versa, and BodyMedia CORE, are built to provide feedback to users about their exercise and sleep habits. For example, my wife's Fitbit provides constant feedback on how much exercise she's gotten. When she makes an adjustment in her routine, it's quickly reflected in the tracker, and she can see how she's doing. That feedback loop allows her to adjust her behavior throughout the day to meet her goals.³

For feedback to be effective at actually helping people change their behavior, it should be:

Timely

Ideally, the feedback should occur while the action is being undertaken so the user can make live adjustments and see the impact.

Clear

The user must understand what the information means.

Actionable

The user must know how to act on the information.

In addition, and this may seem obvious, the user must *care* enough about the feedback to change behavior. Being intrigued and entertained by feedback is not enough. The user must want to, and be able to make the adjustments necessary to, improve performance. The user must also be paying attention. In this combination of motivation, attention, and the ability to act, we find many of the same issues we've already discussed about designing for behavior change.

Common Mistakes

Here are two of the common mistakes I've seen at this stage.

It is easy!

The first common mistake is to be satisfied when it's easy for *you* to do. Too often, especially in my prior political advocacy work, I've seen campaigns that expected people to come to a night-long vigil, write a letter to their representatives from scratch about an issue the author knew little about, or organize a local group of activists on their own. These are each daunting, complex tasks to someone who has never done them before—though relatively straightforward to those who have already built up the expertise.

It's quite difficult to step outside of your own experience (if only because once you think of the action, your System 1 immediately activates the relevant prior experiences, and it really does *feel* easy on an intuitive level). If there's any doubt, though, run the proposal by someone who has never taken the action before.

Hard work builds commitment

Another common mistake is to believe that work makes commitment and to allow the user's tasks to be difficult *on purpose*. This view is half-right, half-wrong. Effort can build commitment to a product and can build a

commitment to continue what you've started. People who complete difficult tasks (successfully and without undue grief) are more committed to continuing further. In psychology and economics, there's the well-known mental blip called the *sunk cost effect*: the more work you put into something, the less willing you are to let go of it—even when it is not in your economic interest.

If users see the target action as a difficult, substantial task—great. *After* the action is completed, they will be committed. The product's job is to get as many people as possible across the finish line. The “make 'em work hard” approach leads to very committed people who finish—but those committed people are only a small subset of the people who could have completed the task; that creates the illusion of effectiveness—because everyone else has already been filtered out!

So building commitment doesn't mean that each step needs to be a pain in the butt or that we should *design* an application to be difficult. For any “hard” behavior (exercising, learning a language, etc.), there will be things that the product can make easier and things that it can't. Make the things that can be made easier easy—and then build up the excitement on the remaining tasks that are hard. Provide users with a sense of accomplishment for the tasks that are truly difficult—and not just badly designed.⁴

Creating Habits

Chapter 1 described the two basic types of habits: habits created out of simple repetition (cue-routine, cue-routine, etc.), and habits that have the added feature of a reward at the end (cue-routine-reward) that drives the person to repeat the behavior. Your product's users could form habits by simple repetition, but then the burden of work and willpower is all on their side. When designing for behavior change, add a reward at the end to help bring people back while the habit is forming.

Either way, though, habits form when the mind takes a repeated action and automates it. In each early iteration, all of the CREATE factors are important; it builds on conscious action, and we need to keep in mind all of

the behavior obstacles that can arise. In time, however, as the behavior becomes automated, the Cue, Reaction (the Routine), and Ability (to actually take the action) are the most important.

Repetition isn't the only aspect that's important, however. To build habits with a product, here is a straightforward recipe:

1. Identify an action that should be repeated dozens of times, without significant variation or thought each time. This is commonly known as a *routine* in the habit literature. For example, maybe your behavioral map includes “run each morning for 30 minutes.” That is a candidate for a habit.
2. Make sure there is a strong and immediate benefit, especially that triggers a strong and positive intuitive Reaction or conscious Evaluation. This is known as the “reward” in the habit literature. An example is receiving congratulations from the app or from a friend.
3. Identify a clear, unambiguous, and single-purpose cue in a person's daily life or in the product itself (an email, an alert, etc.). An example is a notification from the product at the right time of day that it's time to run.
4. Make sure the user knows about the cue, routine, and especially, the reward.⁵
5. Make sure the user wants to and can undertake the routine using the rest of the CREATE framework.
6. Deploy the cue.
7. Track whether the routine occurs.
8. Have the product *immediately* reward the user once the routine has occurred. That allows dopamine in the brain to reinforce neurons associated with the cue and routine before the memory fades.

9. Repeat steps 6–8, tracking completion times and rates and adapting the process until it's right.

There's a lot of nuance there, of course.

First, the *cue* really needs to be single-purpose and unambiguous (i.e., after the habit is formed, the cue is linked to the specific routine and nothing else), because you want to avoid the mind having to think about what to do when the cue occurs. Fogg and Hreha (2010) argue that the triggers (i.e., cues) can be:

- Directly tied to another event (e.g., looking at the bathroom mirror first thing in the morning is connected to picking up your toothbrush)
- At a specific time of day every day or every week

The trigger/cue can be internal (boredom or hunger) or external (seeing the clock first thing in the morning or getting an angry email). Internal triggers are great, since they are inherent in the human condition; however, lots of other things in one's life compete for the same triggers (which makes them not single-purpose and thus ambiguous). External triggers can be just as effective, if wisely constructed.

Second, while the *routine* must be structured so that it can occur effectively without thought. It need not be “stupid” or “simple.” Good driving, for most people, is a (complex, impressive) habit. Remember how hard it was to learn to drive? Remember all of the thought that was required just to start the car and get it going? Yet, after learning, we avoid getting too close to other cars while on the road, we coordinate what our eyes see with what our hands do to steer, and so on. The reason is that driving uses a set of hierarchical habits—large, complex habits built out of thousands of small, routinized behaviors that are cued from the environment and linked to one another in succession. Each piece is structured so that it can be consistently executed after the cue without conscious thought.

KEEPING HABITS OVER TIME

Habits are tremendously powerful once formed, but they can also be fragile. They depend on having a stable cue, in a stable context to trigger. For an app on a phone, that cue might be the sight of the app on the home screen or a push notification. If that cue is lost, like when the person changes phones and hasn't installed the app, the habit no longer triggers.

I use a popular and free app called YouVersion to read the Bible on my phone. YouVersion knows that their users might get a new phone around the Christmas holidays, so they send out an email reminding people to install the app on their new phone—and thus support their habits of spiritual reading. **Figure 11-1** shows the email I received at the end of December last year.

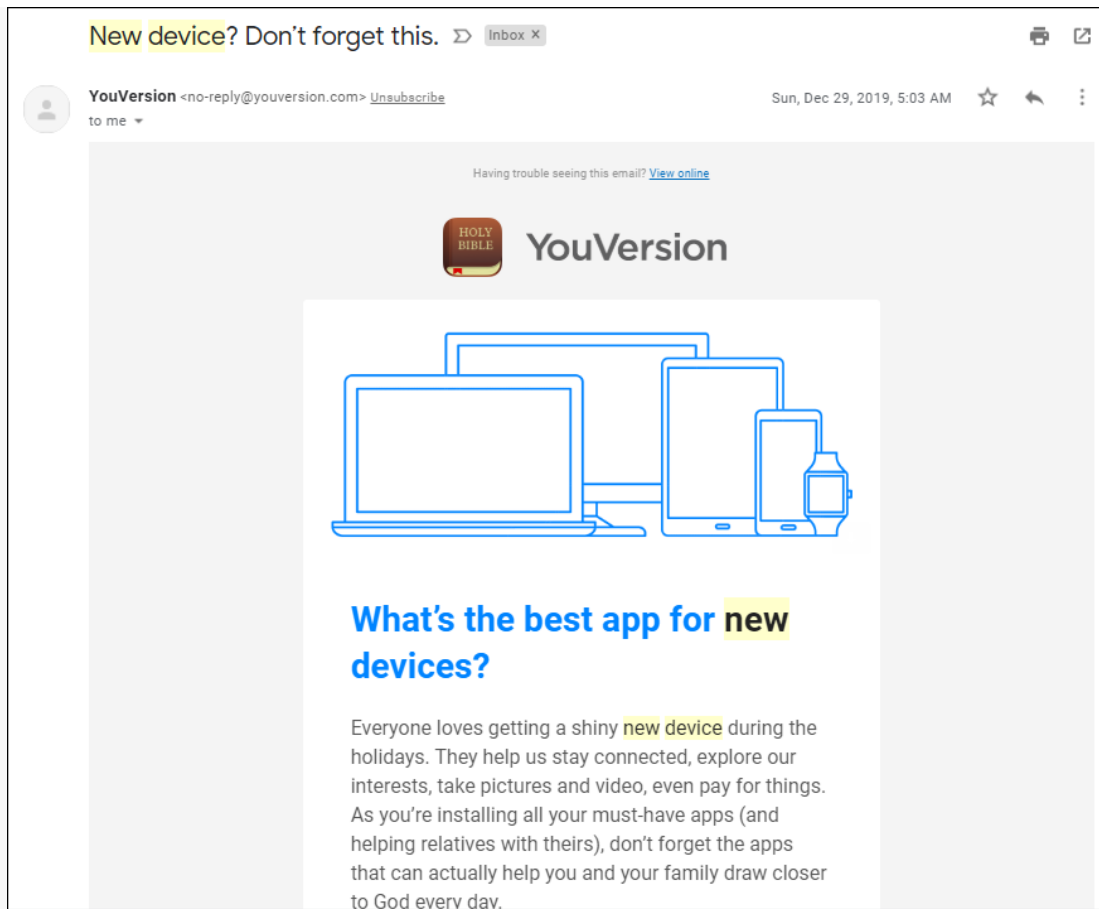


Figure 11-1. An email I received to reinstall the YouVersion app on my new phone—a smart way to ensure that once a habit is formed, the Cue isn't disrupted

Routines that can be made into a habit often will have a strong and clear feedback loop (i.e., after the action is taken, the reward is immediate and unambiguously tied to the success). Habit formation is not a conscious event, though we can consciously put ourselves in situations where we'll learn them.

Third, the *reward* need not be offered every time, as long as it is still clearly tied to the routine. Random rewards are quite powerful in some circumstances. In the operant conditioning literature, habits with random reinforcement take the longest to *form* but also take the longest time to *extinguish* once the reward is no longer given. Gambling provides the ultimate random reward—and once you have the bug, it's difficult as all heck to get rid of. One reason that random reinforcement is so powerful is

that our brains don't really believe in randomness. We look for patterns everywhere. So part of the desire driving a random reward is our brains trying to find a pattern (ever talk to a gambler who has "a system"?).⁶

Finally, a key part of using products to build habits is experimentation and fine-tuning. Your product is probably going to get it wrong the first time—the cue won't be clear or won't grab the user's attention, the user may stop caring about the reward, or the context for the routine might change and conscious thought is required.⁷

The cue-routine-reward process used for *forming* habits is depicted in **Figure 11-2** (remember the reward is optional, but useful). For example, seeing the scale in the morning triggers the exercise routine. The immediate reward is a pleasant muscle burn. Charles Duhigg popularized the process in the *Power of Habit* (Random House, 2012), building on an old tradition in applied behavioral analysis.⁸

Habit Loop

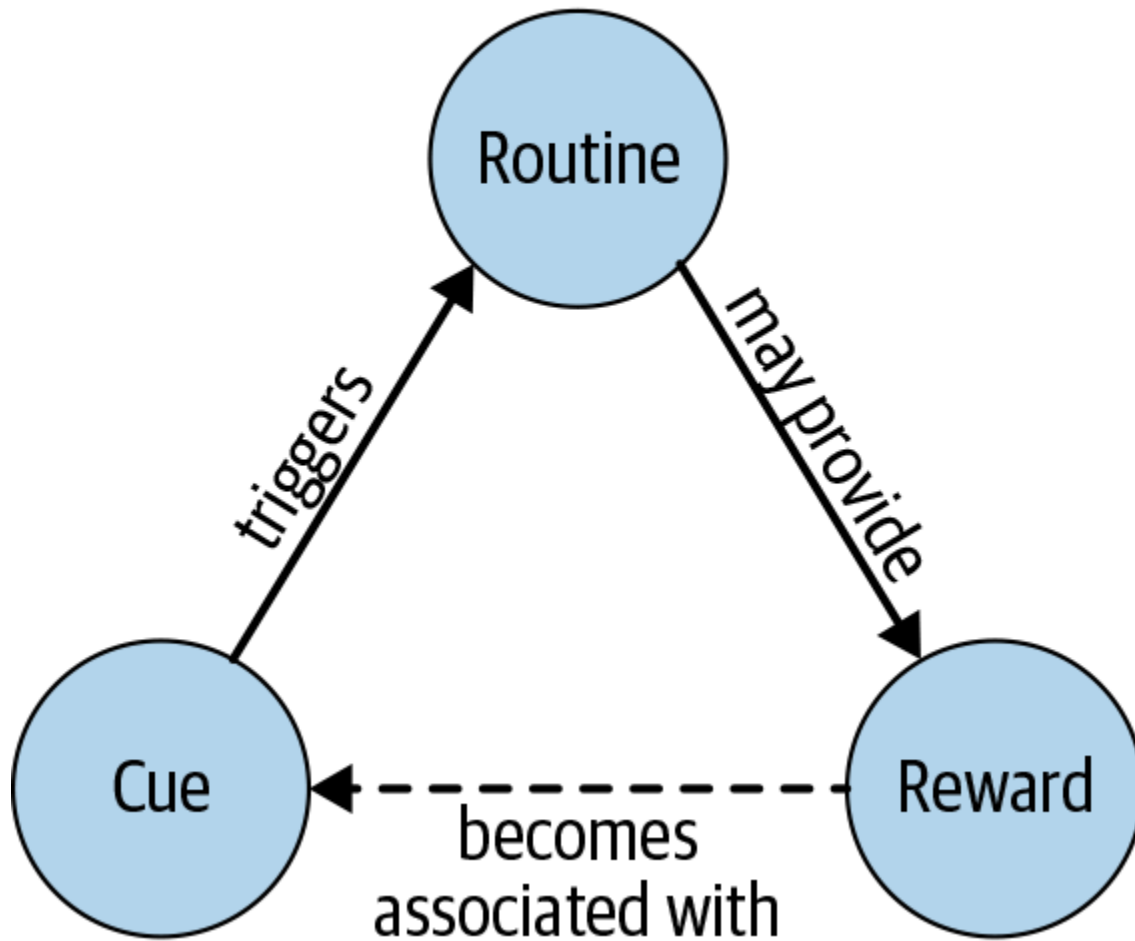


Figure 11-2. The cue-routine-reward process described by Duhigg (2012)

Hindering Action

Believe it or not, we've already covered most of the information we need to hinder action, back in Chapters 3 and 7. That's because hindering an action is far simpler, conceptually, than enabling it. In our diagnosis, we identified how CREATE supports the negative behavior: what is the current cue?

What causes the person to have a positive Reaction and Evaluation? What makes the person Able to act immediately and prioritize it as Timely over other things? And in Chapter 3, we looked at specific ways to hinder habits.

In short, we can add an obstacle at any relevant point in the CREATE funnel. We start by asking the question (as we did in the initial diagnosis, in [Chapter 7](#)): is the behavioral habitual? If it is habitual, then we can focus on the Cue, Reaction, and Ability (C-R-A). If it isn't, then we use the full CREATE. The practice of *creating* obstacles is less researched in behavioral science than the practice of removing them, but often we can find simple solutions that are the opposite of what we've covered already.

Habitual Actions

While stopping a negative behavior in general isn't well researched, stopping habits is. When we first took a look at negative actions in [Chapter 3](#), we discussed many of the core interventions one can use to shape them. Here's a reminder:

1. Cue: Avoid the cue. Since habits are automated, learned behavior triggered by a cue, the most straightforward simplest way to stop them is to avoid the cue—hide it (like a phone), avoid places where one would see (like avoiding bars for drinkers).
2. Reaction: Replace the routine by hijacking the reaction. Old habits don't go away, but another routine can be triggered faster. For example, when boredom in line strikes, load up Duolingo instead of Twitter.
3. Ability: crowd out the old habit with new behaviors. While one can directly attack the routine, the time and ability to act on it can be crowded out indirectly by filling that time with other activities. And, one obviously can't smoke if they don't have (e-)cigarettes—but that isn't designing for behavior change; that is (probably warranted and useful) coercion.
4. In addition, while habits run off of Cue, Reaction, and Ability, there's been success with a conscious approach.
5. Evaluation: Cleverly use consciousness to interfere, including using mindfulness to avoid acting on the cue. While it's frustrating

and tiring to consciously override habits, mindfulness practice has been shown to teach people how to notice, but let pass, the urge to respond to the cue.

Ideas for Hindering Other Actions

Table 11-1 shows our list of techniques to enable action from Chapters 9 and 10, reenvisioned to *hinder* action, after removing some items that aren't relevant in this context.

Table 11-1. Techniques to hinder action

Component	To Start	To Stop
Cue	Relabel something as a cue	Unlink the action from other behaviors that flow into it
	Use reminders	Remove reminders
	Make it clear where to act	Make the cue more difficult to see or notice
	Remove distractions	Add distractions and more interesting actions
	Align with people's time	Move the cue to a time the person is busy, or make the person busy at the existing time
Reaction	Narrate the past	Narrate the past to highlight prior successes at resisting the action
	Associate with the positive	Associate with action with negative things the person doesn't like
	Deploy social proof	Deploy social disproof (show that other people shun it) and social support for change (AA meetings)
	Use peer comparisons	Use negative peer comparisons (show that most other people resist it, don't do it)
	Be authentic and personal	Be authentic and personal in your appeal to stop
	Make it professional and beautiful	Make the appeal to stop professional and beautiful, <i>or</i> make the context of taking the negative action ugly or unprofessional
Evaluation	Make sure the incentives are right	Increase the costs, decrease the benefits
	Leverage existing motivations	Unlink the action from existing motivations
	Test different types of motivators	Don't assume people will be sufficiently motivated to stop, <i>even if the action is hurting them</i> . Instead, as with starting an action, test different types of motivators.
	Leverage loss aversion	Leverage loss aversion

Component	To Start	To Stop
	Use commitment contracts	Use commitment contracts
	Pull future motivations into the present	Pull future motivations to stop into the present
	Use competition	Use competition to stop (e.g., quit competitions, AA chips, Biggest Loser)
	Avoid cognitive overhead	Add to cognitive overhead
	Avoid choice overload	Add to choice overload
Ability	Remove unnecessary decision points	Add small pauses and frictions
	Default everything	Require choices, remove defaults
	Elicit implementation intentions	Elicit implementation intentions (on how to avoid tempting situations)
	Deploy (positive) peer comparisons	Deploy positive peer comparisons—examples of other succeeding at stopping (same!)
	Help them know they'll succeed	Help them know they'll succeed (same!)
	Look for physical barriers	Add physical barriers (no keys to the car, etc.)
Timing	Frame text to avoid temporal myopia	Frame text to avoid temporal myopia (same—for benefits of stopping)
	Remind of prior commitment to act	Remind of prior commitment to act (same, for commitment to stop)

Component	To Start	To Stop
	Make commitments to friends	Make commitments to friends (same, to stop)
Experience	Use fresh starts	Use fresh Starts
	Use story editing	Use story editing
	Use slow-down techniques	Use slow-down techniques

Since stopping a behavior usually means stopping a repeated behavior (why try to stop a behavior in the past that won't recur?), feedback loops are especially important. As we discussed in “**Multi-Step Interventions**”, we want feedback loops to be Timely (feedback is given as quickly as possible after the negative action), Clear (it's unambiguous that the person is off the mark), and Actionable (they can do something about it, right then, and know how to).

Sadly, there are many examples of companies and governments seeking to stop behaviors against the person's wishes or explicitly to harm them. *Nudge* coauthor Richard Thaler refers to the misuse of nudges for ill as “sludge”: applying *friction* with *ill intent*.⁹ From decreasing voter participating by de-registering people to offering rebates on products and then making it difficult to receive them (requiring that someone complete an onerous rebate process), they aren't hard to find. Many of the techniques listed here can (and indeed are) used in sludge settings; whether a behavioral technique is positive or not depends on the person and their desires, more than the technique itself. As Thaler notes “the goal...is to help people make better choices ‘as judged by themselves.’”¹⁰ Thus, when seeking to stop a behavior, the same ethical rules apply as for any other behavioral technique, as we looked at **Chapter 4**: are you being transparent? Is it voluntary? Is it something the person has asked for or wants?

Putting It into Practice

In this chapter, we covered three different advanced issues in crafting behavioral interventions: working with multi-step actions, building habits, and hindering unwanted actions. Let's look at the lessons for each in turn.

For stopping action, here's what you need to do:

- For habitual actions, avoid the cue, replace the routine by hijacking the reaction, crowd out the old habit with new behaviors. See [Chapter 3](#) for more information.
- For non-habitual actions, look to create friction, decrease benefits, or remove attention: that is, to use CREATE in reverse.

How you'll know there's trouble:

- When you're seeking to hinder an action that people want to take
- When you don't know if the action is habitual or not

Deliverables:

- A set of interventions to hinder negative behavior

And for multi-stage actions, here's what you need to do:

- Break the target action into discrete steps that the user needs to complete. These pieces should be:
 - Simple and straightforward (easy to understand)
 - Easy to complete (both appearing easy to the user and being easy in practice to execute)
 - "Small" (so that the user sees clear progress after each step)
 - Meaningful enough to reward after completion
 - Straightforward to see when they are complete (so they the user will clearly know if the action was successful or not, immediately after making an attempt)

How you'll know there's trouble:

- When team members can't clearly and simply convey what the user is supposed to do
- When no one outside the company has been asked how difficult the action is

Deliverables:

- An updated behavioral map, with simplified, more feasible micro-behaviors

Exercises

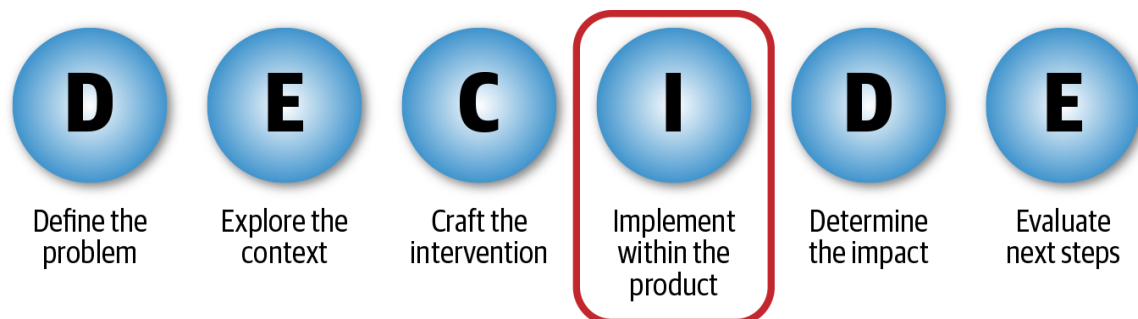
You'll continue to use “**Worksheet: Evaluating Multiple Interventions with CREATE**” for hindering action. In that case, however, you're looking to *add* obstacles. In addition, the table of suggested interventions is reproduced in the workbook for your ease of use.

-
- 1 ideas42 is a large nonprofit behavioral science organization, based in New York City.
 - 2 Chatzky (2009)
 - 3 The **quantified-self movement** has brought rightful attention to feedback loops and their power to both inform and change behavior.
 - 4 When automation of the whole process is possible—something I strongly recommend—then commitment to the action is a real issue. For example, people who are automatically enrolled in a 401(k) without their real commitment are likely to cash out the money and use it for something else. But if your product isn't doing automation and you can still make *everything* that the user needs to do easy, that's a nice, high-class problem to have.
 - 5 In studies of classical conditioning with animals, you actually don't need to link the cue, routine, and reward beforehand. You can build a habit around simple trial and error. However, with us humans, and especially with *voluntary* behavior change, you can skip the trial-and-error part and tell people what's going on.
 - 6 But where we expect a strong pattern, and don't get it, we're angry. Would you be happy if you went to Starbucks for your morning coffee and some days the coffee was terrible and other days it was great? That would be random reinforcement.
 - 7 There's much more that one could say about designing habits, but my goal is not to exhaustively cover them here. Numerous books have been written about forming and breaking

habits in different contexts; Duhigg (2012), Dean (2013), and Eyal (2014) are three good places to start. In addition, BJ Fogg has developed a hands-on approach to creating habits in your own life; see [Tiny Habits](#) and Fogg (2020). For now, though, my goal is to give enough of a foundation that a product team can make a solid product plan and then learn what really works for them in their particular context.

- 8 This cue-routine-reward is a clearer presentation of the antecedent-behavior-consequent (ABC) model used in rational emotive behavior therapy and applied behavioral analysis. See Miltenberger (2011) for one application of the ABC model.
- 9 See, for example, Thaler (2018): “Nudge, not Sludge.”
- 10 Ibid.

Chapter 12. Implementing Within the Product



Safaricom is the largest telecommunications provider in Kenya, with a diverse range of products—from phone service to mobile money to music streaming. They hired the Busara Center, a behavioral economics consultancy based in Nairobi, Kenya, and with projects across the developing world, to help them explore a new education insurance product. The product would help students and their families prepare for school costs—both by saving for it and by ensuring that school fees would be paid even if disaster struck, such as the death of the parent. After talking with Safaricom and conducting qualitative research in the field, they identified the key obstacles and developed a suite of potential interventions.

Often behavioral teams may have multiple interventions they want to test and aren't sure if any will be effective and profitable enough to warrant a full-scale rollout. Busara handled this problem by iteratively developing the product over time, starting with a series of low-cost tests in their lab using members of the target audience. They “built” the product first as a set of wireframes and product marketing materials and tested the audience's willingness to use them. In this low-cost format, they could test multiple

interventions: how the product was framed, how often people contributed, how much they contributed, and whether other features were offered. From day one, they built metrics of success into the process to ensure they received accurate data to drive future iterations.

They then built a “dummy product” that offered key features of the education savings product, but did so in a domain where it’s easier to get immediate feedback: insurance against extra education costs due to rain (a frequent problem in Nairobi). They deployed the simple, but real, product in the field and had impressive interest and take-up, allowing them to more accurately calculate the true interest and impact of the full product. With these results in hand, they have begun to iterate on a minimal viable product that can address the user’s core needs and drive uptake.

Designing for behavior change doesn’t require or benefit from a fundamentally new process to implementing your intervention in the product. The action, as it were, is all in preparing the behavioral maps and diagnosis, in designing the intervention, and after it is built, in measuring the real impact on people’s behavior and outcomes. Many companies use an iterative product development process, as Busara did in this example, to de-risk the process and assess real market interest. That iterative process allows teams to assess the impact of different interventions along the way as well, which is quite valuable. It is not essential, though. Others use a waterfall process, in which case the product, and its behavioral interventions, are all implemented and rolled out in the field in one big bang.

Regardless of the process used to implement the product itself, there are a few pointers along the way that can help the behavioral aspects of the project. In particular, it’s important at this stage to double-check that your incentives and intervention plan are ethically sound, plan to track user behavior and results from the outset, and ensure that thoughtful planning doesn’t get in the way of creative solutions. Let’s look at each of those in turn.

Run the Ethical Review

Once you've determined your intervention(s), the final step before building it and deploying it with real users should be an ethical review. While you should consider ethics from the beginning—asking how a particular behavioral design will benefit the user, and whether they *want* that help—it's only when the intervention has been selected that the full ramifications become clear. In other words, yes, be ethical throughout, but make sure there's also a final review.

The process of the review matters as much as the guidelines themselves. As we discussed in [Chapter 4](#), we all have the temptation and the tendency to blur the lines under the right circumstances. So we should apply behavioral science to our own decision-making environment to make it harder to do so. That starts with ensuring our own incentives are clear and aligned with the user's benefit, that we have an independent review body checking our plans, and that the guidelines, whatever they are, are clear and difficult to misinterpret. For more on the theory, check [Chapter 4](#). For a practical worksheet to help with the process, see [“Putting It into Practice”](#).

Leave Space for the Creative Process

The behavioral interventions we developed over the past few chapters are fundamentally *functional* specifications; that is, they are ideas or approaches to solving a user's behavioral obstacle. They may come with associated tips on how those ideas might be implemented in practice, but behavioral science just doesn't offer mockups or graphic design. It doesn't offer over the best page layout or information architecture. And it certainly doesn't offer the right programming language, the right software engineering pattern, or deployment architecture.

I've seen behavioral designers, myself included at times, try to weigh in on these issues, to take expertise in one area and try to port it over into another. There's a natural tendency to over-specify behavioral interventions, to try to dictate how the product should look and operate, in addition to what it

should do. In reality, most of the behavioral research to date hasn't validated a specific look and feel—and it certainly hasn't validated a specific look and feel that translates and is effective across diverse contexts and user experiences. Instead, our research tests underlying concepts that inform those designs; it helps us inform the design process (UX, UI, graphic design, etc.), but it isn't the same thing as that process.

Part of the challenge is that there's an inherent tension between structured analyses of behavioral obstacles and creative design. Thus far, we've used a rather planning-heavy process—thinking through what the user should do to accomplish the goal, what the context of action is, and how to shape that context. It's just too tempting to think that the *things we want* users to do, and the *way we want* them to behave, is how they'll *actually* behave, when, clearly, there's a lot more required.

Users have to want to change their behavior. When there's a product helping them change, that means they have to want to use the product. The product must not turn them away; no product can be effective at changing behavior if people never use it. The desire to change won't pull most people through an ugly, uninteresting product. Developing a product that people enjoy using takes more than a top-down behavioral map, diagnosis, and intervention—it takes creativity, and it takes the team's development and design expertise.

So, how can you integrate planning and creativity? Part of the solution is philosophical: remembering that everything we've done so far has specified desired functionality, and not how the product should accomplish it.

Another, subtle, part of the solution is to avoid a strong reference point by intentionally separating behavioral design from the process of designing the interface or communication. The goal is to avoid using the behavioral map and intervention as an implicit, unintentional starting point for what it should look like: one that would strongly shape what the final design looks like.

How can you open up the design process and avoid getting a cookie-cutter product? Don't try to push an interface design under the cover of behavioral

science and behavioral interventions. Keep them focused on the what, and not the how. Then, let the design team (or members of the product team who are fleshing out the requirements) put aside the behavioral diagnosis and intervention for a bit, and brainstorm ideas on how the product might look; sketch them out. Prototype them. The design process should be honored and given the space it needs.

People on the team may wear more than one hat—a product manager may also be a behavioralist; it’s the same with the designer. The key is to take off the behavioral hat when seeking creative designs.

A Cautionary Tale: My Exercise Band

A few years ago, a new piece of wearable technology hit the market. The product combined exercise and sleep tracking with a band you wear at night and one you wear during the day. It sounded great, and I preordered mine—I received it right before Christmas. The product turned out to be an excellent example of what happens when a company makes something that is strongly focused on behavior change but forgets to build a good product. (I won’t mention their name, since the company is continuing to revise and improve its products, and it isn’t the only company to build an exercise band that had problems in the early stages!)

My wife and I made the product into a twofer gift: I would use the exercise band and she would use the sleep tracker. So, on Christmas day, I installed the app and started trying out the exercise band.

The product did a lot of things right from a behavioral perspective. It automatically tracked sleep and exercise, which are difficult to do by hand. It had a simple, clean user interface. It helped me set reasonable goals to start with and then provided constant feedback and nice rewards (little icons on the screen) as progress was made.

The next day I went into the office. After a long day of sitting on my duff, I checked how much exercise I’d had. I saw the screen shown in **Figure 12-1**.



Figure 12-1. My daily exercise—while sitting at the computer

Believe it or not, I didn't walk 38 miles that day while sitting at my desk. I wish I had. Disappointment number one: clearly a bug in the tracking system.

As I got ready to go home, I put on my jacket. The wristband came off. The small magnetic clasp that held it together wasn't strong enough. It accidentally fell off multiple times over the following days; I'm lucky I didn't lose it. Disappointment number two: industrial design problem.

The next day, I forgot it on the counter of my desk. Shortly after lunch, I saw the screen shown in **Figure 12-2**.

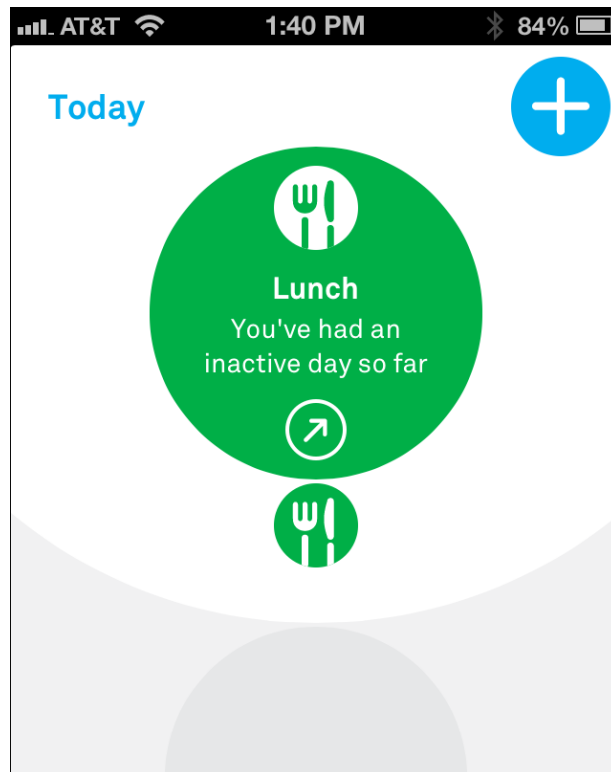


Figure 12-2. A little judgmental, aren't we?

Not the best experience. If the product had been absolutely correct and I had been inactive, maybe I would have taken the message better. But from its lack of knowledge about me, it inferred that zero data meant a problem in my behavior rather than in its knowledge. A good interaction designer should have caught this.

Ah well. I returned it quickly. I'm still excited about the concept and look forward to new wearable technology as it comes out. But for now, this is a cautionary tale about what happens when you focus exclusively on behavior change and not enough on building a good product first. Remember that behaviorally effective products must *also* be interesting and usable.

Build in Behavioral Metrics from Day One

While applied behavioral science doesn't specify how a product should be built, it does very strongly indicate one of the elements that must be included: metrics of success. These are not an afterthought, something to be added after the product is built. I've tried that route across many products

and marketing campaigns, and I know many others have as well. It'll really difficult to retrofit metrics into a product or campaign that's already launched—it's difficult technically and it's difficult psychologically, since the team has already moved on to thinking about other things. Instead, behavioral metrics are part of the product itself, not an add-on for later.

To do this well, you need to make sure those metrics are well-defined and that they're easy to gather in the product.

What You Should Already Have

The first step in measuring the impact of your product is to be absolutely clear on the impact you care about (i.e., the intended outcome of the product). That should have been established in [Chapter 6](#) (and refined in [Chapter 7](#)). In particular, you should have:

- A clearly defined, tangible, and measurable outcome that you seek, with a metric (way to measure it)
- A clearly defined, tangible, and measurable action that drives that outcome, with a metric
- A threshold for each metric that defines success and failure

If you don't have this in hand, please go back. If you don't know how your product or campaign is going to be judged then, by definition, *it's not going to succeed*.

Implementing Behavioral Tracking

You know what you want to measure. Now, you need to measure it by instrumenting the product or otherwise gathering data about your users' behavior and the target outcome. How you gather this data depends on the type of product and behavior change you're working with, whether the target behavior is within, or outside, of the product.

Measuring behaviors and outcomes within the product

If your behaviors and outcomes are part of the product itself, you're in luck. There are tools to help you gather the data. For example, let's say your application aggregates user contacts and helps users keep in touch with them regularly, like **Contactually**. The behavior change problem entails helping your users figure out how to best organize their contacts within the app.

You can code your product to automatically record the actions that users take (organizing contacts) to see if they are successful. In that case, you can code your product to automatically record the action and outcome or push events out to a third-party platform like **Kissmetrics** or **Mixpanel** (Contactually has used them, for example). That's the ideal. When your product is online, you can even gather the data in real time and see what's going on immediately.

Measuring behaviors and outcomes outside of the product

It can be much more challenging if the behavior change problem is outside of the product. First, look for ways to pull in existing real-world data. At HelloWallet, one of our primary goals was to help people save money for the future. But they couldn't actually do that within our application—they moved money into their savings accounts through their bank. Early on in our product development, we realized that we needed to ask our users for read-only access to their bank account information. With the bank account information, we could provide them with better guidance, and, very importantly, we could tell if our guidance was actually working or not.

You'll need to be creative and search for datasets you can pull in. Oracle's Opower, described at the beginning of **Chapter 13**, is a piece of paper—a physical mailer sent to utility customers. There's no way to reliably measure people's real-world behavior with that mailer. But they have built relationships with the utility companies to access utility records on how much energy people actually use. And, with that data, they can reliably tell what impact their mailers are having on behavior.

Your company may need to *consider adding functionality to the app to make real-world measurement possible*. Let's say you have an app that

helps people eat healthier. It provides meal plans for easy and healthy home cooking, so users don't need to eat out as much. That's great, but how do you know if the product is successful? Creating a meal plan isn't enough. You need to know if people are actually acting on that advice. One way to measure behavior outside of the product (actual use of the meal plan) would be to add a feature to link to the person's grocery store loyalty card. The grocery store knows what the user is buying and has a financial incentive to have people buy more there, instead of eating out. Users can be rewarded for following the meal plan and get greater insight into what they are eating.¹ There's a benefit for the user, for the grocery store, and for you—since you'll be able to measure impact.

Sometimes, however, there simply isn't a dataset you can draw upon, or the dataset is too imperfect or infrequent to use. For example, let's say your application encourages people to vote. The act of voting is outside of the product, and it takes months to get official data on whether someone voted or not.

In such cases, where you really don't have a way to regularly gather real-world data, there's a three-part strategy for benchmarking your product's impact:

1. Benchmark the impact your product has on an intermediate user behavior that you can measure regularly, even though it isn't the final real-world outcome you really care about.
2. Determine how to accurately measure the real-world outcome *at least once*.
3. Build a bridge between the intermediate user behavior you measure regularly and the real-world outcome you care about.

The bridge is basically a *second* benchmark—connecting the regularly measured behavior (usually in the app) and the irregularly measured real-world outcome. We'll discuss this option in more detail in [Chapter 14](#).

Implementing A/B Testing and Experiments

In the next chapter, on Determining impact, we're going to talk in detail about experiments and other ways to determine the impact of your product based on the metrics you've just implemented. Here's the short version: experiments are your best route to determining whether you've had the impact you seek, when they are possible. So just like the metrics themselves, you should plan to implement the *ability* to run experiments as part of the product or communication itself. Otherwise, you'll have a hard time retrofitting them.

If you're not familiar with them, A/B tests take a randomly selected group of users and show them one version of the product (version A), and show another randomly selected group another version (version B). The mechanics of A/B testing and multivariate testing are discussed in the next chapter; let's look now at what you'll need to implement in the product, though, to make that possible.

Tools for Behavioral Tracking and Experiments

To measure your product's impact, you'll need to look beyond basic tools that track page views and conversions. Often the impact you're looking for isn't just a simple event in the application, like a page view. For example, if your product helps users form the habit of updating their budget each month, measuring the habit means more than the pages they've seen. Second, you'll need access to raw, per-person data for statistical modeling. Third, in order to assess changes in impact, you'll probably need to run A/B tests.

Most behavioral tracking tools for websites and applications provide aggregate information by default: what segments of people are doing or how many people are following a particular path through your app. Google Analytics, for example, is in this vein; it's really useful to figuring out what's going on in your app, but it doesn't tell you what individual people are doing and thus doesn't allow you to do more fine-grained analyses and experimental tests.

Getting individual-level data—what each person is doing in the system—requires more horsepower; the basic version of Google Analytics doesn't provide that. As of the publication of this book, the Google Analytics 360 package *does* provide individual-level data, as do Heap, Adobe Analytics, and other packages like it via API calls or export files. Heap is currently my personal favorite because it does some wonderful magic behind the scenes to track most activities in a Heap-enabled program, even before you specify what you're looking for (in GA and other tools, you have to specify what you're looking for before it starts tracking). Once you request that activity, it provides historical data all the way back to when Heap started in the app—which is immensely useful.

There's an open source version of Google Analytics that does what GA does (though a few versions behind) and provides that individual-level data: **Matamo**. It can be a bit clunky, but it gets the job done if you know how to analyze the raw database records.

When working with email, you'll get behavioral tracking out of the box for almost all modern email packages—as long as you use a commercial tool and don't send through your Outlook or Gmail account, basically. It's the same thing for mass text message (SMS, WhatsApp, etc.) software like **Twilio**, **Bandwidth**, and **Vonage**.

Tools that support A/B testing or its cousin, multivariate testing, will advertise that fact. There are a variety of tools that can handle the A/B testing for you. As of this writing, tools for app- or web-based testing include Google Optimize 360, Adobe Target, Optimizely, VWO, and Mixpanel. Many support mobile, as do specially targeted ones like **Apptimize** (now part of Airship). In the text message world, A/B testing is also supported **out of the box for big providers**. And again, in mass email, don't even think of using an email package that doesn't support A/B testing (if you can find one!).

The economics of third-party trackers and experiment tools have become so favorable that it's unlikely you'll need to build your own. But, if needed, companies can also readily implement their own per-person tracking by

pushing the events that occur within the system out to a database for later analysis. It's something I had to do many times over the years, but I now rely on third-party tools.

Putting It into Practice

Applied behavioral science, and the process of designing for behavior change, doesn't have a lot to say about the development process or software architecture you should use to implement your product. However, there are two areas that are essential, which we've covered here: separating behavioral design from interface design and ensuring that metrics are a core part of the product itself, and not an afterthought.

Here's what you need to do:

- The behavioral science team (or role) should take a breather between the planning process (covered in Chapters 6 and 11) and implementing the product itself, which we cover here. Chapters 6–11 provide a functional spec—how to help users overcome behavioral obstacles. They can't give the team a visual design for how the product or communication or look or specify exactly how it should interact with users. That's a designer's role. Collaboration is great, but don't let behavioral design crowd out visual design
- Make metrics part of the *first* version of the product or communication. Both behavioral tracking and experiments (A/B tests) should part of any MVP. It's a royal pain to retrofit later.
- Thankfully, there are many third-party packages you can use to cover these needs—you rarely need to build it yourself anymore.

How you'll know there's trouble:

- The behavioral design is being used as an interface design (or communications design)

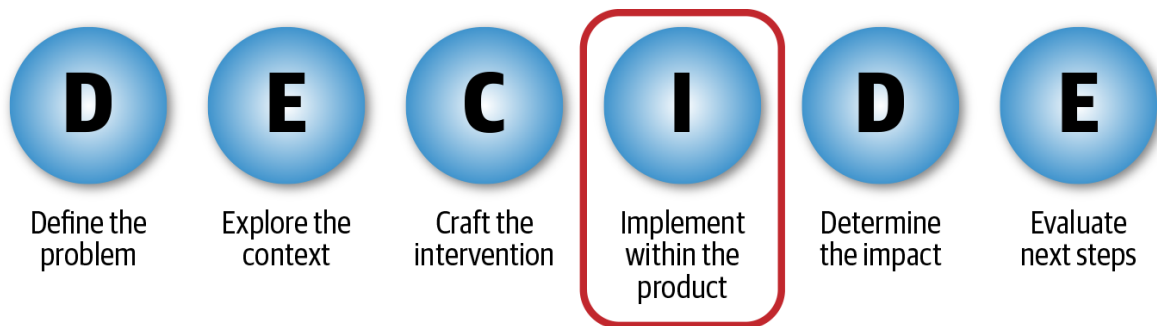
- You're building the product, and tracking outcomes isn't part of version 1.

Deliverables:

- The product, with behavioral metrics built in!

As with all of the “Putting It into Practice” worksheets, we'll use the example of an app meant to help people get exercise through running. A completed worksheet is here, and the blank form is available in the [accompanying workbook](#). Some of the fields can be copied directly from the project brief, or the project brief can be attached for reference.

Worksheet: Ethical Checklist



If your organization has an IRB or a process modeled on the IRB, then you should use the templates and processes from that review board. If you don't have an IRB process, here is a worksheet to get you started, modeled on tools we've starting using on my team. Some fields can be copied directly from the project brief or the project brief can be attached for reference.

Practitioner: Steve Wendel

Project: Flash app

Date: 1 Jan 2020

Description and Purpose

1. Please describe the product, feature, or communication (henceforth: “product”) to be developed: Flash app is a mobile app designed to encourage exercise. It is sold to employers and distributed to employees as a benefits program.
2. What specific behavior (action) does the project seek to change, and does it support or hinder it? The primary behavioral goal is to encourage users to go to the gym and

exercise.

3. What behavioral intervention(s) does the product use to support that change? The app uses multiple interventions at different stages of interaction with the product. First, to encourage downloading the app, it uses testimonials from who uses it (social proof). Second, within the app it uses a social competition to encourage participation. It also employs small wins to encourage people to continue using the app and progressing toward their goals over time.
4. Who is the target population (actor)? White-collar employees of existing clients who don't exercise regularly.
5. How, if at all, will this benefit that population (user outcome)? The target outcome is to help users decrease common sources of pain: back pain, neck pain, joint pain.
6. In what ways might this intervention cause notable harm to the individual, in the short or long term (e.g., products that seek to addict their users to its use)? There don't appear to be serious downsides to these interventions or the product, beyond encouraging individuals who should not exercise (due to heart conditions, etc.) to exercise: we leave them free to opt out without penalty from their employer.
7. How will this benefit your organization or team? This will increase revenue from our corporate wellness program clients.
8. What financial or personal interest do you have in this project succeeding? My career is at stake—if this doesn't succeed, I may lose my job or miss out on opportunities for promotion.

Transparency and Freedom of Choice

1. Does the target population want to accomplish this outcome? Do they want to change this behavior? Yes, in our qualitative research and in surveys of the population, many employees want to accomplish the outcome and see the specific action as an appropriate way to accomplish it.
2. Does the target population know that you are seeking to change their behavior? And, if not, will they be upset when they become aware of it? Yes, the app is clearly described as seeking this goal. Users should already be aware.
3. Is the user defaulted-out, defaulted-in, or is it a condition of use for the product to interact with these interventions? Can the user opt out in a straightforward and transparent manner? Defaulted out. After choosing to participate, the user can opt out at any time.
4. What steps will be taken to minimize the possibility of coercion? The main risk of coercion is that employers (our clients) require use of this program, or penalize non-participants with higher medical premiums. We have a clause in our contracts with the employers barring such penalties and requirements.

Data Handling and Privacy

If data privacy issues aren't addressed elsewhere in the company's standard product development process, this is a good place to cover those issues. If they are covered, skip these.

1. What personal information does this product gather? User email address, first name, personal exercise goals, and exercise history. Location history is also gathered as the person uses the app.
2. How is that data handled to ensure privacy of the users? The directly identifiable information, name and email address, are masked with a salted, one-way hash. All information is stored on a secured server, following the company's standard data handling process.

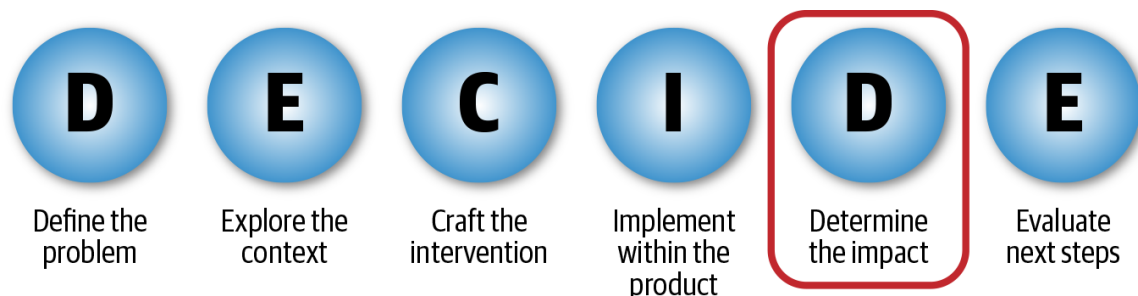
Final Review

This project has been independently reviewed and approved by: FlashCorp Ethical Standards Board

Date: 3/3/2020

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- 1 Most of us forget or don't even think about what we're eating. See Riet et al. (2011) for a summary.

Chapter 13. Determining Impact with A/B Tests and Experiments



You should take the approach that you're wrong. Your goal is to be less wrong.

—Elon Musk

Opower, an energy-efficiency and customer engagement unit owned by Oracle Utilities, has run some of the largest experiments in the world about how products can change behavior. Millions of people have participated in their studies simply by opening a letter from their utility company or fiddling with their thermostat.

Opower is best known for delivering monthly reports to utility customers that show them how their energy usage stacks up against their (anonymous) neighbors. It's a well-studied technique in social psychology called peer comparisons, which we discussed a bit in [Chapter 9](#). [Figure 13-1](#) shows an example of one of its comparisons.

A host of government, private, and academic publications have shown that Opower's simple comparisons help consumers cut their energy bills by roughly 2% on average.¹ That may seem small, but it adds up to over 2.6

terawatt-hours of electricity—enough to power 300,000 homes for a year, or roughly \$300 million in consumer savings on energy bills.²

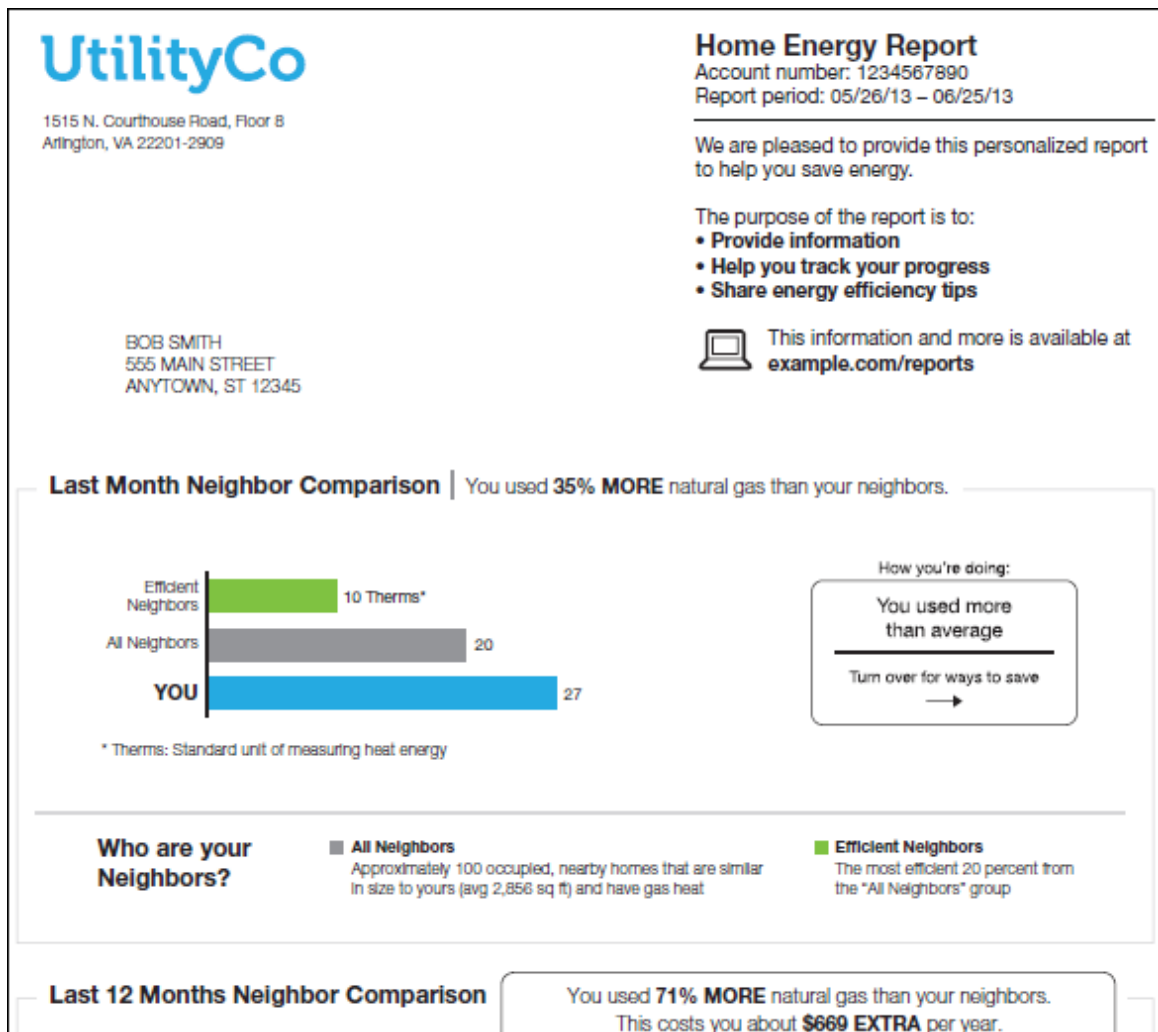


Figure 13-1. An Opower energy report, comparing the reader's home heating usage to that of their neighbors

Opower has repeatedly run experiments to measure their impact and test ways to improve it further. Rigorous measurement and testing have been a key factor in their success.

In designing for behavior change, that unambiguous signal is vital. Our good intentions—and even our thoughtful statistical analyses—aren't enough. Human behavior is so complex and context-dependent that it's all too easy to fool ourselves into thinking that our products work as planned. And there should be no confusion or doubt:

A randomized control trial is simply the most effective and rigorous tool out there for determining the impact of your product or communication on behavior. It provides the clearest and most unambiguous signal of impact.

Randomized control trials aren't the domain only of world-class researchers, however. You've probably run one type of randomized control trial yourself: the humble A/B test. You don't need a big team to do A/B tests; often, you don't need anyone else but yourself to measure impact in this rigorous and highly effective way. And so, this chapter provides a broader understanding of randomized control trials, including A/B tests, and how to make sure that your tests are effective, rapid, and clear.

ANYONE CAN MEASURE IMPACT

When you read the title of this chapter, do you imagine arcane symbols and inscrutable formulas? That's not what you'll find. Instead, you'll find commonsense explanations of how you can measure your product's impact.

For software products, there are numerous powerful and user-friendly tools that handle the underlying math and statistics for you. For most impact experiments, that's all you really need. You usually don't need an econometrician to understand if your product is working or not and how to improve it.

Some techniques are more advanced, though; I'll mention that up front and explain what is going on in nontechnical terms. If you don't have a statistical background and you decide that you need those techniques, that's a point at which you need to get some expert assistance. For readers with a statistical background, those sections will quickly tell you which tool to pull from your toolbox so you can get to work.

The How and Why of Randomized Control Trials

When you want to know whether a product or communication actually does what it's supposed to do, randomized control trials are the most trusted and rigorous tool. In fact, they are the gold standard in science; the same tool is used to measure how effective medicines are at curing a disease, for example.

Here's how they usually work, using the example of an exercise app:³

1. *Write out what you're trying to do.* Start by writing out three things, which we've already covered in the previous chapters:
 - a. The *outcome* you care about (e.g., decreased back, neck and other joint pain—measured by a decrease in doctors and physical therapy visits)
 - b. The *intervention* that may cause that outcome (e.g., going to the gym)
 - c. The *target audience* (e.g., white collar, sedentary male workers, aged 35 to 45)
2. *Randomly assign the audience.* Randomly sample enough people from the target audience, and randomly assign them to two groups, which we'll call a *control* group and a *treatment* group; for example, five hundred men assigned to the control group and five hundred to the treatment.
3. *Deploy the intervention.* Offer the intervention to the treatment group and not to the control group. For example, give the treatment group the app and give the control group basic information without the app's features. (We discuss how to make this a *double-blind* study later in this chapter.)
4. *Measure what happens.* After enough time, measure the outcomes for each group. For example, measure the number of physical

therapy visits for each person in the treatment and control groups.

5. *Calculate the impact.* Use the following simple formula to measure the difference between the two groups' outcomes:

$$\text{Impact of intervention} = \text{Average outcome for the treatment group} - \text{Average outcome for the control group}$$

6. *Make a call.* If the impact is large enough, you can conclude that your intervention is practically and statistically meaningful. For example, the app decreases physical therapy visits by 10%, and that means a real benefit for your users and your company's business.

That's it. There are details we'll have to examine, especially in terms of what *enough* means in each case, but getting the concept down is important. Assuming the right conditions are met, experiments are simple, powerful, and highly informative.

Different types of experiments go by many names, such as randomized control trials (RCTs), split tests, and A/B tests. In the software world, the most common term is A/B test, and in this chapter we'll focus on the particular opportunities (and constraints) of running digital experiments in software. Another popular technique that we'll look in more detail at later, multiarmed bandits, is another type of digital experiment.⁴ To reinforce the fact that mathematically they are the same, and for simplicity, we'll use the more general term *experiments* here.

Why Experiments Are (Almost) Better Than Sliced Bread

Let's say we wanted to know whether our new exercise routine helps people lose weight. A common, but flawed, approach to answering that question would be this: compare people who use the exercise routine versus those who don't, and see whether those in the first group lose more weight than the second. The problem is that if we discover that people in the first group lose more weight, do we really know why that happens? It might be

because of the exercise routine. Or, it might be because people in the group are simply more motivated to lose weight. Or, people in the first group might have faster metabolisms.

Instead of comparing the people who use the exercise routine versus those who don't, let's say we ran an experiment in which people were randomly assigned to each group; that changes everything. The experiment would remove each of these potential explanations and any other that you might imagine. We could then say with strong empirical support that people in the first group lost weight *because* of the exercise routine.

The way that experiments accomplish this magic is through randomization. By randomly assigning a set of people into two groups, those two groups are statistically identical. People's metabolisms in each group are statistically the same (the number of people with fast or slow metabolism, for example, is statistically the same). People's levels of motivation in the two groups are statistically the same. And so is their distribution of ages, incomes, gender, political preferences, and everything else that you might imagine about the two groups. Most importantly, their likelihood of losing weight in the future is also statistically identical.

When we take these two statistically identical groups and treat one of them differently by assigning one group to the exercise routine and the other not—any resulting difference in the degree to which they lose weight must be due to that intervention and nothing else. A properly designed experiment allows us to say, within statistical limits, that an intervention actually *causes* an outcome. That's why experiments are awesome.

Experimental Design in Detail

While the concept of experiments is straightforward and easy, the details really do matter. There are a few rules we need to carefully follow to ensure that an experiment is designed and executed correctly. The first question that new experimenters generally ask is: how many people are “enough”?⁵

How Many People Are “Enough”?

When you’re measuring the impact of your product or communication, the quality of that measurement depends on how many people are in your study. Here’s a simple example to show why. Let’s say your product is an app that encourages people to eat less ice cream. You try it with one person. Dang it! They ignore it and keep binging on ice cream! But you keep going. The second person cuts back a lot. The third person also does, but not quite as much, and so on. That’s actually a pretty normal range of variation across people.

Overall, the product is a success. It helps people cut their (unwanted) ice cream habit by 50%. If you’d only looked at the first person, though, you’d have thought it was a failure. If you only looked at the first two people, you’d be really confused. If you looked at the first four, then the picture would be clearer—overall, it seems to help, but there are exceptions. Adding people makes the picture clearer and clearer. At some point, adding people doesn’t matter; you already have a very good idea of what the product’s impact is. The number of people you need to have a (statistically) clear picture is known as the minimum sample size.

You determine the minimum sample size using a function called a *power calculation*. You can use an online power analysis calculator or any statistical package like R, Python, G*Power, or Stata. If you don’t know where to start, try **G*Power**,⁶ which is free and extremely powerful, but requires some documentation reading. If the product’s outcome can be expressed as a number (like weight, height, or the number of cigarettes smoked in the month), then you’ll use a calculator that can handle the average value for the population. If the product’s outcome can be only one of two things, like the patient is either alive or dead, then you’ll use a calculator that handles percentages, proportions, or rates. In R, you can use the functions `power.t.test()` when working with average values and `power.prop.test()` when dealing with percentages. In Python, you’d use the slightly more complicated **StatsModel package**.⁷

Here are the specific numbers you’ll need:

- The average outcome for people who don't have the product (or who don't have the new feature or communication you want to test). That's the *baseline*. For example, the average weight of the target audience for a weight-loss app is 245 pounds.
- The variance in outcomes (difference from the average) among people without the product. That's the *noise*. Note that for outcomes that are rates (click rate, etc.), this is built into the calculation; you don't need a separate measurement of noise. For example, the standard deviation of our target audience's weight is 35 pounds.
- The *impact* you expect from the intervention on the outcome. Be conservative: ideally the product will have a much *larger* impact, but here you want to make a reasonable lower estimate of impact. For example, we expect a decrease of 10 pounds.

The calculator will also ask for two parameters that indicate how sensitive the test should be:

- Confidence level or, equivalently, an alpha error level (alpha error level = $1 - \text{the confidence level}$). Usually the default confidence level is 95% (alpha of 5%). That means you can expect to incorrectly say there *is* an impact when really there *isn't* one, 5% of the time. *That's a false positive.*
- Statistical power, or, equivalently, a beta error level (beta error level = $1 - \text{statistical power}$). A good default statistical power level is 90% (beta of 10%); that means you can expect to incorrectly say there *isn't* an impact when really there *is* one 10% of the time. *That's a false negative.*

Using these default values for our exercise app, we'd need 258 people per group—or 516 people total.⁸

These default parameter values are built around the assumption that you want to be very careful to avoid claiming that there's an impact when there

really isn't (a false positive). In our exercise app example, that means it's especially important for us to avoid claiming that the app is successful at decreasing people's weight when it isn't. It's important, but not as important, to avoid missing an impact that really is there (a false negative).

Each experiment is different, but those are pretty good assumptions to use when you're testing whether your product works or not. It will be very embarrassing (and costly for the engineering team) to claim you've found an answer, pursue it, and then find out it was a mirage. So, generally, keep these defaults.

You may also be confronted with a question of whether you want a *one-sided* or *two-sided* test. In a two-sided test, you're looking to see whether the product causes any change, positive or negative, in outcomes. In a one-sided test, you're assuming that if the product has an effect, it will be positive (or maybe negative!). If the effect is actually in the opposite direction, the test won't work correctly. There's always debate around this issue, but I prefer to take the more open-minded route and stick to two-sided tests that understand that the product might make things better or worse.

Once you enter these values into the calculator, it will tell you how many people you need in the treatment and control groups. If you have more, that's great. If you have less, look for more people! If you have just the number of people it shows but have the *ability* to collect more, please do so; there's a chance that some participants will have invalid data, leave the process early, and so on.

How Long of a Wait Is “Enough”?

Another common question concerns how long to run the experiment. If you have a fixed set of people, such as an existing email list, the answer is straightforward. Run the test for as long as you think the intervention needs to have its effect (plus a little bit longer, just to be safe).

If you don't have a fixed set of people and instead have a stream of people coming to a website or product over time, things become a bit trickier. The

common temptation is to run the experiment until it looks like there's a "clear winner."

Using the ice cream app described earlier, we can quickly see why this is problematic. Let's say your app has a steady stream of people signing up for it and using it. The first few people break their unwanted ice cream habit very nicely—far better than the control group. However, as more people sign up for the app that strong initial result seems to fade. After a hundred users, it looks like it's not much more effective than the control group after all. If we had stopped the test after the first few people, we would have gotten a false signal about the app's impact. In fact, at *any given moment*, either the treatment or control will often look like it's doing better than the other. That doesn't mean that it has a meaningful impact; the difference may be due simply to noise and will fade (or even reverse) over time. If you constantly check the results of the experiment, waiting for something that looks promising, you'll increase the likelihood that you'll say there's a difference between the two groups when there actually isn't one.⁹

Many well-meaning product managers have been stung by the rush to call an experiment "finished" simply because the results look convincing. The way to avoid this problem is threefold:

1. Run a power calculation, as described earlier, to determine how many people would be required to detect the effect we expect.
2. Determine how long it will take for that number of people to interact with the product, given the stream of people over time. When you expect to reach the target number of people, stop adding more people.¹⁰ Regardless of what we see in the meantime, and how tempted we are to stop the test early, the end date is when the test completes. For example, if we need one thousand people for a website test, and we get two hundred people a day, we keep adding people for five days.
3. After the test is complete, run a significance test—a procedure that is related to the power calculation—to determine whether the

results are statistically significant. We'll talk about statistical significance a little bit later.

Using Business Importance to Determine “Enough”

Thus far, we've talked about the standard academic version of an experiment: take a reasonable estimate of how much of an impact you expect, and figure out how many people you needed based on that impact. In a business environment, there's another way to approach this problem. Instead of considering what impact might happen, consider instead what impact is *important*. Specifically, the experimenter would follow this process:

1. Determine the minimum meaningful effect (MME). What is the smallest change in the outcome that you would consider meaningful or important to the company or its users? There is no universal rule for determining this—instead, what matters are a company's particular business priorities and opportunity cost.¹¹
2. Calculate the sample size based on that MME. In other words, do the power calculation described before, but use the MME in the equation instead of the “expected impact.”
3. If there are plentiful users available all at once, or there's a continual stream of people over time from which to draw from, then the number of people in each version (or “arm”) of the test should be *at least* as many people as calculated in step 2.
4. If, instead, there is a fixed and limited number of people available for an experiment, then:
 - a. Determine the largest viable effect (LVE). What is the *largest* change in the outcome that you'd expect the intervention to have?
 - b. Calculate the sample size based on the LVE. In other words, run a power calculation using the LVE as the

expected impact from the intervention.

- c. Use the illustration in **Figure 13-2** to determine whether to run the experiment at all, given the available sample size available for the test.

Sample Size	Decision	Next Steps
 Less than LVE	 Stop! Don't bother running the test, it can't tell you anything.	- Move on to other tests
 More than LVE, less than MME	 Think carefully! The test might show an impact, but if it doesn't, you can't conclude that the approach failed.	- Optional: Decrease the number of variations - Take the risk and go for it - Get more people - Look for other, clearer tests
 More than MME	 Stop! You can (relatively) conclusively see if your approach works or doesn't work.	- Go for it, run the test! - Optional: Decrease the number of people, down to the MME size - Optional: Add variations until each one is the MME size

Figure 13-2. Before you run a test, use this table to figure out whether the results will be meaningful

This approach provides a tremendous benefit to companies with limited time and resources (i.e., almost all companies!). By focusing on the MME and using that to determine the sample size of the test, then no matter the outcome of the test, the company gets a clear signal on what to do next. Namely:

- If the test comes back showing a statistically significant positive impact: great, full steam ahead.
- If the test comes back showing a statistically significant negative impact: don't launch the new version of the product or communication. You've just saved yourself lots of pain down the line.

- If the test comes back showing no statistically significant impact at all: wonderful. It means that there's no meaningful impact to be found and no further research is required. If you haven't finished building it and deploying it to users, then stop. There's no business value there. If you have, it really doesn't matter if you launch it; it will have no meaningful business impact.

Thus, all experiments run in this context are meaningful and useful for the company. They either tell the company that yes, a new product or communication is impactful, or that it should be abandoned as irrelevant. There is no wiggle room of “well, maybe another test with more people would find an impact.” Any such hidden impact is irrelevant to the business.

From a technical perspective, what's new here is that the minimum meaningful effect allows the company to determine a clear plan of action based on null results. Why doesn't everyone do that? It's because experiments are still primarily used and thought about in an academic context, where the incentives are different. For academics, the goal is to publish a paper, and that (usually) means getting statistically significant results. Thus, academic researchers have a tendency to look for very large populations that can show a statistical significance even if the expected impact is laughably small from a practical perspective. Statistical significance in no way implies practical significance. And businesses care about practical significance.

Points to Remember in Designing an Experiment

In addition to these basics of experimental design, there are a few other rules to keep in mind:

Random selection isn't always easy

When people are drawn from the population, they have to be randomly selected, and not selected based on a convenient (but not truly random) criteria. For example, in testing the impact of a new product, you can't

rely on volunteers to be part of the experiment and expect them to accurately represent the views and outcomes of the overall population.

You need random assignment as well

When assigning people into groups, the assignment must be truly random. For example, if the experimenter has two groups of people and doesn't know where the two lists came from, the two groups can't be used. Or if a group of people is divided based on something that seems random like the first letter of their last name, the two resulting groups are truly random and identical (last names, and their place in the alphabet, are strongly tied to ethnicity, for example). If you have an existing list of people and it *looks* random, it almost never actually is—there's some ordering there, but you can't know how it influences the results. Use a random number generator, and generate a new number for each person. As a sanity check, you can verify that the random assignment process was done correctly by checking whether the two groups have similar averages on things that the product couldn't impact—like age or gender. If so, great. If not, it wasn't a true random assignment.

Check that the groups are drawn from the same population

For example, a design in which a randomly selected group of men is compared to randomly selected group of women is not a true RCT; there's an obvious difference between the two groups that has nothing to do with the intervention.

Make sure you're only varying one thing

Only the product (or communication) should vary across the two groups. Carefully review the experience that each group will face to ensure that they will see and interact with the exact same things, except for the desired change and product or communication. “One thing” doesn’t mean that the change in the product has to be simple; it could be an entirely new feature or even an entirely new product. “One thing” means instead there is one conceptual entity that you’re changing and analyzing (which, again, can range from the text on a button to the entire functionality of the product).¹² Also, you *can* test multiple versions of the product at once with an “A/B/C test” or a multivariate test, as we’ll discuss later.

This section has provided a short introduction to designing an experiment. In many cases, your email or website targeting package (Eloqua, Marketo, Optimizely, Google Analytics 360, Adobe Target, etc.) will handle the mechanics of the experiment itself for you if you need an experiment and know what to watch out for when using the software. That’s been the goal here: to arm you with the right questions to ask. However, if you’d like to learn more about the design of experiments, two good resources are Shadish, Cook, and Campbell (2001) and List, Sadoff, and Wagner (2010).

Analyzing the Results of Experiments

After you’ve designed a great experiment and deployed it in the field, you’ll want to look at the results to see whether the intervention you developed in your product or communication actually had the effect you expected. As already mentioned, the core idea is quite simple: as long as the effect is large enough, the intervention’s impact is just the difference in the average outcomes for the two groups. “Enough” has a very specific meaning—let’s take a look at how you ensure your analysis is solid.

Is the Effect Large “Enough”? Determining Statistical Significance

One of the most common mistakes with digital experiments is that people look to see if there is a difference between the two groups and don't look at whether that difference actually means anything. They can get really excited about unreliable results, and, most damagingly, they take the wrong lesson from the experiment and waste time building the wrong thing.

Always run a test of statistical significance. That's how to save yourself from that fate. A statistical significance test is the after-test version of a power-calculation (which is what we run before the test); it tells us, if, in fact, we have a large enough effect to comment on.

Just as with the power calculation, there are two main versions depending on how you measure the outcome. If the outcome is only two outcomes per participant—such as people log into the application or not—then you'll run a test on the percentage of people in each group that have the outcome (i.e., the rate among each group). If the outcome is a real number (like the amount someone spends), then you'll determine statistical significance based on average outcomes.

In R, you can use the `prop.test()` function for outcomes with only two options (i.e., each group has a single rate or “proportion”), and the `t.test()` function, or a regression, for real numbers. In Python, you'd use the StatsModel package for both. If the outcome is ordinal (the possible values are in order, but the spacing between them may be irregular and they aren't directly comparable), things are a bit trickier and a skilled stats person should be consulted.

Other Considerations

In addition to determining statistical significance, here are a few other rules that apply to experiments:

Go double-blind when you can

In a double-blind experiment, neither the participant nor the experimenter knows who is in the treatment or control group (in the A or B arm). If the experimenter knows, it can influence their interpretation of the data, or the experimenter's behavior can (unintentionally or intentionally) affect the behavior of the participant.¹³ If the participant knows, they're likely to behave differently. Thankfully, many A/B tests are double-blind accidentally because the software used to execute them often doesn't make it easy for experimenters to identify and directly interact with participants. Participants receive a seamless experience as part of the product or website and wouldn't know what is being tested.

Measure the same way

If the outcome is measured differently across the two groups, then apparent changes may be because of the product or because of how the outcomes were measured. To fix this, find a way to measure the outcome outside of the product consistently (for both groups) or offer a trimmed-down product (or feature) for the control group that just tracks the relevant outcome and doesn't do anything else that should drive.

Compare results for everyone

Make sure that when you compare the two groups, you compare all of the people in each group. In the treatment group, for example, there will be some people who are offered the product but don't actually use it. Count the non-users too; otherwise, the results mix up the effect of the product with the effect of who chooses to use it or not.

Generalize outcomes to the same population

Researchers and product teams shouldn't assume that the results apply to everyone after they've done a test. For example, after running a test with college students in the US, you shouldn't use those results to comment on the behavior of European pensioners. The results *may* in fact generalize well to that population—but it takes more than a single experiment to determine that. It takes a series of replications of an experiment that change only the underlying population, to determine the *structure of portability* of the results (for whom and under what conditions the results apply to a different population).

Types of Experiments

Experiments come in many flavors in terms of how they are designed and executed and in terms of the particular problem or purpose they are meant to address. Let's look at some of the options.

Other Types of Experiments

Thus far, we discussed two of the most common types of experiments: one in which the second group receives nothing (aka a null control), and one in which the second group receives a different version of the product or communication (aka an A/B test). There are a few other designs one often sees in digital experiments. Here's a more comprehensive list:

Simultaneous impact

In these experiments, also known as an A/Null tests, one randomly selected group receives a new feature or product, and the other group

does not receive it. The result of the experiment tells you the *absolute impact* of that new feature or product.

Simultaneous comparison

Also known as an A/B test, the randomly selected groups receive two different versions of a product or feature. The result of the experiment tells you what the *difference* in outcomes is between those versions. It makes it easy to compare their impacts.

Multiarm comparison

A simple extension of an A/B test looks at more than two versions at the same time. Each version gets its own randomly assigned and selected group of participants, all from the same pool of people. This is called an A/B/C (/D etc.) test, for obvious reasons.

Staggered rollout

In a staggered rollout, each group eventually receives the intervention. However, one randomly selected group receives the intervention *earlier* than the other. At the moment when the other group receives the intervention, the experimenter compares the outcome variable for the two groups. The only difference between those two groups should be the prior exposure to the intervention. This type of experiment is nice in that it's basically a *pilot study* (something that product companies and clients are quite used to). By randomly selecting the members of the pilot, it is a statistically valid experiment with the full power to determine causal impact.¹⁴ A clever way to do a staggered rollout is to ask for people to pre-commit to buy or receive the product when it's

released. Then, use a rolling schedule for the release—only make enough units of the product, or give out enough access credentials, to supply to a randomly selected subset of the enrollees. Then, later, supply it to the rest of the people who signed up.

Attention treatment

In this version, both groups have access to the intervention at the *same time*. However, the experimenter has reason to believe that people are unlikely to find and interact with that intervention on their own; for example, when the product has many features and users are unlikely to find a new one on their own. The experimenter draws *attention* to the intervention for one of the randomly selected groups. Any difference in outcomes between the two groups is caused by the difference in awareness of the intervention and subsequently the use of the intervention. Attention treatments are appealing because they mean that no one is denied the intervention—like a new feature within a product. We can still measure the experimental impact of that feature simply by drawing selective attention to it.

Multivariate experiment

A multivariate experiment entails running multiple simultaneous experiments at the same time. The key to multivariate experiment is that each component experiment is independently and randomly assigned. A simple example of a multivariate experiment would be with a landing page; the experimenter tests two different versions of the headline and two different versions of the main image on the page. There are thus

four different versions of the page that people experience: Headline A/Image A, Headline B/Image A, Headline A/Image B, etc.

Multiarmed bandit

In a multiarmed bandit, the percent of people going to each version of the product (or “arm” of the experiment) changes dynamically based on how well the arms are performing. The arm that’s performing the best receives more people. In addition, the arm that the experimenter believes will be more effective is given more traffic to start. The benefit of this approach is that if the experimenter is right, the “better” version receives more traffic, and thus the impact can be identified with a smaller sample.¹⁵ The downside is that if the experimenter guessed wrong, it takes more observations to find out which treatment is actually more impactful. The temptation with multiarmed bandits, especially, is to turn it off once it *looks like* one arm is the winner—even if statistically the results are meaningless noise that’s likely to mislead the team.

As you can see, experimental designs can differ in a variety of dimensions and still be experiments. The rollout of the new intervention can be simultaneous or staggered. It can be compared against a null control or compared against another version of the product or feature. The assignment can be based on a static ratio between the arms or a dynamic one as in the multiarmed bandits. None of these changes violates the core rules, and they can each deliver valid and meaningful experiments.

Experimental Optimization

In many business contexts, the mere idea of running an experiment strikes many product managers (and marketers) as wasteful since at least one group of people receives the “wrong” version of the product or communication. Putting aside the issue that experiments are often the best tool to determine which version is wrong, there’s an important element of truth in that concern: that the primary goal for a digital product (or marketing campaign) in a business context is not to answer a theoretically interesting question, but rather to increase impact over time.

Companies can structure a series of experiments to *optimize* that impact over time. This is especially the case when there are multiple interventions that the company wants to try.

How Experimental Optimization Works

When you do experimental optimization, you deploy interventions in waves over time (a staggered rollout), with different interventions across each wave. The lessons from prior waves are applied to subsequent ones, to optimize the combined package of interventions over time. Step by step, it’s like this:

1. *Write out the changes* you want to test. You probably have a pretty good idea about what you think will improve the impact of the product. Design your best guess as one coherent version of the product or feature: that’s the baseline intervention. Then, draw up a list of the other changes you’re not sure about but probably will help in order of how impactful you think they’ll be.
2. *Set the threshold*. Figure out how much of an *improvement* you need to see in the intervention’s impact for it to be worthwhile to re-apply in the field. This process doesn’t seek to prove out small “nice to have” interventions; it searches for big ones and ignores everything else. Thus, this target impact is usually greater than the minimum meaningful effect we discussed before. We’ll call that impact X.

3. *Chop up the list.* Calculate the number of participants you need to distinguish that impact in an experiment. This threshold will often be much higher than is used in academic studies; thus, it can radically *decrease* the required sample size you need. If you have a predefined list of participants, chop them up. If there's a stream of people who interact with the product over time, then automatically batch up each group of people as they come in. You now have N groups of people, where each of them is enough to measure an impact of X .
4. *Test your initial baseline.* With your first group of people, take your “best guess” version and test it against no change at all—just to make sure what you're starting with isn't worse than nothing. If it has *no* impact, that's OK—you've learned you really need to optimize more. If the best guess makes things worse, start again. You now have $N - 1$ groups. Either way, whichever one wins becomes your new baseline.
5. *Run a comparison test.* Starting with change that you believe will have the next greatest impact, compare it against your current baseline. Run an experiment with one of the remaining groups. *You can actually run multiple comparisons at once*; these multivariate tests can be more efficient, but more complex.
6. *Adapt the intervention.* If the “new” version was better, update the baseline intervention to use that lesson for future waves.
7. *Keep doing 5 and 6 until the population is exhausted.*

To put some numbers around this, imagine that you are doing a very simple optimization with only two versions of an intervention that leads people to take action in some important way, like saving for retirement. One of them leads to 20% of people taking action; with the other only 10% take action. The problem is you don't know which one will work best. If you don't iteratively test, you can expect either 10% or 20% to take action. A big difference, and a big risk. If you do iteratively test, using a quarter of the

population in the first wave, you are nearly certain to get 19% of the population using the system, with almost no risk.

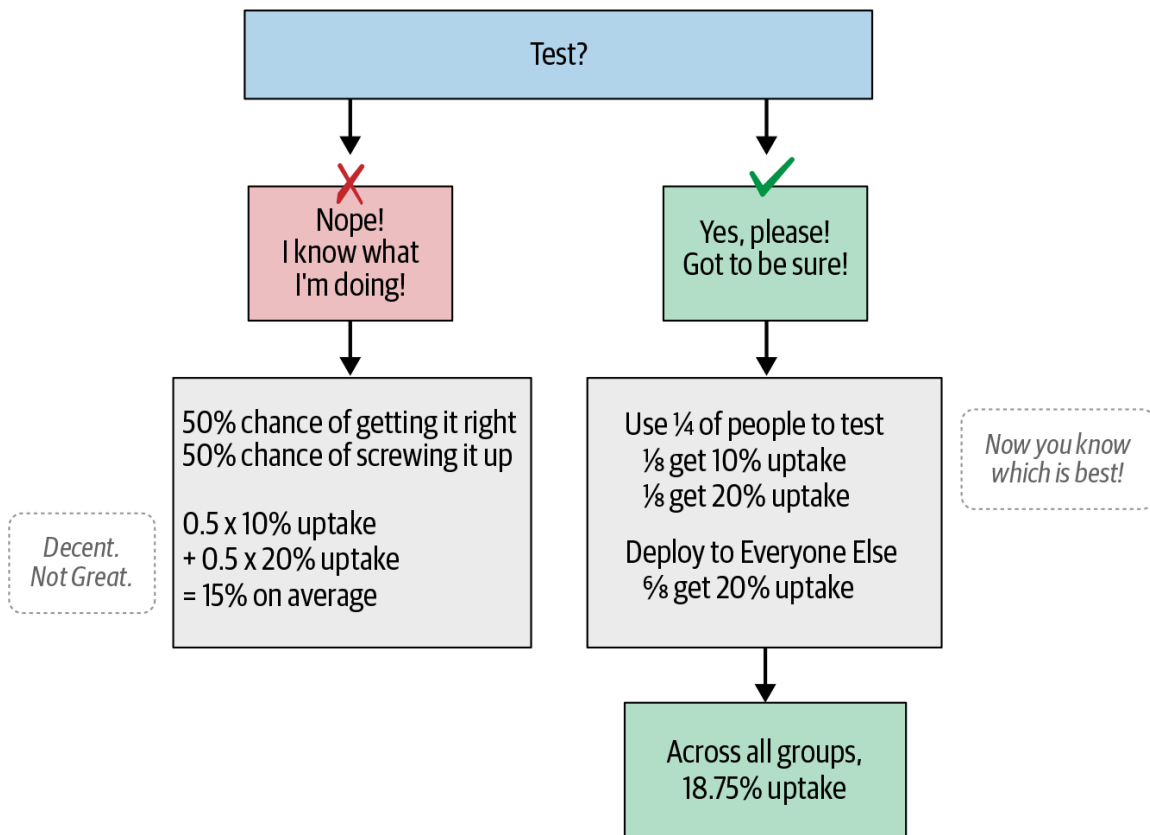


Figure 13-3. Don't take chances with your product—optimize

If you don't test, you have a 50% probability of picking the “good” intervention, with 20% uptake, and a 50% probability of the picking the intervention with 10% uptake. If you do test and you have sufficient sample size to accurately tell which intervention is “good” based on your first test, then in the first wave you have 12.5% of the population receiving the “good” intervention, and 12.5% receiving the “less good” one. In the second wave, all 75% remaining people receive the good intervention. The result is an expected uptake of 18.75%. Good job!

When you have more options (changes you think will improve the product), and more waves, the optimization process further shrinks the risk and increases the expected impact of the intervention.¹⁶

On the practical side, in order for this process to work, you'll need a few pieces in place:

- A large sample size. While we're looking for big impacts (and thus smaller sample sizes per test), running many tests still means having lots of participants.
- The ability to rapidly deploy experiments.
- The ability to quickly track outcomes and determine the winner—either with the true outcome of interest or with a reliable *leading indicator*. For example, someone's 401(k) contribution rate is a reliable (if imperfect) leading indicator of how much their 401(k) balance will grow over time.
- The ability to seamlessly adapt the intervention based on those experiments.

This approach is well suited for optimizing impact where the target population is large, and a digital context facilitates the necessary data instrumentation and product adaption. For many digital products and communications, this is exactly what we have: large numbers of users (or prospects), a platform for measuring impact, and many ideas about what could help improve our impact.

When and Why to Test

Most experiments are about impact and determining whether something works or not. That measurement occurs *after* the product or communication is built and provides a final, rigorous assessment. That's not the only reason, however; in general, there are four main reasons we do experiments (and points in the product development cycle when we use them):

Measuring impact

First and foremost, experiments measure the real impact of a product or feature on people's actual behavior. This is usually after the product is

built and deployed with real users.

Vetting ideas

Quick, simple experiments with competing prototypes or product concepts can vet big ideas at low cost. This is before the product is built; because you're interested in large differences, you don't need many participants for a statistically valid test.

Optimizing impact

You break a product rollup into groups of people to provide real-time feedback about the product to maximize its impact; we covered that in **“Experimental Optimization”**.

Assessing drift and regression to the mean

Some interventions work simply because they are new—like reminders or novel features. You can re-run an initial experiment to check if it's lost its effectiveness. It's better to know than to blindly assume a product or communication is still works.

Putting It into Practice

Designing for behavior change can and does succeed. In both the practitioner and academic communities, we have seen impressive successes, helping people with everything from sleeping better to learning a new language. But what happens in the aggregate isn't a good guide to each individual effort. Most individual efforts have *no effect*; our successes are few, and our failures are many. Thus, at the start of any behavior change effort *we should assume that we will fail*. That's the safest bet, and empirically, the most accurate one. And so we need a tool that can

accurately, and quickly, tell us when a particular effort has succeeded or failed, given the tremendous complexity of human behavior.

Whether you're helping workers save for retirement or encouraging people who want to exercise more, experiments are the best tool we have to do that. They are the best measurement tool we have to make our efforts more effective. Experiments can help you measure the impact of your work, test assumptions, and better understand what drives client behavior. An experiment's power to show a *causal* impact depends on how it's designed. A badly designed experiment can't tell you anything, but thankfully a good experimental design doesn't need to be complex.

Here's what you need to do:

- When you have clearly defined metrics and the technology to support experiments (covered in Chapters 6 and 12), A/B tests and other experiments don't have to be difficult. The key idea is that a group of randomly selected people receive your behavioral intervention at the same time that another group of randomly selected people doesn't.
- Analyzing the results of these straightforward tests requires a single line of code in most stats packages and is built into many modern technical tools for marketing and behavioral tracking (covered in Chapter 12).
- Before you run your test, you determine how many people, or how much time, you need for the experiment. That avoids many common errors and stopping a test too soon.
- It can be helpful to structure an A/B test as a staggered rollout (in which everyone gets the new feature or communication, but some people get it earlier than others) or as process of optimization (in which once you learn what works best, you apply it immediately to the rest of the population and move on to the next set of tests).
- Multiarmed bandits and multivariate tests are just other ways of structuring experiments—they each have their upside and

downside, but the basic math is the same.

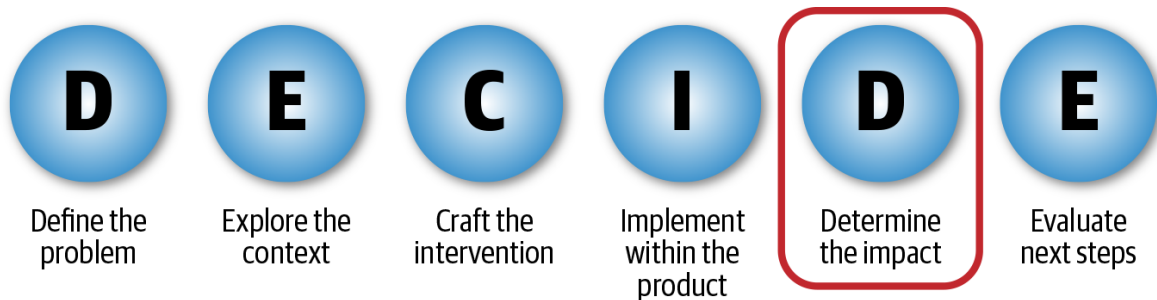
How you'll know there's trouble:

- The team can't decide on a reliable, accurate metric of the product's outcome or doesn't define success and failure by that metric. If that's the case, return to **Chapter 6**.
- The team is so certain of their new feature or product that they aren't willing to measure that impact.
- When you don't have enough users to run a formal test yet—in which case, you can use weaker statistical methods and gather quality qualitative feedback in the meantime (covered in the next chapter).

Deliverables:

- An impact measurement: you know if your product works to change behavior and drive outcomes!

Worksheet: Design the Experiment



Experiments (e.g., A/B tests) are often the best way to assess whether your product is having the desired effect. This worksheet will walk you through the process of designing one. We'll use the example of the Flash app, once again. In particular, we'll examine how to test the effectiveness of two different email campaigns at driving uptake of the app.

Step 1: What's being tested?

Control:

☒ Do nothing (no email reminding them to sign up for the Flash app)

☐ Existing version

Variation 1: Email reminder focused on creating peer comparison and competition

Variation 2: Email reminder focused on reaching individual goals, investing in yourself

What's the outcome metric? Number of sign ups for the app (short-term outcome of the email campaign); Decreased physical therapy visits (long-term outcome of the app)

Is it measured the same way for both versions?

☒ Yes (Continue to Step 2)

☐ No/Not Sure (Stop!)

Step 2: What are the extreme outcomes?

What's the baseline value? (What should the control group have?)

Baseline: 35% of eligible employees are currently signed up for the app based on our existing outreach efforts.

What's the MME? (That is, the smallest change in the outcome that means you've been successful.) No idea? Enter the smallest delta seen before.

MME: 2.5% increase in enrollment

What's the LVE? (That is, the largest change in the outcome that you'd ever expect to have.) No idea? Enter twice the largest delta seen before.

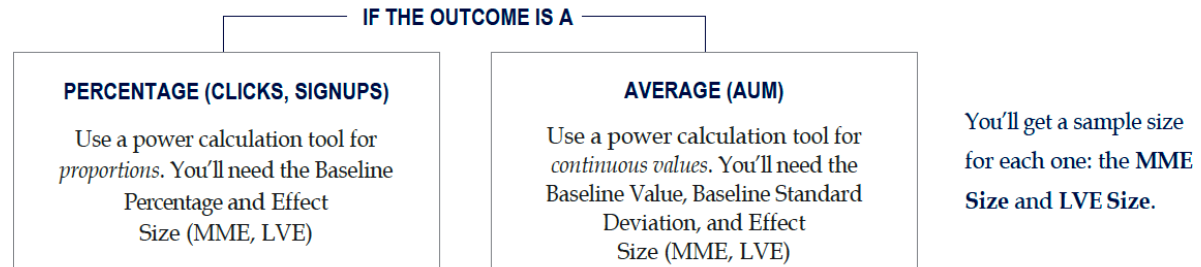
LVE: 10% increase in enrollment

Step 3: Calculate sample size at the extremes

For the MME and LVE:

STEP 3: CALCULATE SAMPLE SIZE AT THE EXTREMES

For the MME and LVE:



* See BehavioralTechnology.co for links to sample tools. Use 0.9 for "power" and .95 for "alpha"

Required sample size, MME: 7,768 in each group (23,304 total across three groups)

Required sample size, LVE: 502 in each group (1,506 total)

Step 4: How many people could you include?

Do you have a fixed list of people?

- ☒ Yes (Use the list size.)
- ☐ No, I have a stream of people over time (What's your timeline? Calculate how many people you'd see by then.)

Number of people available: 30,000

Step 5: How many people should be in each group?

Divide the number of people you have (step 4) by the number of variations (step 1) to calculate the number of people (sample size) per version

Sample size: $30,000/3 = 10,000$

Step 6: Do you have what you need?

Refer to Figure 13-2, reproduced here, which shows how to interpret these values.

Sample Size	Decision	Next Steps
 Less than LVE	 Stop! Don't bother running the test, it can't tell you anything.	- Move on to other tests
 More than LVE, less than MME	 Think carefully! The test might show an impact, but if it doesn't, you can't conclude that the approach failed.	- Optional: Decrease the number of variations - Take the risk and go for it - Get more people - Look for other, clearer tests
 More than MME	 Stop! You can (relatively) conclusively see if your approach works or doesn't work.	- Go for it, run the test! - Optional: Decrease the number of people, down to the MME size - Optional: Add variations until each one is the MME size

What do these results mean? The available sample size per group is larger than both the required amount for the MME and the LVE: *Proceed*. This test will tell us:

- Whether this particular test has the desired impact, and it will also tell us
- If nothing is found on this test, that it is unlikely that further testing will find a sufficiently large impact for the business to care.
- In other words, either way, no further testing will be needed from a business perspective, on this intervention after this experiment.

¹ See Allcott (2011), for example.

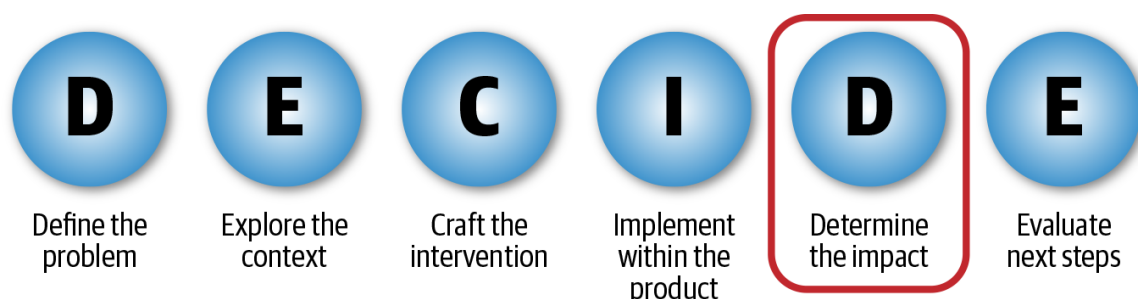
² Oracle (2020)

³ My thanks to Jesse Dashefsky for spurring me to clarify this section.

⁴ See Hopkins et al. (2020) for a nice summary of the different types in practice. Much of what we're talking about here might go under the term "Nimble RCT," coined by Dean Karlan as a contrast to long-term impact-focused RCTs. They are all experiments, though.

- 5 This section and the next draw upon an internal article for Morningstar written by Steve Wendel. Used with permission.
- 6 Or just search for “G*Power”—thankfully, it’s an unusual name.
- 7 See also the [pyDOE2 package](#) for fancier designs to do the same.
- 8 Values calculated using R’s `power.t.test()` function.
- 9 [List et al. \(2010\)](#). My thanks again to Katya Vasilaky for the reference and description of the problem.
- 10 If the effect of the intervention is immediate, that’s the end date of the experiment. If it takes time to have an effect, then the end date is the date at which you stop adding people (because you have “enough”) plus the time it takes to have its effect.
- 11 There are criteria in the academic work for identifying whether the size of an effect is “small,” “medium,” or “large.” These may be good starting points to set your MME if you don’t have a threshold for your business. For example, the oft-used effect size when comparing two groups is Cohen’s *d*. This statistic tells you how many standard deviations (index of a sample’s spread around its mean) two mean values are from one another. The criterion for a “small” *d* is 0.20, or 1/5th of a standard deviation. In our exercise example (the Flash app), if the standard deviation of our sample is 2.5 physical therapy visits, then a “small” effect would be a difference of one-half visit ($2.5 \times .2$) between the treatment and the control group. Many thanks to Stan Treger for adding this section.
- 12 A test of the full product has a particular meaning: an impact assessment of the combined package. If the package is effective, then one can follow up with analyses to separate out the particularly effective components.
- 13 Thanks to Fumi Honda for highlighting the problem of observer-expectancy.
- 14 This is also useful when you have a product that your users want and you can’t withhold it from them. This happens in many international development projects, where the funder strongly believes in the success of the project before it is tested and feels it would be morally wrong to withhold the product from potential recipients (see Karlan and Appel 2011).
- 15 See [Hanov \(2012\)](#) for an enthusiastic, if a bit too optimistic, description of multiarmed bandits. See [Gupta \(2012\)](#) for a nice summary of the pros and cons of A/B testing and multiarmed bandits.
- 16 From a scientific perspective, this process makes a number of assumptions about the population and the impact of the intervention. These assumptions actually underlie many RCTs in the field currently, but aren’t explicitly stated. Namely, (a) the experimental results are generalizable over time, (b) the experimental results are independent of one another, or additive, and (c) the researchers don’t already, and reliably, know what is “best” (i.e., the experiments add knowledge and don’t merely document it).

Chapter 14. Determining Impact When You Can't Run an A/B Test



Recently I had the good fortune to attend an event featuring many of the most prominent and prolific behavioral scientists in the world. They had banded together to address one of the frontiers of behavioral science: long-term behavior change.

The researchers were testing whether they could move the needle on gym attendance. They ran nearly 20 simultaneous studies, with each researcher pursuing their “best guess” of what would work. This event was the first time that everyone in the group would hear the results.

What happened? Not a single study was effective at long-term behavior change. The best researchers in the world had each taken a shot and had fallen short of their target.

Something that is often hidden, at least to those outside of the research community, is that in our space the successes are few and the failures are many. That’s normal. That’s expected. It’s because the most interesting applications of behavioral science are often focused on difficult and

*seemingly intractable problems, like exercise. Even the best researchers in the world try again and again, before they find a breakthrough that generates headlines and a best-selling book.*¹

*The lesson here isn't merely to "fail fast" (a motto that is deeply engrained in product development, at least in Silicon Valley and similar tech hubs) or that embracing failure means embracing iteration.*² *Rather, it's a bit more nuanced. How did the researchers know that they'd failed? It's because they looked for failure. In their case, they used randomized control trials to receive an unambiguous signal that things hadn't turned out as planned, without wiggle room to lie to themselves or try to put a positive spin on the results.*

Teams can't always run experiments, but the need for rigorous measurement doesn't go away. The less wiggle room—the less one can explain away the results or hide from flaws in the product and its ability to behavior change—the better. We want to find problems early and not let them turn into expensive messes. So let's look at other ways to measure impact.

Other Ways to Determine Impact

Experiments take care of all of the nasty details of figuring out whether the application, or something else, changed the user's behavior and outcomes. The random assignment process, properly done, ensures that nothing is different between the two groups except for what they want: the product itself. So any difference in outcomes is caused by the application.

As an academic, I could make the case that experiments are really the only way to measure *causal impact* because of these benefits. But in real-world products, that's unrealistic and too restrictive. If you aren't using experiments, then you have to face the nasty details of estimating the causal impact of the application head on. It can certainly be done, but it should be done with open eyes.

The easiest and most common way to look at impact is a pre-post analysis.

A Pre-Post Look at Impact

In a pre-post analysis, you look at user behavior and outcomes before and after a significant change. For example, if users on average walked 500 steps a day before using your product and 1,500 steps a day after using it for one month, then the product *might* have increased their walking by 1,000 steps a day.

In a pre-post analysis, you take the difference you see and try to adjust it for all the other things that could have caused the change that weren't part of your product. This can be done informally or formally. The formal version requires running a multivariate statistical analysis like estimating a regression model. The informal version means carefully thinking through what else could have impacted the users and their behavior.

Personally, while I was trained in the formal, econometric approach, I find that starting with an informal analysis is immensely valuable (even if you later do the econometric analysis as well). Also, you'll probably need a stats person to handle the econometric analysis, but anyone can do an informal analysis to help gauge how important further analysis is, and as a reality check on the stats. So here's how to run an informal analysis of a pre-post study.

You have measurements of user behavior and real-world outcomes before and after a change, either when you gave the users the product for the first time or added a new feature, etc. Subtract the *pre* from the *post*: that's your working impact number. You also should have a sense of how big of a change you need for you to care. If you get people to walk two more steps a day, is that relevant? No. Maybe you only care if the product can get people to walk at least a hundred more steps a day—it's not much, but at least it's something to build upon. That "when do I care?" number is your threshold.

Now, look for non-product-related things that would have caused the impact you're seeing. With pre-post studies, there are a few very common factors. I'll use the example of an exercise tracker to make things concrete:

Time

Would the time of year, day of month, day of week, time of day, etc., matter for this outcome? For example, if you saw that users were walking more in the spring than in the dead of winter, would that surprise you? No. So it's unlikely that your *product* would have caused a change you see in walking between winter and spring.

Experience

Let's say you launched your product last month. You've just added a new feature that puts a smiley face on the tracker when users do well. In a pre-post study, it will be difficult to know if the smiley face caused increased exercise or if users just gained experience with the product from its initial launch and slowly exercised more because of that experience (rather than the smiley face). *Gradual changes over time* are often caused by experience; *sharp changes* in behavior are more likely to be caused by a change in the product or another external "event" you can search for.

Data availability or quality

Let's say that in the new release of the tracker, you added a smiley and someone in the engineering department fixed some bugs in the analysis of accelerometer data. Walking is up! Hmm. That could be because of the smiley or because you're simply getting better data about the users. I've found that data quality issues in particular are often invisible and therefore often misleading—someone changed something and didn't think it was important or didn't want to admit the previous problem. Like product changes, data quality and data availability changes are

sharp, sudden changes, so they are very hard to distinguish in a pre-post study.

Composition of the population

Let's say that with the new release of the tracker, you've added lots of new features and made a big announcement. You see that average walking is up! Excellent. But that may be because the product caused people to walk more, or it may be that the announcement of the new features caused new users to join who were already walking more—and the new users brought up the average. This occurs in sudden ways (like product announcements) or through the slow addition and attrition of users over time. You should counteract this by looking at a specific group of people before and after the change.

In each case, you're looking for a gut check—is this a big deal? Measuring walking behavior in the dead of winter versus in spring is a big deal. Measuring it from one Tuesday to the next usually isn't (barring holidays). A big deal is anything that looks like it's going to have a large impact on behavior *relative to what you're seeing pre-post and relative to the threshold at which you care*. If the combination of many small things pushes the likely impact of your product below the threshold at which you care, then you can usually stop—and move on to something more promising.

If the pre-post impact is so large that nothing else seems to explain it other than the product, excellent. You should check your work with a statistical model and prepare to be surprised; but if you can't, at least you've gotten an initial estimate of impact.

This informal analysis feeds into formal statistical modeling. Each of the factors you identify that might be important become variables in the model—things that you are trying to control for in order to isolate the unique

impact of your product. You'll need to identify data that measures them and run the model itself. That's beyond the scope of this work—but a good stats person can help.

Seem complex? It can be. That's why experiments are useful, because they remove these complexities. But we can't always run them or get enough users into the system to get a solid result. And so we sometimes must use pre-post analyses. When the product has a big effect, and there aren't many other things going on that confuse the result, that's enough to get a signal for further product development. Another option is cross-sectional multivariate analysis, which is up next.

A Cross-Sectional or Panel Data Analysis of Impact

In a cross-sectional analysis, you look for differences among groups of users at a given point. You want to see how their usage of the product impacts their behavior and outcomes, after taking into account all of the other things that might be different about the users. For example, you might look at the impact among frequent users of the application versus infrequent users. As with pre-post analyses, I usually start with an informal, logical analysis and then feed that understanding into a formal statistical or machine learning model if it looks like there's enough of an impact from the product to care.³ Cross-sectional analyses usually pull together diverse groups of people; in order for the analysis to be valid, you'll need to control for all of the factors that make those groups different *other than the product*.

As before, there are some common differences you need to take into account. Most importantly is this: why are some people more frequent users than others? Age, income, prior experience with the behavior, prior experience with the product's medium (mobile versus web), self-confidence, sufficient free time, etc.—all of these are factors that affect the users' behavior above and beyond the product.

If there aren't obvious candidates that explain the difference in behavior across users, then take the list of factors you generate and plug them into a

statistical model. Again, it's beyond the scope of this work, but a good stats person can help.

In addition to cross-sectional and pre-post analyses, one can (and should) also look at models that examine changes in behavior and outcomes among many users over time. These models, using *panel* datasets (or time-series cross-sectional datasets with many people, but shorter time frames) provide a much more fine-grained look at behavior. They can pull out impacts of the product that pre-post and cross-sectional models can't because they can control for other differences across the individuals. However, they require much more data and statistical knowledge.

Unique Actions and Outcomes

Experiments are the best general-purpose and accurate way to measure the impact of your product. But there's an important case in which they aren't needed to accurately gauge impact. That's when there is no conceivable way that the outcome would occur without the product being there.

For example, imagine a new and highly effective cancer treatment. A team is developing a product to make people aware of it; the target outcome is for people to use the new cancer treatment. Without that awareness, no one would know. There's no comparison group needed—any impact that occurs is because of the product.

Similarly, it's easy to measure the *baseline* impact of the product when the action only exists in the product itself—which often occurs where the behavior change process entails the user learning to use a new product. That's the benchmark you can use to compare against future changes. After that baseline has been established, you'll still need to run experiments (or use other means) to gauge the impact of new features and other changes to the application to distinguish the impact of the feature from that of the existing functionality.

What Happens If the Outcome Isn't Measurable Within the Product?

You can safely skip this section if users take action directly in your product and you can easily measure the outcome there.

As we briefly discussed in [Chapter 12](#), sometimes the target outcome, and even the target action, may not be directly measurable in the product. For example, think about a website that helps users set up an urban vegetable garden with video tutorials. The target outcome is more vegetable gardens; the target action is that users set them up (rather than, for example, contractors being paid to set them up).

Each person who uses the urban garden site is tracked with a cookie or authenticated login. Each step in the “how to set up an urban garden” tutorial is tracked. When users complete the tutorial, are they “done”? Did they complete the action? No. The action that the company wants to drive is setting up vegetable gardens, not completing a tutorial about setting up vegetable gardens. The difference between those two could be slight, or it could be massive if no one actually follows through. Without further information, there's no way the company can know if the product is successful at driving behavior change. Similarly, it has no way of knowing that the product has caused more vegetable gardens to be set up than there otherwise would have been.

So what can a company do? If the action or outcome is not directly measurable with the product, then a *data bridge* is needed. A data bridge is something that convincingly connects the real-world outcome with behavior within the product. There are two basic strategies for building a data bridge:

Build it yourself

Find a reliable way to measure the target action and outcome. Then, build a model of what behavior in the product relates to that action and outcome.

Cheat

Find an academic researcher who has already established the link between something you can reliably measure in the application and the real-world outcome. For example, there are numerous studies that document “overreporting” (lying) about voting when people actually don’t vote.⁴ If there isn’t an existing research paper on the topic, work with researchers to generate one (ideas on how to partner with researchers are discussed in [Chapter 17](#)).

For the rest of this discussion, I’ll assume that you haven’t been lucky enough to find an existing research paper or interested researcher to do the work for you—and so you have to build the data bridge yourself.

Figure Out How to Measure the Outcome and Action by Hook or by Crook (Not by Survey)

For our sample company that’s encouraging users to set up urban vegetable gardens, it will need to measure the number of vegetable gardens, plain and simple. An obvious route would be to ask the participants with a survey. Not ideal. Surveys are good for gathering facts when people have an incentive to actually answer the survey but don’t have an incentive to lie. Imagine that the vegetable garden company asked users of its website, after a week, whether they set up a garden. Most people wouldn’t answer—especially those who didn’t set one up. Some would answer truthfully. And some would answer with, “What will be truthful when they get around to it” (i.e., they’ll tell a white lie). The company can’t really know which is which, at least, not without doing additional field research to verify if people aren’t telling the truth.

If the company asked users about their *intention* to set up a garden, that would be even worse. Users would be sorely tempted just to give the answer that’s expected of them (“Yes, of course!”); that’s called the social

desirability bias in surveys.⁵ Or, people might honestly believe that they will set up a garden, but never get around to it. You can try to reduce this bias in surveys with carefully worded questions, but it's difficult to know the success of that effort without verification.

Direct observation is often the best option: either observing the number of vegetable gardens themselves or other things that uniquely indicate that the action was taken, like the number of people buying special vegetable garden supplies within a city. The company doesn't need to measure every single action or every time the outcome changes—it just needs to be able to measure a few times so it can understand the relationship between the product, the action, and the outcome. So a small pilot study where an intern goes out and manually counts the number of vegetable gardens in an area is fine.⁶

Building a data bridge follows the same rules as creating a benchmark for the product, described earlier in this chapter. In this case, you're looking for the causal relationship between something easily measurable in the application and a real-world outcome that's hard to measure, but what you really care about. If the real-world outcome is unique to the product (i.e., if *no one* normally creates vegetable gardens in the area you care about), then you can do a simple observation of the real-world outcome, after people use your product, as your metric. If the real-world outcome has multiple possible causes, you'll need to use an experiment, statistical model, or pre-post analysis.⁷

In either case, there are three factors to keep in mind when measuring the real-world outcome itself. These determine how sturdy the resulting data bridge will be:

Representativeness

You want to observe cases that are representative of what “normally” happens; if you decide to count vegetable gardens in Portland (very rainy), and most of your app's users are in Phoenix (rather dry), that won't help you generalize much about vegetable garden creation. The

most solid results come from taking your user base and randomly selecting some of them to directly observe.

Getting enough data points

You need to ensure you have enough information to get a solid signal about what the real-world outcome really is. For example, if you make only one observation of whether people make a vegetable garden after saying they will in the app, that's not going to tell you much about whether *other* people will. You need the general percentage of people who create gardens when they say they will. So how many observations are enough? There's no hard-and-fast rule—it depends how important the accuracy of the estimate is to the company. For experiments, we discussed in detail how you compute sample sizes. If you're not using experiments, you can use online tools for computing *confidence intervals*,⁸ which tell you how confident you can be in your estimate; if you build a statistical model of the relationship, that will also provide you with confidence intervals.

Getting a baseline

Sometimes things happen in the real world that have nothing to do with your product. I know, it's hard to believe. Some people will create vegetable gardens on their own, even without the vegetable garden app. So when you're observing your real-world outcome, include some cases in which people don't use the app. This is important if you're doing a simple model in Microsoft Excel or if you're running a full experiment to build your data bridge.

If these options fail, and there's really no way to measure the product's real-world outcome, then the rest of this discussion about impact can't help. That signal—what's actually happening in the world—is essential for keeping the whole process honest.

Find Cases Where You Can Connect Product Behavior to Real-World Outcomes

Now you have measurements of actions taken within the product and of real-world outcomes (though perhaps imperfect measurements). How can you connect the two? You can connect them at the individual level or at an aggregated level. At the individual level, for example, the urban gardening app could ask for users' names and addresses to connect their behavior in the product to whether they actually have a vegetable garden (send the intern to their home and mark it down in the record). Getting data about individual users is the ideal—as long as the data meets the standards (representative, sufficient in size, and with a clear baseline).

Alternatively, the action and outcome can be measured as an aggregate, a known geographic area or a known group of people. If you know that a certain set of users in the product correspond to the known area or group (even if you don't know who is who) and you can measure the actions and outcomes reliability in that area or group, you're in business. As we'll see, it'll be more challenging to figure out exactly what is going on, but you can do it.

Build the Data Bridge

A data bridge brings together something you know and can measure frequently (user behavior within the application) with something you have measured only a few times (the impact of the product on the real-world target outcome). It allows you to estimate how much the target outcome has probably changed based on behavior within the product. You'll estimate that relationship by running a pilot project that gathers both datasets:

1. Take a circumstance in which you can reliably connect user behavior in the product with the real-world outcome or action, as we've just described.
2. Measure the causal impact of the product on the real-world outcome or action using an experiment (ideal), statistical model, pre-post analysis, etc.
3. Analyze the various user behaviors that occur within the application and identify one or more that is strongly related (correlated) to the application's causal impact. If a statistician is available, use a mediation analysis.
4. When the indicative user behavior occurs within the product, build a model (in Excel or in a statistical package) of how much that changes the target outcome. That's the data bridge.⁹
5. In the future, whenever you see the behavior in the product, use your model to estimate the likely impact on the target outcome.

For example, the urban gardening site runs a pilot study where it takes two sets of randomly selected people and offers its program to one group and not the other. Some of the people in the first group completed the training program; some did not. An intern visits the homes of everyone in the study and measures the truth. The company finds that 65% of people who were offered the program created a garden, and 90% of those who were offered the program *and completed their training within the application* created gardens. Meanwhile, 15% of those who weren't offered it nevertheless created a garden. Those three stats provide a basic understanding of how to interpret user behavior on the website in the future.

The company would improve the chance that a person will set up a garden by 50 percentage points (from 15% to 65%) if it *offers* the person its training program. It will improve the chance that the person will set up a garden even more if it can convince them to complete the training program.¹⁰ The company can get a precise estimate of that impact using what's known as a mediation analysis on the experiment.

In short, if your target outcome is something outside of the product and not directly measurable, then you'll need to build a data bridge. The easiest way to do that is to find an existing research study that documents the relationship you're looking for—like between the intention to plant a garden and the actual act of doing so. If not, look for a case in which your team can directly observe the users' behavior and compare the things they do or say in the product to what they actually do in the real world. That's your data bridge. In the future, you can use that relationship to estimate how much of an impact you're having based on what you see in the application and iteratively improve your product for greater impact.

Putting It into Practice

Here's what you'll need to do:

- In a pre-post analysis, look for discontinuities in behavior and outcomes at the moment the new feature or communication was deployed. The sharper the change and the fewer alternative explanations there are for that change, the more confidence you can have that your intervention was the cause.
- In a cross-sectional or panel-data analysis, look for other people in as similar a situation as possible who didn't receive the intervention to compare against. Again, the goal is to remove alternative explanations for any differences you see in behavioral outcomes.
- The text describes the logic behind measuring impact without an experiment, and if you have a unique action or outcome, that can be enough. Usually, however, you need a trained stats person to carefully analyze the data and statistically control for (eliminate) alternative explanations.

How you'll know there's trouble:

- There's no clear definition of success and failure for the product's attempt to change behavior. If so, return to **Chapter 6**.
- Many other things changed within your product or user base at the same time as your new feature or communication—making it difficult to eliminate alternative explanations for behavioral outcomes.
- You have a complicated environment, no experiment, and no stats person to analyze the data. Don't try to wing it and look at bar charts or line graphs over time—behavior change is just too complicated.

Deliverables:

- A clear measurement of the product's impact!

1 Now, before you dismiss this experience as one limited to a particular set of studies, a recent review of research where the authors preregistered (wrote down and published) their hypotheses before they conducted a study found that over 50% of studies in biomedicine and psychology did not show the results that researchers expected (Warran 2018). Johnson et al. (2017) estimate that 90% of research efforts in psychology overall (i.e., including those that aren't preregistered) had null or negligible results. And, again, before you dismiss this as a researcher's problem, remember that famous researchers like Edison and Dyson iterated hundreds or thousands of times before generating successful products (e.g., Syed 2015).

2 Pontefract (2018)

3 There are often many possible changes to the product you want to analyze—so focusing too long on features that don't appear to change behavior in practically significant ways means you're wasting time that could be used more valuably elsewhere. This is a difference from academic social science work in that researchers usually devote a significant amount of time to a single question; because of a lack of data, they usually don't have a long list of alternative questions that can be explored immediately.

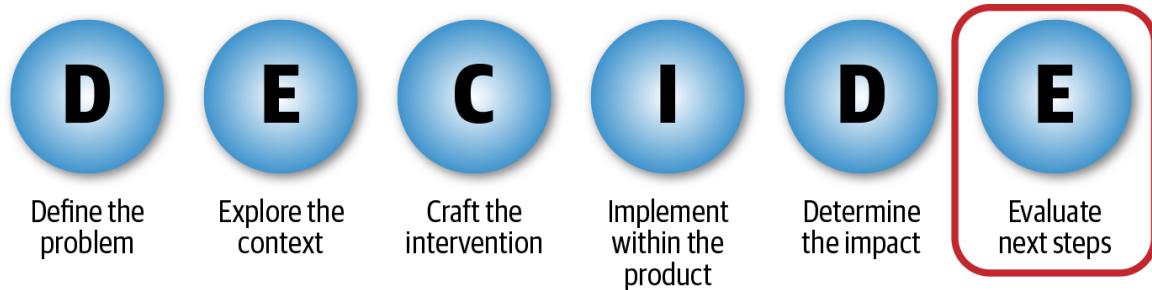
4 Silver et al. (1986)

5 Fisher (1993)

6 By the way, if the area is large, I imagine that the best way to do this would be access government or commercial satellite imagery. Professional geographers have worked out amazing algorithms to automatically detect vegetation cover, and even the type of vegetation. The GeoEye satellite that is used by Google Earth, for example, *measures down to increments of 16 inches*.

- 7 To clarify—at this point we’re just talking about how to measure the real-world outcome. That forms half of the data you need to run an experiment, do a pre-post analysis, or build a statistical model of the relationship between the real-world outcome and user actions in the application. That process is what actually creates the data bridge and is covered later. But it helps to plan ahead for the *type* of analysis you will be running to ensure you’re gathering the right data you need when measuring the real-world outcome.
- 8 For example, you can use for calculating confidence intervals of proportions (percent of people creating vegetable gardens) and for calculating confidence intervals of quantities (number of pounds lost after an exercise program). Penn State has a nice summary of **the underlying math**.
- 9 In the simplest case, you might look at the simple linear relationship between the real-world impact and user behavior in the product. But there’s no reason to limit the analysis to a linear relationship. You want to build a model that most accurately describes the relationship between behavior in the product and outcomes in the real world.
- 10 Exactly how much additional improvement might occur would require additional analysis, to separate out the self-selection into the program from the program’s causal impact.

Chapter 15. Evaluating Next Steps



Throughout the first decade of the 21st century, microcredit was hailed as a broad solution to poverty worldwide. Low-income people, especially in developing countries, were offered small, often unsecured loans, at rates ranging from a few percentage points to over 100% a year. Many of these programs focused on enabling poor women to become entrepreneurs, providing loans to groups of women at a time who would support and hold each other accountable. The idea was that if motivated but cash-poor women simply had access to capital, they could start or grow their businesses and lift themselves out of poverty.

As U2's Bono said of the microcredit, "Give a man a fish, he'll eat for a day. Give a woman microcredit, she, her husband, her children, and her extended family will eat for a lifetime."¹ One of the most prominent microcredit organizations, Grameen Bank in Bangladesh, and its founder Mohammad Yunus, in fact won the Nobel Prize in large part for their efforts to alleviate poverty through microcredit. In 2008, for example, Grameen Bank had over seven million borrowers, 97% of whom were women, with over \$500 million USD in loans outstanding.

When I was in graduate school in 2005, microcredit was a shining example of innovative work for social good: many of my friends went into careers in microfinance (as the broader field is known), as nonprofits, private companies, and governments all around the world were excited about its potential. Microfinance was the place to be if you wanted to devote your life to helping others on a massive scale.

That picture soon changed.

*A series of studies found that the impact of microcredit was not nearly as universal and transformative as many had hoped. Dean Karlan and Jacob Appel summarize these lessons in their book *More Than Good Intentions* (Penguin, 2011), citing a study in India, for example, in which “it looked like people were, on the whole, no wealthier than before” and that “the most common reason for borrowing was to pay off other debt.”² Karlan and Appel share how the impacts of microcredit were simply much more nuanced than had been portrayed; some people (especially existing business owners) could benefit, but perhaps in nonobvious ways (helping them cut costs, not hire new people); in other cases, people were simply left with debt they couldn’t afford to pay off. “They wound up looking more like characters from a cautionary tale about, say, credit card debt, than the inspiring figures of microcredit literature.”³*

In a highly cited 2015 paper, Banerjee, Karlan, and Zinman presented six randomized control trials on the impact of microcredit and concluded, “We note a consistent pattern of modestly positive, but not transformative, effects.” In that paper and in related writings,⁴ they found that while the broad effects were overblown, relatively small changes to microcredit programs could make them far more powerful, like adding flexibility in repayment periods. In other words, they used the power of rigorous measurement not simply to cut down and critique—but to refine.

If we think about microlending as a product, one that sought to accomplish a specific purpose for its users (lifting them out of poverty, often with the assumption that would occur through debt-financed entrepreneurship), it didn’t hit the mark with its first iteration. However, by digging into the data

about who used it, who benefited from it, and under what specific circumstances, we can learn how to better target users and make it more effective overall.

That is what we'll explore starting in this chapter: how to gather the data you need, and identify areas to improve, to increase the impact of your product.

Before we do so, however, one final note: *More Than Good Intentions* is one of the three books I recommend to every person who asks me about exploring behavioral science. Sadly, the promise and perils of microcredit is just one of many examples the authors give of well-meaning people putting their hearts and souls into untested products that they thought were wonderful—until they discovered otherwise. Since I read it many years ago, it's served as a haunting reminder that my own intentions to do good simply aren't enough. We all need to rigorously measure, to humbly look at our products, and to ask: how can we do better?

Determine What Changes to Implement

At the end of each cycle of product release and measurement, the team will have gathered a lot of data about what users are doing in the product and potential improvements to it. Continued obstacles to behavior change are only one source of those product improvements. Business considerations and engineering considerations must also be reviewed. It's time to collect the potential changes from these diverse sources and see what can be applied to the next iteration of the product. I think of it as a three-step process:

1. *Gather* lessons learned and potential improvements to the product.
2. *Prioritize* the potential improvements based on business considerations and behavioral impact.
3. *Integrate* potential improvements into the appropriate part of product development process.

Gather

First, look at what you learned in the two preceding chapters about the current impact of the product and obstacles to behavior change. What did users struggle with? Where was there a significant drop-off among users? Are users returning to the application or only trying it once or twice? Why does that appear to be happening?

1. Start by picking the low-hanging fruit. List the clear problems with a crisp follow-up action; for example, no one knows how to use page Y.
2. Then, write down the lessons that are more amorphous; for example, users don't trust the product to help them to change behavior. Maybe the team has started thinking about potential solutions, but there's more work to be done. The next step is to further investigate what's going on and settle on a specific solution to resolve the problem.
3. Next, gather lessons about the core assumptions of the product:
 - Does the target action actually drive the real-world outcomes that the company seeks? For example, maybe walking a bit more each day isn't enough to reduce heart disease among the target population, and a stronger intervention is needed.
 - Are there other actions that appear to be more effective? Could the product pivot to a different action that is more effective?
 - Are there major obstacles in the user's life, outside of the product, that need to be addressed? Looking again at the causal map, what major factors that are currently outside of the product's domain are counteracting the influence of the product? If exercising more leads the person to also drink more alcoholic beverages (as a "reward"), is that

defeating the product's goals? To design for behavior change, we care about the net impact of the product, not just the intended consequences. Is there anything the product can do about that countervailing force, or is it just a fact of life?

4. Finally, look beyond the specific behavioral obstacles and impact studied in the previous two chapters. The team has probably generated numerous ideas for new product features or even new products. Collect them. Other parts of the company will also suggest changes to the product as well: changes designed to increase sales, improve product branding, resolve engineering challenges, and so on. Behavioral considerations are just one (vital!) element in the larger review process.

Lessons and proposed improvements can come at different times during the product development cycle—from early user research to usage analysis after the product is released. Some lessons will only come at the end, during a formal sprint review or a product postmortem. I suggest creating a common repository for them, so that ideas don't get lost. That can be someone's email box, a wiki, or a formal document of lessons. In an agile development environment, they should be placed in a project backlog.

Prioritize

In any product-development process, there is a point at which the team needs to decide what to work on in the future. The prioritization process should estimate the behavioral impact of major changes to the product: how will the change affect user behavior, and how will that affect the product's outcomes in the real world? Since the product is designed for behavior change, these behavioral impacts will likely have knock-on effects on sales or the quality of the company brand. Naturally, the prioritization will also incorporate business considerations (will the change directly drive sales or company value?), usability considerations (will it make the users happy and

reduce frustrations, hopefully driving future engagement and sales?), and engineering considerations (how hard will it be to implement the change?).

The team should ground its assessment of behavioral impact in real data; the drop-off numbers at each step of the user's progression and the behavioral map allow the team to make a quick estimate of how large of an impact a change in the product should have. That helps the team answer: how big of a problem does this change really address? What *very rough* change in the target behavior and outcomes do we expect from it? Even if the proposed change to the application wasn't driven by a behavioral concern—for example, if it came from a client request during a sales conversation—it should be evaluated for its possible behavioral impact. It may have the added benefit of helping improve user success at the target behavior, or *it may distract the user and undermine the product's effectiveness*.

The weight of each of these considerations—business, behavioral, engineering, etc.—in the company's prioritization will vary, and there's no hard-and-fast rule.

Integrate

Your company has a prioritized list of changes to the product (including open questions that need to be answered) and a sense of how difficult each piece is to develop. Now, separate out changes that require adjusting core assumptions about the product and its direction from less fundamental changes that keep the same direction. If the change entails targeting a different set of users (actors), a different target action, or, especially, a different real-world outcome, then those go into the first bucket. If the change entails a new product or new product feature, where there are major unknowns, it also goes in the first bucket. Everything else can go into the second bucket.

Here's one of the few places I take a strong stand on the product-development process—items in the first group, with changes to core assumptions or major new features, need to be separately planned out by the

product folks before they are given to the rest of the team. Even in an agile development process, core product planning shouldn't be done in parallel with the rest of the process; that's the same dictum that Marty Cagan, in *Inspired* (SVPG Press, 2008), gives in his analysis of product management. It's just too much to determine what to build and how to build it at the same time.

When designing for behavior change, core changes to the product require updating the behavioral map. They may also require updating the product's outcomes, actions, and actors. In other words, they require another cycle of the full discovery or design process, starting with Chapter 6 or 7 of this book. Everything else can go directly into crafting new interventions, starting with Chapter 9.

Each time the core assumptions—actor, action, and outcome—are changed, they should be clearly documented, as described in Chapter 6. Then the behavioral map should be updated. This formalism helps the team pull problems into the present—rather than let them lurk somewhere in the future, only to be found after significant resources have been developed. Making the assumptions and plan clear up front is intended to trigger disagreements and discussion (if there are any underneath). It's better to find those disagreements sooner rather than later.

When a core problem—like a particular step in the sequence of user actions—is confusing users, there's a natural tendency to settle on a proposed solution and just get it done (i.e., a “fix” is often implemented before vetting and testing the problem). But human psychology is tremendously complex, and trying to build a product around it is inherently error prone. There's no reason to think that the *proposed solution* is going to be any less plagued by unexpected problems than the previous solution. The discovery process—documenting the outcome, action, and actor, and then developing the behavioral map—is one way to draw out the unexpected and provide opportunities to test assumptions early. It'll never be perfect, but it's a whole lot better than just shooting from the hip.

Measure the Impact of Each Major Change

Each major change to the product should be tested for its impact on user behavior; measuring changes in impact should become a reflex for the team. It's not always easy to stomach, but it's necessary. That way, the team is constantly learning and checking its assumptions about the users and the product's direction. As we've seen, small changes in wording and the presentation of concepts can have major impacts on behavior; if we're not testing for them, we can easily and unintentionally undermine the effectiveness of our product. But without a reflex to always test, testing the marginal impact of changes can raise all sorts of hackles and resistance.

Let's look at some of the issues that can arise and how to handle them:

Most tests will (and should) come back showing no impact

Many people get frustrated at test results that come back with no clear difference between the versions they're testing and call that "failing." Assuming you designed and ran the test correctly, a "no difference" result should be celebrated. It tells you that you weren't changing something important enough for the test to have found a difference. Either be happy with what you have, or try something more radical. It saves you from spending further time on your current approach to improving the product.

What's a well-designed test? It's one where you've defined success and failure beforehand. It's not one where you go searching for statistical significance (or a "strong" qualitative signal). For example, let's say you have a potential new feature/button color/cat video. How much of an impact does it need to have before you care? If you improve impact by 20%, is that your threshold for success? Is it worthwhile to work on this further if you're only getting a 2% boost? That definition of success and failure, along with the amount of noise in the system, determines how many people you need in the test. If you get a result of "no difference" from the test, that doesn't necessarily mean there's no effect; it means there's no effect *that you should care about*. You can move on.

A/B tests in particular seem to mean you're showing some people a "bad" version

If you have a good UX team, then most of the time, no one really knows if a *change* in the app will improve it. You can't accurately predict whether the new version will be better or worse. You usually are showing a "bad" version; the problem is that you don't know which one it is! Our seemingly solid hunches are usually random guesses, especially when we have a good design team. There are two reasons why.

First, a good UX team will deliver an initial product that is well designed and will deliver product improvements that are also well designed. We all make mistakes, but a good design team will get you in the right ballpark with the first try. By definition, further iterations are going to have a small impact relative to the initial version of the product. Don't be surprised that new versions have similar results (impact, etc.) to earlier versions—celebrate the fact that the earlier version was a good first shot.

Second, human behavior is just really confusing. As we've seen repeatedly throughout this book, we just can't forecast exactly how people will react to the product. In familiar situations, we can and should use our intuition about a set of changes to say which one is likely to be better—like when we're applying common lessons we've learned in the past. But when you have a good design team, the common lessons have already been applied. You're at the cutting edge, and so your intuition can't help anymore. That's why you need to test things, and not rely (solely) on your intuition.

Does planning for tests imply you're not confident in the changes you're proposing?

This is another issue I've heard, and it's a really tricky one. You naturally expect that any changes that you're planning to make to the product will improve it. But that's often not the case (since it's hard to

make a good product better, and human behavior is inherently complex).

That sets up a problem of cognitive dissonance, though. It's very uncomfortable to think that some of the changes you've carefully planned out, thought about, and *decided will help* are actually going to do nothing—and you don't know which ones those are! It would be like you're admitting a lack of confidence in the changes that you've already advocated. So a natural (but dangerous) response is to plow ahead and decide that testing is not needed.

There's no simple solution to address this situation—the need to confidently build something you shouldn't actually be confident in. The best approach that I've come across is to move the testing process out of the reach of that cognitive dissonance. Make testing part of the culture of the organization; make it a habit that's followed as standard procedure and not something that the organization agonizes over and debates each time a new feature is added.

Alrighty. Those are three of the major issues I've confronted as teams explore testing incremental changes to their product. Thankfully, it's not difficult to actually measure incremental impact. If you created a benchmark of the product's impact in [Chapter 12](#), then all you need to do is to reapply the same tools here: experiments, pre-post analyses, and statistical models.

Qualitative Tests of Incremental Changes

I didn't mention qualitative research in [Chapters 13](#) and [14](#), when we were establishing a benchmark of the impact of the product on user behavior and real-world outcomes. That's because it's difficult to generate a repeatable, reliable ROI metric of real-world impacts using most qualitative methods. But qualitative research can be quite valuable when you want to quickly judge how users are responding to a change in the application.

Put the revised application in front of users during user interviews, user testing (speak out-loud methods), or even focus groups. If you get a clear signal about whether the change has caused problems, you've just saved a lot of time. You can get feedback and insight in a fraction of the time it would take to test the product change with an experiment or pre-post analysis. While I am a big proponent of experiments (and statistical modeling), the benefit in terms of speed and depth of understanding from qualitative testing is too much to ignore. Of course, the team should have already performed a round of qualitative testing on the prototypes before the change was made to the product itself too.

When Is It “Good Enough”?

Ideally, the outcome of any product development process, especially one that aims to change behavior, is that the product is doing its job and nothing more is needed. When the product successfully automates the behavior, builds a habit, or reliably helps the user make the conscious choice to act, then the team can move on. There are always other products to build. And, for commercial companies, there are always other markets to tap. So, how can the team tell when it is good enough?

Return to the product's target outcome and try to stop thinking about the product itself. What's the target level (or change in the target) that the company decided would count as success? If the product currently reaches that threshold, wonderful. Forget the product's bugs. Forget the warts in the design. Move on to address other challenges. If the current product doesn't yet meet that threshold, what is the best alternative use of the team's time? If the alternative is more beneficial to the target outcome and can be achieved with similar resources, the team should switch its focus.

Putting It into Practice

Here's what you'll need to do:

- Gather all of the proposed product changes—changes to improve the behavioral impact of the product and other changes suggested by sales, marketing, or other parts of the company.
- Prioritize the changes based on the company’s and user needs and their likely impact on the user behavior.
- Measure the impact of each major change to the product, using the same tools outlined in Chapters 13 and 14. Make incremental measurement part of the culture of the company.

How you’ll know there’s trouble:

- Major changes are planned for the product without assessing their likely impact on user behavior.
- The team is afraid to test the new feature, because the tests usually come back negative or testing would imply a lack of confidence.

Deliverables:

- A new and (ideally) improved product!

1 h/t Karlan and Appel (2011) for the quote.

2 Banerjee et al. (2015)

3 Ibid. 81.

4 For example, Karlan et al. (2016).

Part III. Build Your Team and Make It Successful

Chapter 16. The State of the Field

The common spaces are flooded with light and greenery, following behavioral research on well-being. The apartments empower their residents to control their layout and reconfigure them as their needs change. There's a single switch by the bed that turns off all of the appliances and electronics at once—removing friction and decision points. The shower heads light up after five minutes to remind users of the costs to themselves and the environment. Welcome to the “Paris Nudge Building,” the Bains Douches Castagnary.

In 2015, Paris's Mayor Anne Hidalgo announced a new architectural competition to help “Reinvent Paris,” looking for “innovative urban projects to build what the Paris of tomorrow might be.”¹ The BVA Nudge Unit partnered with French real-estate firm OGIC, sustainable development firm e-Green, and others to apply behavioral research to redesign an existing apartment building, the Bains Douches Castagnary.

To design the building, the team drew on a growing body of work on behavioral science and the built environment: how architecture influences our well-being, emotions, and behavior. They interviewed residents of the area to better understand what was important to them—such as being in community, and respecting the environment—and how they sometimes struggled to fulfill these goals. They then designed, field tested, and ultimately implemented these ideas throughout the building.

In 2019, residents started moving in, and the team is measuring the impact of their interventions over the next decade, to see whether they have been successful or not.

By now, you've probably gotten a sense of just how broad the application of behavioral science has become, from helping young mothers at risk of

homelessness to dark patterns used to manipulate customers in troubling ways to an entire building designed around behavioral economics principles to help its occupants live healthy lives. In this chapter, we'll review scope of behavioral science around the world—the good and the bad—thanks to a new survey of behavioral teams.

If you already have a team in place, this chapter can help you learn from what other likeminded groups are doing. If you don't have a team in place, this chapter and the next one are meant to get you started on the career path. The book thus far has focused on the *process* of applied behavioral science; the next two chapters focus on the *organizational structure* that enables applied behavioral science.

What We Did: A Global Survey of Behavioral Teams

Eight years ago, only a small set of finance, health care, and Silicon Valley technology companies were applying behavioral science in their work. Now, the picture is quite different. At least four hundred companies and nonprofits organizations now have behavioral scientists on staff, in addition to long-standing groups of traditional psychologists working in the private sector. These teams are diverse. For example:

- A 12-person behavioral analytics and experimentation group at Uber, which studied the transportation behavior of millions of people
- Two powerhouses of data-driven international development, J-PAL and IPA, applying behavioral science to the everyday problems of sanitation and health and safety all around the world
- Numerous single-person behavioral consultancies, especially within the realm of marketing
- Google's People Analytics group, including in-house behavioral scientists working to improve employee benefits and well-being

To better understand the range of teams out there, and their experiences, I helped organize the largest known survey of behavioral science teams in the world. We received detailed responses from more than two hundred distinct organizations across 54 countries ranging from the United States to Kenya to Saudi Arabia. In this chapter, we'll dive into the results to better understand the current state of the field.

The Behavioral Teams survey was a joint project between two nonprofit organizations in the field—the Behavioral Science Policy Association (BSPA) and the Action Design Network (ADN)—and myself. Starting in June 2019, we drafted the initial survey and then tuned and distributed it in the field. We promoted it through our own direct contacts, social media, and email lists in the industry. The survey launched on July 23, 2019; the last response included here was entered on December 23, 2019. The survey was written and promoted only in English.

The survey's target population was people who work on teams applying behavioral science to the development of products, communications, or policies, in other words, the same audience as for this book, plus policymakers.

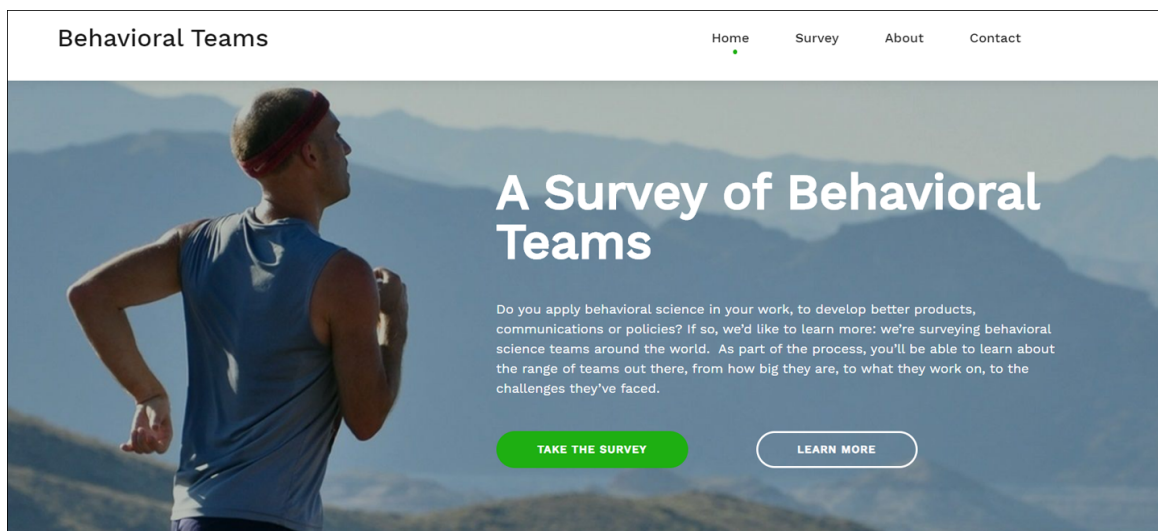


Figure 16-1. The *Behavioral Teams* home page, inviting people to participate in the international survey

The survey consisted of three main sections:

- Section 1: Contact information and basic information about the team, to support a public directory of behavioral science teams. This includes the type of organization (company, nonprofit, academic, or government organization), the number of people on the team, their primary location, and their formal training in behavioral science (if any).
- Section 2: Questions on the work of the team, such as the types of behaviors they seek to change, the techniques they use, their manner of validating results, and the roles of each member on the team.
- Section 3: Questions about the challenges and successes of each team.

An archived copy of the survey, and the public directory of teams around the world, can be found at www.behavioralteams.com.

After cleaning the data and removing invalid responses, the resulting dataset provides detailed information about 231 organizations, across 253 individual respondents.²

Currently, there is no exhaustive list of organizations applying behavioral science to their work. Thus, it's not feasible to conduct a statistically representative survey of the field or to know the precise coverage of the survey relative to the entire field. However, two other, independently created lists of behavioral science teams can help us estimate that coverage. Ingrid Melvær Paulin, senior behavioral scientist at Rally Health, and Faisal Naru, head of strategic management and coordination at the executive director's office of the OECD, each maintain a list of organizations, focusing particularly on private-sector companies and on government organizations, respectively. We combined the organizations from each list and found that the resulting directory covered 529 unique organizations, approximately 44% of whom responded to the detailed Behavioral Teams survey.³

In the following sections, we'll offer the most comprehensive look yet at the scope of the field (based on the directory of organizations) and detailed look at the makeup, tactics, and operations of these teams (based on the survey).

Who's Out There?

Using the directory of behavioral science teams mentioned before, we find that teams applying behavioral science to the development of products, communications, and policies are heavily concentrated in three countries (among the 526 teams with known locations): the United States (217), the United Kingdom (77), and the Netherlands (30).

Many of these companies are international, with offices all around the world: for example, Walmart, Coca Cola, and Ipsos all have behavioral science teams and a global presence. For survey respondents, we asked them to provide the location of their behavioral science team. Where no location was given (e.g., for data pulled from other nonsurvey lists of organizations), we used the company's headquarters location.

While behavioral science took off in the last two decades in the US and UK, together they hold only a slim majority of the teams out there. Today, the range of teams includes:

- The Busara Center, based in Kenya and with offices in eight countries, which combines academic research with consulting for major companies across the developing world.
- Nudge Rio, a small behavioral science unit in the provincial government of Rio, Brazil.
- The Reinsurance Group of America, pioneering a “**Behavioral Approach to Insurance**”.⁴

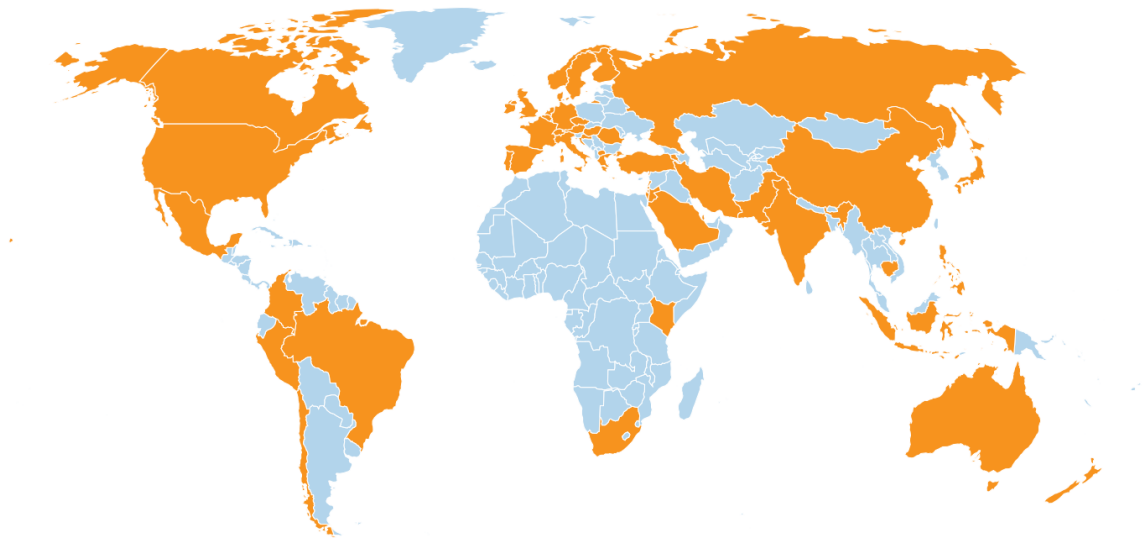


Figure 16-2. The distribution of behavioral science teams around the world

What types of teams are there? The majority of behavioral teams are within companies—333 of them, in fact (64%, $N = 520$ with known type); 81 different academic institutions have been identified, as well as 60 government institutions and 46 nonprofit organizations.

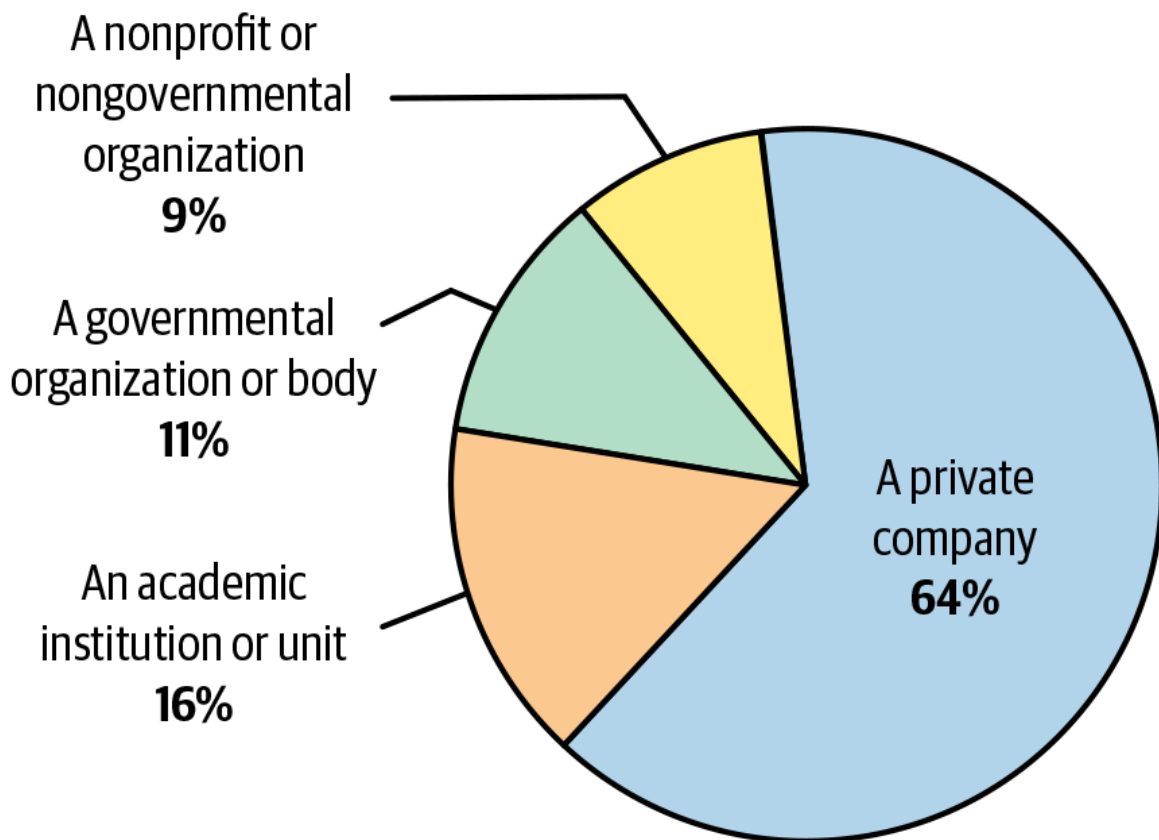


Figure 16-3. The proportion of behavioral teams, by type

Relative to the directory of organizations, the Behavioral Teams survey had an overrepresentation of US-based teams (43% of respondents, versus 41% in the directory), an underrepresentation of teams in the UK (8% of respondents, versus 15% in the directory),⁵ and an underrepresentation among private companies (58% of respondents versus 64% in the directory).

For the rest of the analysis, we will focus on the respondents to the Behavioral Teams survey since that is where we have detailed data. However, we should keep in mind that the broader field is at least twice this size, with a strong concentration in government organizations not covered here.

This survey best represents behavioral teams in companies and nonprofit organizations, and unless otherwise noted, we'll restrict the analysis to the primary respondent⁶ at each company or nonprofit organization: 161 out of the 229 organizations that completed the survey with a known organizational type. Putting this in the context of the combined worldwide list, 42% of all known companies or nonprofits with behavioral teams completed the survey.

SPANISH RESOURCES FOR APPLIED BEHAVIORAL SCIENCE

Is your native language Spanish, and you're looking to learn more about designing for behavior change? The **Instituto Mexicano de Economía del Compartamiento** is a great place to start. Based in Mexico City, the nonprofit organization was founded in 2015 and offers an extensive collection of free resources in Spanish, including a three-volume guide to behavioral economics in public policy, consumer behavior, and behavior finance (more than 750 pages in all), a directory of behavioral science university programs, a guide to major books and TED talks in the field, and more. They run a behavioral science summer school and courses in behavior design and have taught more than seven hundred students already. It's an impressive organization.

Thus far, the vast majority of training programs, books, and events in the applied behavioral science community have been in English. Thanks to Instituto Mexicano de Economía del Compartamiento, and groups like it, that's finally starting to change.

Where the Interest Lies

The Behavioral Teams survey focused on dedicated groups of behavioral scientists (or other behavioral designers) within organizations. That's an important segment to understand, but we can see hints that the interest in and application of behavioral science is actually much broader. Let's take a look at these dedicated teams and the potential landscape beyond them in turn.

The Dedicated Teams

How big are dedicated behavioral teams within companies and nonprofit organizations? There is a wide variety, but most of the respondents came from small teams: median team size is four; the largest team was under two

hundred in our survey (N = 153). Well over half (59%) of these organizations say that behavior change is explicitly part of the organization's goals and mission—often because the behavioral change team *is* the organization. For example, many small behavioral science-focused consultancies have popped up over the years, from **The Behaviorist** in Canada to **Habitude** in Spain.

How big is the field? The respondents represented teams with 1,216 members and indicated that another 815 individuals applied behavioral science on other teams within their companies.⁷ Combining these figures and assuming that the survey represented 42% of the worldwide total (see above), we can very roughly estimate the total worldwide employment within companies and nonprofits. These teams, specifically identified as applying behavioral science, appear to employ roughly 4,840 people.

That number may feel surprisingly low, given the high-profile teams at Walmart, Pepsi, and other major brands. However, we should be wary of generalizing from these teams, if nothing else because of the availability heuristic. Even within these companies, the teams are generally small. Further, the largest dedicated behavioral teams in the world, including the Behavioural Insights Team in the UK and ideas42 in the US, employ fewer than two hundred people each.⁸ The largest-known development agencies focusing on applied behavioral science, the Abdul Latif Jameel Poverty Action Lab (J-PAL) at MIT and Innovations for Poverty Action (IPA) at Northwestern and Yale, are not much larger—and many of their staff members are not directly applying behavioral science in a meaningful way or are academic professors.⁹

Given the newness of the field, this isn't too surprising. The following graph shows when each of the behavioral teams started. With the exception of a few pioneers in the field like Paul Slovic's Decision Research in 1973,¹⁰ the real growth only started in 2013; 2% of teams started before the year 2000 and 87% started on or after 2013.

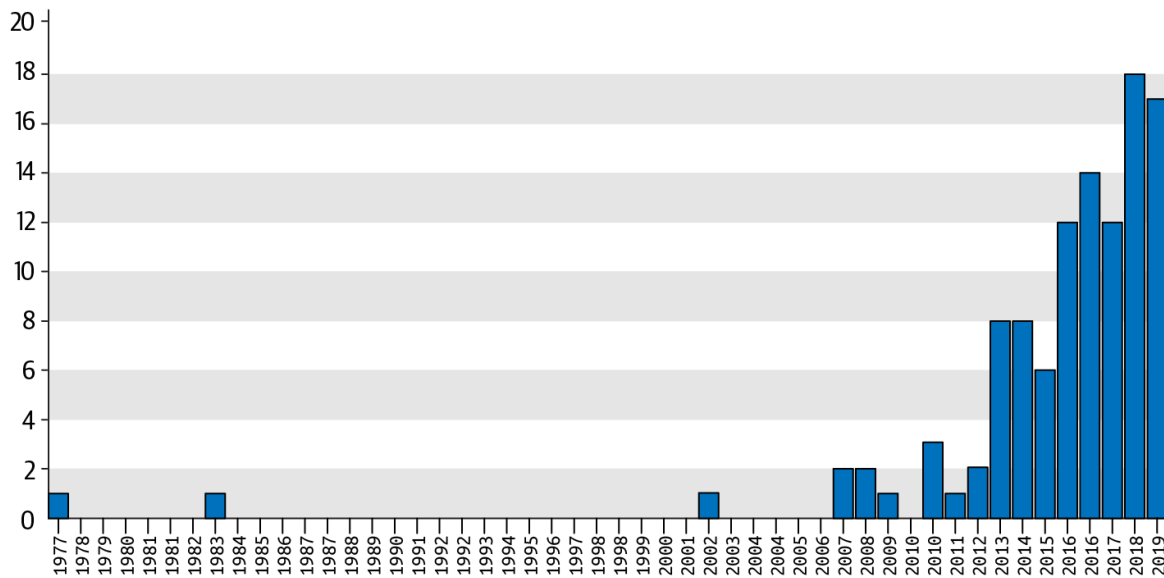


Figure 16-4. Starting date of behavioral teams

That said, we can expect continued new entrants, and growth among existing teams. In the next year, the median behavioral team expects to expand by 25% (average increase is 52%), which if it held true would entail a growth of 1,208–2,515 roles next year. We should take such projected hiring with a large dollop of salt, but nevertheless even these optimistic numbers would result in a larger but generally small field. To put these numbers in perspective, there are roughly 200,000 psychologists in the US alone.

The Nondedicated Teams

The dedicated teams covered by the survey should be thought of as a small portion of the total population of people interested in applying behavioral science to their work. A few anecdotes can help us make this distinction between dedicated teams, and broader interested population, clear. The Action Design Network, a small nonprofit organization that I founded in 2013 to promote behavioral science, has more than 16,000 people signed up for our events around North America. Similarly, the first edition of *Designing for Behavior Change* sold many times more than our estimate of total employment in the field. And to be frank, this book is for an interested practitioner audience; it's not something that a general audience would normally buy (sadly, for my publisher!). Nir Eyal's first book, *Hooked*,

which could appeal to a somewhat larger audience but still was squarely focused on the psychology of product development, was picked up by approximately 300,000 people.¹¹

It is likely within the broader design, product management, marketing, and to a lesser extent, human resources communities where the interest lies. These communities are huge, with over half a million graphic designers alone.¹²

There are thus two significant gaps: the number of people actively interested in applying behavioral science to product and communications development is much larger than formal employment in the sector. Similarly, the field of people who could be interested (other designers, product managers, etc.) is much larger than those who have expressed an interest (at least according to these anecdotal numbers).

What does this mean for someone looking to enter the field? The short lesson is that joining an existing, dedicated team will be difficult. Instead, one should look at either starting a new behavioral practice within a company or applying these lessons as part of one's larger work, especially as a product manager, designer, or marketer.

With that, let's look at how these teams get started and what we can learn from them.

A Broad Range of Application

We're now seeing behavioral science teams around the world and in a wide range of disciplines. Let's look next at how these teams are formed, their business model, placement within their organizations, and the specific behavioral problems they are looking to solve.

Origins

How do behavioral teams get started? There's no single path. Our respondents described a mix of bottom-up and top-down approaches—from

starting a new small company specifically geared toward behavioral science (29%) to a CEO or department head driving it (22%, 18%) to individual contributors making it part of their work and growing from there (17%). What was *uncommon*, however, was someone outside the company convincing the company to start one (3%); that is, behavioral (organization) change comes from within.¹³ That is how it happened in my case, at both HelloWallet and Morningstar; I was already an employee of each company and started our teams from within.

Business Model

Among the 161 dedicated teams in companies and nonprofits who filled out the survey (and among the 379 organizations in the overall directory that we identified as such, using web searches and prior knowledge), there is a considerable diversity. Roughly speaking, though, we can divide them into two big categories: consulting companies (28%) and companies that apply behavioral approaches to their own products and services (72%). The vast majority of employment in the field, according to our survey at least, is in consulting, specifically in consulting companies in the United States, the United Kingdom, and the Netherlands. Three of the top five largest teams in the directory are all nonprofit consulting organizations: the UK's Behavioural Insights Team, ideas42 (in the US), and the Busara Center (based in Kenya).

Placement

In terms of where teams are located within the organization, and putting aside those who are in external consulting (33%), the most common placement was data science (26%), followed by product (20%), design (18%), and marketing (14%).

When it comes to the individuals on the team, 52% said that they had a formal degree in behavioral science ($N = 155$). Among the others, 85% learned through books, 80% on the job, or through formal coursework

(41%) or informal online classes (59%) that did not result in a degree in the field.

Focus Area

What types of behavior do these teams seek to influence? Some teams are focused on particular outcomes for the individual—the most common being financial behavior like saving, spending, and investing (57%), health behaviors (49%), education (42%), and energy use (36%). Many also spent time on company-driven outcomes of product usage (60%) and sales (51%). Respondents selected all that applied to them, and many of the companies and nonprofits in the sample consult for a range of clients.

The teams use a range of techniques, but at 83%, by far the winner is social influence (social norms, social proof, etc.). The next most popular item was directing attention (79%) and shaping the choice set (78%). The often-discussed approach of forming habits was used by 62% of respondents (see [Figure 16-5](#)).

In most cases, the target audience for these interventions *did not know about them*—something that can raise ethical red flags, especially where the behavior is directed for the organization's benefit rather than that of the individual; 40% of respondents said that virtually no users know about the use of the behavioral interventions, 20% said a few do, and only 20% said most people did or everyone did (20%) ($N = 143$).

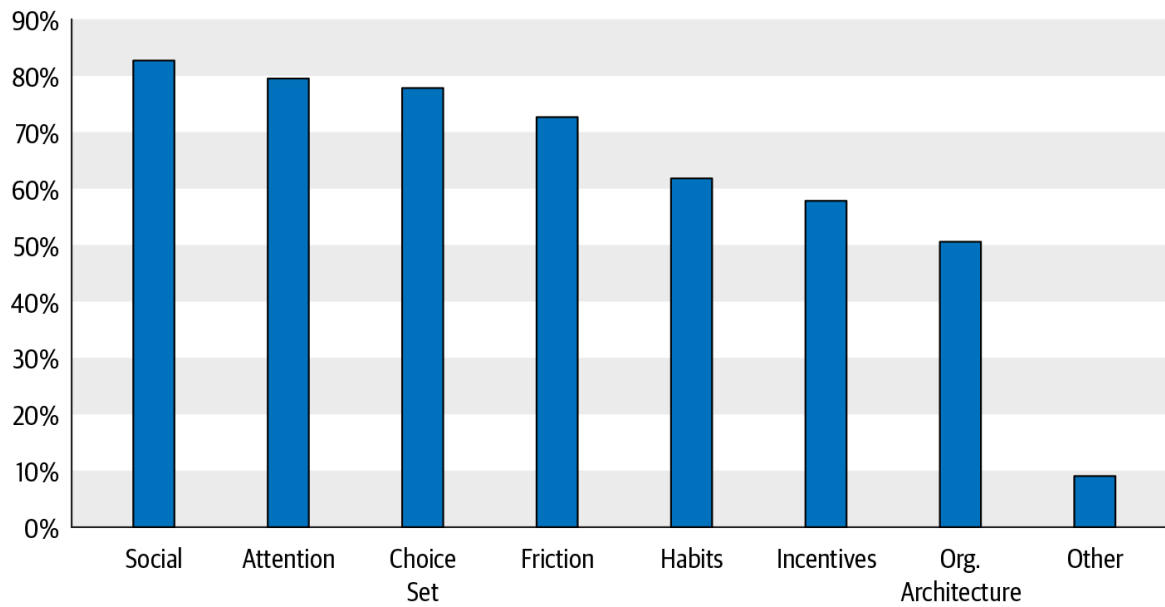


Figure 16-5. Techniques used by behavioral teams (respondents could choose more than one option)

Respondents reflected upon how important various aspects of their work were. Direct behavior change, not surprisingly, was consistently most important, as was sharing the results internally. Most teams did not value (or perhaps did not have the opportunity) to share their results externally or to seek to influence policy. Again, this analysis is limited to the companies and nonprofits in the sample: the picture for academics and government agencies would be quite different.

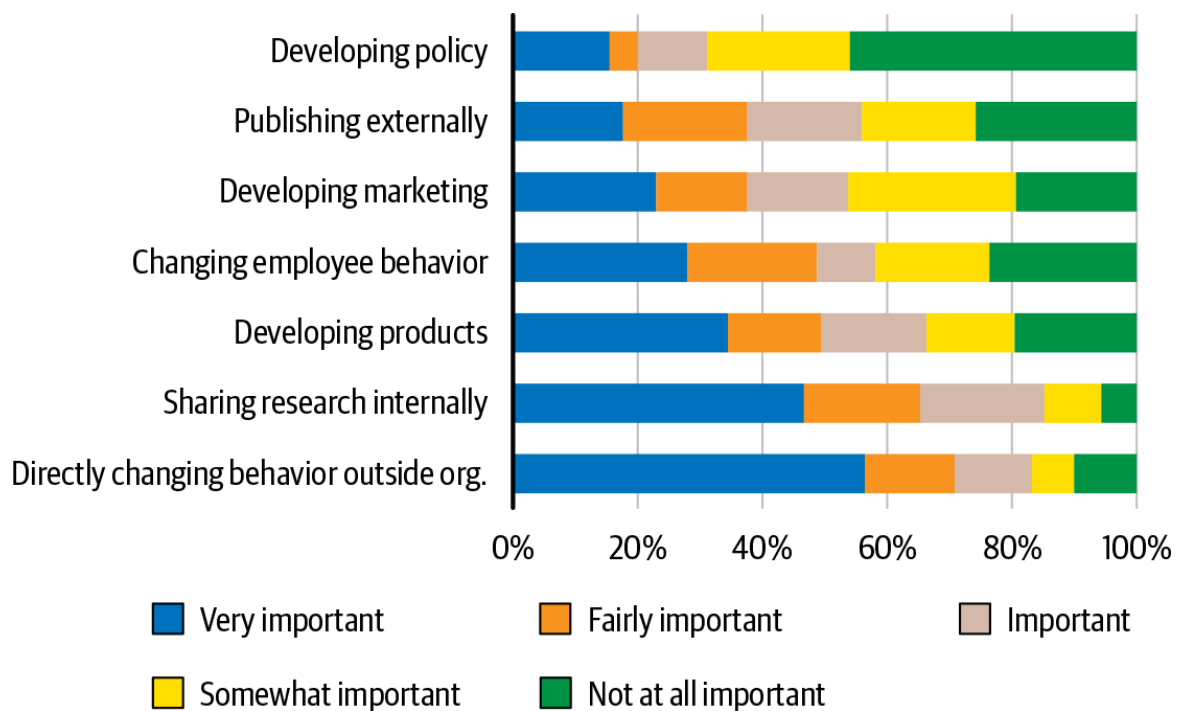


Figure 16-6. How important is each activity to the teams?

The Challenges

The field is growing rapidly—both in area of application, in geography, and in the number of organizations with behavioral teams. There are three main problems facing the field, however: practical problems of setting up and running a team, replication, and ethics. We already discussed ethical challenges in detail in [Chapter 4](#). Let's look at the other two challenges now.

The Practical Challenges of Running a Team

The single biggest challenge that teams faced was getting their interventions implemented in practice (43%), or measuring their impact (41%); generating ideas for interventions was rarely a problem (11%). In the comments and subsequent interviews for this book, respondents similarly mentioned that some of their key challenges were implementation and impact measurements.

The challenges of implementation appear to come from two main sources. First, I've often heard behavioralists who serve as external consultants complain that after they write their report, they move on to another project and they doubt that the client ever implements it. And, the same complaint from internal consultants, except they *know* their advice wasn't really taken. The other reason implementation is a problem, based on conversations with peers in the field, stems from the normal challenges of product development: many good ideas simply aren't put into practice, whether they are behaviorally informed or not. There isn't space on the road map, there isn't buy-in at the appropriate levels of the company, etc.

Ironically, the opposite problem also occurs: companies rush to implement without measuring impact. Numerous interviewees among the survey respondents talked about their clients or their companies acting too quickly. Once the team had presented an idea, the response was "OK, well, let's do to it then—why would we waste time with the test?" This is an issue we've faced at Morningstar as well, and other researchers in the field have written about it too:¹⁴ it's difficult to simultaneously say that you have a potential solution to a known problem and that you're not sure it would work. Stakeholders simply aren't used to hearing from their experts that an approach might not work! The reality is that all solutions, derived from behavioral science or not, could fail to have an effect or, worse, backfire. It's just that behavioral teams are generally more comfortable saying so, and that can be misinterpreted. These impact measurements are a vital part of behavioral science and point us to the next problem: the replication crisis.

The Replication Crisis in Science

Across many fields of science—especially psychology and medicine—we are experiencing a decade-long crisis of replication. Some of the most famous early studies in the fields, and many others, have been found not to replicate—i.e., subsequent researchers attempted to replicate the results of the initial authors, under the same circumstances, and were unable to. In many cases, the later researchers found results that were either not

statistically meaningful or significantly smaller than had been previously reported. John Ioannidis, clinical researcher and meta-scientist (a scientist who studies science), summed up the problem pointedly and succinctly in the title of his 2005 paper, “Why Most Published Research Findings Are False.”

Since then, a wide range of studies from marketing to sports science have failed to replicate. Behavioral science, which draws heavily on psychology, is not immune to this problem. Some prominent psychological studies used in the field that were later undermined include:¹⁵

- The resource model of willpower (ego-depletion), popularized by Roy Baumeister, in which engaging in a difficult task would make you more likely to give in to subsequent temptation.
- The use of subtle primes on behavior, studied by Bargh and others. For example, exposure to words related to old age would (supposedly) make participants walk more slowly, without their knowledge.
- The power pose associated with researcher Amy Cuddy, in which a robust stance would increase testosterone and cortisol, and lead to riskier behavior.

Despite popular TED talks, books on the topics, and literally hundreds of research studies in these areas alone, follow-up analyses found that they simply could not be trusted. Estimates vary, but regularly one sees meta-analyses (studies of multiple research studies in the same research domain) report that 20%–40% of the studies did not replicate as the original authors had published.

All of this can and should be concerning—not simply to social scientists and other researchers but to us as we try to apply behavioral research to product development and to the broader public whose lives and are in many ways shaped by the results of these studies.

The replication crisis is, however, a healthy and good thing. It is the process of identifying and removing rot from our base of knowledge, and the

alternative is far worse. The alternative is simply to be blind to our ignorance.

As a researcher, the failure to replicate so many prominent studies tells us that to truly advance knowledge is extraordinarily difficult. And, in a behavioral science context, that behavior change is likewise difficult. It's not that researchers design foolish and ineffective interventions (though some do, certainly)—rather it is that the research process, and especially the replication process, exposes which interventions are solid and which ones aren't. Without that process, we'd still have just as many bad, ineffective interventions; we just wouldn't know which ones are which. The history of medicine is rife with examples of treatments that either did nothing or did more harm than good (like trepanation and bloodletting), as is international development (like the example of microfinance in [Chapter 15](#)). Without rigorous measurement and replication, we'd still have these scourges.

And so, what do we do as people applying behavioral science? First, we look to apply research that is well-founded. That's what I've tried to do throughout this book, though certainly some interventions we've talked about will later be shown to be less effective than hoped—that's to be expected. Second, we measure and test ourselves. We perform our own replications. Every time we run an A/B test or other rigorous measurement of impact for an intervention, we replicate the results of others. And more importantly, we gain confidence that we've found something that is effective for the particular context of our products and our users. This is one of the reasons why impact measurement isn't optional when designing for behavior change: it's an absolutely essential part of the process, even if it isn't as exciting and interesting as the interventions themselves.

Finally, we can also seek to learn from bad research. Where did prominent studies fail to replicate? Where there were small impacts, small numbers of people, with incomplete randomization or no experimental control at all, where null results were ignored or not published, and where the results appeared to be “too good to be true.”¹⁶ We've tried to address many of these issues in [Chapter 13](#) by focusing our attention on impacts that are practically meaningful, and getting enough people in an impact

measurement to assess that impact; by harping on the power of experiments versus pre-post tests and other less rigorous techniques; and by setting rules for when we can use a null result to stop any further analysis.

That brings us back to the Behavioral Teams survey and Chapters 13 and 14, and how behavioralists in the field measure impact. Given the challenges in implementation and measuring impact, it is noteworthy that 70% of respondents said their teams measured their success in terms of A/B tests or other forms of RCT. We should be cautious: the median number of experiments the teams conducted in the last twelve months was only four. While many of the teams are relatively new, that indicates either that A/B tests are not as widely spread as the response might indicate or that the teams have implemented them sparingly.

In addition to RCTs, 71% used pre-post analyses; 50% looked for direct feedback from users to gauge the effectiveness of behavioral interventions: two technique that can be immensely valuable to gain understanding about *why* an intervention worked or didn't, but often aren't up to the task of measuring the impact itself effectively. Interesting, 25% used statistical or machine learning techniques (beyond A/B tests). We'll discuss the integration of statistical methods and machine learning and behavioral science in the next chapter, keeping in mind that this combination is still relatively new and not widespread.

Putting It into Practice

Here's what you need to know:

- The field of applied behavioral science has grown rapidly since 2013: 87% of teams started in the last seven years.
- While the US and the UK still hold a majority of teams and jobs, that is changing rapidly as new teams across the globe are proliferating.

- Much of the action is occurring in consulting companies, both nonprofit and for-profit.
- Current teams especially struggle with implementing their interventions and measuring impact.
- While diverse techniques are used, the most common approach teams use are social nudges.

If you want to start your own team in a company or nonprofit, look for places where behavioral science adds to the value of existing roles, and not where behavioral science would be a role onto itself; the former are far more common. The roles, often, are in consulting for external clients, product design and management, research and analytics, and marketing. In the next chapter, we look in more detail at the types of people and “pitch” you can use to start the team.

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- 1 This case study comes from a phone interview and subsequent email exchanges with Scott Young, head of the BVA Nudge Unit in London. Find more information about the Nudge Building at Singler (2018).
 - 2 Here is the data cleaning procedure. Respondents were first asked whether their teams fit the desired criteria and were offered a copy of the resulting report even if they did not meet those criteria (i.e., removing an incentive to complete the survey with invalid data). Roughly 5% of respondents filtered themselves out at this stage. Second, responses with fewer than 25 values (answers to questions or subquestions) were filtered out. Third, multiple responses by the same individual were grouped and only the most complete set were kept. Finally, multiple responses by individuals at the same organization were grouped and in sections 1 and 2 above, only the most senior individual at that company (by title) was included; in section 3, the responses are about an individual’s perspective, and all valid responses were included. The resulting dataset had 231 organizations across 253 respondents. On the survey, respondents indicated the type of organization their team belonged to; we determined that the Other and Independent Research Organization options were interpreted differently by respondents, and we manually mapped these responses to the other options: company, nonprofit, government, or academic.
 - 3 From the OECD list, we used the handful of nongovernmental and private organizations (since the government agencies were often clients of existing teams without a team of their own), and we manually verified that each had a team.
 - 4 See also “[Behavioral Science and Insurance](#)” by the Reinsurance Group of America.
 - 5 While the survey was written in English, it does not appear that there was significantly less coverage in countries where English is not an official language. In other words, relative to the

directory, survey responses do not appear strongly biased by country. However, the process of creating the directory itself could have been biased toward English-speaking countries or groups and there is no clear external standard by which to measure that problem, if it exists. My thanks the Anne-Marie Léger for raising the issue.

- 6 See prior footnote on data cleaning. In some cases, multiple respondents from the same organization completed the survey. Unless otherwise noted, we use the primary respondent.
- 7 After removing entries that were clearly inaccurate and verified as such manually.
- 8 As of October 24, 2019, the UK's [Behavioural Insights Team](#) lists 181 employees, only a portion of whom are actually applying behavioral science in their work, and [ideas42](#) lists 126 employees.
- 9 As of October 24, 2019, the J-PAL listed 294 employees worldwide, including academic professors, grant writers, and so on.
- 10 These early pioneers are in fact, real, based on manual verification of the underlying data.
- 11 Statistic provided by Nir Eyal.
- 12 For estimates on the size of the graphic-design community, see [this report from IBISWorld](#). As of January 2020, they placed the field at 534,680 people.
- 13 In the survey, 12% of respondents replied Other and did not provide information that readily fit these categories.
- 14 For example, Wallaert (2019)
- 15 Willpower: [Engber \(2016\)](#); priming: [Doyen et al. \(2012\)](#); power pose: [Dominus \(2017\)](#). There are many resources online about the replication crisis; one list of problematic studies in psychology can be found at [Jarrett \(2016\)](#).
- 16 See the discussion in [Resnick \(2018\)](#).

Chapter 17. What You'll Need for Your Team

Over the past ten years, a range of researchers across the Inter-American Development Bank (IADB) have applied behavioral lessons in their development projects or conducted field studies of their own. Starting in 2015, the IADB wanted to have a more coordinated approach, however.

In working with the bank leadership, Carlos Scartascini created a unified group to support and grow the diffuse efforts already in place. The core team composed of fellows and visiting scholars provides training for government officials in member countries and for IADB specialists (they've trained more than two hundred staff members thus far), disseminates best practices, and consults on individual projects as needed.

Beyond the central team, they work with more than 20 dedicated behavioralists in other departments and numerous others who apply behavioral science across the bank, from education to sanitation. Nearly every department has someone in-house who combines their area of expertise with some knowledge of behavioral science for development.

For the IADB, with \$1 billion lent for development projects each year, this structure helps them manage the massive scope and diversity of their work. It allows them to combine deep knowledge within each department, on the specific needs of healthcare or infrastructure projects, for example, with knowledge of the latest findings and techniques in behavioral science.

Behavioral science teams don't have a single design or structure; they often grow organically out of existing programs and departments, where people in those departments find that behavioral science can aid their work. A few large groups, like at the IADB, have evolved into a distributed model of central support and diffuse practitioners. Here, we'll look at what it takes to build your own team, in your company.

From What They've Done to What You'll Do

Let's dive into how you can get involved, especially what's needed to start applying behavioral science at your company. As we learned from the Behavioral Teams survey, despite the rapid growth in the field, the best job and career advice for people looking to enter the field is to build up from within your existing organization.

Clearly, you'll need a set of tools to think about and actually use behavioral science in products and communications. That's what the rest of this book covers. What else is needed? Let's look at two areas in particular: a strong case for other stakeholders within the company, and the right people and skills for the job.

Making the Case

The best way to argue for the value of designing for behavior change isn't to argue at all—but rather to demonstrate it. As Matt Wallaert nicely puts it: “Ideally, you wouldn't talk about behavioral science at all in the beginning. Instead, you'd do your job really well for a year, earn respect, and then tear off the mask and say ‘Ta-da! It's because of behavioral science!’”¹

In other words, do as many individuals in the field are already doing: read the books, take the online courses, do a formal program if you can. But first and foremost, don't look for “behavioral science jobs.” Instead, look for how behavioral science can improve the career you already have: how it can improve your design practice (like design companies like Mad*Pow have done), how it can provide better consulting solutions to your clients (as with Saudi Arabia's first unit—see sidebar), or how it can improve your product development.

Sometimes you don't have the luxury to build behavioral science into your work on your own, or you've started to, but need additional resources to grow and expand that effort. At that point, it's time to make the business case for behavioral science.

THE FIRST BEHAVIORAL SCIENCE TEAM IN SAUDI ARABIA

Wiam Hasanain started the behavioral practice at a Saudi Arabian consulting company in 2015. She was a partner at the firm, focusing on social benefit projects. In addition to her professional work, she decided to do an executive master's in behavioral science at the London School of Economics to expand her skills. As she did, "things started to align perfectly: with my master's and with a particular client who wanted to pioneer the first behavioral unit in Saudi." From there, she proved the value and has been able to build out a new capacity and team within the company. The company's core competency was marketing and communications, and Wiam found that she could weave behavioral science into her work: building on the base she had, to innovate and do more for her clients and career.²

Thinking Through the Business Model

Helping users voluntarily change their behavior can directly support the core business goals of a company, but these types of products do raise special business considerations.

Recall from the **Preface** that there are two main (ethical) targets for behavior change:

- Behaviors that users want to change *within their daily lives*
- Behaviors that occur *within the product itself* and are required to use the product

In the latter group, the business impact is clear and straightforward. Let's say the behavior change task is learning how to better organize email in an email client. If you help users do this more effectively, they are more likely to (a) buy later editions of the product in the future (or renew, if the product is on a subscription model) and (b) recommend it to others. In both cases, behavior change means more revenue.

For behaviors in the user's daily life, the company's business model will depend on whether the behavior is frequent or one-time. Let's say the target behavior is exercising: it's repeated. Standard business models work perfectly well for this type of product and align user success with business success:

- *Revenue increases with the time users spend using the product*
 - *Subscriptions and reoccurring fees.* For example, a product that helps people set, track, and compete with friends over exercise goals. If the product is successful, both exercising and revenues are sustained. A similar model exists for products that automate user tasks altogether—like target date funds in the investment space, which charge an ongoing fee for automating the process of asset allocation.
 - *Advertising.* For example, the same exercising tracking/competition product could be free to the user but advertise fitness products. If the product is successful, it continues to deliver an interested, engaged audience for advertising.
- *Revenue increases with new sales*
 - *Market penetration.* For example, an exercise tracker from Fitbit. People don't really need two exercise trackers. But if the product is successful at changing behavior, then current customers refer others, and sales grow. If the market becomes saturated, that's a nice problem to have, and the company can move on to new product lines (up-sell).
 - *Cross-sell and up-sell.* For example, the company has multiple exercise products (a tracker, shoes, clothing, etc.). If one product is successful at changing behavior,

then the customer is more inclined to buy other products from the company.

- *Revenue increases with the success of users at changing their behavior*
 - *Incentive-aligned third-party support.* For example, a third-party benefit from the same thing that users benefit from: behavior change. The third party would then pay for the product on the user's behalf. Employers benefit from their employees being physically healthy and pay for products like Keas to help employees exercise. Energy companies benefit from their customers being energy efficient, and pay for products like Oracle's Opower.

Again, these are same business models as any other product—plus the somewhat unusual option of third-party support.

Products that change a one-time or infrequent behavior have a more difficult time aligning user success with business success. Consider, for example, getting a mortgage, and products that help people shop for a better one. Thankfully it's not something that most users do very often. Standard business models become problematic:

- *Revenue increases with the time users spend using the product*
 - *Usually not relevant.* Customers just want to find a good mortgage; they won't pay more if it takes longer to find one.
- *Revenue increases with new sales*
 - *Cross-sell and up-sell.* This is feasible if the company offers related services—like banks that offer mortgages and checking accounts. However, customers have a hard time assessing the quality of one-time services like mortgage-buying assistance; that uncertainty leaves room

for chicanery. In the mortgage market, there's a significant temptation (financial incentive) to shift from providing unbiased decision-making support to pushing mortgages that generate high, lead-generation fees.

- *Market penetration.* Generating new sales based on high-quality decision support for mortgages is possible. However, because customers have difficulty assessing quality, the same temptations exist for the companies that seek to “help” users make that decision. Hence, businesses use gimmicks that made customers excited up front but leave them loaded with significant fees and risk, like balloon mortgages.
- *Revenue increases with the success of users at changing their behavior*
 - *Incentive-aligned third-party support.* A one-time change in behavior would need to be very significant to attract (incentive-aligned) third-party support. In the case of decision-making support, Fannie Mae and Freddie Mac have appropriate incentives for end users to receive optimal mortgages, but few other entities have incentives aligned with the end user for such a product.

And, of course, there are behavior change business models that clearly are not aligned with the interests of the user. Gyms are the classic example of businesses that profit because their users fail to change their behavior. Many gyms rely on fees paid by users who plan to exercise but never do.

The Skills and People You Need

From the Behavioral Teams survey, we learned that there's no single path into applied behavioral science. Recall that 52% of respondents in the private sector and nonprofits had a formal degree in the field; 80% of the

others had learned some of their skills on the job. However, having the *appropriate skills on hand* isn't optional. You can think about them as falling into three categories.

Skillset 1: The Non-Behavioral Basics

While some behavioral teams are centrally located centers of excellence, according to our survey, the vast majority aren't; they are embedded in product, design, marketing, analytics, or other functions. And, for these groups, the first skills that are needed are those used in the core work of the team. If you're applying behavioral science to product development, that means design or product management, etc. If you're working on communications and marketing, it means knowing communications and marketing.

Behavioral science isn't a substitute to these core disciplines; it's an added skill and understanding of the mind. I've seen a number of behavioralists, especially newly minted behavioral scientists coming out of school and without a professional background, act as if they are designers or marketers simply because their skills can *help* design or marketing. It doesn't end well for them: without a foundation in that specific field, they make mistakes that are obvious and avoidable to anyone trained in the field. Worse, they can cause an allergic reaction in their departments by trying to take over someone else's job (who is trained), they end up being locked out and distrusted.

The potential exception is consulting. When a consultant is hired into a company or department, they have a certain license to give advice, but good consultants know to listen first, and learn how to adapt their language and approach to the context. And even then, behavioral science alone isn't a skillset that makes one a good consultant: instead, it can help consultants understand and solve behavioral problems, after they already know how to work with clients.

Skillset 2: Impact Assessment

As I've argued throughout this book, and as casual observation has probably told you too: people are really complex. Human behavior is really complex. The interaction between people's environment and their behavior is really complex. And for all of these reasons, we can't assume *anything* we do to change behavior—whether informed by behavioral science, design thinking, or meditating on a mountaintop for great ideas—is going to have its desired effect.

To be honest to ourselves, to be honest to our clients and to stakeholders across our organizations, we need to rigorously assess causal impact. It's an important set of skills that generally *isn't* covered in Skillset 1 (domain expertise in marketing, product development, design, etc.).

Where can one learn impact assessment? I'm afraid this is one of the few things that isn't easily learned on the job: there's a foundation of statistical knowledge that one needs that comes from careful study, usually at a university (I do believe in autodidacts—they just aren't as common here).

Note that I didn't say that one needs to know how to do randomized control trials. That's for two reasons. First, while randomized control trials are absolutely and without question the best tool for the job of causal impact assessment, they just aren't always feasible and they are, like everything, **imperfect even when they are feasible**. Sometimes we need to use a statistical model to triangulate causality and a broader statistical skillset is vital. Second, because I've found that if someone has a broad statistical skillset, then they can learn how to analyze (first) and design (second) randomized control trials. The reverse isn't always true.

Even 10 years ago, it used to be very difficult to find solid statistical skills in many organizations. With the rise of data science, that is changing. While data science and behavioral science are really quite different (see next section), data science team members can have the foundation of technical rigor that one can build on and adapt to analyze RCTs.

Skillset 3: A Deep Understanding of the Mind and Its Quirks

Lastly, we come to what most laypeople consider behavioral science: a knowledge of the mind's quirks and of nudges that can affect behavior. In my work, I've found that given the right toolset and a fundamental understanding of the mind, people can discover creative ways to work around behavioral obstacles. Having a list of existing nudges on hand is perhaps the *least* important skill in applying behavioral science; again, every intervention should be tested anyway.

It's where these nudges come from that matters: knowing that people have limitations, they use shortcuts to economize, and these shortcuts can go awry depending on the context of a decision and action. That's why I covered this foundation in Chapters 1–3, instead of offering a list of biases and fun stories about how people make mistakes. The fundamentals covered there are a good start. Further reading in behavioral science that emphasizes models and underlying mechanisms (not cute nudges) can help immensely as well.

These are all skills that can be learned on the job and through individual reading and coursework. However, beware of online courses that focus on what I call a “bag of tricks” approach to behavior science: all biases, no underlying theory that connects them. It's flashy, but won't provide the real direction and understanding required to help your users overcome behavioral obstacles.

What's Not Listed: A PhD

I'm a strong believer that formal study, in particular a PhD, is not required. I do have a PhD myself, and I don't think it's required for this job. A PhD provides many valuable skills, including statistical rigor to assess impact (Skillset 2), and sometimes domain expertise, especially in marketing (Skillset 1). Those are nonnegotiable and your team needs them with or without PhDs. Also, at its best, the training for a PhD provides people with a love of learning, an openness to exploration and being proven wrong, the ability to take messy ideas and shape them into a coherent and predictive theory, and an abiding skepticism about the correctness of any such theory or model. These are also nonnegotiable, I believe they are essential for

doing thoughtful, effective applied work. These traits aren't unique to PhDs, however (or master's degree holders, either, for that matter). Thankfully, smart, curious, and thoughtfully skeptical people can be found in many places—not just in PhD programs.

How You Combine These Skills on a Team

Many behavioral teams consist of a single person, and that person has all of these skills. Betterment's Dan Egan is one such powerhouse. He consults with groups across the company, but in the end, he's playing each of these roles. Other teams, especially those embedded in marketing, product, or design, work side by side with their more traditional and domain-trained colleagues and don't need to have as much of the domain expertise themselves. There doesn't seem to be a single model, nor one that is necessarily better than the others. The skills are necessarily—whether they are direct reports of a central chief behavioral officer or partners across the company doesn't seem to matter as much.

In case you're curious, on our team, we do a mix of both. I currently have 13 people who roll up to me, another 13 who are part of our behavioral “group” (reporting to other managers but dedicated to our projects), and many others that we have the pleasure of consulting with and partnering with on projects on a more ad-hoc basis. Our product work started out very much as pure consulting to help existing product teams; now we have engineers and others working directly with us, for example. While I certainly have lessons from each of the models we've experienced, I think the particulars of a given situation are probably more relevant, to be frank.³ We now have a wonderful team of in-house researchers (PhD and non-PhD), but early on when it was just a few of us, we found it immensely valuable to partner with outside academics to bolster our skills and run projects we otherwise couldn't muster. Let's look at how to do that next.

Getting Help from Outside Researchers

Experimental testing, especially for outcomes that are outside of the product, can be an intimidating endeavor. Believe it or not, academic researchers would probably love to help test your product's impact. Many of them can't be "hired" in a traditional sense—because they have full-time jobs in academic institutions and for professional reasons can't accept consulting contracts. But you can build partnerships of mutual benefit if you have enough users of your product to support a scientifically valid study and know how to navigate the process.

Sound impossible? This happened to me, and it's how I first became deeply involved in applied behavioral science. I worked full-time, and I was studying on the side. I had no internal team to call upon, and no budget. However, I found behavioral researchers who were doing interesting work that related to our product, and I approached them at conferences or cold-emailed them. I was fortunate to find great researchers who wanted to work together to solve our users' problems. Over the years, I learned immensely from these partnerships—and in many ways they helped me become an effective applied behavioral scientist myself.

Here's what I learned through my own experience and subsequently by helping other companies build these research partnerships:

1. *Find researchers in your field.* Start with [Google Scholar](#) to search for the topics you work on, and see whose names are in the most commonly cited articles (the results are sorted by the number of citations). Academic conferences on your topic are another good way to start, but they take more time and energy. The [Behavioral Science Policy Association](#) also seeks to be a central hub, connecting interested companies with interested researchers.
2. *Contact them,* asking for suggestions about academics who might be interested in studying the topic with your user base. In particular, ask who might be interested in testing the impact of interventions. Follow up and contact the suggested researchers; the researchers you initially contacted may be interested, of course.

3. *Discover what they need to benefit their work.* Describe your product and user base. Ask them what they need. Depending on the person, their field, and where they are on the academic career path, that will vary. Here are common options:

- *Access to unique data.* In some disciplines, it is extremely costly and difficult to obtain detailed information about individual users, from demographic information to observed preferences, and especially about changes in behavior over time.
- *Access to a large user base.* The power of scientific tests increases with more users, but it is costly for most researchers to gather a large enough test population on their own. If your product already has them, excellent.
- *Access to funding.* Financial support from companies to academic social science researchers can be highly problematic; it can taint researchers' independence and undermine their ability to publish the results of a study with the company. However, if the company is looking for expert advice, and not academic publication of the results, some researchers certainly can be hired for paid consulting arrangements. Other options include providing grants for research on the topic, and supporting grant proposals submitted by the researchers to third-party grant agencies.

4. *Develop a shared research plan.* With a small, trial project, try working together. Set clear expectations on access to data, funding (if any), staffing (on both sides), and timing of the study and analyses.

5. *Don't try to restrict the results or ideas.* Academic research runs on innovative ideas and the data to back them up. Companies can't lock down the ideas learned from the study—they must be shared

between the company and researchers. Similarly, if there is a whiff of restriction on the publication of negative results, that will undermine the reception of the study and turn away most researchers. Paid consulting arrangements work differently, of course.

6. *Respect the need for specific testing protocols.* In order for a study to be scientifically valid, researchers will have to execute the study according to specific rules. For example, they may require very specific wording for questions asked of users. That's part of the bargain; and, it will often help the company formulate much more solid conclusions than if it had informally developed slapdash surveys.

Thus, if the testing process seems overwhelming, don't despair. Your first option is to look to off-the-shelf testing tools, as discussed in [Chapter 12](#). If those aren't enough, look for professional research partnerships with the academic community. If your company is doing innovative things to help users change their behavior (which of course you are!), researchers may be interested in working with you.

Data Science and Behavioral Science

Do you need data scientists on the team? Are they the same thing as behavioral scientists? I've found that many people aren't clear on the differences between behavioral science and data science. Even within my company, I can't tell you how many times someone has asked me something like, "So, can your team run a report on product usage for us?" Product usage reports are certainly a useful thing to do; they just aren't really in the behavioral science wheelhouse (nor probably for many data scientists either).

So what's the difference between the two, and where is each needed? To understand this, let's take a step back from the discussion of team composition, and look more deeply at the types of questions each practitioner community seeks to answer.

Data science often seeks to understand how something works and *predict* the future. Behavioral science seeks to *change* the future, particularly through changing human behavior. Or as Sarah van Caster puts it in her article “**Data Science Meets Behavioral Science**”:

Data science is the discipline that allows us to analyze the unseen—and with machine learning, it allows us to look at large sets of data and surface patterns, identifying when past performance is indicative of future results. For instance, it lets us forecast what products are most likely to be sold and which customers are most likely to buy. But what if you not only want to understand potential outcomes, what if you want to completely change outcomes, and more specifically, what if you want to change the way in which people behave? Behavioral economics tells us that to make a fundamental change in behavior that will affect the long-term outcome of a process, we must insert an inflection point.

Because of these two different purposes, data scientists and behavioral scientists often use different statistical methods. Data scientists can predict the future very accurately and thoughtfully using variables that are correlated with the outcome of interest. They use regressions, decision trees, neural networks, and such to find hidden relationships between context and outcome.

Behavioral scientists, when possible, use experiments since they are the best tool to measure our ability to *cause* a change in behavior or outcomes. Analyzing experiments, when properly designed, doesn’t require advanced statistics at all: a simple comparison of means (e.g., T-test) is often enough. Behavioral scientists do also use regression and sometimes machine learning techniques, but we do so in the service of understanding the causal relationship between context, behavior, and outcomes.

Because of these two goals—predicting outcomes versus causing changing behavior—data and behavioral scientists also differ in how they use *theory*: an explanation of why something works the way it does. Strictly speaking, one needn’t understand *why* there’s a relationship between A and B in order to accurately predict B, given A. Many data scientists do have a theoretical

understanding of what they study, and that helps them with feature selection and data analysis, but it's not actually required.

In behavioral science, a positive result from an experiment tells us that, at the particular moment of the experiment and for the particular people involved, A caused B. It is our underlying explanation of why that happened (which we seek to test in the experiment) that allows us to try generalize to other situations and contexts. Behavioral theory makes it easier to provide advice on how *other people, in other situations*, could change behavior and outcomes.

Table 17-1. Comparing data science and behavioral science

	Data science	Behavioral science
Theory:	Helpful, but not required	Not meaningful without it
Training:	Computer science or social science	Social science
Number of Jobs:	Many	Few

Leveraging Data Science When Designing for Behavior Change

While data science is distinct and different from behavioral science, the rise of data science and the tools that come with it have opened up new possibilities for understanding, and designing for, behavior change.

The first and most obvious way in which data science can improve behavior change work is that it can help identify behavioral obstacles by analyzing the various paths in a behavioral map (or marketing funnel) to look for blockages. And, similarly, to look for the naturally occurring segments of the population for whom a particular approach might be needed and, after an intervention has been deployed, to analyze its segment-specific impact. Behavioral scientists have *started* to look at these questions with techniques called moderation and mediation analysis, but frankly most work is still very much at the “try this, and see if it works for everyone level.” Data scientists either within the team or helping out from other teams, have a much more developed skillset in that area.

In addition, one of the big limitations of behavior change work—and the experiments that come with it—is that most behavioral changes that we create and measure are short term. There just aren't that many experiments that measure behavioral changes for longer than six months (and most are far shorter). So we really don't know what the long-term impacts of our changes are. Data science and *computational behavioral science* can help solve this: by projecting the effect of behavioral interventions into the future. This projection can be through closed-form equations or simulation models, especially when the individuals are expected to interact with each other and one person's behavior change may affect another person over time.

Do these projections tell us clearly and definitively what will happen in the future, if people use our behavioral products or interventions? No. Actually, they'll often help us see how *we're wrong*. They'll make predictions about future behavior change that are simply unrealistic, and obviously so. Computational behavioral science helps us more fully vet or test what we know about behavior change: where those ideas are already vetted, and we've found that they don't make crazy and unreasonable predictions. They help us model the long-term effects of our products and interventions for our users, for our business, and for society.

Putting It into Practice

If you're not currently on a team that applies behavioral science in product or communications development, understand that official job opportunities in the field are small, at least compared to the people who could be interested. The real opportunity lies in designing for behavior change as part of one's existing career, especially in design, product management, marketing, and HR.

In these areas, behavioral science provides a powerful and unusual skillset that peers in the field are unlikely to have.

Whether for you individually, or if you're looking to start a team, three skills are essential:

- Domain expertise (in product, design, etc.)
- Statistical skills to assess the impact of your efforts on behavior
- A fundamental understanding of how the mind works and makes decisions

The first two generally require formal training, or at least previous experience. The last can be learned on the job, with thoughtful study. This book can help you and other members of your team, as can online courses from Duke University, the University of Toronto, and others.

A PhD is not required—it's one good way, but not the only way—to gain these skills.

The skills and activities of data science are often confused with behavioral science. In short, data science seeks to predict outcomes (whether or not we *cause* them); behavioral science seeks to causally change behavior and outcomes. The two groups can and should work together—but respecting the different background and goals of each.

1 Interview with Matt Wallaert (2019).

2 This sidebar was developed from an interview with Wiam Hasanain (2019) and subsequent email exchange.

3 If you'd like to see another detailed breakdown of a behavioral team's staffing, check out [Clover Health's](#).

Chapter 18. Conclusion

Throughout this book, we've looked at how to deliberately, and ethically, develop products that help your users change their behavior, from exercising more to drinking less.

To do so, we've tried to develop three major conceptual tools:

- An understanding of how people make decisions and act in their daily lives
- A model of what's required for someone to take action relating to your product in a given moment
- A process for applying that knowledge to the practical details of product development

Let's briefly review the big ideas that support each tool.

How We Make Decisions and Act

In [Chapter 1](#), we summarized the breadth of behavioral research with five core lessons:

- *We're limited beings*: we have limited attention, time, willpower, etc. For example, there is a nearly an infinite number of things that your users could be paying attention to at any moment. They could be paying attention to the person who is trying to speak to them, the interesting conversation someone else is having near them, the report on their desktop that's overdue, or the notification on your app. Unfortunately, researchers have shown again and again that people's conscious minds can really only pay proper attention to one thing at a time.

- *Our minds use shortcuts* to economize and make quick decisions because of our limitations. Your users have a myriad of shortcuts (aka heuristics) that help them sort through the range of options they face on a day-to-day basis and make rapid, reasonable decisions about what to do. For example, if they don't know what to do in a situation, they may look to what other people are doing and try to do the same (aka *descriptive norms*). Similarly, *habits* are a powerful way people's minds economize and allow them to act quickly by immediately triggering a behavior based on a cue. Unfortunately, these shortcuts can go awry with ingrained and self-destructive habits (over-drinking) or heuristics that are applied in the wrong context (like herd behavior). Misapplied heuristics are one cause of *biases*: negative tendencies in behavior or decision making (differing from an objective standard of "good"). Often because of these biases, there's a significant gap between people's intentions and their actions.
- *We're of two minds*: what we decide, and what we do, depends on both conscious thought and nonconscious reactions like habits. This means that your users are often not "thinking" when they act; or at least, they're not choosing consciously. Most of their daily behavior is governed by nonconscious reactions. Unfortunately, their conscious minds *believe* that they are in charge all the time, even when they aren't. We're all "strangers to ourselves"; we don't know the causes of our own behavior and decisions. Thus, your users' self-reported comments about a problem in your product or what they plan to do in the future aren't necessarily accurate.
- *Decision and behavior are deeply affected by context*, worsening or ameliorating our biases and our intention–action gap. What your users do is shaped by our contextual environment in obvious ways, like when the architecture of a site directs them to a central home page or to a dashboard. It's also shaped by nonobvious influences, like the people they talk and listen to (the *social environment*), by what they see and interact with (their *physical environment*), and

by the habits and responses they've learned over time (their *mental environment*).

We can *cleverly and thoughtfully design a context* to improve people's decision making and lessen the intention–action gap. And that is the point of *Designing for Behavior Change*.

Shaping Behavior with Your Product: The CREATE Action Funnel

In Chapters 2 and 3, we looked at the preconditions for action: what your product team needs to either facilitate or support action (or to remove or hinder negative actions). It started with a seemingly simple question: from moment to moment, why would your users undertake one action and not another? Six factors must align, at the same time, before someone will take conscious action. Behavior change products help people close the intention–action gap by influencing one or more of the following preconditions: Cue, Reaction, Evaluation, Ability, Timing, and Experience. For ease of remembering, they spell CREATE, because that is what's needed to create action.

To illustrate these six factors, let's say your user is sitting on the couch watching TV. Your app that helps them plan and prepare healthy meals for their family is on the phone; they downloaded it last week. When and why would they suddenly get up, find their mobile phone, and start using the app?

We don't often think about user behavior in this way—we usually assume that somehow our users find us, love what we're doing, and come back whenever they want to. But researchers have learned that there's more to it than that because of the mind's limitations and wiring. So, again imagine your user is watching TV. What needs to happen for them to use the meal planning app right now?

1. *Cue*. The possibility of using the app needs to somehow cross their mind. Something needs to cue them to think about it; maybe they're hungry or see a commercial about healthy food on TV.
2. *Reaction*. They'll intuitively react to the idea of using the app in a fraction of a second. Is using the app interesting? Are other people they know using it? What other options come to mind, and how do they feel about them?
3. *Evaluation*. They might briefly think about it consciously, evaluating the costs and benefits. What will they get out of it? What value does the app provide? Is it worth the effort of getting up and working through some meal plans?
4. *Ability*. They'll check *whether* it's actually feasible to use the app now. Do they know where their mobile phone is? Do they have their username and password? If not, they'll need to solve those logistical problems before using the app.
5. *Timing*. They'd gauge *when* they should take the action. Is it worth doing now or after the TV show is over? Is it urgent? Is there a better time? This may occur before or after checking for the ability to act. Both have to happen, though.
6. *Experience*. Even if logically using the app is worth the effort and makes sense to use it now, they'd be loath to try again if they'd tried the app before (or something like it) and it made them feel inadequate or frustrated. Idiosyncratic personal experiences can overwhelm any "normal" reaction a person might have.

These six mental processes are gates that can block or facilitate action. You can think of them as "tests" that any action must pass: your user must complete them successfully in order for them to consciously, intentionally, engage in the target action. And, they all have to come together at the same time. For example, if they don't have the urgency to stop watching TV and act now, they could certainly do it later. But when "later" comes, they'll still face these six tests. They'll reassess whether the action is urgent at that

point (or whether something else, like walking the dog, takes precedence). Or maybe the cue to act will be gone and they'll forget about the app altogether for a while.

So, products that encourage people to take a particular action have to somehow cue their users to think about the action, avoid negative intuitive reactions to it, convince their conscious minds that there's value in the action, convince them to do it now and ensure that they can actually take the action. We can think about these factors as a funnel: at each step, people could drop off, get distracted, or do something else. The most common outcome in behavior change work, and the one we should expect, is the status quo. We seek to nudge that status quo into something new.

If someone already has a habit in place and the challenge is merely to execute that habit, the process is mercifully shorter. The first two steps (cue and reaction) are the most important ones, and, of course, the action still needs to be feasible. Evaluation, timing, and experience can play a role, but a lesser one, because the conscious mind is on autopilot.

DECIDE on the Behavioral Intervention and Build it

Behavioral science helps us understand how our environments profoundly shape our decisions and our behavior. It shouldn't come as a surprise that a technique that was tested in one setting (like a research laboratory) doesn't affect people in the same way in a real life. To be effective at designing for behavior change, we need more than to understand the mind: we need a process that helps us find the right intervention, the right technique, for a specific audience and situation.

What does this process look like? I like to think about it as six steps, which we can remember with the acronym DECIDE; that's how we decide on the right behavior-changing interventions in our products and communications.

First, Define the problem. Who's the audience, and what is the outcome we're trying to drive? Second, Explore the context. Gather qualitative and

quantitative data about the audience and their environment. If possible, reimagine the action to make it more feasible and more palatable for the user before we build anything.

From there, Craft an intervention—a behavior-changing feature of the product or communication. You craft on both the conceptual design (figuring out what the product should do) and the interface design (figuring out how the product should look). As we prepare to Implement the intervention, we consider both the ethical implications and how to instrument the product to track outcomes.

Finally, test the new design in the field to Determine its impact: did it move the needle or did it flop? Based on that assessment, Evaluate what to do next. Is it good enough? Since nothing is perfect the first time, we'll often need to iteratively refine it.

In shorthand:

1. *Define* the problem
2. *Explore* the context
3. *Craft* the intervention
4. *Implement* within the product
5. *Determine* the impact
6. *Evaluate* what to do next

It's important to emphasize that the process is inherently iterative. That's because *human behavior is complicated, and thus stuff is hard!* If one could simply wave a magic wand and other people would act differently, we wouldn't need a detailed process for designing for behavior change (this would be very disturbing). Instead, there's a cyclical process of learning about our users and their needs, and our efforts to address them if they miss the mark. The most overlooked, yet the most essential part of the process, isn't great ideas and nifty behavioral science tricks: it's careful

measurement of where our efforts go awry, and the wiliness and tools to learn from those mistakes.

Other Themes

Stepping back from these three conceptual models, there are some underlying themes that I'd hoped to express here. These themes provide additional color and guidance on how to design for behavior change:

We've learned a great deal about how the mind makes decisions, but our knowledge is still limited

The behavioral social science literature provides the foundation for this book, and there are numerous field experiments that illustrate how our decision making works. The research on biases and heuristics is well established, but research on how products can help change behavior is still in its infancy. This book pulls together the best of what's known currently, but our knowledge is certainly not complete.

First, understand your users

Changing behavior is a highly personal affair. Our products should meet users where they are in their daily lives and help them take action—that starts with understanding their needs and interests, constraints, prior experiences, and levels of expertise.

There are no magic wands for behavior change

There's no sure-fire way to *make* people take an action. Even if we fully understood the decision-making process, we each have unique personal histories and environments that shape our behavior. While we can't dictate behavior, we can facilitate it. If someone wants to act, well-designed products can help them do it.

Be intentional

If the goal is to help people change behavior, then do it. Don't hide behind wishy-washy recommendations or "raising awareness" among users; that leads us to design and build products that are internally conflicted and ineffective. Be up front and proud of products that help users change something important about their lives.

A behaviorally effective product must first be a good product

The product must be well designed, pleasant to use, and solve a user need. Any considerations about changing behavior need to build on top of that foundation—otherwise, people will simply choose not to use the product.

Avoid user work

Look for technical solutions to avoid user work; it's much easier to engineer a solution than it is to change behavior. We can and should celebrate user triumphs—but those triumphs should occur where hard work was unavoidable, and not where the product was poorly designed.

We don't need to get fancy

Look for the minimum viable action that a user can take to reach their goal, and remember simple, obvious lessons about the psychology of design: we like beauty, simplicity, familiarity, and following our peers.

We should assume we're (partially) wrong

No matter what we design and build, we'll get some things wrong. A dose of humility in face of the vast complexity of human behavior is a good thing. It spurs us to test the real, quantifiable impact of our products, and test every major change we make to them.

There are certainly other lessons throughout the book, but why I think about the underlying philosophy of how to approach designing for behavior change, these points start to capture them.

Frequently Asked Questions

Now, let's turn to questions that I've often encountered in this field and that you may have or face yourself. There's no strict ordering or relationship here; these are just some of the most common issues that people have wanted to better understand.

How Do the Preconditions for Action Vary from Day to Day?

For people to take action, the six CREATE preconditions must align for the specific action, the specific environment, and the specific person (the actor). Both the person and the environment vary over time, and thus so do the preconditions for action:

Environment

As a person moves through the day, the environment generally changes. The different environments shape whether the person will take a particular action or not. For example, someone may have internet access while waiting for a meeting and can use mobile apps, but not while driving. The person may see reminders to exercise at home, but not while at the bar.

Actor

Over the course of the day, *people* also vary in themselves. Their wakefulness varies, their tendency to be distracted varies, and of course, so do their emotions. An action that seems interesting and worthwhile to a rested, well-fed person can seem overly complex, too much work, and not urgent to someone who is hungry and tired.

This variation is a mix of happenstance and structure. Day to day, a person's work schedule may be too irregular to identify stable moments of opportunity for engagement with the product. But the commuting schedule may be stable and clear. Where feasible, the product should align with the

moments and situations of opportunity, in which some of the preconditions for action are already naturally in place, and then fill in the gaps.¹

From the perspective of product design, look for structure within this day-to-day (and moment-to-moment) variation, and build on it to encourage action. Are there particular times and places when the user of the product is least distracted? Or most motivated to act? Where are the cues to act in the person's daily life?

You have two main options to harness the natural variation in people's days. You can do user research and try to find structured opportunities for engagement that are common across the user base. Alternatively, you can prompt users to self-identify the best times for them—when they want to receive reminder messages, when they want to be motivated with uplifting stories from other users, etc.

What Changes as a User Gains Experience with the Product?

There are many ways people can change as they use a product. Rather than a proscriptive pathway, there are different states that users can be in, relative to the product. In each of these states, the dynamics of the CREATE Action Funnel (Cue, Reaction, Evaluation, Ability, Timing, Experience) are somewhat different:

- *Not a user of the product*
 - *Dynamics.* For someone who has never used the product before, its very newness is a benefit and a curse. Cues are likely to be noticed. Newness can increase motivation to explore. But newness can make the user unsure about the logistics of actually using it and that person's ability to succeed. Also, our intuitive reaction to the idea is based on our experiences with similar activities, which could be good or bad.

- *What the company should do.* Ensure the user knows about the product at all (cue). Make sure the value is clear (evaluation), and associate the product with familiar, pleasant things (reaction).
- *A one-time user of the product who had a positive experience*
 - *Dynamics.* Once we gain experience with a product, our future intuitive reactions take that into account. If we liked the product the first time we used it, excellent. We gain knowledge of how to use it (decrease costs, increase logistical ability).
 - *What the company should do.* Keep cueing the user; it's unlikely that the user has built a strong association between the product and an existing cue in their environment. Highlight the positive experience (evaluation and reaction). Use knowledge gathered during the first use to try to align the product with times and situations when the user isn't distracted and can take action (ability and timing).
- *A one-time user of the product who had a negative experience*
 - *Dynamics.* If the first experience was negative, we have two obstacles to overcome: the allure of newness is gone, and we have an intuitive negative reaction. It is much harder to bring these users back.
 - *What the company should do.* Honestly, focus attention on other users. This is a tough group to win back. And focus attention on improving the experience of first-time users in the future.
- *A user who's returned to the product one or more times*

- *Dynamics.* When a user has successfully returned to the product, that's a sign that the conditions are ripe for future use. Something in the user's context has pulled them back. However, it's not clear yet how *stable* that something is—it might be a temporary assignment at work that makes the user think about or need the product. It might be a core desire. At this point, minor disruptions to the user's context (a different routine at work, etc.) can easily stop them from returning.
 - *What the company should do.* Keep cueing the user until there's a strong association between the product and an existing environmental cue. Highlight prior positive experiences and successes. With user research, try to understand the user's context: are there temporary factors pushing them to use it for which the product must find a substitute?
- *A user who regularly uses the product*
 - *Dynamics.* When the user returns to the product repeatedly, there's a stable context that pulls them back. Only major changes in the user's context are likely to disrupt continued use—the user changes jobs, gets divorced, takes on significant new activities that pressure their time, or the product suddenly drops the particular functionality they love.
 - *What the company should do.* Don't screw it up. Be very careful when changing functionality that's driving usage. Look for regular users who drop off—if the context disruption is temporary, there's a very good chance you can win them back (but don't take it for granted). If you are working with a potentially habitual behavior, make sure the cue and routine are constant over time.

- *Habitual user*
 - *Dynamics.* If the user returns on a regular basis and responds automatically to an environment cue, you've successfully built a habit of use. (You can see whether it's a habit by looking at the usage pattern in the data and through user research.) Since the behavior is largely on autopilot, it's highly resistant to change. Only major changes in the cue or a lack of ability to act are likely to break it.
 - *What the company should do.* Don't mess with the cue or change the fundamental learned routine that users enact.

Nir Eyal talks about how to leverage the initial usage of an application to build future interest and make it easier or more valuable for users to return. He refers to it as the “investment” step that occurs right after an individual has had a positive experience with the product. He focuses on investment in the formation of habits, but it's an insight that can be applied for nonhabitual behaviors as well.²

As users gain experience with a product, the type of support they need changes. The most critical periods for adopting a new product to change behavior is the first experience (when curiosity and newness is replaced by an evaluation of the product) and in the transition from a returning user to a regular user (when minor changes in circumstance can disrupt the new behavior). Does the product provide an excellent first-time user experience? Does it provide a compelling, ongoing reason to return to the product—especially on a regular schedule with a stable cue to facilitate habit formation?

How Can You Sustain Engagement with Your Product?

Products that seek to change long-term behavior can't do that if people stop using the product. Sustained engagement requires an act of behavior change

just like the target behavior itself. At a high level, *driving continued usage of a product follows the same rules outlined throughout this book.*

Too often, companies just focus on providing value and wonder why users don't come back. Value is important—users have to want to use the product or they won't do so voluntarily. But there's much more involved, and much more that companies can do.

The six CREATE preconditions are need for *reengaging* with a product, and it's the cue that I think has received the least attention and thought in product development circles—at least outside of growth hacking. Perhaps it's because we think that if we build the greatest product in the world, people will naturally come and use it.

To cue individuals to frequently reengage with a product:

Continue to provide value

Absolutely, this is essential. If users don't think your product is worthwhile, you aren't going to be able to grab their attention repeatedly (they'll change their environment to avoid you). So, value is the first step. But it's only the first step.

Uniquely become part of the person's environment

One way to remind people to use a product is to ensure its seen—by placing the Fitbit by the side of the bed or by making your application the home page on a browser. By *uniquely*, I mean that there aren't other shiny things the person is seeing at the same time; that's a real problem with tweets and emails, for example—they are extraordinarily crowded channels.

Uniquely become part of the person's expected routine

At a particular time of day (or situation), train the user to uniquely think of the application as a way to do something or relieve boredom.

“Training” isn't happenstance: ask the user to plan out a particular time to use the product (i.e., as part of implementation intentions). If you're providing value, then help users form a habit around getting that value.

Be so darn cool and memorable that people think of you, on their own

We all aspire to this, and we think that a beautiful product will make users dream about us. I don't know about you, but I've seen a lot of beautiful artwork in museums. I don't dream about them, and neither do I dream about more than a handful of products. So, invest in other ways to get attention.

Build strong associations with something that is part of the user's environment or daily routine

If you can't get in front of users' eyeballs directly or reserve a slot on their daily calendar, build on what's already there. For example, whenever there's a crisp spring weekend day in DC (an existing aspect of my environment), I think of the bike trailer my kid loves riding in and head to the garage to get it.

Be useful each and every time they see you

Don't train the user to ignore you. Does your carpet catch your attention? No. It's in your line of sight, and you step on it as part of your daily schedule. But it does nothing for you most of the time. Same thing with sending lots of emails to users. *Don't be the carpet.*

Be new and different each time (or at least appear so!)

One way to avoid being the carpet is to make sure that each attention-grabbing piece of content contains something new (or potential for newness)—social network notifications do this beautifully with their teaser emails about friends doing stupid stuff. It's more than a random reward for logging into Facebook or Twitter—the attention grabber itself is different each time.

To reiterate, you can think about behavior change as a two-stage problem. First, there's the behavior of engaging with your application. Second, there's the behavior that the application seeks to support. The same tools

taught throughout this book for the latter behavior are also applicable to the former.

What Happens Before People Take Action the First Time?

The CREATE Action Funnel provides the six preconditions for action, and the DECIDE process ensures those preconditions are in place for a product. But how do people normally progress from inaction to action *over time*?

There's no consensus in the literature about how individuals move from inaction to action on their own. There are many pathways by which people become active, each of which brings together the preconditions for action in their own way. The intentional design of products to support action, described here, is just one of those pathways.

Consider the example of a person who wants to clean the house. They have a positive intuitive reaction and decide that the benefits (partner won't leave them) outweigh the costs (an afternoon of TV watching lost) after a conscious evaluation. But there are always more pressing things to do (no time pressure), they forget to do it (no cue), or they don't know how to do it effectively and don't have the right cleaning supplies (lacking ability).

Here's a quick overview of different pathways by which this person could move from failing to take the action to successfully doing it:

Self-directed action

They think about cleaning a few times and it doesn't happen. When they're at the grocery store, they remember to pick up cleaning supplies but don't set aside time to use them. One day, they say enough is enough and plan out a specific day and time to do it, set a reminder, and remove other distractions.

Intentional help from another person

A friend hears them complaining about their dirty house and decides to push a bit to get it done. The friend reminds them at the grocery store to

get supplies and makes them commit to cleaning on a particular day.

Directed help from a product

They download a task-management app based on David Allen's book *Getting Things Done* (Penguin Books, 2001). They record all the things they need to do (cleaning and otherwise) and, with the help of the app, works out the logistics and timing to finally clean up the house.

Sudden change in environment

They want to clean the house but just don't get it together for other reasons (they change jobs and move). Their new house is much smaller and they get rid of most of their stuff. It's also much easier to clean, and they find it's easy to continue a habit of keeping the house clean once they've started.

Sudden change in life circumstances

The user gets married. Their spouse won't put up with the messiness and lays down the law. The user shapes up.

Social drift

For various reasons, they get to know new people at work and some of them become friends. The user visits their coworkers' houses, which are pleasantly clean. When they go home, they look at their house differently. This happens a few times, and they're ashamed of bringing other people over. The user gets up the gumption and does it.

These are common pathways, but of course there are many more. The point is that there simply isn't a single route by which people change their behavior over time. Sometimes there's a single, culminating event that forces change (marriage); at other times, things move more slowly (social drift).

While these examples point to multiple *possible* routes, anecdotes aren't science. Thus far, researchers haven't established clear, generally accepted

rules for how individuals move from inaction to action over the course of time. There's a great deal of activity in the addiction and health space, with models such as the Transtheoretical Model³ and Health Belief Model,⁴ but their general applicability is unclear. We know a lot more about what happens at the moment of action (a cue, a conscious choice to act, etc.) than we do about what happens before that action.

Looking Ahead

Designing for behavioral change has gone mainstream. Even a few years ago, it was rare to find organizations with dedicated in-house behavioral science teams; now we can count over 450. Major companies as diverse as Walmart and Weight Watchers are designing for behavior change, as are numerous small consultancies and scrappy NGOs. Since 2013, behavioral teams have sprouted all around the world, and there appears to be no slowdown on the horizon.

The benefits of this work are tremendous: from saving the lives of anemic mothers to helping people save for the future. As the field grows, these wonderful societal benefits will continue and expand. Significant challenges, however, loom ahead: the most important of which is ethical. Simply said, there are shady practices in our field that deceive and hurt everyday people: from pushing gig economy drivers to drive past safe limits to tricking people into giving up their detailed location data.

That is the nature of designing for behavior change today: exciting growth and good being done side by side with troubling abuses. I believe that to foster the former, we must confront the latter. The path isn't clear, but by applying behavioral science on ourselves and our job sites, we may be able to more deeply design ethical behavior into our work, just as we seek to (beneficially) shape others' behavior and outcomes.

We're at turning point in our field. We have a nuanced understanding of how the mind works and how people take action, one that will continue to grow and be adjusted over time, but one that provides us with guidance and insights today. We have, effectively, a blueprint for behavior change—one

that many teams working in parallel around the globe have developed on their own, which I've summarized here with the DECIDE framework. Designing for behavior change is here to stay. What shape our field will take exactly, and what mistakes and successes we'll have, depends on what we do with the foundation we've built over the last few years and how seriously we face the challenges ahead.

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- 1 The concept of fluctuating environmental and social factors that create opportunities for action is a core concept of the Political Opportunity Structure tradition in political sociology (e.g., McAdam et al. 2001). BJ Fogg also models fluctuations in motivation with his concept of a Motivation Wave over time. [Fogg \(2012\)](#).
 - 2 Eyal (2013, 2014)
 - 3 Prochaska and Velicer (1997). Unlike my preceding assertion, this model proposes that there's a *single* pathway to action. But it's just not one that I see as generally applicable or true.
 - 4 Janz and Becker (1984)

Glossary of Terms

This book introduces many new concepts and terms, such as the *behavioral map* and the CREATE Action Funnel. This glossary provides definitions of those terms, for your convenience. For the sake of brevity, though, many of the common behavioral economics terms referenced in the book (e.g., *implementation intentions*) are not included here.

A/A test

an “experiment” in which there is no difference between the arms of the test. An A/A test allows you to verify that the randomization process and analysis code is working correctly. If you see a difference in outcomes across the two groups, there’s something wrong!

A/B test

a method of comparing two versions of something against each other to determine which one performs better. Two variants are shown to users at random, and a statistical analysis is used to determine which variation performs better for a given outcome metric. An A/B/C... test is an extended version, with 3+ versions.

A/Null test

a method of testing that takes the baseline intervention (aka, version of the product, feature, etc.) and tests it against no intervention at all. It provides the clearest and simplest measure of that intervention’s impact. It is also valuable to ensure that the intervention doesn’t actually make the user *worse* off (which is something that is missed with many A/B tests).

ability (per the CREATE Action Funnel)

a stage in the CREATE Action Funnel when the user evaluates whether or not she can take the target action now. The ability to take the target action has four criteria: knowing logistically how to take the action, having the resources necessary to act, having the skills necessary to act, and having a sense of self-efficacy or belief in success. See *CREATE Action Funnel*.

actor (aka target actor)

the person who takes action because of the product. When the *actor* takes the *action*, it causes the product's *outcome*. See *(target) action*, *(target) outcome*.

action (aka target action)

the behavior that the design process seeks to engender. When the *actor* takes the *action*, it causes the product's *outcome*. Note: "action" and "behavior" are used as synonyms throughout the book. See *(target) actor*, *(target) outcome*.

behavioral bridge

a description that links a behavior that the user already knows and is comfortable with to a new, unfamiliar behavior. For example, an appeal to users that describes the (unfamiliar) act of running a race as similar to the (familiar) running around the office they normally do.

behavioral map

a detailed "story" of how the user progresses from being a neophyte to accomplishing the action while using the product. The "story" can take many forms, such as a journey map, written narrative, or a simple list of actions. The behavioral map provides the conceptual design for the product, providing its functional requirements. See *conceptual design*.

behavioral persona

a stylized individual used to represent a group of users who are likely to respond similarly to an appeal to change their behavior. For example, the persona “active Jake” could represent users of the product who are active in their daily lives and would be likely to join a competition to see who exercises the most. See *persona*.

behavioral strategy

a high-level strategy for changing behavior with a product. This book discusses three strategies: supporting the conscious choice to take the target action, building (or changing) habits, and “cheating.”

behavioral tactic

a low-level technique for changing behavior in a product. For example: showing a peer comparison, highlighting loss aversion, or priming a particular mindset. See also *behavioral strategy*.

cheating (per behavioral strategies)

a strategy for changing behavior in which the burden of work is shifted from the user to the product, and the user need only give consent for the action to occur on his behalf. See *behavioral strategy*.

company objective

what the company seeks to achieve, for itself, by building the product.

conceptual design

a set of documents or illustrations that indicate what the product should do (i.e., what functionality the product should provide) at a conceptual level. When designing for behavior change, the behavioral map fulfills this role. See *behavioral map*.

context (of action)

the three factors that shape a user’s decision whether or not to act. Namely, the *user* himself, the *environment* the user is in, and the *action*

the user is deciding upon.

CREATE Action Funnel

a stylized model of how the mind makes the conscious decision to act. Once the mind detects a cue, it has an intuitive reaction, which may bubble up into conscious awareness for evaluation of the merits of action, and an assessment of whether the user has the ability and right timing to act, tempered by prior experience. If all of these mental processes pass successfully without the user getting distracted or deciding against the action, the user will execute the action. Together, they spell the acronym CREATE. See cue, reaction, evaluation, ability, timing, and experience.

cue (per habits)

something that causes a habit to occur. The cue can be something the person sees, hears, smells, or touches in the environment (an external cue) or an internal state like hunger (internal cue) that initiates the habitual routine.

cue (per the CREATE Action Funnel)

the first stage in the CREATE Action Funnel when something first makes the user think about taking the target action. The cue can be something the person sees, hears, smells, or touches in the environment (an external cue) or an internal state like hunger (internal cue) that starts the process of taking action. For habitual actions, the cue alone can be enough to cause the behavior to occur. See *CREATE Action Funnel*.

data bridge

a mathematical relationship or statistical model that relates a target outcome outside of the product to behavior within the product. For example, “60% of the time that users indicate in the application that they will create a vegetable garden, they actually create one.”

designing for behavior change, DECIDE

a six-stage process of designing a product with the specific purpose of changing user behavior. The six stages are: *Define* the problem, *Explore* the context, *Craft* the intervention, *Implement* within the product, *Determine* the impact, *Evaluate* what to do next.

dual process theory(ies)

a family of related theories in psychology that posit that the mind effectively has two independent decision-making processes: one deliberative and one intuitive. The deliberative, or “System 2,” process is associated with intentional conscious thought. The intuitive, or “System 1,” process is associated with automatic, emotional reactions or implicit, “subconscious” behaviors.

environment (of action)

one of the three parts of the decision-making context (along with the user and the action itself) that shapes a user’s decision to act. The environment consists of the product that the person is interacting with and the physical environment surrounding the person when he decides whether or not to act. See *context*.

evaluation (per the CREATE Action Funnel)

the third stage in the CREATE Action Funnel, when the user consciously evaluates the value of taking the target action, often considering its costs and benefits. See *CREATE Action Funnel*.

experience (per the CREATE Action Funnel)

the sixth stage in the CREATE Action Funnel, which reminds behavioral designers of the tremendous power of an individual’s past experience. Even if “most” people have a particular reaction or bias, a particular person’s past experience can override that response. See *CREATE Action Funnel*.

external cue

something in our environment that causes us to think about or take a certain action. See *cue*, *internal cue*.

extrinsic motivation

the desire to achieve a particular outcome, such as receiving a reward for it (like money or winning a competition). See *intrinsic motivation*.

habit

a repeated behavior that's triggered by internal or external cues. A habit is automatic: the action occurs outside of conscious control, and we may not even be aware of it happening. Habits can be formed through simple cue-routine repetition or can include a reward that becomes associated with the cue and encourages the user to repeat the behavior. See *cue*, *routine*, *reward*.

interface design

a set of documents or illustrations that says how the product should look and interact with the user.

internal cue

a prior thought or bodily state (like hunger) that leads us to think about or take a certain action. See *cue*, *external cue*.

intrinsic motivation

comes from the inherent enjoyment of the activity itself, without considering any external pressure or reward. See *extrinsic motivation*.

mindset

a mental mechanism for interpreting and responding to the world, which shapes how we act. Mindsets as facets of the mind, which are built up in different contexts. Here, the term is used to encompass a range of

psychological mechanisms that guide our behavior in ambiguous contexts, including schemas and activated frames.

Minimum Viable Action (MVA)

the shortest, simplest version of the target action that users can be asked to take, with which the company can still test whether the product has the desired impact on behavior. See *target action*.

Multivariate Test

a technique for testing a hypothesis in which multiple, independent interventions are simultaneously tested. The goal being to determine which combination or variations performs the best, out of all of the possible combinations.

outcome (aka target outcome)

the real-world impact that the company seeks to have because of the product. A measureable change in the world that happens when the product succeeds in changing behavior. When the *actor* takes the *action*, it causes the product's *outcome*. See *(target) actor*, *(target) action*.

persona

in the user experience field, a persona is a stylized individual used to represent a group of similar users, usually based on a particular demographic profile. See *behavioral persona*.

reaction (per the CREATE Action Funnel)

the second stage in the CREATE Action Funnel when the user has an automatic, System 1 reaction to the idea of taking action. That reaction renders an intuitive verdict (whether the action is interesting or not) based on prior associations with the action or similar experiences. It also activates thoughts about other possible actions the person could take. See *CREATE Action Funnel*.

reward (per habits)

something that gives us a *reason* to repeat a behavior. It might be something inherently pleasant, like good food, or the completion of a goal we've set for ourselves, like putting away all of the dishes.

routine (per habits)

the habitual action that a person takes when exposed to the habit's cue. For example, buying Starbucks coffee whenever a person sees the Starbucks sign next to her office at 9 a.m. in the morning.

self-narrative

how we label ourselves, and how we describe our behavior in the past.

small win

a feeling of accomplishment after a (relatively small) action is taken.

timing (per the CREATE Action Funnel)

the fifth stage of the CREATE Action Funnel, in which the user determine *when* to act. See *CREATE Action Funnel*.

user story

A term used in product development (especially agile development) for a plain-English statement about what the user needs. It captures the “who,” “what,” and “why” of a product requirement. For example, “As a user, I want to [take an action], in order to [reason for action].” See [Wikipedia](#) for more information.

vision (aka product vision)

why the product is being developed, at a high level.

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Index

Symbols

401(k) plans

autoenrollment/autoescalation, [Automate it](#)

financial literacy seminars, [The Value and Limitations of Educating Your Users](#)

A

A/A test, [Glossary of Terms](#)

A/B tests

defined, [Other Types of Experiments](#), [Glossary of Terms](#)

(see also experiments)

for determining impact, [Determining Impact with A/B Tests and Experiments-Determining Impact with A/B Tests and Experiments](#)

implementing, [Implementing A/B Testing and Experiments](#)

tools that support, [Tools for Behavioral Tracking and Experiments](#)

A/Null test, [Other Types of Experiments](#), [Glossary of Terms](#)

Abdul Latif Jameel Poverty Action Lab (J-PAL), [The Dedicated Teams](#)

ability (CREATE Action Funnel stage), [Ability](#), [The User's Ability to Act-Look for "Real" Obstacles](#)

defined, [Glossary of Terms](#)

eliciting implementation intentions, [Elicit Implementation Intentions](#)

increasing the power of other behaviors to stop negative behavior,
Ability: Increase the power of other behaviors

knowing user will succeed, **The Other Side of the Wall: Knowing You'll Succeed**

peer comparisons and, **Peer Comparisons Can Help Here Too**

real obstacles, **Look for "Real" Obstacles**

removing friction/channel factors, **Remove Friction and Channel Factors-Set appropriate defaults**

accountability, **Make Commitments to Friends**

action (target action)

automating, **Automate it**

changing the context of, **Reach Out of the Screen**

clarifying, **Clarify the Action**

defined, **Glossary of Terms**

determining who should take action, **Who Takes Action?**

diagnosing why people don't start an action, **Diagnosing Why People Don't Start-Diagnosing Why People Don't Start**

documenting initial idea of, **Document Your Initial Idea of the Action-Look for the Minimum Viable Action**

doing action for user, **Do It for Them When You Can-Cheating at the Action Funnel**

(see also cheating)

examples from various domains, **Examples from Various Domains**

making incidental to other action, [Make it incidental-Make it incidental](#)

metric for, [A Metric for Action](#)

minimum viable, [Look for the Minimum Viable Action](#)

obvious action as least likely to work, [The Obvious Is Our Enemy](#)

reevaluating assumptions about desired action, [Is There a Better Action for Them to Take?-Updating the Behavioral Personas](#)

selecting the ideal action, [Select the Ideal Target Action-Select the Ideal Target Action](#)

stopping negative actions (see negative actions, stopping)

techniques for generating alternate actions, [Techniques for Generating Ideas](#)

Action Design Network (ADN), [How This Book \(and This New Edition\) Came About, What We Did: A Global Survey of Behavioral Teams, The Nondedicated Teams](#)

action metrics, [A Metric for Action](#)

action, creating, [Creating Action-A Short Summary of the Ideas, From Problems to Solutions](#)

(see also CREATE Action Funnel)

action, taking, [Deciding and Taking Action-A Short Summary of the Ideas, Quirks of Action](#)

actor (target actor), [When Product Teams Don't Have a Clear Problem Definition, Who Takes Action?, Glossary of Terms](#)

addictive products, [Addictive Products-Addictive Products](#)

ADN (Action Design Network), [What We Did: A Global Survey of Behavioral Teams, The Nondedicated Teams](#)

advertising, [We Use Shortcuts, Part II: Habits](#)

Affordable Care Act (Obamacare), [The Behavioral Map: What Micro-Behaviors Lead to Action?](#)

anchoring, [We Use Shortcuts, Part I: Biases and Heuristics](#)

Appel, Jacob, [Evaluating Next Steps](#)

Appiah, Kwame, [The Behavioral Science of Ethics](#)

Apple, [Digital Tools, Especially, Seek to Manipulate Their Users](#)

Applied Behavioral Science Team (Walmart/Sam's Club), [A Summary of the Process](#)

Ariely, Dan, [Pull Future Motivations into the Present](#)

attention

avoiding cue to stop negative action, [Attention: Avoid the Cue](#)

increasing with mindfulness, [Evaluation: Increase attention with mindfulness](#)

limitation, [We're Limited](#)

attention economy, [Addictive Products](#)

attention treatment, [Other Types of Experiments](#)

authenticity, [Be Authentic and Personal-Be Authentic and Personal](#)

authority, speaking with, [Display Strong Authority on the Subject](#)

automation

defaulting and, [Automate the act of repetition](#)

of action on user's behalf, [Automate it](#)

of repetition of action, [Automate the act of repetition](#)

availability heuristic, [We Use Shortcuts, Part I: Biases and Heuristics, Quirks of Decision Making](#)

B

B4Development, [Stopping Negative Actions](#)

Bains Douches Castagnary, [The State of the Field](#)

Banerjee, Abhijit, [Evaluating Next Steps](#)

baseline, [How Many People Are “Enough”?](#)

beauty, of site, [Make the Site Professional and Beautiful](#)

behavior change, [Behavior Change...](#)

alternatives to CREATE framework, [A Simple Model of When, and Why, We Act](#)

behavioral science and, ...[And Behavioral Science](#)

commonalities among various processes, [The Process Is a Common One](#)

deliverables and outputs at each stage of process, [The Details Do Matter-The Details Do Matter](#)

design and, ...[And Behavioral Science](#)

practical guides/worksheets, [Since We’re Human Too: Practical Guidelines and Worksheets](#)

putting into practice, [Since We’re Human Too: Practical Guidelines and Worksheets](#)

six steps of DECIDE process, [Understanding Isn't Enough: We Need Process-Understanding Isn't Enough: We Need Process](#)

summary of process, [A Summary of the Process-Workbook Exercises](#)

behavioral barriers, assessing, [Assess Behavioral Barriers](#)

behavioral bridge, [Associate with the Positive and the Familiar](#), [Glossary of Terms](#)

behavioral map, [The Behavioral Map: What Micro-Behaviors Lead to Action?-The Behavioral Map for Stopping Behaviors](#)

building, [Building the Behavioral Map-Building the Behavioral Map](#)

defined, [Glossary of Terms](#)

drawing/adding behavioral detail, [Write or Draw It Out, and Add Behavioral Detail-Write or Draw It Out, and Add Behavioral Detail](#)

for existing product, [New Products or Features Versus Existing Ones](#)

for new products/features, [New Products or Features Versus Existing Ones](#)

for stopping behaviors, [The Behavioral Map for Stopping Behaviors](#)

reevaluating assumptions about desired action, [Is There a Better Action for Them to Take?-Updating the Behavioral Personas](#)

behavioral metrics

as integral to product, [Build in Behavioral Metrics from Day One-Tools for Behavioral Tracking and Experiments](#)

implementing A/B testing and experiments, [Implementing A/B Testing and Experiments](#)

implementing behavioral tracking, [Implementing Behavioral Tracking-Measuring behaviors and outcomes outside of the product](#)

(see also behavioral tracking)

starting point for, [What You Should Already Have](#)

behavioral personas, [Behavioral Personas-Behavioral Personas](#)

defined, [Glossary of Terms](#)

updating in light of revised target action, [Updating the Behavioral Personas](#)

behavioral research

Behavioral Teams Survey, [What We Did: A Global Survey of Behavioral Teams-What We Did: A Global Survey of Behavioral Teams](#)

current state of, [The State of the Field-Putting It into Practice](#)

behavioral science

basics, [Behavioral Science 101: How Our Minds Are Wired-We Can Design Context](#)

behavior change and, [...And Behavioral Science](#)

context, [We're Deeply Affected by Context-We Can Design Context](#)

deliberative versus reactive thinking, [We're of Two Minds](#)

ethics of (see ethics of behavioral science)

human limitations and, [We're Limited-We're Limited](#)

quirks of action, [Quirks of Action](#)

quirks of decision making, [Quirks of Decision Making-Quirks of Decision Making](#)

shortcuts: biases and heuristics, [We Use Shortcuts, Part I: Biases and Heuristics-We Use Shortcuts, Part I: Biases and Heuristics](#)

shortcuts: habits, [We Use Shortcuts, Part II: Habits-We Use Shortcuts, Part II: Habits](#)

Behavioral Science Policy Association (BSPA), [What We Did: A Global Survey of Behavioral Teams, Getting Help from Outside Researchers](#)

behavioral scientists, as team members, [Data Science and Behavioral Science](#)

behavioral strategy, [Glossary of Terms](#)

behavioral tactic, [Glossary of Terms](#)

behavioral teams (see teams)

Behavioral Teams Survey, [What We Did: A Global Survey of Behavioral Teams-What We Did: A Global Survey of Behavioral Teams](#)

business model for teams, [Business Model](#)

challenges for teams, [The Challenges-The Replication Crisis in Science](#)

creation of survey, [What We Did: A Global Survey of Behavioral Teams-What We Did: A Global Survey of Behavioral Teams](#)

dedicated teams, [The Dedicated Teams-The Dedicated Teams](#)

focus areas, [Focus Area](#)

geographic distribution of teams, [Who's Out There?-Who's Out There?](#)

nondedicated teams, [The Nondedicated Teams](#)

origins of teams, [Origins](#)

placement of teams, [Placement](#)

practical challenges of running a team, [The Practical Challenges of Running a Team](#)

replication crisis in science, [The Replication Crisis in Science-The Replication Crisis in Science](#)

size of teams, [The Dedicated Teams](#)

types of teams, [Who's Out There?](#)

behavioral tracking

implementing, [Implementing Behavioral Tracking-Measuring behaviors and outcomes outside of the product](#)

measuring behaviors/outcomes outside of the product, [Measuring behaviors and outcomes outside of the product](#)

measuring behaviors/outcomes within the product, [Measuring behaviors and outcomes within the product](#)

tools for, [Tools for Behavioral Tracking and Experiments](#)

Behavioural Insights Team (UK), [The Process Is a Common One, Crafting the Intervention: Cue, Reaction, Evaluation, The Dedicated Teams, Business Model](#)

Benartzi, Shlomo, [Frame Text to Avoid Temporal Myopia](#)

Berman, Kristen, [Creating Action](#)

Berra, Yogi, [When Product Teams Don't Have a Clear Problem Definition](#)

Betterment, [Rushed Choices and Regrettable Action, How You Combine These Skills on a Team](#)

bias, [We Use Shortcuts, Part I: Biases and Heuristics-We Use Shortcuts, Part I: Biases and Heuristics, Quirks of Decision Making](#)

cognitive, [Behavioral Science 101: How Our Minds Are Wired](#)

confirmation, [Cue](#)

fundamental attribution, [Remember the Fundamental Attribution Bias](#)

present, [We Use Shortcuts, Part I: Biases and Heuristics](#)

recency, [Quirks of Decision Making](#)

status quo, [We Use Shortcuts, Part I: Biases and Heuristics](#)

blinking text, [Bonus Tactic: Blinking Text](#)

Bogost, Ian, [Addictive Products](#)

Bono, [Evaluating Next Steps](#)

Brignull, Harry, [Ethics of Behavioral Science](#)

BSPA (Behavioral Science Policy Association), [What We Did: A Global Survey of Behavioral Teams](#), [Getting Help from Outside Researchers](#)

Buddha/Buddhism, [We're of Two Minds](#), [Evaluation: Use conscious interference](#), [Handling Prior Experience](#)

Burroughs, Augusten, [Quirks of Action](#)

Busara Center, [Implementing Within the Product](#), [Who's Out There?](#), [Business Model](#)

BVA Nudge Unit, [The State of the Field](#)

C

Cagan, Marty, [Integrate](#)

Cambridge Analytica, [Digital Tools](#), [Especially](#), [Seek to Manipulate Their Users](#)

Chabris, Christopher, [Cue](#)

Chatzky, Jean, [Provide "Small Wins"](#)

cheating (behavioral strategy), [Do It for Them When You Can-Cheating at the Action Funnel](#)

automating a repetition of action, [Automate the act of repetition](#)

automating an action, [Automate it](#)

defined, [Glossary of Terms](#)

justification for, [But Isn't Cheating, Well, Cheating?](#)

multi-step interventions and, [Again, Cheat If You Can](#)

one-time actions, [Strategies to Cheat at One-Time Actions-Make it incidental](#)

repeated actions, [Strategies to Cheat at Repeated Actions-Automate the act of repetition](#)

Choi, James, [The CREATE Action Funnel](#)

choice architecture, [...And Behavioral Science](#)

choice overload, [Avoid Choice Overload](#)

Cialdini, Robert, [Digital Tools, Especially, Seek to Manipulate Their Users](#)

Clover Health, [Exploring the Context](#)

cognitive bias, [Behavioral Science 101: How Our Minds Are Wired](#)

cognitive capacity, [We're Limited](#)

cognitive overhead, [Avoid Cognitive Overhead](#)

COM-B Behaviour Change Wheel, [A Simple Model of When, and Why, We Act](#)

commitment contracts, [Use Commitment Contracts and Commitment Devices](#)

commitment devices, [Use Commitment Contracts and Commitment Devices](#)

commitment, effort and, [Hard work builds commitment](#)

company objectives, [Working with Company-Centric Goals](#), [State the company's objectives](#), [Glossary of Terms](#)

company-centric goals

examples of desired outcomes/target actions, [Examples from Various Domains](#)

objectives of company, [State the company's objectives](#)

target outcomes and, [Working with Company-Centric Goals-Define the user outcomes](#)

user outcomes defined for, [Define the user outcomes](#)

vision for product, [State the vision](#)

competition (among behaviors)

removing distractions, [Remove Distractions: Knock Out the Competition](#)

strategies for counteracting, [Remove Distractions: Knock Out the Competition](#)

competition (social motivator), [Use Competition](#)

computational behavioral science, [Leveraging Data Science When Designing for Behavior Change](#)

conceptual design, [Understanding Isn't Enough: We Need Process](#), [Glossary of Terms](#)

confidence level, [How Many People Are "Enough"?](#)

confirmation bias, [Cue](#)

conscious (System 2) thinking, [We're of Two Minds](#)

conscious evaluation, [The Conscious Evaluation-Pull Future Motivations into the Present](#)

(see also decision making)

avoiding choice overload, [Avoid Choice Overload](#)

avoiding cognitive overhead, [Avoid Cognitive Overhead](#)

avoiding direct payments, [Avoid Direct Payments](#)

commitment contracts/commitment devices, [Use Commitment Contracts and Commitment Devices](#)

competition and, [Use Competition](#)

incentives and, [Make Sure the Incentives Are Right](#)

leveraging existing motivations before adding new ones, [Leverage Existing Motivations Before Adding New Ones](#)

leveraging loss aversion, [Leverage Loss Aversion](#)

making sure instructions are understandable, [Make Sure Instructions Are Understandable](#)

pulling future motivations into the present, [Pull Future Motivations into the Present-Pull Future Motivations into the Present](#)

slowing down the process, [Slow Them Down](#)

techniques for, [A Few Notes on Decision Making](#)

testing out different types of motivators, [Test Different Types of Motivators](#)

context (of action), [Behavioral Science 101: How Our Minds Are Wired, We're Deeply Affected by Context-We Can Design Context](#)

defined, [Glossary of Terms](#)

elements of, [Reach Out of the Screen](#)

exploring (see exploring the context)

redesigning, [We Can Design Context](#)

coronavirus (COVID-19), [What Types of Behaviors This Can Help With](#)

Covenant Eyes, [Attention: Avoid the Cue](#)

crafting the intervention (DECIDE stage)

advanced topics, [Crafting the Intervention: Advanced Topics-Exercises](#)

conscious evaluation, [The Conscious Evaluation-Pull Future Motivations into the Present](#)

and CREATE Action funnel, [Crafting the Intervention: Cue, Reaction, Evaluation](#)

creating habits, [Creating Habits-Creating Habits](#)

cueing the user to act, [Cueing the User to Act-Bonus Tactic: Blinking Text](#)

Fitbit example, [Crafting the Intervention: Ability, Timing, Experience](#)

getting timing right, [Getting the Timing Right-Make a Reward Scarce](#)

handling prior experience, [Handling Prior Experience-Check In Again: You're Not Interacting with the Same Person](#)

hindering action, [Hindering Action-Ideas for Hindering Other Actions](#)

intuitive reaction, [Narrate the Past to Support Future Action-Make the Site Professional and Beautiful](#)

making it clear where to act, [Make It Clear Where to Act](#)

multi-step interventions, [Multi-Step Interventions-Hard work builds commitment](#)

New Moms example, [Crafting the Intervention: Advanced Topics](#)

UK Behavioural Insights Team example, [Crafting the Intervention: Cue, Reaction, Evaluation](#)

user's ability to act, [The User's Ability to Act-Look for "Real" Obstacles](#)

CREATE Action Funnel, [From Problems to Solutions, The CREATE Action Funnel-A Short Summary of the Ideas, Shaping Behavior with Your Product: The CREATE Action Funnel](#)

ability stage (see ability)

alternative frameworks to, [A Simple Model of When, and Why, We Act](#)

as means of understanding negative behavior, [Stopping Negative Actions](#)

basics, [A Simple Model of When, and Why, We Act](#)

changes as user gains experience with product, [What Changes as a User Gains Experience with the Product?-What Changes as a User Gains Experience with the Product?](#)

changing nature of repeated actions, [The Funnel Repeats Each Time the Person Acts and Each Time Is Different](#)

changing the context of action, [Reach Out of the Screen](#)

cheating at, [Cheating at the Action Funnel](#)

cheating strategy, [Cheating at the Action Funnel](#)

cognitive/practical barriers in, [The Funnel Repeats Each Time the Person Acts and Each Time Is Different](#)

conscious evaluation, [The Conscious Evaluation-Pull Future Motivations into the Present](#)

cue stage (see cue)

cueing the user to act, [Cueing the User to Act-Bonus Tactic: Blinking Text](#)

defined, [Glossary of Terms](#)

diagnosing context problems with, [Diagnosing the Problem with CREATE-Diagnosing Why People Don't Stop](#)

diagnosing why people don't start a positive action, [Diagnosing Why People Don't Start-Diagnosing Why People Don't Start](#)

diagnosing why people don't stop a negative action, [Diagnosing Why People Don't Stop](#)

evaluating multiple decisions with, [Worksheet: Evaluating Multiple Interventions with CREATE](#)

evaluation stage (see evaluation)

experience stage (see experience)

interaction of stages with one another, [The Stages Can Interact with One Another](#)

interventions to overcome obstacles, [Crafting the Intervention: Cue, Reaction, Evaluation](#)

intuitive reaction, [Narrate the Past to Support Future Action-Make the Site Professional and Beautiful](#)

looking beyond motivation, [Look Beyond Motivation](#)

progression of users from inaction to action over time, [What Happens Before People Take Action the First Time?-What Happens Before People Take Action the First Time?](#)

reaction stage (see reaction)

relative nature of each stage, [Each Stage Is Relative](#)

sustaining engagement with product, [How Can You Sustain Engagement with Your Product?](#)

techniques for hindering actions, [Ideas for Hindering Other Actions](#)

timing stage (see timing)

using to add obstacles to negative actions, [Using CREATE to Add Obstacles to Action](#)

value and limitations of educating users, [The Value and Limitations of Educating Your Users-The Value and Limitations of Educating Your Users](#)

variations in preconditions, [How Do the Preconditions for Action Vary from Day to Day?](#)

credit card companies, [Where Things Have Gone Wrong: Four Types of Behavior Change](#)

cross-sectional analysis, [A Cross-Sectional or Panel Data Analysis of Impact](#)

cue (CREATE Action Funnel stage), [Cue-Cue](#)

aligning with user's spare time, [Align with When People Have Spare Time](#)

asking the user to act, [Ask Them](#)

avoiding to prevent negative actions, [Attention: Avoid the Cue](#)

blinking text, [Bonus Tactic: Blinking Text](#)

crafting the intervention, [Cueing the User to Act-Bonus Tactic: Blinking Text](#)

defined, [Glossary of Terms](#)

going to where user's attention already is, [Go Where the Attention Is](#)

making it clear where to act, [Make It Clear Where to Act](#)

product strategies for external cueing, [Cue](#)

relabel existing feature as cue, [Relabel Something as a Cue](#)

reminders and, [Use Reminders](#)

removing distractions, [Remove Distractions: Knock Out the Competition](#)

cue (habits), [We Use Shortcuts, Part II: Habits](#), [We Use Shortcuts, Part II: Habits](#)

defined, [Glossary of Terms](#)

external, [Cue](#), [Glossary of Terms](#)

internal, [Cue](#), [Glossary of Terms](#)

Curtis, Dustin, [Ask Them](#)

D

Dai, Hengchen, [Use Fresh Starts](#)

dark patterns, [Ethics of Behavioral Science](#)

data bridge

building, [Figure Out How to Measure the Outcome and Action by Hook or by Crook \(Not by Survey\)](#), [Build the Data Bridge](#)

defined, [What Happens If the Outcome Isn't Measurable Within the Product?](#), [Glossary of Terms](#)

data science, [Leveraging Data Science When Designing for Behavior Change](#)

"Data Science Meets Behavioral Science" (van Caster), [Data Science and Behavioral Science](#)

data scientists, [Data Science and Behavioral Science](#)

de Bono, Edward, [Techniques for Generating Ideas](#)

debiasing, [Rushed Choices and Regrettable Action](#)

Deceptive Experiences To Online Users Reduction (DETOUR) Act, [Ethics of Behavioral Science](#), [Use Legal and Economic Incentives as Well](#)

DECIDE process, [The Details Do Matter](#), [DECIDE on the Behavioral Intervention and Build it](#)

(see also specific stages: Defining the problem; Exploring the context; Crafting the intervention; Implementing within the product; Determining impact; Evaluating next steps)

basics, [Understanding Isn't Enough: We Need Process-Understanding Isn't Enough: We Need Process](#)

defined, [Glossary of Terms](#)

putting into practice, [Since We're Human Too: Practical Guidelines and Worksheets](#)

six steps of, [Understanding Isn't Enough: We Need Process-Understanding Isn't Enough: We Need Process](#)

decision making, [Deciding and Taking Action-A Short Summary of the Ideas](#), [How We Make Decisions and Act](#)

avoiding choice overload, [Avoid Choice Overload](#)

avoiding cognitive overhead, [Avoid Cognitive Overhead](#)

handling prior experience, [Use Techniques to Support Better Decisions](#)

making sure instructions are understandable, [Make Sure Instructions Are Understandable](#)

quirks of, [Quirks of Decision Making-Quirks of Decision Making](#)

removing unnecessary decision points, [Remove unnecessary decision points](#)

slowing down the process, [Slow Them Down](#)

techniques for, [A Few Notes on Decision Making](#)

decision points

defined, [Ability](#)

removing unnecessary, [Remove unnecessary decision points](#)

decision-making process map, [A Map of the Decision-Making Process-A Map of the Decision-Making Process](#)

defaulting, automation and, [Automate the act of repetition](#)

defaults, setting appropriate, [Set appropriate defaults](#)

defining the problem (DECIDE stage), [Defining the Problem-Worksheet: The Behavioral Project Brief](#)

action's relation to outcome, [Reminder: Action != Outcome](#)

determining who should take action, [Who Takes Action?](#)

documenting initial idea of the action, [Document Your Initial Idea of the Action-Look for the Minimum Viable Action](#)

examples from various domains, [Examples from Various Domains](#)

hypothesis for behavior change, [A Hypothesis for Behavior Change-A Hypothesis for Behavior Change](#)

nailing down the target outcome, [Nail Down the Target Outcome-A Quick Checklist](#)

(see also outcome)

putting lessons into practice, [Putting It into Practice-Worksheet: The Behavioral Project Brief](#)

starting with product's vision, [Start with the Product's Vision](#)

when product teams lack a clear problem definition, [When Product Teams Don't Have a Clear Problem Definition-When Product Teams Don't Have a Clear Problem Definition](#)

deliberative (System 2) thinking, [We're of Two Minds](#)

descriptive norms, [We Use Shortcuts, Part I: Biases and Heuristics](#)

Design Council, [The Process Is a Common One](#)

Design of Everyday Things, The (Norman), [Make It Clear Where to Act](#)

designing for behavior change, DECIDE (see DECIDE process)

Designing with the Mind in Mind (Johnson), [Make It Clear Where to Act](#)

determining impact of product or communication (DECIDE stage),
[Determining Impact with A/B Tests and Experiments-Determining Impact with A/B Tests and Experiments](#)

A/B tests and experiments for, [Determining Impact with A/B Tests and Experiments-Determining Impact with A/B Tests and Experiments](#)

(see also experiments)

cross-sectional analysis, [A Cross-Sectional or Panel Data Analysis of Impact](#)

data bridge construction, [Build the Data Bridge](#)

nonexperimental approaches, [Other Ways to Determine Impact-Unique Actions and Outcomes](#)

panel data analysis, [A Cross-Sectional or Panel Data Analysis of Impact](#)

pre-post analysis, [A Pre-Post Look at Impact-A Pre-Post Look at Impact](#)

unique actions/outcomes, [Unique Actions and Outcomes](#)

when outcome isn't measurable within the product, [What Happens If the Outcome Isn't Measurable Within the Product?-Build the Data Bridge](#)

when you can't run an A/B test, [Determining Impact When You Can't Run an A/B Test-Putting It into Practice](#)

DETOUR (Deceptive Experiences To Online Users Reduction) Act, [Ethics of Behavioral Science, Use Legal and Economic Incentives as Well](#)

disclosure rules, [Rushed Choices and Regrettable Action, The Value and Limitations of Educating Your Users](#)

distractions, removing, [Remove Distractions: Knock Out the Competition](#)

Don't Make Me Think (Krug), [Make It Clear Where to Act](#)

Double Diamond, [The Process Is a Common One](#)

double-blind experiments, [Other Considerations](#)

dual process theory(ies), [We're of Two Minds, Glossary of Terms](#)

Duhigg, Charles, [We Use Shortcuts, Part II: Habits, Reaction: Replace the routine by hijacking the reaction, Creating Habits](#)

E

EAST Framework, [A Simple Model of When, and Why, We Act](#)

economic incentives, [Use Legal and Economic Incentives as Well](#)

education insurance, [Implementing Within the Product](#)

education of users, [The Value and Limitations of Educating Your Users-The Value and Limitations of Educating Your Users](#)

Egan, Dan, [Rushed Choices and Regrettable Action, How You Combine These Skills on a Team](#)

Emanuel, Dana, [Crafting the Intervention: Advanced Topics](#)

eMBeD (Mind, Behavior, and Development Unit), World Bank, [Crafting the Intervention: Cue, Reaction, Evaluation](#)

engagement, sustaining, [How Can You Sustain Engagement with Your Product?](#)

environment (of action)

defined, [Glossary of Terms](#)

effect on ethical behavior, [The Behavioral Science of Ethics](#)

ethical review, [Run the Ethical Review](#)

ethics checklist, [Remind Ourselves with an Ethics Checklist](#)

ethics of behavioral science, [Ethics of Behavioral Science-A Short Summary of the Ideas](#)

addictive products, [Addictive Products-Addictive Products](#)

changing the environment to support ethical uses of behavioral science, [A Path Forward: Using Behavioral Science on Ourselves-Use Legal and Economic Incentives as Well](#)

environment's effect on ethical behavior, [The Behavioral Science of Ethics](#)

four types of behavior change, [Where Things Have Gone Wrong: Four Types of Behavior Change-Addictive Products](#)

poisoning the water, [Poisoning the Water](#)

profit motive and, **We'll Follow the Money Too**

user manipulation by digital companies, **Digital Tools, Especially, Seek to Manipulate Their Users-Digital Tools, Especially, Seek to Manipulate Their Users**

why designing for behavior change is especially sensitive, **Why Designing for Behavior Change Is Especially Sensitive-Why Designing for Behavior Change Is Especially Sensitive**

evaluating next steps (DECIDE stage), **Evaluating Next Steps-Putting It into Practice**

determining what changes to implement, **Determine What Changes to Implement-Integrate**

determining when product is good enough, **When Is It "Good Enough"?**

gathering information about current impact of product and potential improvements, **Gather**

integrating potential improvements to product, **Integrate**

measuring impact of each major change, **Measure the Impact of Each Major Change-Qualitative Tests of Incremental Changes**

prioritizing of potential improvements to product, **Prioritize**

qualitative tests of incremental changes, **Qualitative Tests of Incremental Changes**

evaluation (CREATE Action Funnel stage), **Evaluation-Evaluation**

conscious evaluation, **The Conscious Evaluation-Pull Future Motivations into the Present**

defined, **Glossary of Terms**

increasing attention with mindfulness, [Evaluation: Increase attention with mindfulness](#)

using conscious interference to stop negative actions, [Evaluation: Use conscious interference](#)

exercise trackers, [Automate the act of repetition](#), [Go Where the Attention Is](#), [A Cautionary Tale: My Exercise Band-A Cautionary Tale: My Exercise Band](#), [A Pre-Post Look at Impact](#)

(see also [Fitbit](#))

exercise, action metrics for, [A Metric for Action](#)

experience (CREATE Action Funnel stage), [Experience](#), [Handling Prior Experience-Check In Again: You're Not Interacting with the Same Person](#), [Glossary of Terms](#)

experiments

analyzing results of, [Analyzing the Results of Experiments-Other Considerations](#)

consistent metrics for outcomes, [Other Considerations](#)

determining how long to run, [How Long of a Wait Is "Enough"?](#)

determining Minimum Meaningful Effect, [Using Business Importance to Determine "Enough"-Using Business Importance to Determine "Enough"](#)

determining statistical significance, [Is the Effect Large "Enough"?](#) [Determining Statistical Significance](#)

double-blind, [Other Considerations](#)

experimental design in detail, [Experimental Design in Detail-Points to Remember in Designing an Experiment](#)

for determining impact, [Determining Impact with A/B Tests and Experiments](#)-[Determining Impact with A/B Tests and Experiments](#)

getting help from outside researchers, [Getting Help from Outside Researchers](#)-[Getting Help from Outside Researchers](#)

implementing, [Implementing A/B Testing and Experiments](#)

minimum sample size for, [How Many People Are “Enough”?-How Many People Are “Enough”?](#)

optimization, [Experimental Optimization](#)-[How Experimental Optimization Works](#)

randomized control trials, [The How and Why of Randomized Control Trials](#)-[Why Experiments Are \(Almost\) Better Than Sliced Bread](#)

reasons for doing, [When and Why to Test](#)

rules for designing, [Points to Remember in Designing an Experiment](#)

tools to support, [Tools for Behavioral Tracking and Experiments](#)

various types of, [Types of Experiments](#)-[Other Types of Experiments](#)

exploring the context (DECIDE stage), [Exploring the Context](#)-[Exploring the Context](#)

beached fish example, [Understanding Our Efforts: A Brief Story About a Fish](#)-[Understanding Our Efforts: A Brief Story About a Fish](#)

behavioral map, [The Behavioral Map: What Micro-Behaviors Lead to Action?](#)-[The Behavioral Map for Stopping Behaviors](#)

diagnosing problem with CREATE, [Diagnosing the Problem with CREATE](#)-[Diagnosing Why People Don’t Stop](#)

learning about your users, [What Do You Know About Your Users?](#)-[Behavioral Personas](#)

putting lessons into practice, [Putting It into Practice-Exploring the Context](#)

understanding our efforts, [Understanding Our Efforts: A Brief Story About a Fish-Exercise: Review the Map](#)

value and limitations of educating users, [The Value and Limitations of Educating Your Users-The Value and Limitations of Educating Your Users](#)

external cue

defined, [Cue, Glossary of Terms](#)

product strategies using, [Cue](#)

external cues, [Cue](#)

extrinsic motivation, [Leverage Existing Motivations Before Adding New Ones, Glossary of Terms](#)

Eyal, Nir, [The Nondedicated Teams, What Changes as a User Gains Experience with the Product?](#)

F

Facebook, [Digital Tools, Especially, Seek to Manipulate Their Users, Where Things Have Gone Wrong: Four Types of Behavior Change](#)

fake news, [Stopping Negative Actions](#)

false negatives, [How Many People Are “Enough”?](#)

false positives, [How Many People Are “Enough”?](#)

feedback loops

for stopping behaviors, [Ideas for Hindering Other Actions](#)

multi-step interventions and, [Generate a Feedback Loop](#)

financial aid for students, [Creating Action](#)

Financial Conduct Authority (UK), [Rushed Choices and Regrettable Action](#)

financial incentives, [Avoid Direct Payments](#)

financial literacy seminars, [The Value and Limitations of Educating Your Users](#)

financial services, [Rushed Choices and Regrettable Action](#)

Fischer, Deb, [Ethics of Behavioral Science](#)

fish, beached, [Understanding Our Efforts: A Brief Story About a Fish](#)

Fitbit, [Narrate the Past to Support Future Action](#), [Leverage Existing Motivations Before Adding New Ones](#), [Crafting the Intervention: Ability, Timing, Experience](#), [Generate a Feedback Loop](#)

Flash (app example), [When Product Teams Don't Have a Clear Problem Definition](#)

behavioral project brief, [Putting It into Practice](#)

defining the problem, [Putting It into Practice-Worksheet: The Behavioral Project Brief](#)

ethical checklist for, [Worksheet: Ethical Checklist](#)

examples of desired outcomes/target actions, [Examples from Various Domains](#)

refining the actor and action, [Worksheet: Refining the Actor and Action](#)

user-centric versus company-centric products, [Define the user outcomes](#)

Flo Health, [Digital Tools, Especially, Seek to Manipulate Their Users](#)

Fogg, BJ

Behavior Model, [The Stages Can Interact with One Another](#)

on habits, [Changing Existing Habits](#)

on triggers, [Creating Habits](#)

Fraser, Scott C., [But Isn't Cheating, Well, Cheating?](#)

Freedman, Jonathan L., [But Isn't Cheating, Well, Cheating?](#)

fresh starts, [Use Fresh Starts](#)

friction

decision points and, [Ability](#)

removing, [Remove Friction and Channel Factors-Set appropriate defaults](#)

removing unnecessary decision points, [Remove unnecessary decision points](#)

setting appropriate defaults, [Set appropriate defaults](#)

fudge factor, removing, [Remove the Fudge Factor](#)

fundamental attribution bias, [Remember the Fundamental Attribution Bias](#)

G

G*Power, [How Many People Are "Enough"?](#)

gambling, [Creating Habits](#)

GDPR (General Data Protection Regulation), [Why Designing for Behavior Change Is Especially Sensitive](#)

German Corporation for International Cooperation (GIZ), [Crafting the Intervention: Cue, Reaction, Evaluation](#)

Get Bad News (psychological vaccine), [Stopping Negative Actions](#)

Gilbert, Daniel, [Use Story Editing](#)

goals

company-centric (see company-centric goals)

savings, [Defining the Problem](#)

Gong, Min, [A Summary of the Process](#)

Google, [Digital Tools, Especially, Seek to Manipulate Their Users](#)

Google Analytics, [Tools for Behavioral Tracking and Experiments](#)

Google Scholar, [Getting Help from Outside Researchers](#)

Grameen Bank, [Evaluating Next Steps](#)

gut feelings, [Reaction](#)

H

habits, [We Use Shortcuts, Part II: Habits-We Use Shortcuts, Part II: Habits](#)

changing existing, [Changing Existing Habits-Ability: Increase the power of other behaviors](#)

creating, [Creating Habits-Creating Habits](#)

crowding out with new behaviors, [Ability: Increase the power of other behaviors](#)

defined, [Glossary of Terms](#)

unintentional actions and, [Quirks of Action](#)

Haidt, Jonathan, [We're of Two Minds](#)

halo effect, [We Use Shortcuts, Part I: Biases and Heuristics](#)

Hamms, Gil, [The Behavioral Science of Ethics](#)

Happiness Hypothesis, The (Haidt), [We're of Two Minds](#)

Hasanain, Wiam, [Making the Case](#)

healthcare, Oregon lottery for, [Be Authentic and Personal](#)

HelloWallet, [How This Book \(and This New Edition\) Came About](#),
[Document Your Initial Idea of the Action](#), Provide “Small Wins”,
[Measuring behaviors and outcomes outside of the product](#)

heuristics, [Behavioral Science 101: How Our Minds Are Wired](#), [We Use Shortcuts](#), Part I: Biases and Heuristics-[We Use Shortcuts](#), Part I: Biases and Heuristics

Hidalgo, Anne, [The State of the Field](#)

hindering negative actions, [Hindering Action-Ideas for Hindering Other Actions](#)

habitual actions, [Habitual Actions](#)

techniques for, [Ideas for Hindering Other Actions](#)

Homer, [Pull Future Motivations into the Present](#)

Hooked (Eyal), [The Nondedicated Teams](#)

Hopkins, Claude C., [We Use Shortcuts](#), Part II: Habits

Hreha, Jason, [Creating Habits](#)

hypothesis for behavior change, [A Hypothesis for Behavior Change-A Hypothesis for Behavior Change](#)

I

IADB (Inter-American Development Bank), [What You'll Need for Your Team](#)

ideas42, The Process Is a Common One, The Process Is a Common One, Crafting the Intervention: Advanced Topics, The Dedicated Teams, Business Model

IKEA effect, We Use Shortcuts, Part I: Biases and Heuristics

impact assessment, Skillset 2: Impact Assessment

(see also determining impact of product or communication)

implementation intentions, Relabel Something as a Cue, Elicit Implementation Intentions

implementing within the product (DECIDE stage), Implementing Within the Product-Final Review

behavioral metrics as integral to product, Build in Behavioral Metrics from Day One-Tools for Behavioral Tracking and Experiments

determining what changes to implement, Determine What Changes to Implement-Integrate

ethical review for, Run the Ethical Review

exercise band example, A Cautionary Tale: My Exercise Band-A Cautionary Tale: My Exercise Band

leaving space for creative process, Leave Space for the Creative Process-A Cautionary Tale: My Exercise Band

Safaricom example, Implementing Within the Product

inattentional blindness, Cue

incentives

conscious evaluation and, Make Sure the Incentives Are Right

using social power to change, Raise the Stakes: Use Social Power to Change Incentives

incremental changes, [Qualitative Tests of Incremental Changes](#)

Innovations for Poverty Action (IPA), [The Dedicated Teams](#)

Inspired (Cagan), [Integrate](#)

Instituto Mexicano de Economía del Compartamiento, [Who's Out There?](#)

instructions, making understandable, [Make Sure Instructions Are Understandable](#)

intention, [Assess Intention](#)

intention–action gap, [Behavioral Science 101: How Our Minds Are Wired, Quirks of Action, The CREATE Action Funnel](#)

Inter-American Development Bank (IADB), [What You'll Need for Your Team](#)

interface design, [Understanding Isn't Enough: We Need Process, Glossary of Terms](#)

internal cue, [Cue, Glossary of Terms](#)

International Review Board (IRB), [Create a Review Body](#)

interventions, crafting (see crafting the intervention)

intrinsic motivation, [Leverage Existing Motivations Before Adding New Ones, Glossary of Terms](#)

Intuit, [Digital Tools, Especially, Seek to Manipulate Their Users](#)

intuitive (System 1) thinking, [We're of Two Minds, Reaction](#)

intuitive reaction, [Narrate the Past to Support Future Action-Make the Site Professional and Beautiful](#)

(see also reaction)

associating with positive/familiar experiences, **Associate with the Positive and the Familiar**

authenticity of appeal, **Be Authentic and Personal-Be Authentic and Personal**

bringing success to top of mind, **Bring Success Top of Mind**

display strong authority on subject, **Display Strong Authority on the Subject**

gut feelings, **Reaction**

narrating past to support future action, **Narrate the Past to Support Future Action**

peer comparisons, **Use Peer Comparisons**

personal appeals, **Be Authentic and Personal-Be Authentic and Personal**

professionalism/beauty of site, **Make the Site Professional and Beautiful**

social proof and, **Deploy Social Proof**

Ioannidis, John, **The Replication Crisis in Science**

iodine deficiency, **Make it incidental**

IPA (Innovations for Poverty Action), **The Dedicated Teams**

Irrational Labs, **Creating Action**

Iyengar, Sheena S., **Avoid Choice Overload**

J

J-PAL (Abdul Latif Jameel Poverty Action Lab), **The Dedicated Teams**

Jive Voice, **Associate with the Positive and the Familiar**

Johnson, Jeff, [Make It Clear Where to Act](#)

K

Kahneman, Daniel, [We're Deeply Affected by Context-We're Deeply Affected by Context](#)

Karlan, Dean, [Evaluating Next Steps](#)

Kosovo, [Crafting the Intervention: Cue, Reaction, Evaluation](#)

Krug, Steve, [Make It Clear Where to Act](#)

L

language instruction, action metrics for, [A Metric for Action](#)

Last Mile, The (Soman), [The Process Is a Common One](#)

learned helplessness, [Narrate the Past to Support Future Action](#)

legal incentives, [Use Legal and Economic Incentives as Well](#)

Lepper, Mark R., [Avoid Choice Overload](#)

Lieb, David, [Avoid Cognitive Overhead](#)

limitations of human cognition/memory, [We're Limited-We're Limited](#)

LinkedIn, [Digital Tools, Especially, Seek to Manipulate Their Users](#)

loss aversion, [Leverage Loss Aversion, Handling Prior Experience](#)

M

marketing swag, [Go Where the Attention Is](#)

Matamo, [Tools for Behavioral Tracking and Experiments](#)

memory, limitations of, [We're Limited](#)

mental environment, [We're Deeply Affected by Context](#)

metrics

action, [A Metric for Action](#)

behavioral metrics as integral to product, [Build in Behavioral Metrics from Day One-Tools for Behavioral Tracking and Experiments](#)

for target outcome, [Define the Metric to Measure Outcomes](#)

ideal characteristics, [Define the Metric to Measure Outcomes](#)

micro-behaviors, [The Behavioral Map: What Micro-Behaviors Lead to Action?-The Behavioral Map for Stopping Behaviors](#)

microcredit, [Evaluating Next Steps](#)

Milkman, Katherine L., [Rushed Choices and Regrettable Action, Use Fresh Starts](#)

Mind, Behavior, and Development Unit (eMBeD), World Bank, [Crafting the Intervention: Cue, Reaction, Evaluation](#)

mindfulness, [Evaluation: Increase attention with mindfulness](#)

mindset, [Glossary of Terms](#)

minimum meaningful effect (MME), [Using Business Importance to Determine "Enough"-Using Business Importance to Determine "Enough"](#)

minimum sample size, [How Many People Are "Enough"?-How Many People Are "Enough"?](#)

Minimum Viable Action (MVA), [Look for the Minimum Viable Action, Glossary of Terms](#)

mission statements, [Start with the Product's Vision](#)

money, as motivator, [Avoid Direct Payments](#)

More Than Good Intentions (Karlan and Appel), [Evaluating Next Steps](#)

Morningstar, [How This Book \(and This New Edition\) Came About](#),
[Defining the Problem](#)

mortgage disclosures, [The Value and Limitations of Educating Your Users](#)

motivation

leveraging existing motivations before adding new ones, [Leverage Existing Motivations Before Adding New Ones](#)

looking beyond, [Look Beyond Motivation](#)

pull future motivations into the present, [Pull Future Motivations into the Present](#)-Pull Future Motivations into the Present

testing out different types of motivators, [Test Different Types of Motivators](#)

multi-step interventions, [Multi-Step Interventions](#)-Hard work builds commitment

cheating strategies, [Again, Cheat If You Can](#)

combining multiple steps into one, [Combine Where Possible](#)

common mistakes, [Common Mistakes](#)

effort and commitment, [Hard work builds commitment](#)

feedback loops, [Generate a Feedback Loop](#)

providing small wins with each step, [Provide “Small Wins”](#)

multiarm comparisons, [Other Types of Experiments](#)

multiarmed bandit, [Other Types of Experiments](#)

multivariate testing, [Other Types of Experiments](#), [Glossary of Terms](#)

Musk, Elon, [Determining Impact with A/B Tests and Experiments](#)

N

Naru, Faisal, [What We Did: A Global Survey of Behavioral Teams](#)

negative actions, stopping, [Stopping Negative Actions-A Short Summary of the Ideas](#)

avoiding the cue, [Attention: Avoid the Cue](#)

behavioral map for, [The Behavioral Map for Stopping Behaviors](#)

changing existing habits, [Changing Existing Habits-Ability: Increase the power of other behaviors](#)

creating obstacles, [Hindering Action-Ideas for Hindering Other Actions](#)

diagnosing why people don't stop, [Diagnosing Why People Don't Stop](#)

replace the routine by hijacking the reaction, [Reaction: Replace the routine by hijacking the reaction](#)

rushed choices and regrettable action, [Rushed Choices and Regrettable Action-Rushed Choices and Regrettable Action](#)

using CREATE to add obstacles to, [Using CREATE to Add Obstacles to Action](#)

New Moms (nonprofit), [Crafting the Intervention: Advanced Topics](#)

noise, [How Many People Are “Enough”?](#)

nonconscious habits, [Behavioral Science 101: How Our Minds Are Wired](#)

nonconscious judgments, [Behavioral Science 101: How Our Minds Are Wired](#)

nonconscious reactions, [Reaction-Reaction](#)

Norman, Donald A., [Make It Clear Where to Act](#)

Norwegian Consumer Council, Digital Tools, Especially, Seek to Manipulate Their Users

Nudge (Thaler and Sundstein), What Types of Behaviors This Can Help With, Ideas for Hindering Other Actions

Nudge Lebanon, Stopping Negative Actions

Nudge Rio, Who's Out There?

O

Obama, Barack, The Behavioral Map: What Micro-Behaviors Lead to Action?

objectives, of company, State the company's objectives

obstacles, Look for "Real" Obstacles

obvious actions, as likely failures, The Obvious Is Our Enemy

Odyssey (Homer), Pull Future Motivations into the Present

OECD (Organisation for Economic Co-operation and Development), What We Did: A Global Survey of Behavioral Teams

one-sided test, How Many People Are "Enough"?

one-time actions, Strategies to Cheat at One-Time Actions-Make it incidental

Opower, Measuring behaviors and outcomes outside of the product, Determining Impact with A/B Tests and Experiments

optimization, of experiments, Experimental Optimization-How Experimental Optimization Works

Oracle Utilities, Measuring behaviors and outcomes outside of the product, Determining Impact with A/B Tests and Experiments

Oregon healthcare lottery, [Be Authentic and Personal](#)

organ donations, [Remove Friction and Channel Factors](#)

Organisation for Economic Co-operation and Development (OECD), [What We Did: A Global Survey of Behavioral Teams](#)

outcome (target outcome)

checklist for, [A Quick Checklist](#)

clarifying, [Clarify the Outcome-What if no one can agree on the product's intended outcome?](#)

defined, [Glossary of Terms](#)

defining the metric to measure, [Define the Metric to Measure Outcomes](#)

determining, [Nail Down the Target Outcome-A Quick Checklist](#)

disagreements on product's intended outcome, [What if no one can agree on the product's intended outcome?](#)

examples from various domains, [Examples from Various Domains](#)

focusing on outcomes instead of actions, [Avoid stating how the product \(might\) do it](#)

importance of clear definition, [Why go through the trouble?](#)

narrowing scope of, [That and nothing else](#)

prioritizing/combining multiple outcomes, [Prioritize and combine](#)

states of mind versus observable outcomes, [Avoid states of mind](#)

working with company-centric goals, [Working with Company-Centric Goals-Define the user outcomes](#)

panel data analysis, [A Cross-Sectional or Panel Data Analysis of Impact](#)

paradox of choice, [We're Limited](#)

Paris Nudge Building, [The State of the Field](#)

Paulin, Ingrid Melvær, [What We Did: A Global Survey of Behavioral Teams](#)

Payne, John W., [Rushed Choices and Regrettable Action](#)

peer comparisons, [Use Peer Comparisons](#), [Peer Comparisons Can Help Here Too](#)

personal appeals, [Be Authentic and Personal-Be Authentic and Personal](#)

personal finance applications, [Automate the act of repetition](#)

personas, [Glossary of Terms](#)

(see also behavioral personas)

physical environment, [We're Deeply Affected by Context](#)

power calculation, [How Many People Are "Enough"?](#)

Power of Habit, The (Duhigg), [We Use Shortcuts, Part II: Habits, Reaction: Replace the routine by hijacking the reaction](#), [Creating Habits](#)

pre-post analysis, [A Pre-Post Look at Impact-A Pre-Post Look at Impact](#)

present bias, [We Use Shortcuts, Part I: Biases and Heuristics](#)

prior experience, [Experience, Handling Prior Experience-Check In Again: You're Not Interacting with the Same Person](#)

(see also experience)

checking in with same person at later time, [Check In Again: You're Not Interacting with the Same Person](#)

handling, **Handling Prior Experience-Check In Again: You're Not Interacting with the Same Person**

making intentionally unfamiliar, **Make It Intentionally Unfamiliar**

story editing, **Use Story Editing-Use Story Editing**

use fresh starts, **Use Fresh Starts**

using techniques to support better decisions, **Use Techniques to Support Better Decisions**

prioritizing

improvements, **Prioritize**

multiple outcomes, **Prioritize and combine**

Prius effect, **Evaluation: Use conscious interference**

privacy rights, **Digital Tools, Especially, Seek to Manipulate Their Users**

professionalism, of site, **Make the Site Professional and Beautiful**

profits, increased, **We'll Follow the Money Too**

ProPublica, **Digital Tools, Especially, Seek to Manipulate Their Users**

Python, **How Many People Are "Enough"?, Is the Effect Large "Enough"? Determining Statistical Significance**

Q

qualitative tests, **Qualitative Tests of Incremental Changes**

quirks of action, **Quirks of Action**

quirks of decision-making, **Quirks of Decision Making-Quirks of Decision Making**

R

R (package)

power calculation function, [How Many People Are “Enough”?](#)

statistical significance testing, [Is the Effect Large “Enough”?](#)

[Determining Statistical Significance](#)

Radicalise game, [Stopping Negative Actions](#)

random assignment, [Points to Remember in Designing an Experiment](#)

random rewards, [We Use Shortcuts, Part II: Habits, Creating Habits](#)

random selection, [Points to Remember in Designing an Experiment](#)

randomization, in experiment design, [Why Experiments Are \(Almost\) Better Than Sliced Bread](#)

randomized control trials (RCTs), [The How and Why of Randomized Control Trials-Why Experiments Are \(Almost\) Better Than Sliced Bread](#)

reaction (CREATE Action Funnel stage), [Reaction-Reaction](#)

defined, [Glossary of Terms](#)

hijacking to stop negative actions, [Reaction: Replace the routine by hijacking the reaction](#)

intuitive, [Narrate the Past to Support Future Action-Make the Site Professional and Beautiful](#)

(see also intuitive reaction)

reactive (System 1) thinking, [We’re of Two Minds, Reaction](#)

recency bias, [Quirks of Decision Making](#)

Reinsurance Group of America, [Who’s Out There?](#)

reminders, as cues, [Use Reminders](#)

repeated actions, [Strategies to Cheat at Repeated Actions-Automate the act of repetition](#)

replication crisis, [The Replication Crisis in Science-The Replication Crisis in Science](#)

retirement savings

401(k) autoenrollment/autoescalation, [Automate it](#)

automating employee contributions, [Look Beyond Motivation](#)

financial literacy seminars, [The Value and Limitations of Educating Your Users](#)

revenue, [Thinking Through the Business Model-Thinking Through the Business Model](#)

review body/review board, [Create a Review Body](#)

reward schedule, [We Use Shortcuts, Part II: Habits](#)

reward substitution, [Pull Future Motivations into the Present](#)

rewards, [We Use Shortcuts, Part II: Habits, Creating Habits](#)

defined, [We Use Shortcuts, Part II: Habits, Glossary of Terms](#)

scarce/time-sensitive, [Make a Reward Scarce](#)

Riis, Jason, [Use Fresh Starts](#)

Roozenbeek, Jon, [Stopping Negative Actions](#)

routines, [We Use Shortcuts, Part II: Habits, Creating Habits-Creating Habits](#)

defined, [We Use Shortcuts, Part II: Habits, Creating Habits, Glossary of Terms](#)

replacing to stop negative actions, [Reaction: Replace the routine by hijacking the reaction](#)

Runkeeper, [Automate the act of repetition](#)

S

Safaricom, [Implementing Within the Product](#)

Sam's Club Applied Behavioral Science Team, [A Summary of the Process](#)

sample size, minimum, [How Many People Are “Enough”?-How Many People Are “Enough”?](#)

Saudi Arabia, [Making the Case](#)

Save More Tomorrow, [Frame Text to Avoid Temporal Myopia](#)

savings goals and, [Defining the Problem](#)

savings lotteries, [Make it incidental](#)

Scartascini, Carlos, [What You’ll Need for Your Team](#)

self-narrative

defined, [Glossary of Terms](#)

narrating past to support future action, [Narrate the Past to Support Future Action](#)

story editing and, [Use Story Editing-Use Story Editing](#)

Simons, Daniel, [Cue](#)

simultaneous comparison experiments, [Other Types of Experiments](#)

(see also A/B tests)

simultaneous impact experiments, [Other Types of Experiments](#)

situational self-control, [Stopping Negative Actions](#)

small wins

defined, [Glossary of Terms](#)

multi-step interventions and, [Provide “Small Wins”](#)

smoking cessation, [The Behavioral Map for Stopping Behaviors](#)

social accountability, [Make Commitments to Friends](#)

social environment, [We’re Deeply Affected by Context](#)

social motivation, competition as, [Use Competition](#)

social power, [Raise the Stakes: Use Social Power to Change Incentives](#)

social proof, [Deploy Social Proof](#)

Soll, Jack B., [Rushed Choices and Regrettable Action](#)

Soman, Dilip, [The Process Is a Common One](#)

Spanish-language resources for applied behavioral science, [Who’s Out There?](#)

spare time, user's, [Align with When People Have Spare Time](#)

staggered rollout, [Other Types of Experiments](#)

Start at the End (Wallaert), [The Process Is a Common One](#)

states of mind, observable outcomes versus, [Avoid states of mind](#)

statistical power, [How Many People Are “Enough”?](#)

statistical significance, [Is the Effect Large “Enough”? Determining Statistical Significance](#)

StatsModel package, [How Many People Are “Enough”?](#)

status quo bias, [We Use Shortcuts, Part I: Biases and Heuristics](#)

story editing, [Use Story Editing-Use Story Editing](#)

structure of attention, [Align with When People Have Spare Time](#)

student loans, [Creating Action](#)

sunk cost effect, [Hard work builds commitment](#)

Sunlight Foundation, [Start with the Product's Vision](#)

Sunstein, Cass, [What Types of Behaviors This Can Help With](#)

swag, [Go Where the Attention Is](#)

System 1 Group, [Digital Tools, Especially, Seek to Manipulate Their Users](#)

System 1 thinking, [We're of Two Minds, Reaction](#)

System 2 thinking, [We're of Two Minds](#)

T

target action (see action)

target actor (see actor)

target outcome (see outcome)

tax reporting, [Crafting the Intervention: Cue, Reaction, Evaluation](#)

teams, [What You'll Need for Your Team-Putting It into Practice](#)
(see also Behavioral Teams Study)

advanced degrees and, [What's Not Listed: A PhD](#)

combining skillsets on a team, [How You Combine These Skills on a Team](#)

data scientists and behavioral scientists on, [Data Science and Behavioral Science](#)

impact assessment, [Skillset 2: Impact Assessment](#)

leveraging data science when designing for behavior change,
[Leveraging Data Science When Designing for Behavior Change](#)

making business case for, [Making the Case-Thinking Through the Business Model](#)

non-behavioral basics, [Skillset 1: The Non-Behavioral Basics](#)

partnerships with outside researchers, [Getting Help from Outside Researchers-Getting Help from Outside Researchers](#)

requirements for, [What You'll Need for Your Team-Putting It into Practice](#)

skills and people for, [The Skills and People You Need-Leveraging Data Science When Designing for Behavior Change](#)

thinking through the business model, [Thinking Through the Business Model-Thinking Through the Business Model](#)

understanding of mind and its quirks, [Skillset 3: A Deep Understanding of the Mind and Its Quirks](#)

teaser rates, [Where Things Have Gone Wrong: Four Types of Behavior Change](#)

temporal myopia, [Frame Text to Avoid Temporal Myopia](#)

testimonials, [Deploy Social Proof](#)

text

blinking, [Bonus Tactic: Blinking Text](#)

framing to avoid temporal myopia, [Frame Text to Avoid Temporal Myopia](#)

Thaler, Richard H., [What Types of Behaviors This Can Help With, Frame Text to Avoid Temporal Myopia, Ideas for Hindering Other Actions](#)

Thorp, Justin, [Make Commitments to Friends](#)

time-sensitive rewards, [Make a Reward Scarce](#)

timing (CREATE Action Funnel stage), [Timing-Timing, Getting the Timing Right-Make a Reward Scarce](#)

defined, [Glossary of Terms](#)

framing text to avoid temporal myopia, [Frame Text to Avoid Temporal Myopia](#)

making a reward scarce, [Make a Reward Scarce](#)

making commitments to friends, [Make Commitments to Friends](#)

reminding user of prior commitment to act, [Remind of a Prior Commitment to Act](#)

Tulchinskaya, Anna, [Relabel Something as a Cue](#)

Tversky, Amos, [We're Deeply Affected by Context-We're Deeply Affected by Context](#)

two-sided test, [How Many People Are "Enough"?](#)

U

Ulysses contract, [Pull Future Motivations into the Present](#)

United Kingdom (see Behavioural Insights Team)

urgency, decision making and, [Timing-Timing](#)

user data, [Why Designing for Behavior Change Is Especially Sensitive](#)

user outcomes, defining, [Define the user outcomes](#)

user personas, [Behavioral Personas-Behavioral Personas](#)
(see also behavioral personas)

user story, [Glossary of Terms](#)

user-centric approach

defining behavioral problems for, [Working with Company-Centric Goals](#)

examples of desired outcomes/target actions, [Examples from Various Domains](#)

users

behavior in daily life, [How Do They Behave in Daily Life?-How Do They Behave in Daily Life?](#)

behavior in existing application, [How Do They Behave in the Application?](#)

behavioral personas, [Behavioral Personas-Behavioral Personas, Updating the Behavioral Personas, Glossary of Terms](#)

learning about, [What Do You Know About Your Users?-Behavioral Personas](#)

progression from inaction to action over time, [What Happens Before People Take Action the First Time?-What Happens Before People Take Action the First Time?](#)

value and limitations of educating, [The Value and Limitations of Educating Your Users-The Value and Limitations of Educating Your Users](#)

V

van Caster, Sarah, [Data Science and Behavioral Science](#)

van der Linden, Sander, [Stopping Negative Actions](#)

vanity metrics, [That and nothing else](#)

vision (product vision)

and company-centric goals, [State the vision](#)

clarifying, [Start with the Product's Vision](#)
defined, [Glossary of Terms](#)

W

Wallaert, Matt, [Addictive Products, The Process Is a Common One, Making the Case](#)

Walmart Applied Behavioral Science Team, [A Summary of the Process](#)

Warner, Mark, [Ethics of Behavioral Science](#)

wearables, [Go Where the Attention Is, Generate a Feedback Loop](#)
(see also exercise trackers; Fitbit)

"Why Most Published Research Findings are False" (Ioannidis), [The Replication Crisis in Science](#)

Wilson, Tim, [Use Story Editing](#)

World Bank Mind, Behavior, and Development Unit (eMBeD), [Crafting the Intervention: Cue, Reaction, Evaluation](#)

Y

Young, Scott, [The State of the Field](#)

YouVersion, [Creating Habits](#)

Yunus, Mohammad, [Evaluating Next Steps](#)

Z

Zinman, Jonathan, [Evaluating Next Steps](#)

Zumdahl, Laura, [Crafting the Intervention: Advanced Topics](#)

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Stephen has authored three books on applied behavioral science: *Designing for Behavior Change* (November 2013), *Improving Employee Benefits* (September 2014), and *Spiritual Design* (October 2019). Outside of work, he is also the Founder and Chair of the nonprofit Action Design Network, which helps more than 15,000 practitioners apply behavioral research to product development with monthly events in 15 cities across North America.

Stephen holds a BA from U.C. Berkeley, a master's from Johns Hopkins-SAIS, and a PhD from the University of Maryland, where he analyzed the dynamics of behavioral change over time. He has a wife and two wonderful kids, who don't care about behavioral science at all.

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Colophon

The cover of *Designing for Behavior Change* hosts a coral reef of mixed species. Corals are marine invertebrates of the class *Anthozoa*, though researchers in the past thought they were minerals or plants. In fact, a coral reef is made up of thousands of tiny sessile creatures called polyps. They thrive in tropical and subtropical waters. The most famous is the Great Barrier Reef off the coast of Australia.

These ecosystems are at least as large, lasting, and biodiverse as Earth's oldest forests. Their characteristic, stunning colors come from algae that inhabit polyp tissues. Corals suffer from rising water temperatures and acidity, endangering their own lives and those they support, including most ocean life and all who rely on oceans. Many of the animals on O'Reilly's covers are endangered; all of them are important to the world.

The cover illustration is by Karen Montgomery, based on a black and white engraving from *Meyers Kleines Lexicon*. The cover fonts are Gilroy Semibold and Guardian Sans. The text font is Adobe Minion Pro; the heading font is Adobe Myriad Condensed; and the code font is Dalton Maag's Ubuntu Mono.